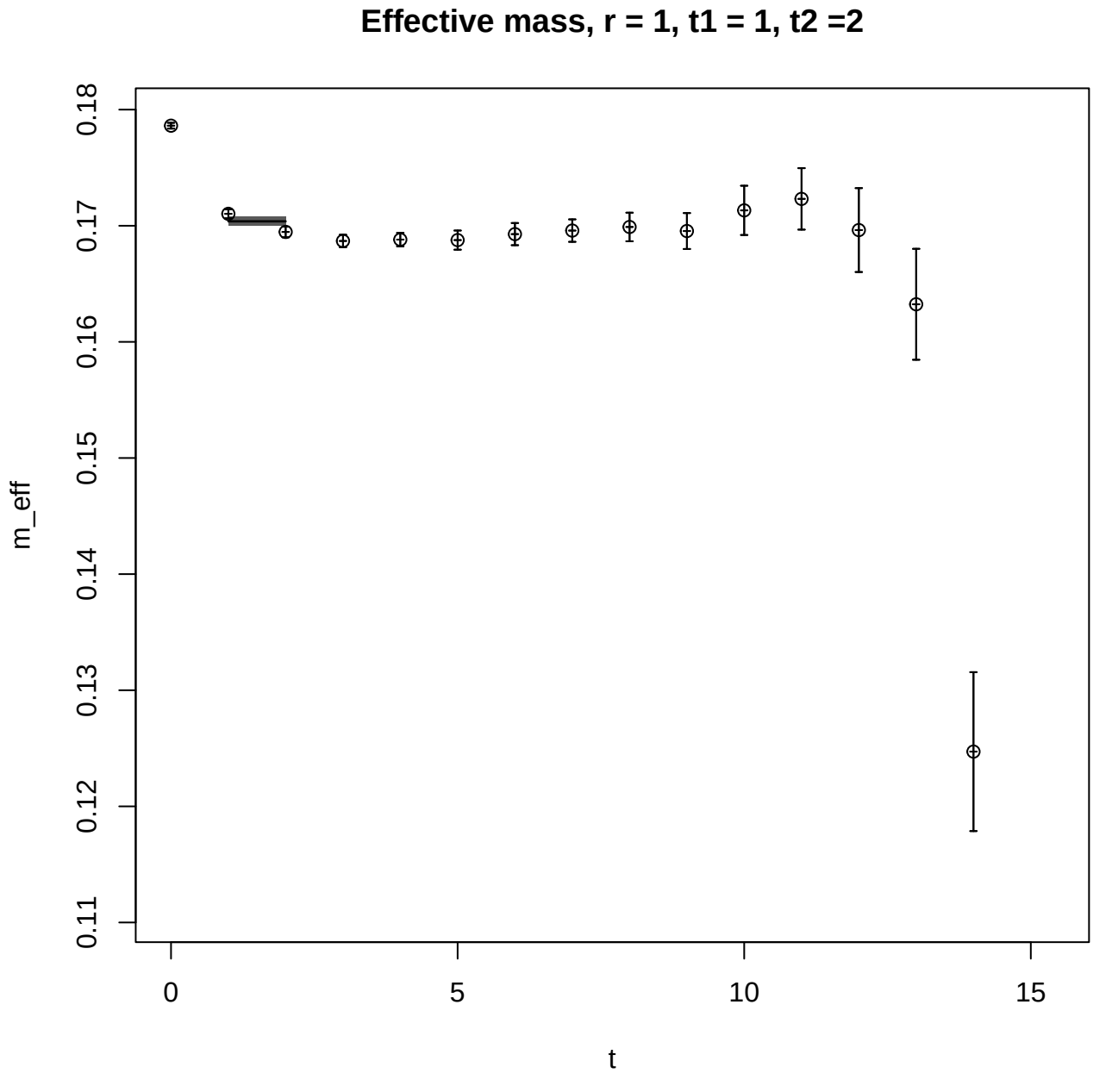
Effective mass, r = 1, t1 = 1, t2 =2
m_eff
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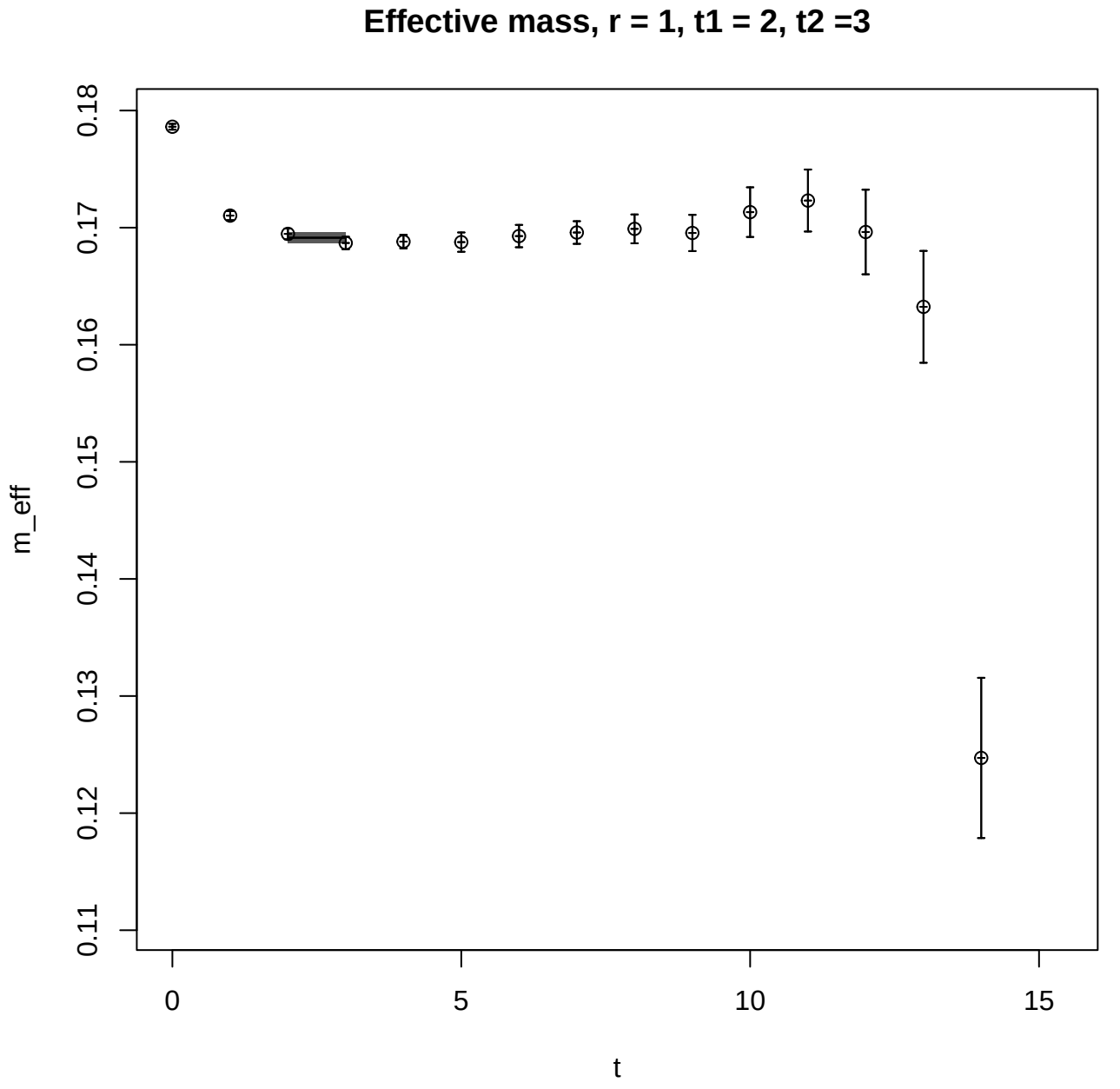
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =3
m_eff
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0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
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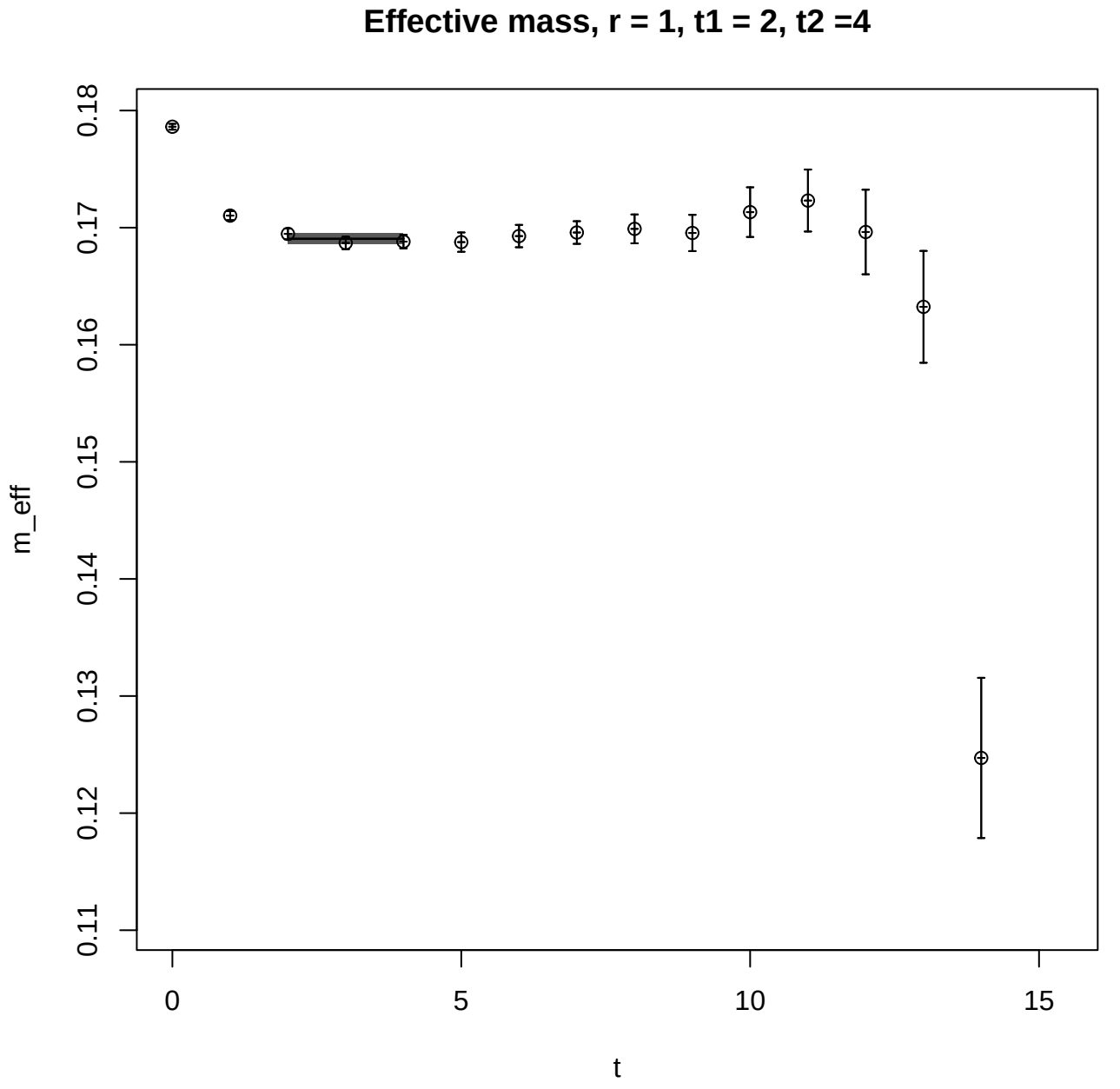
15
t


Effective mass, r = 1, t1 = 2, t2 =4
m_eff
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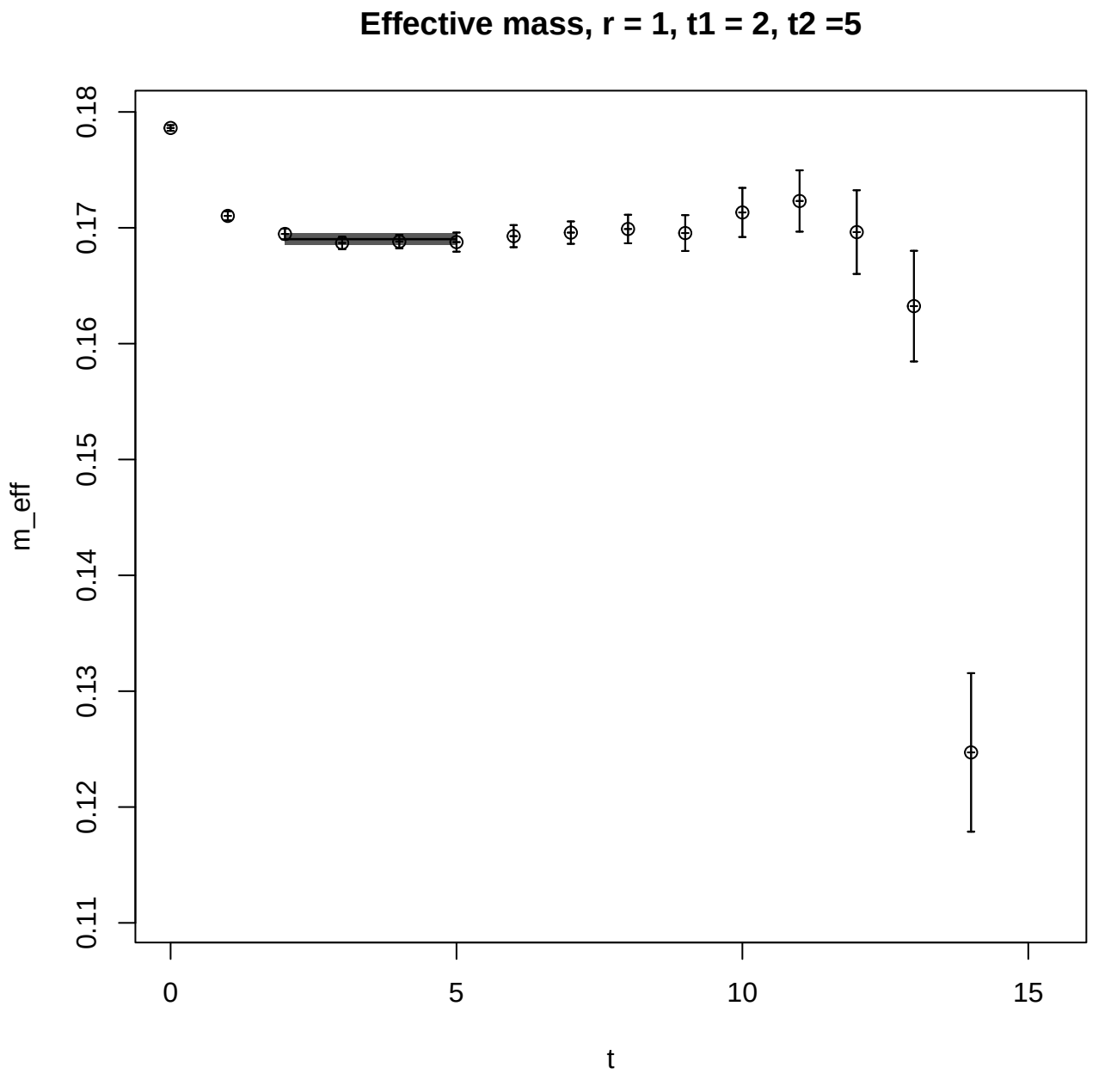
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =5
m_eff
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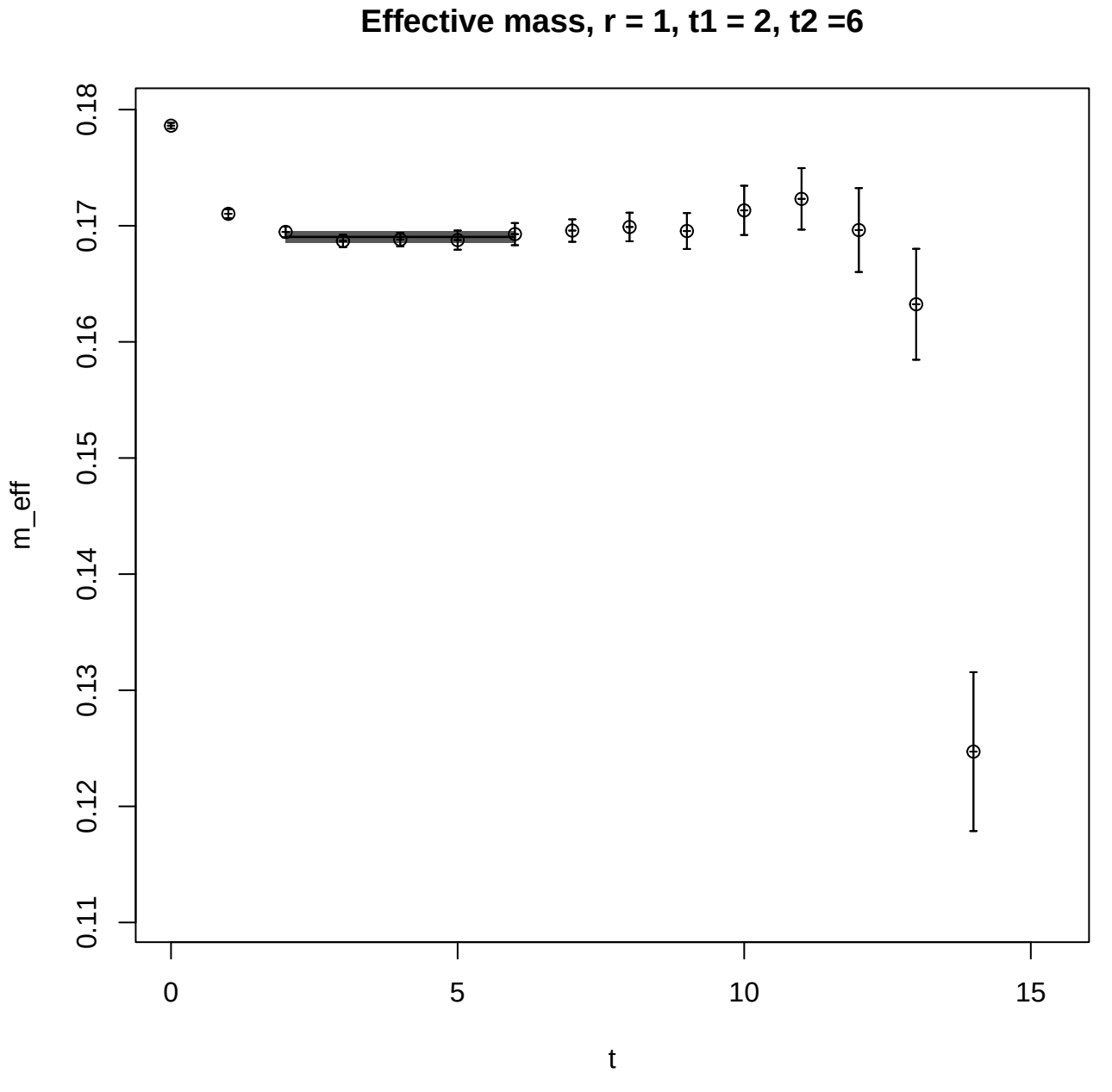
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =6
m_eff
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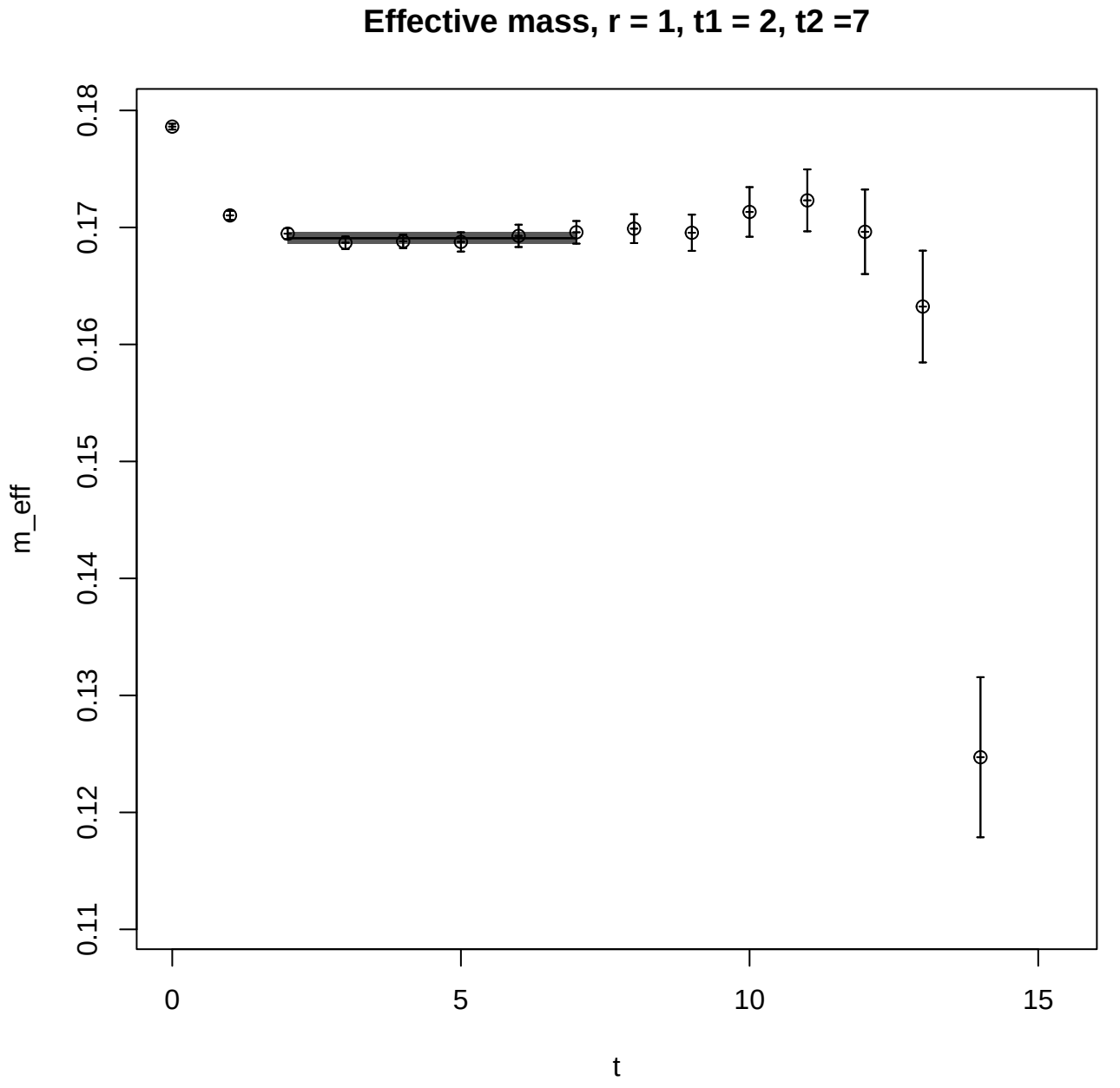
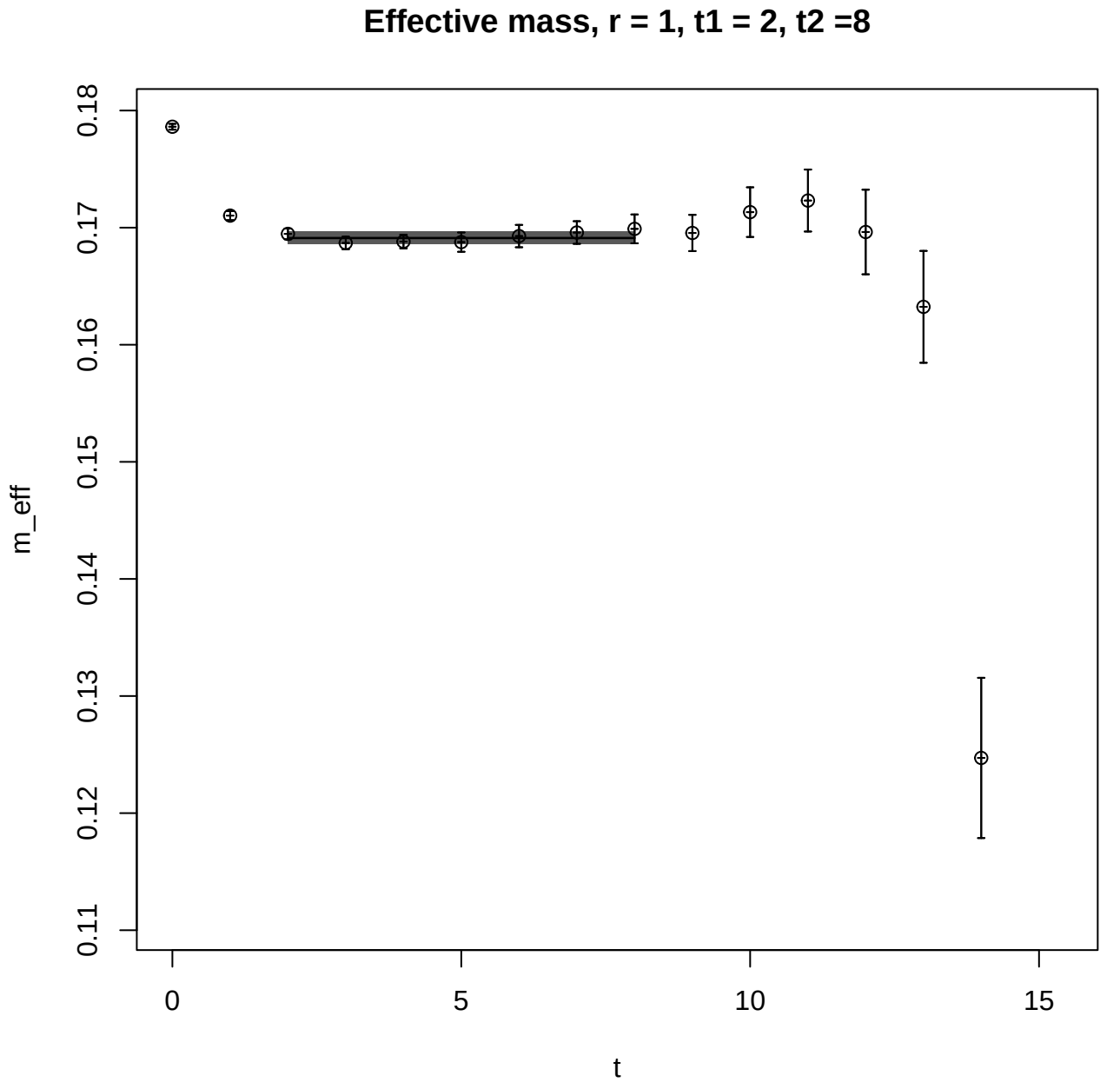
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =7
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =8
m_eff
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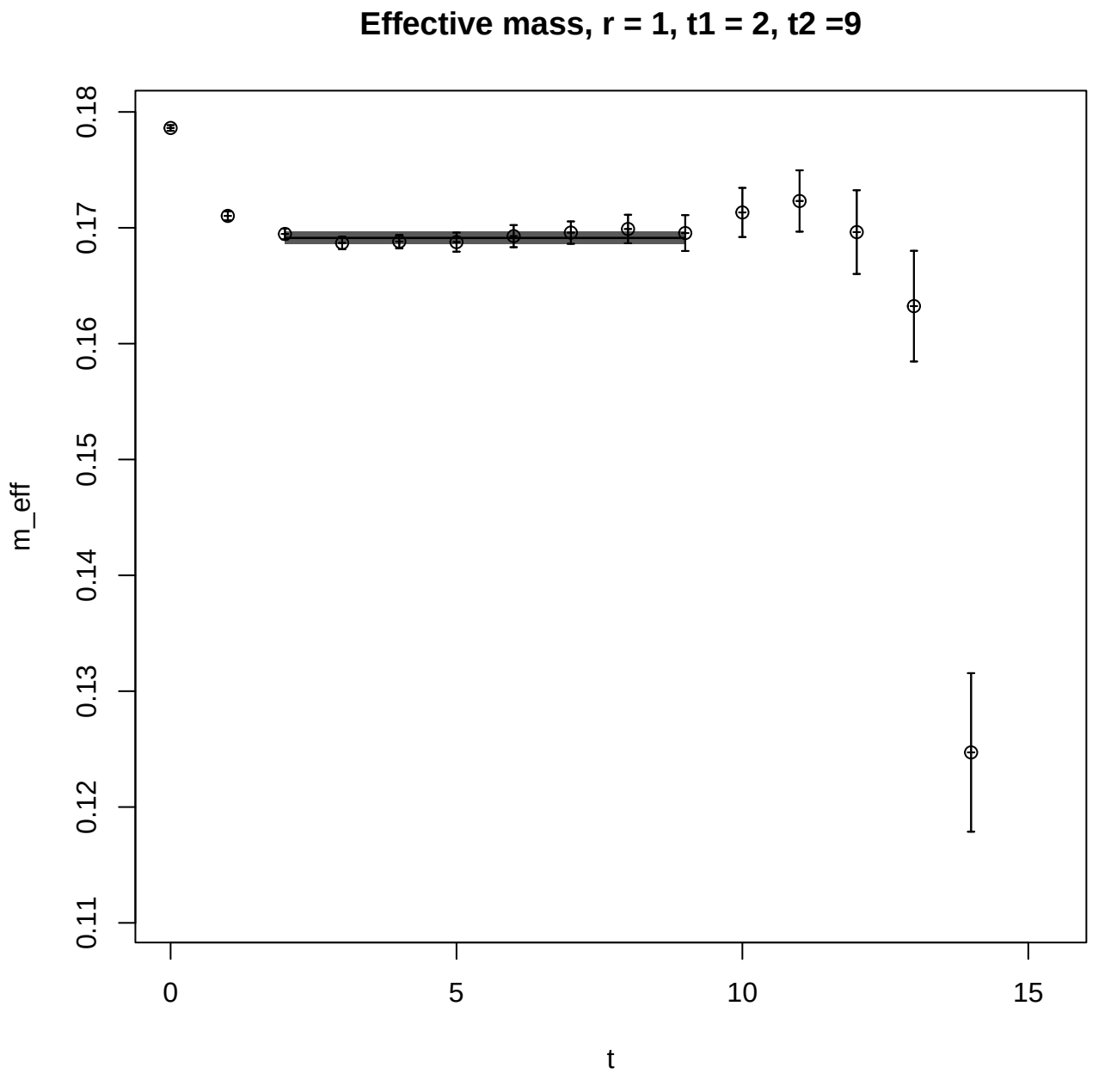
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =9
m_eff
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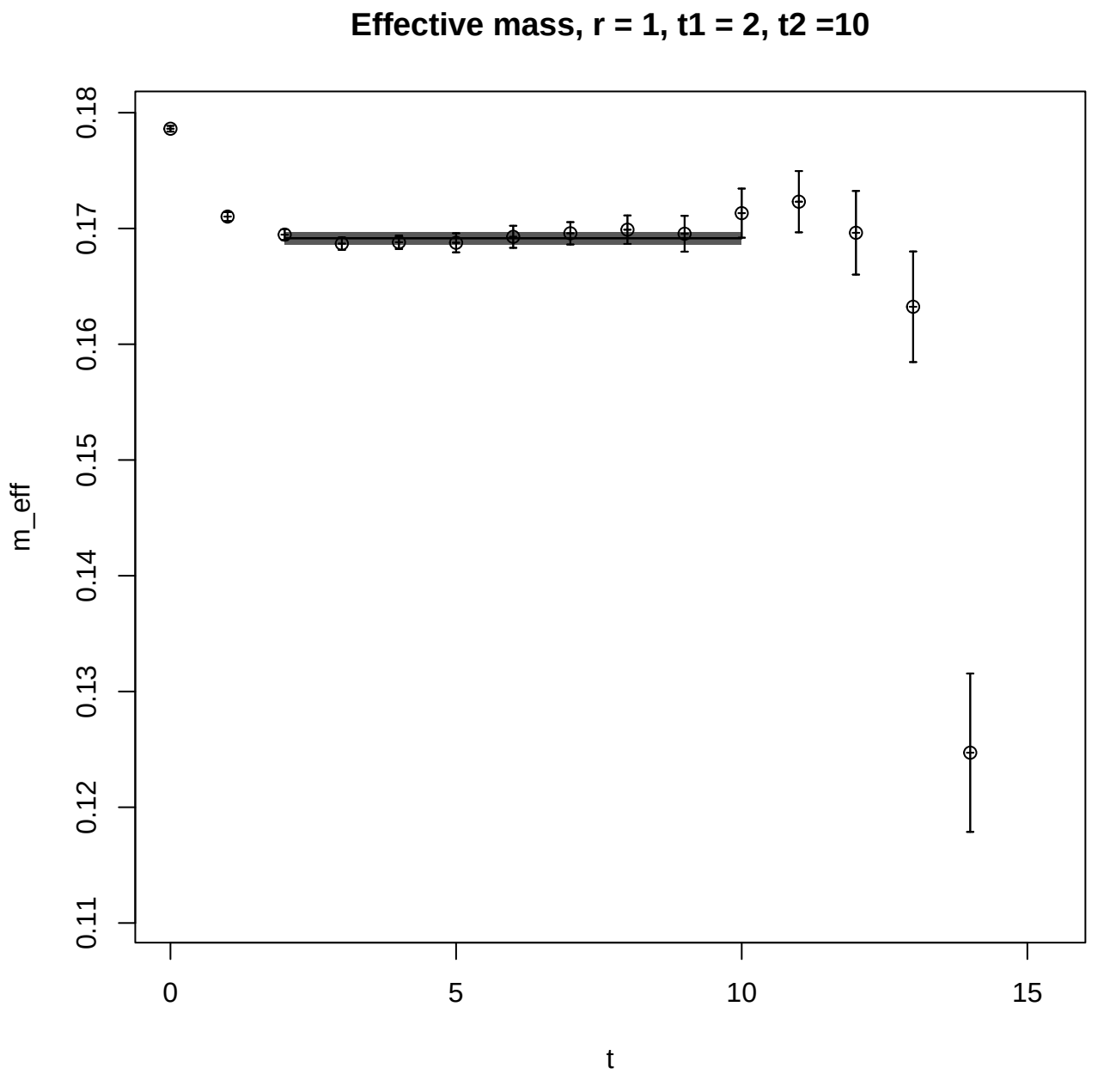
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =10
m_eff
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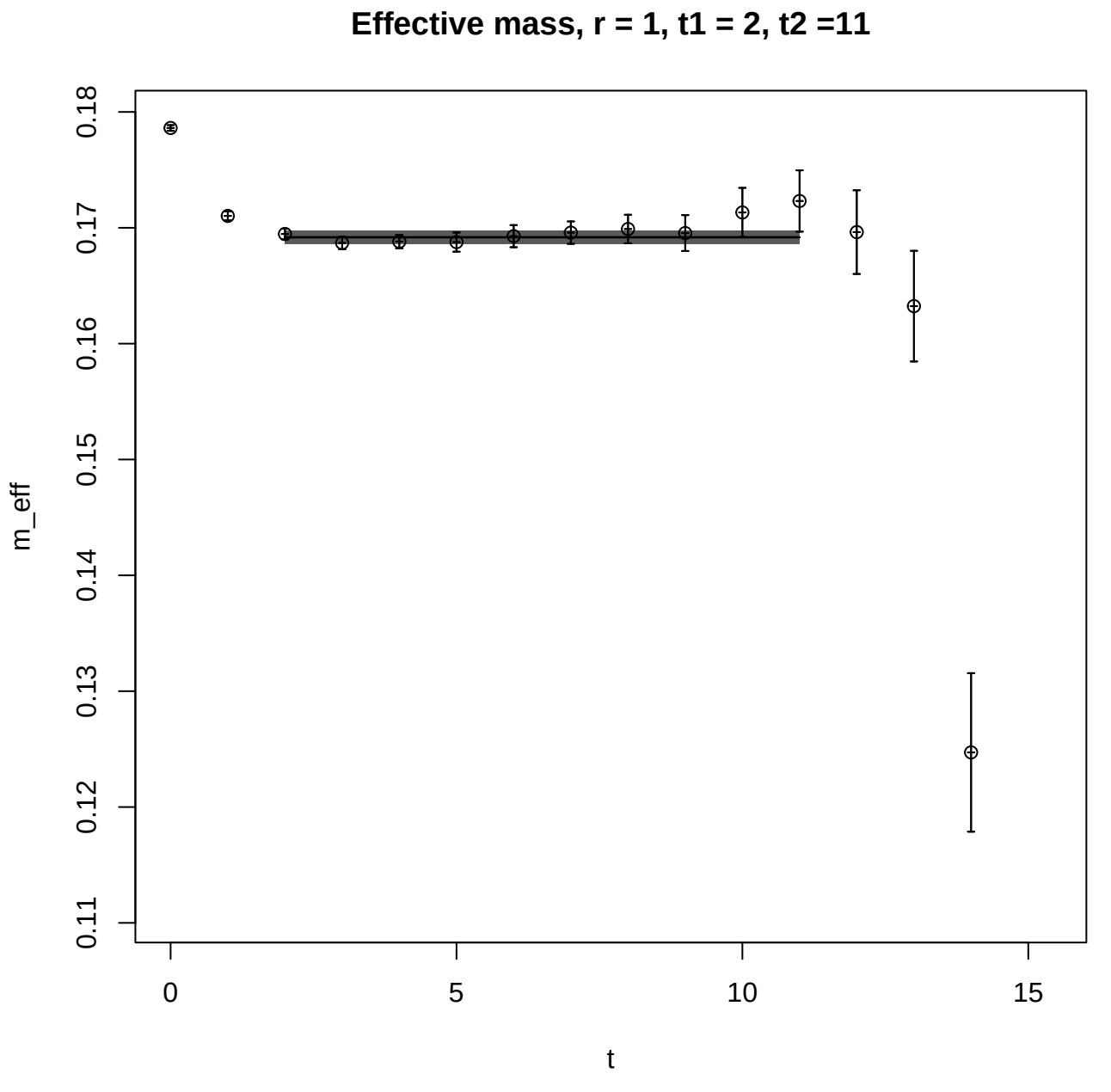
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =11
m_eff
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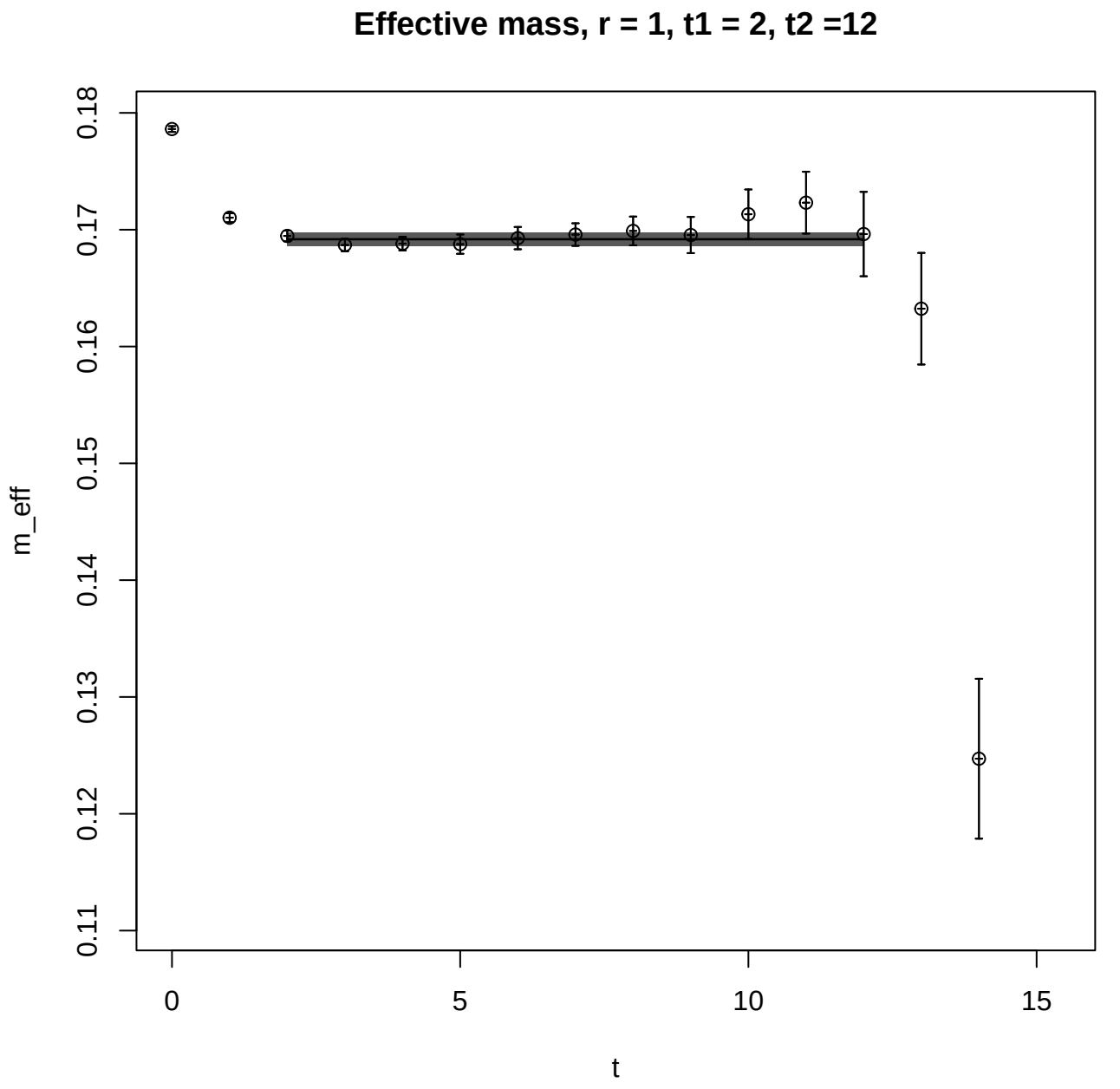
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =12
m_eff
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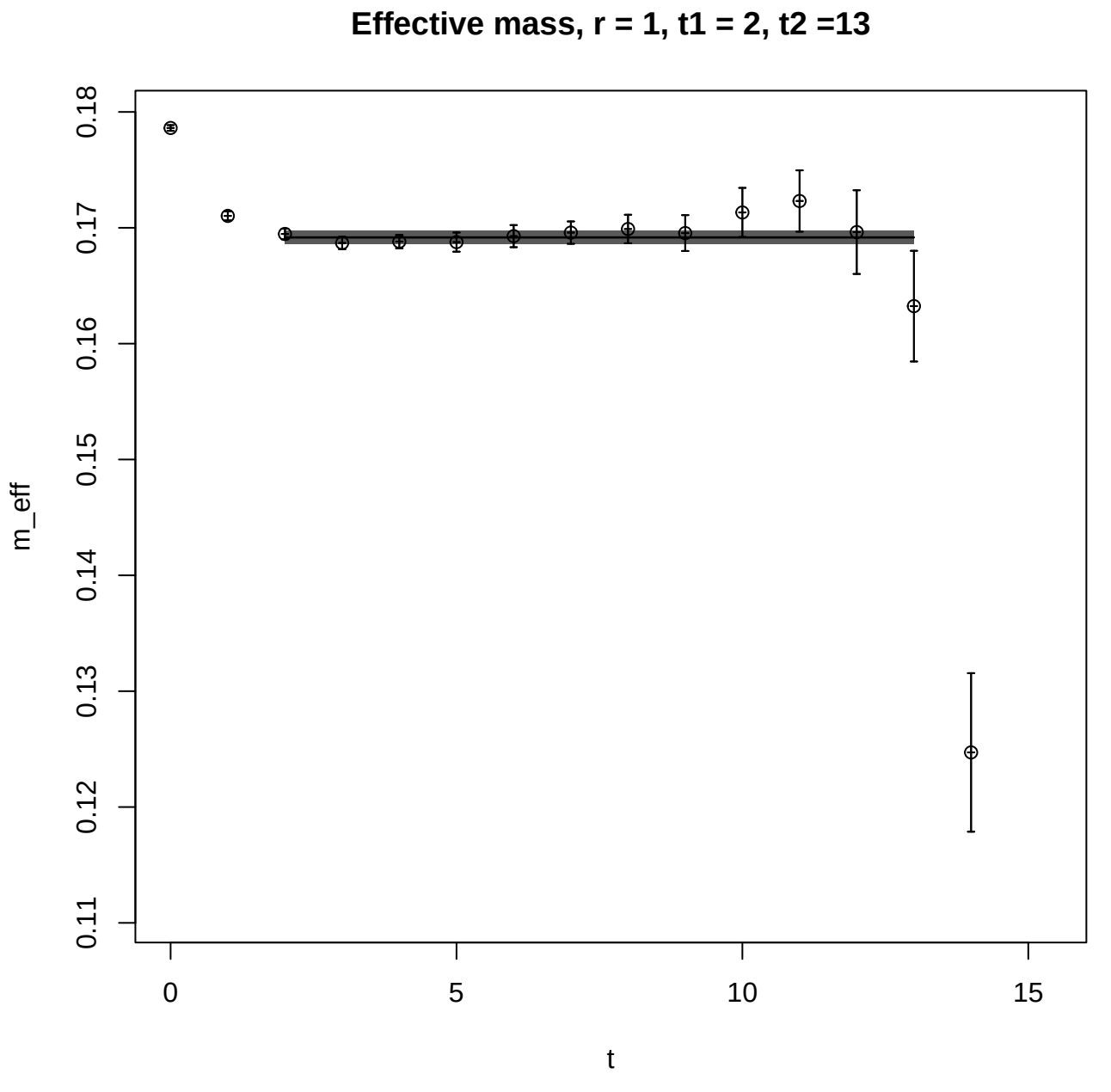
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 2, t2 =13
m_eff
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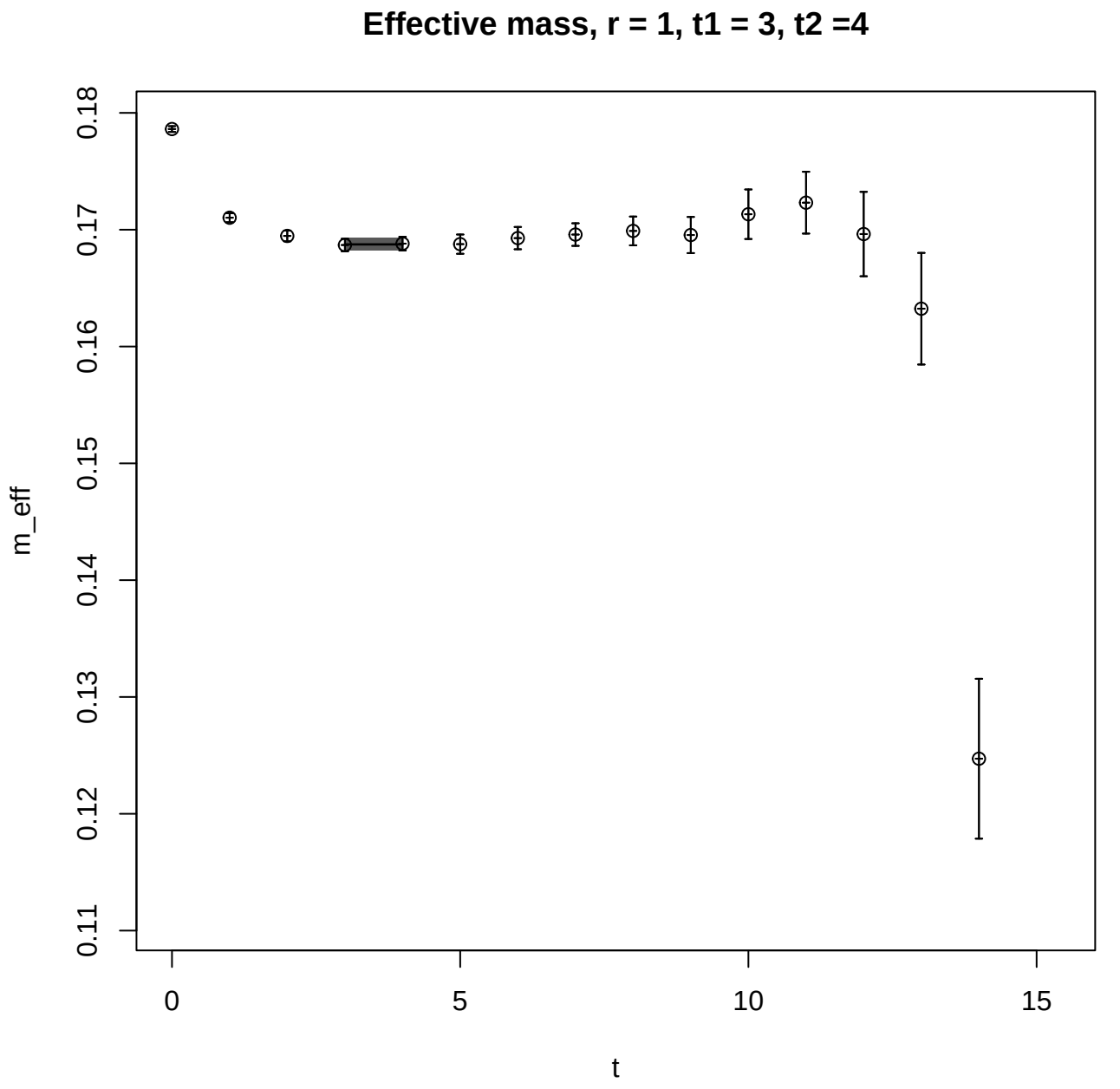
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =4
m_eff
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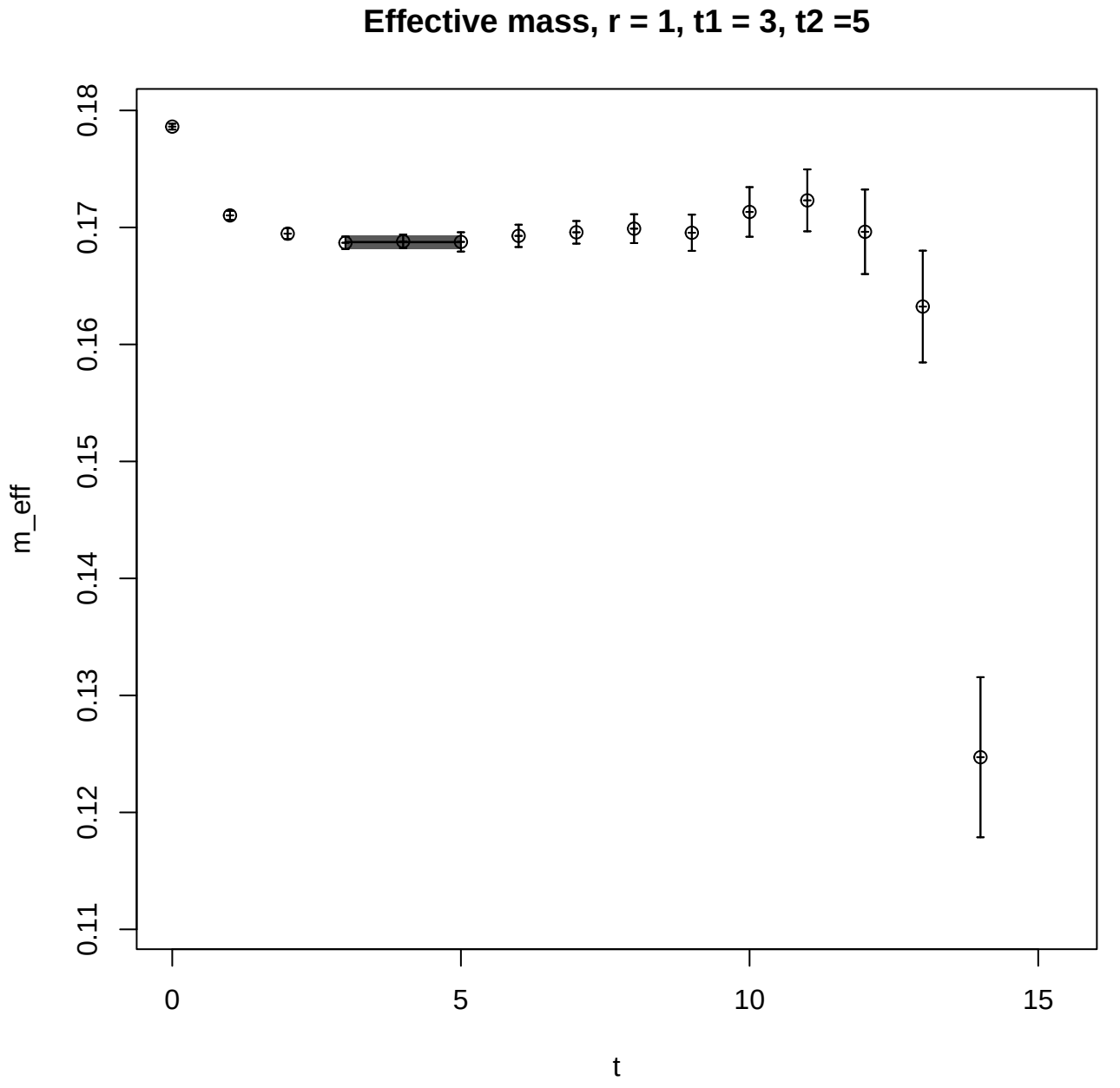
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =5
m_eff
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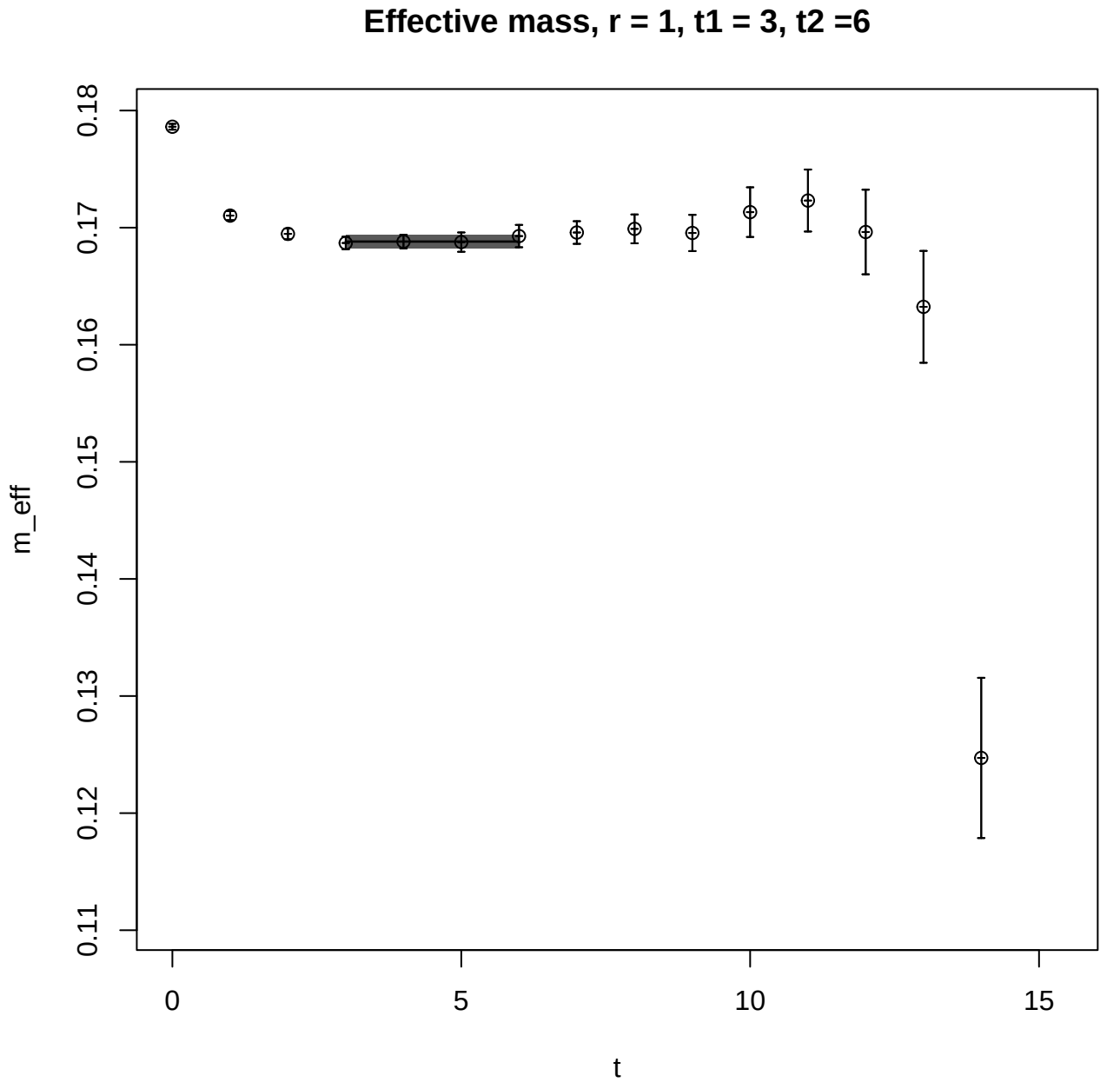
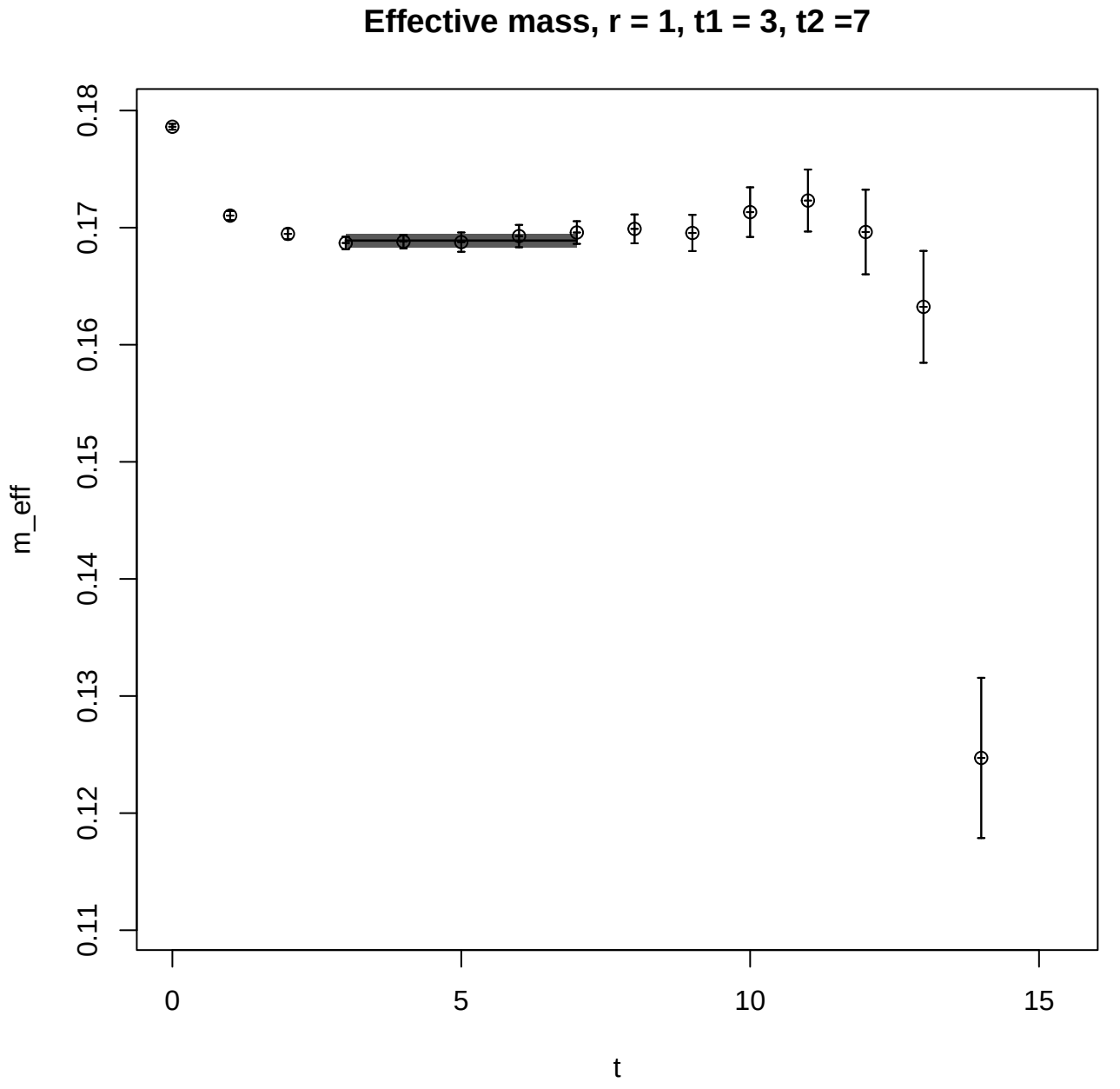
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =6
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =7
m_eff
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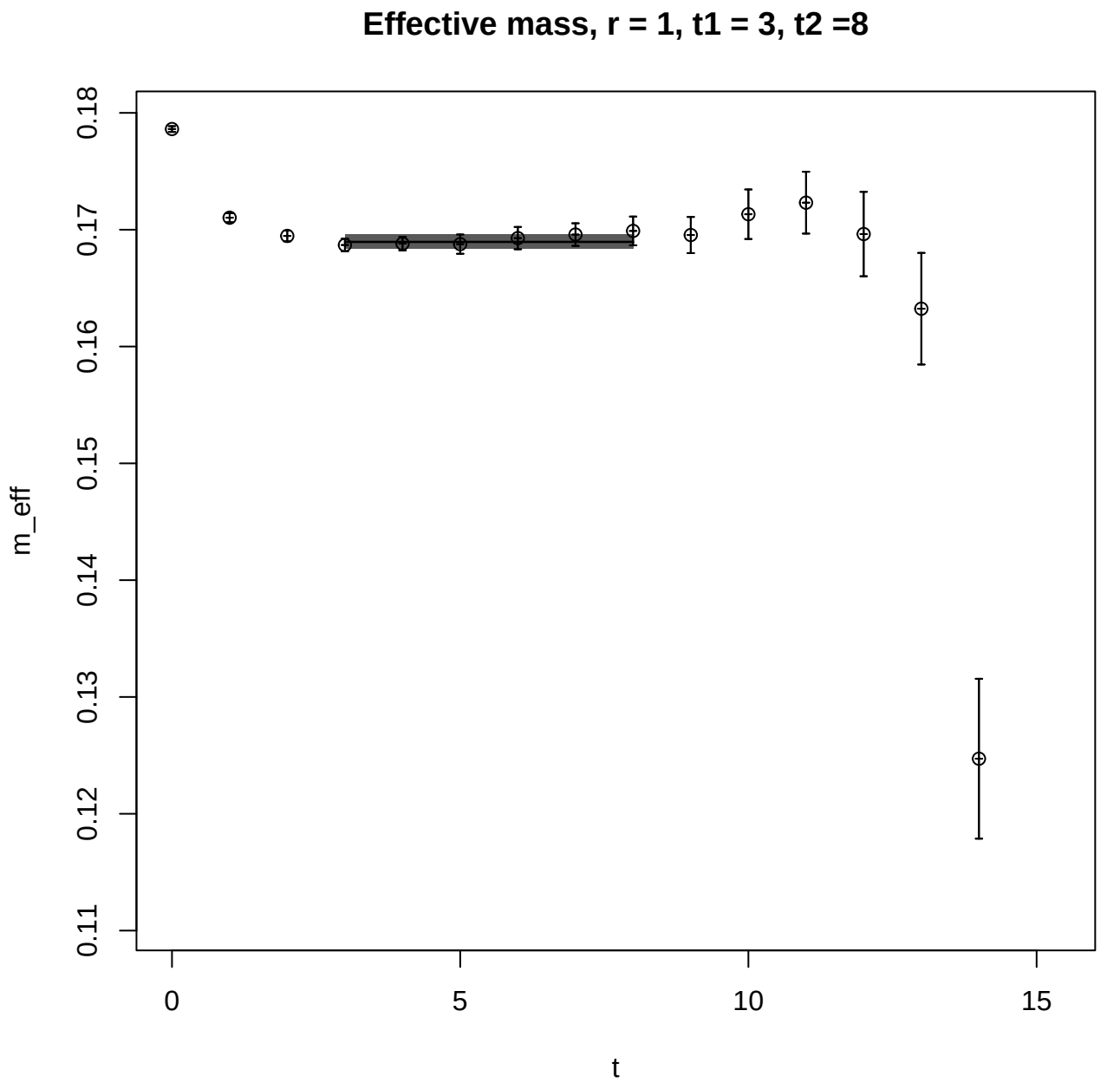
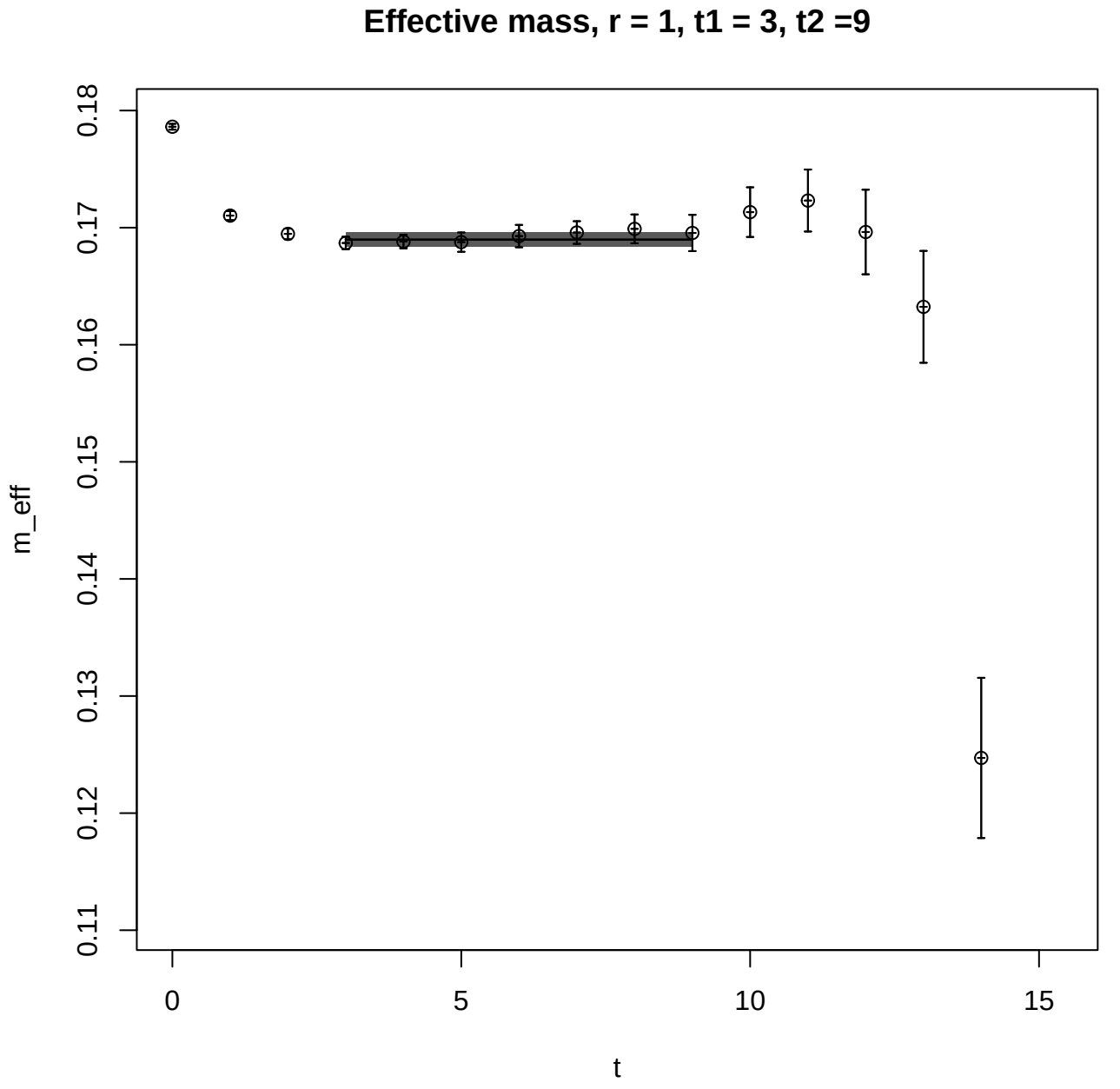
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =8
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =9
m_eff
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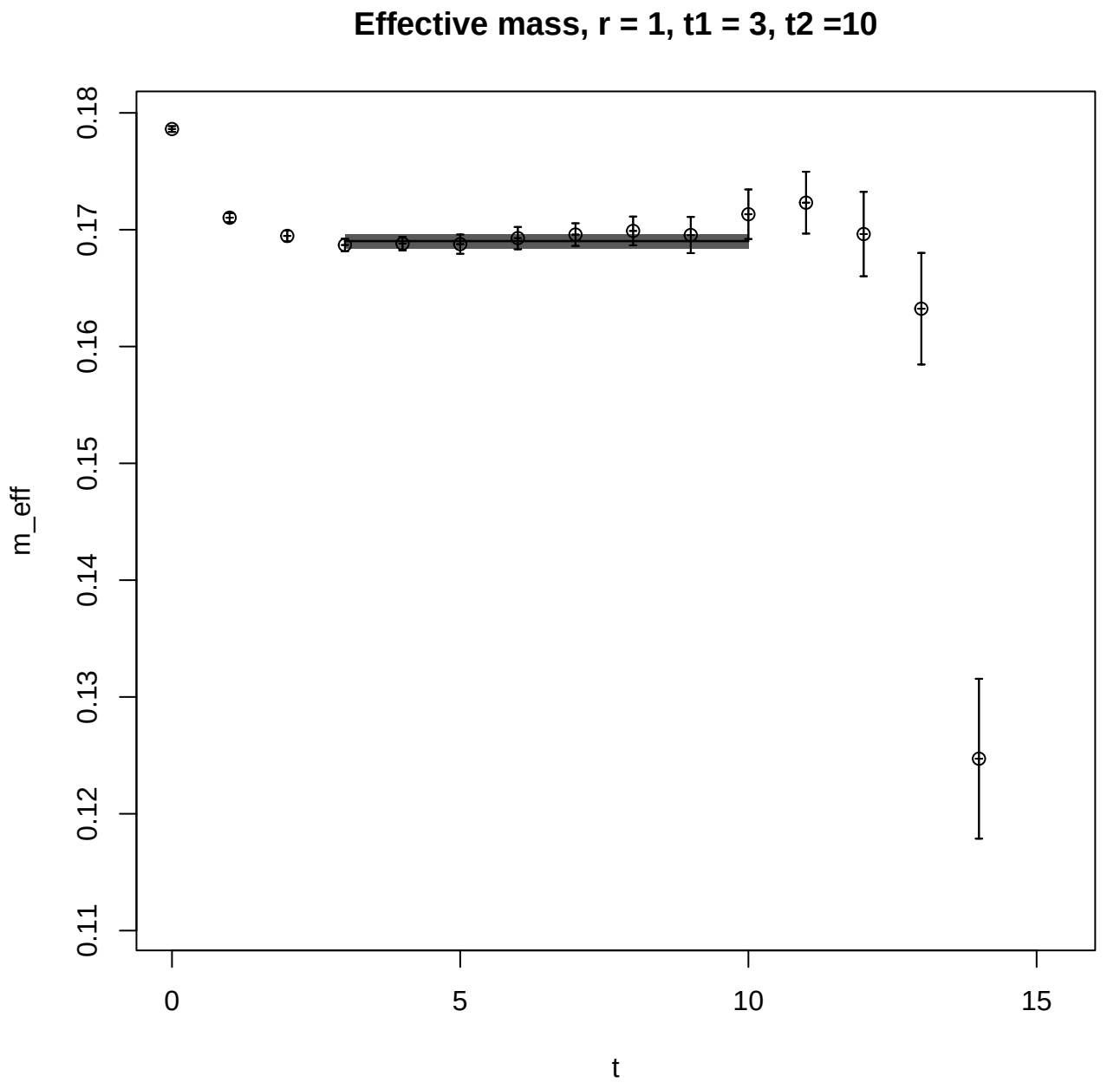
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =10
m_eff
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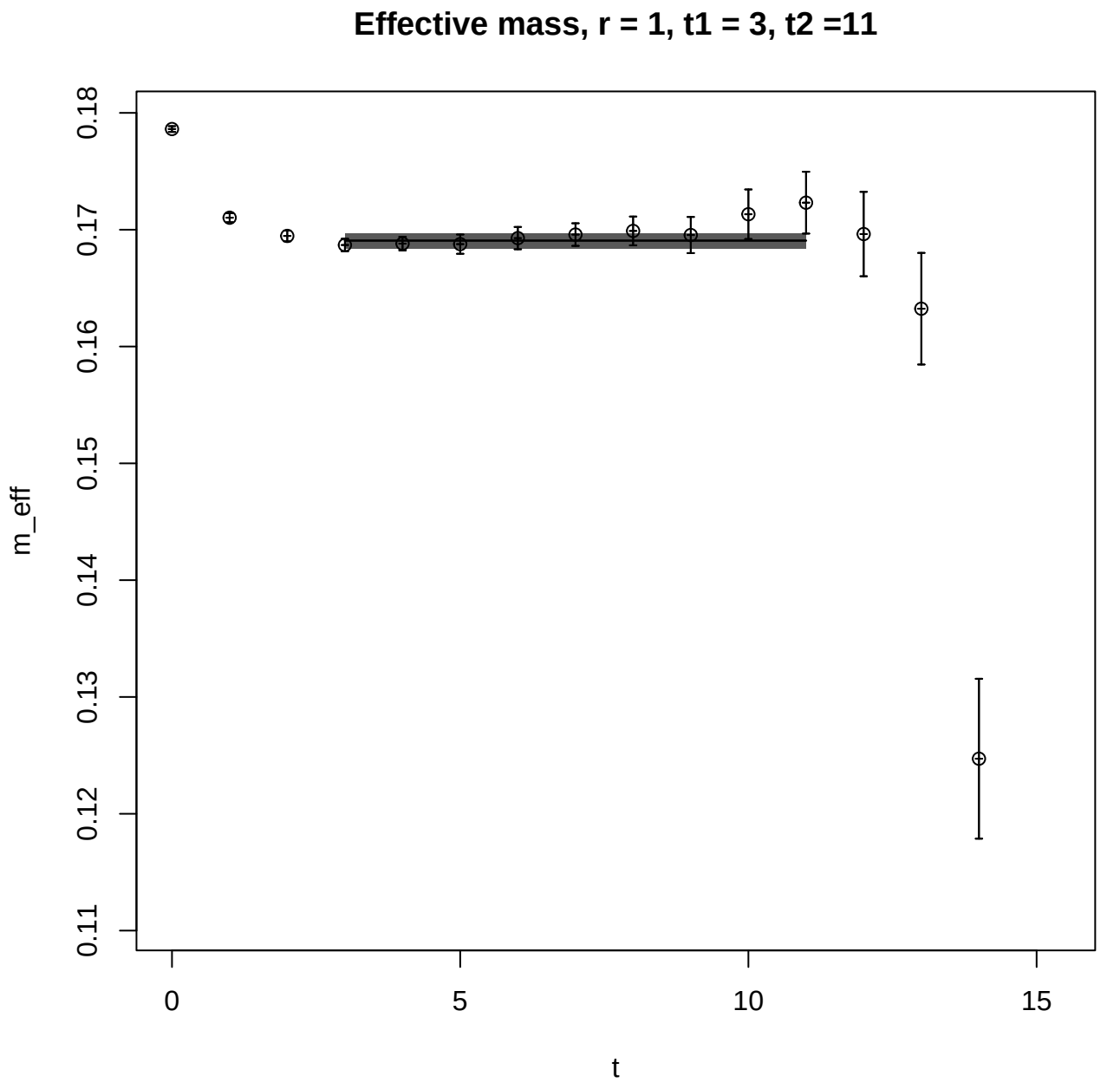
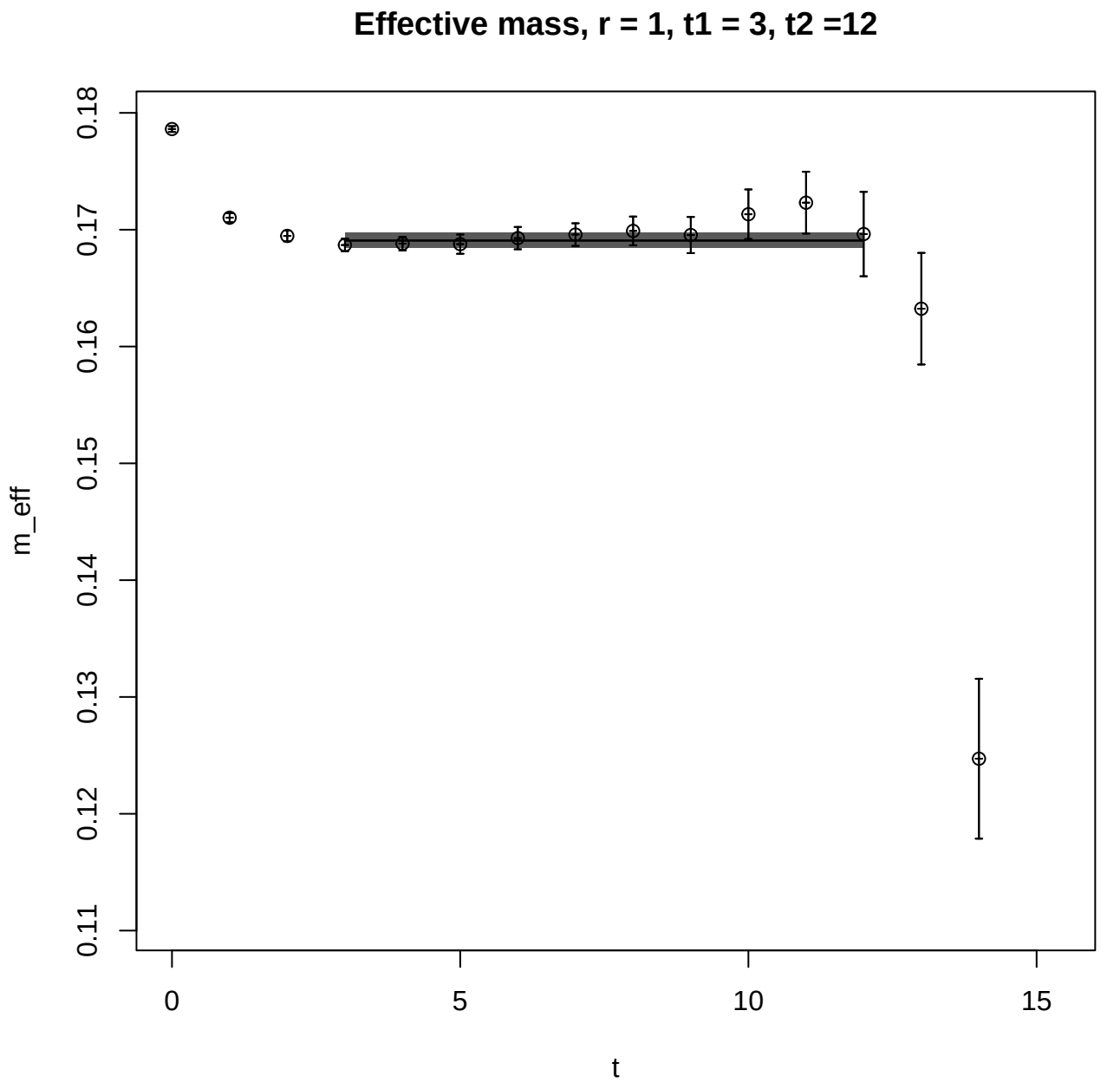
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =11
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =12
m_eff
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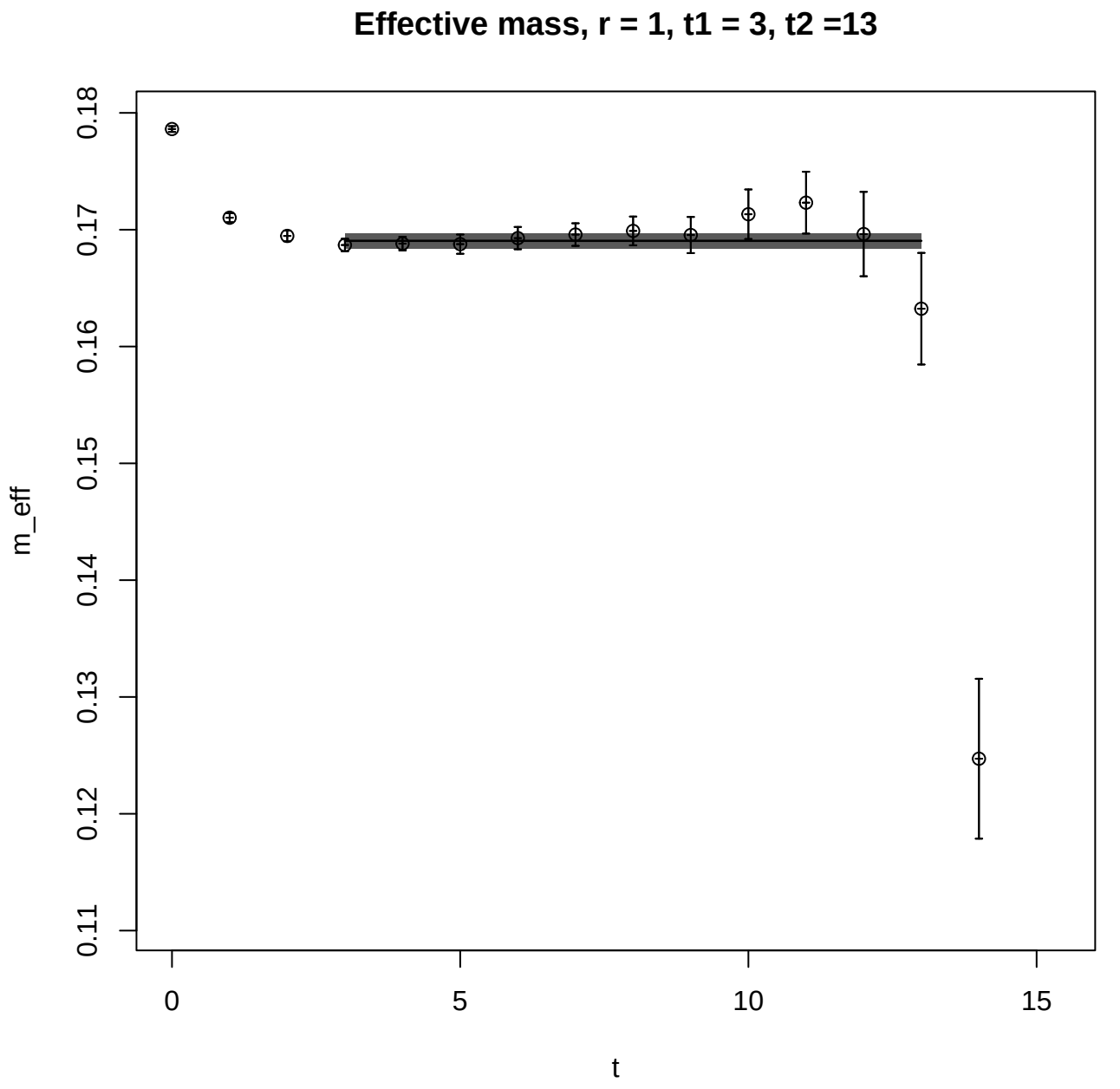
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 3, t2 =13
m_eff
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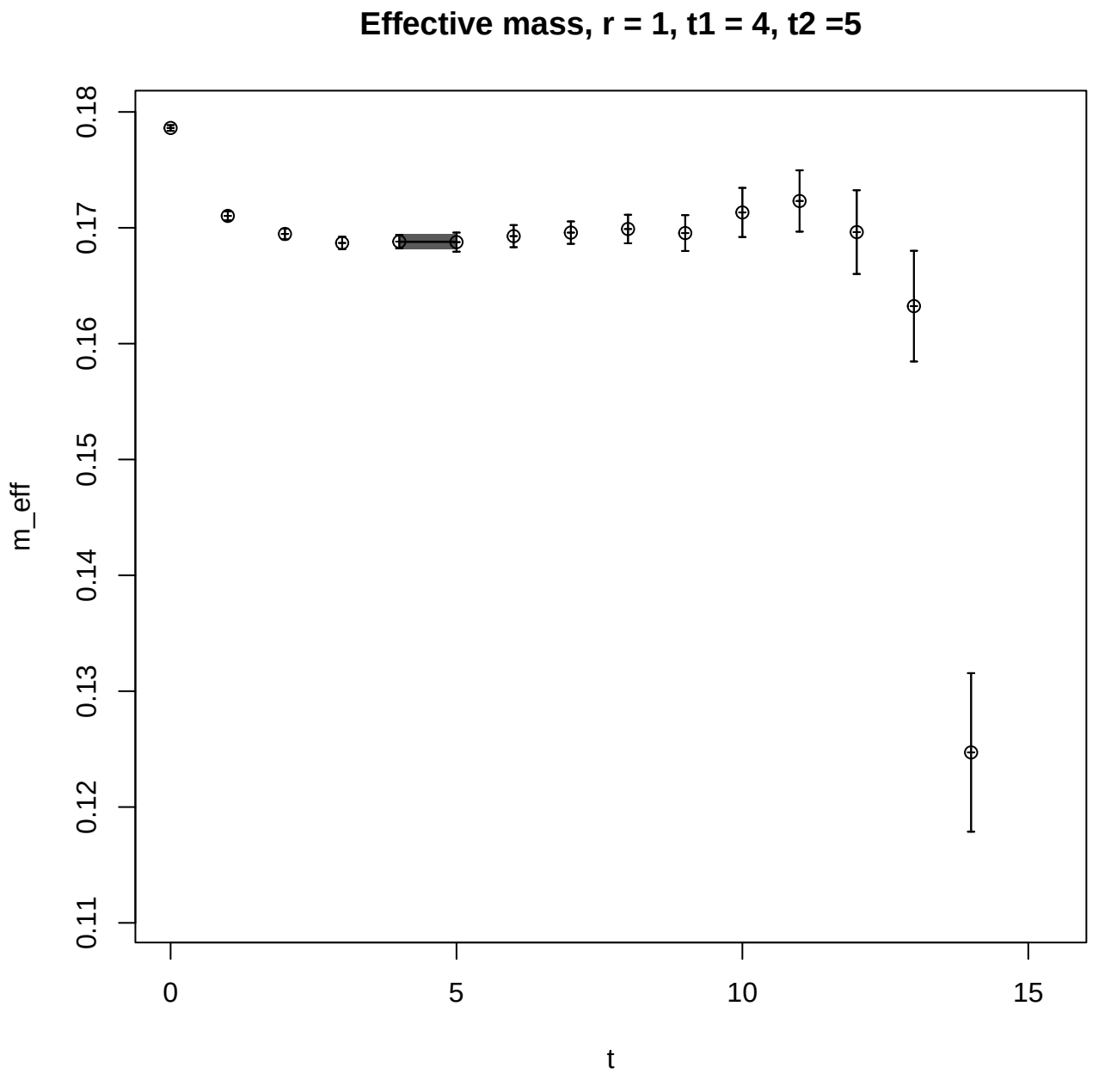
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =5
m_eff
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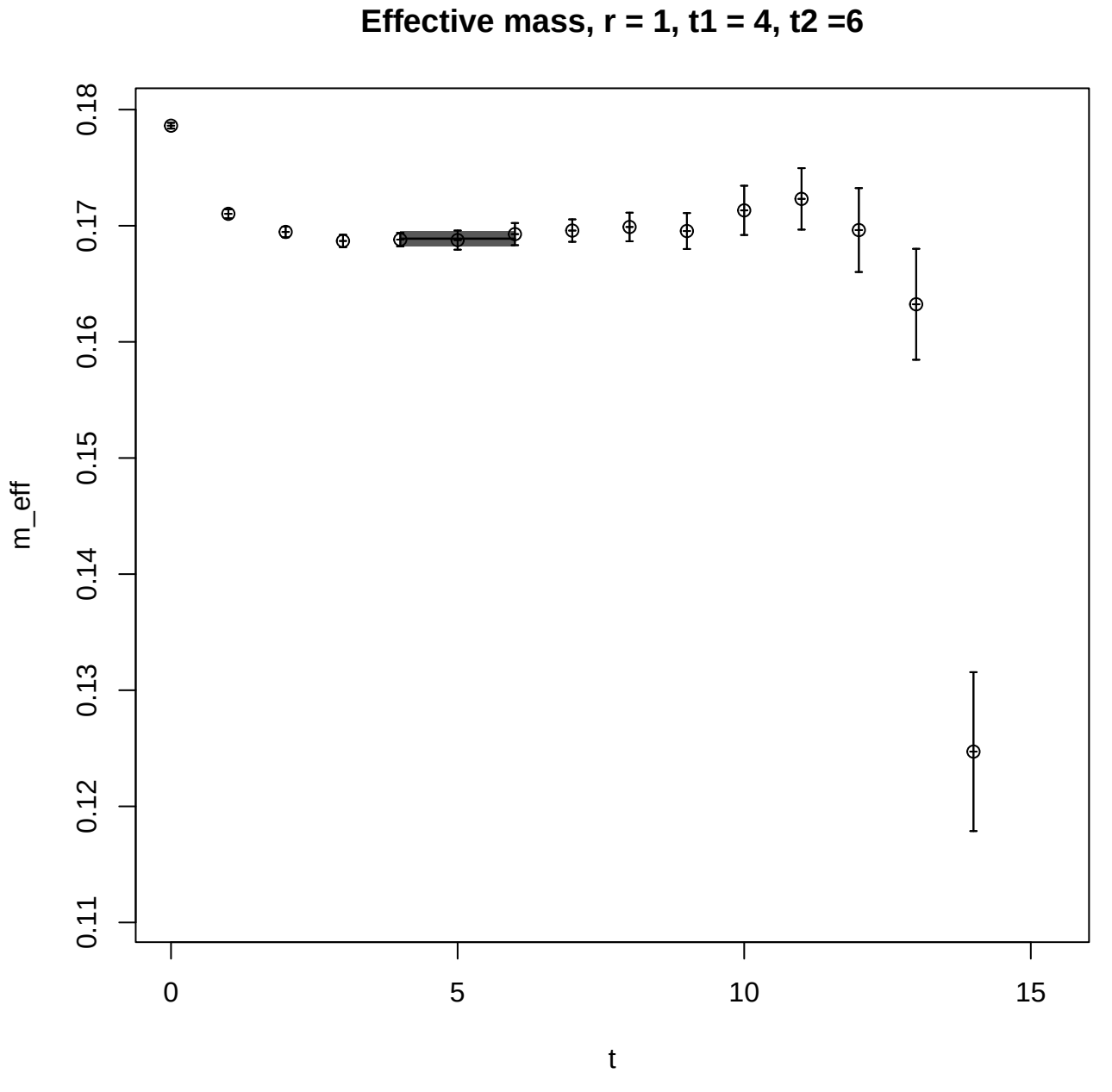
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =6
m_eff
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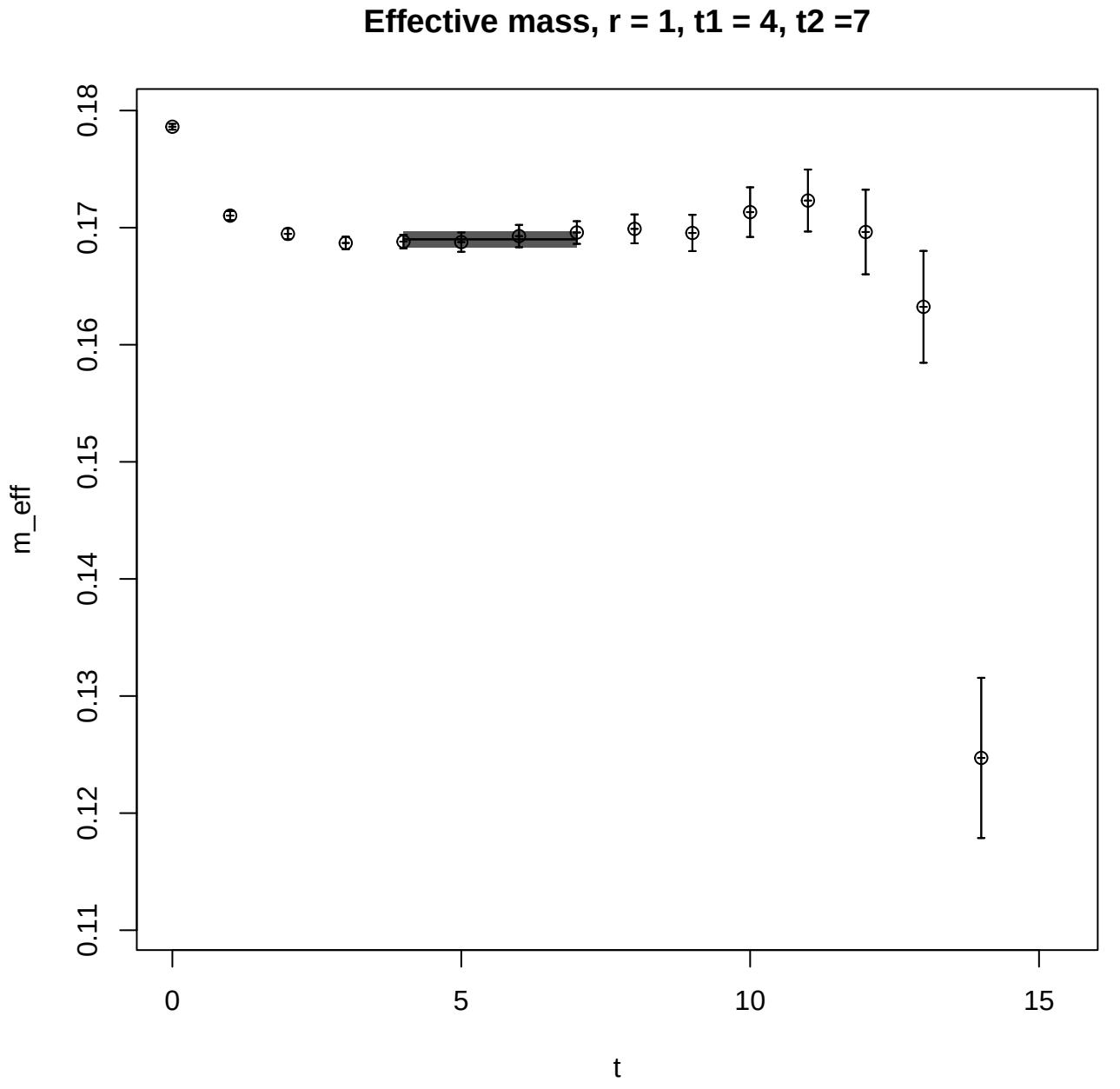
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =7
m_eff
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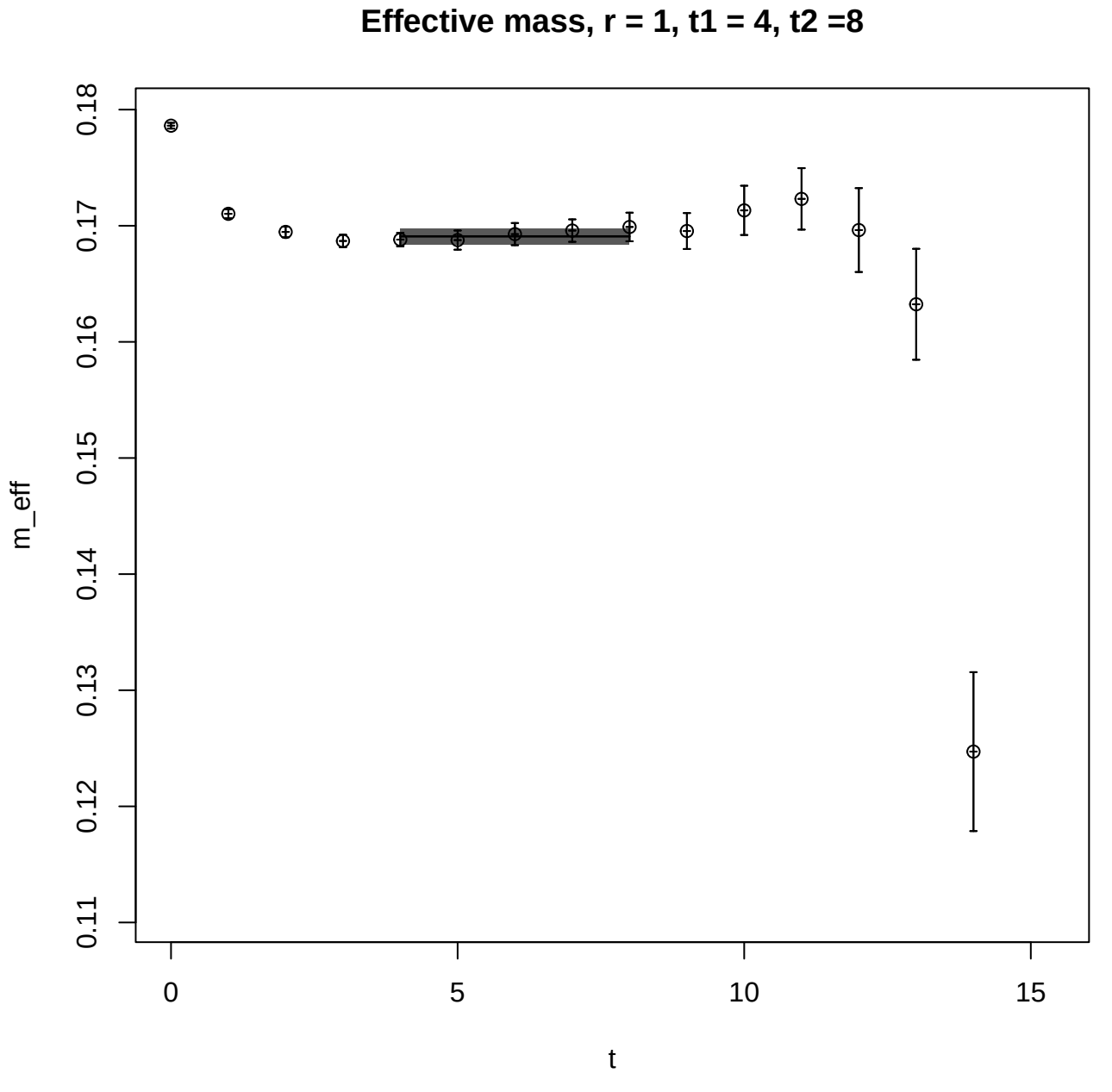
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =8
m_eff
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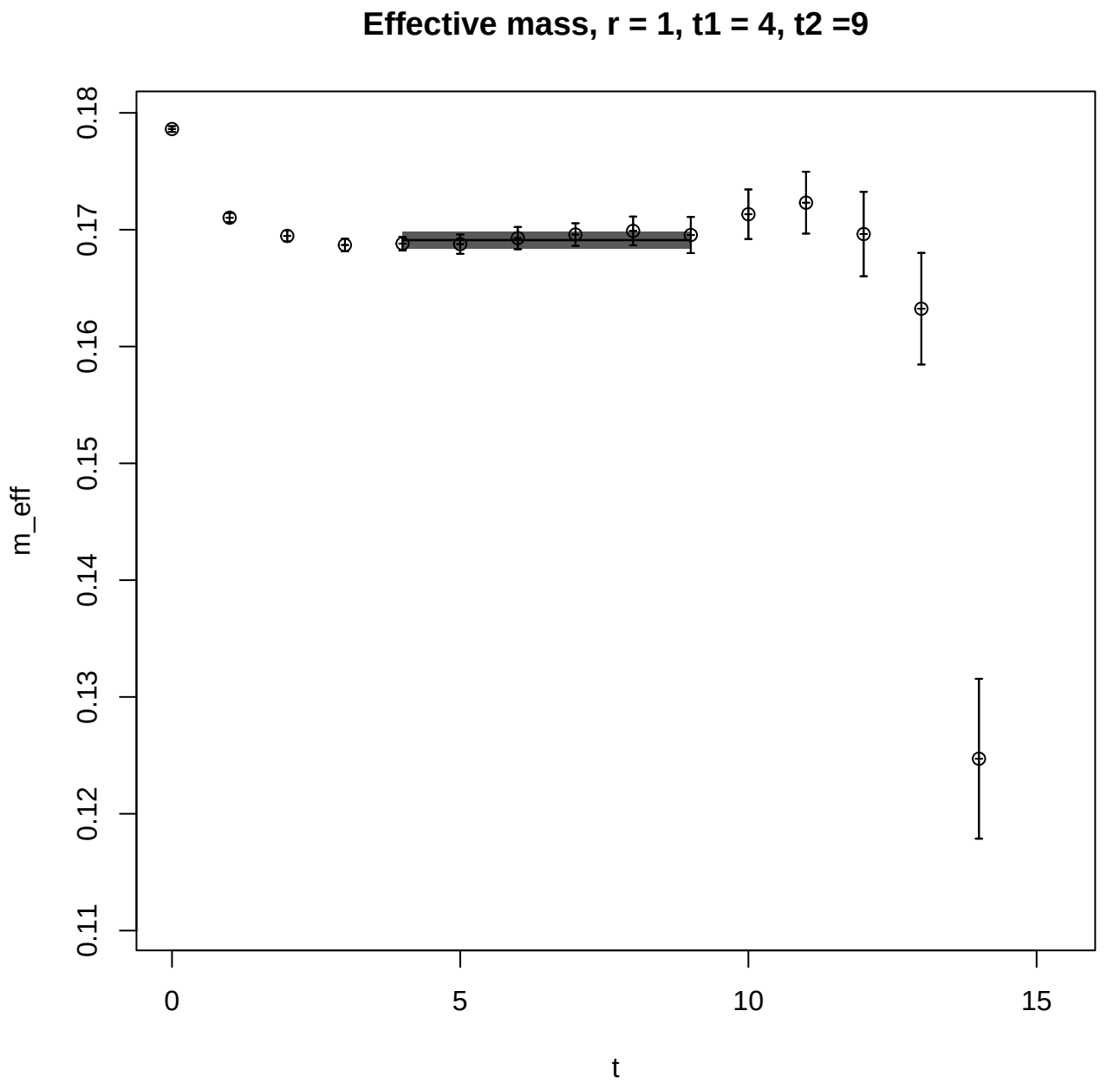
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =9
m_eff
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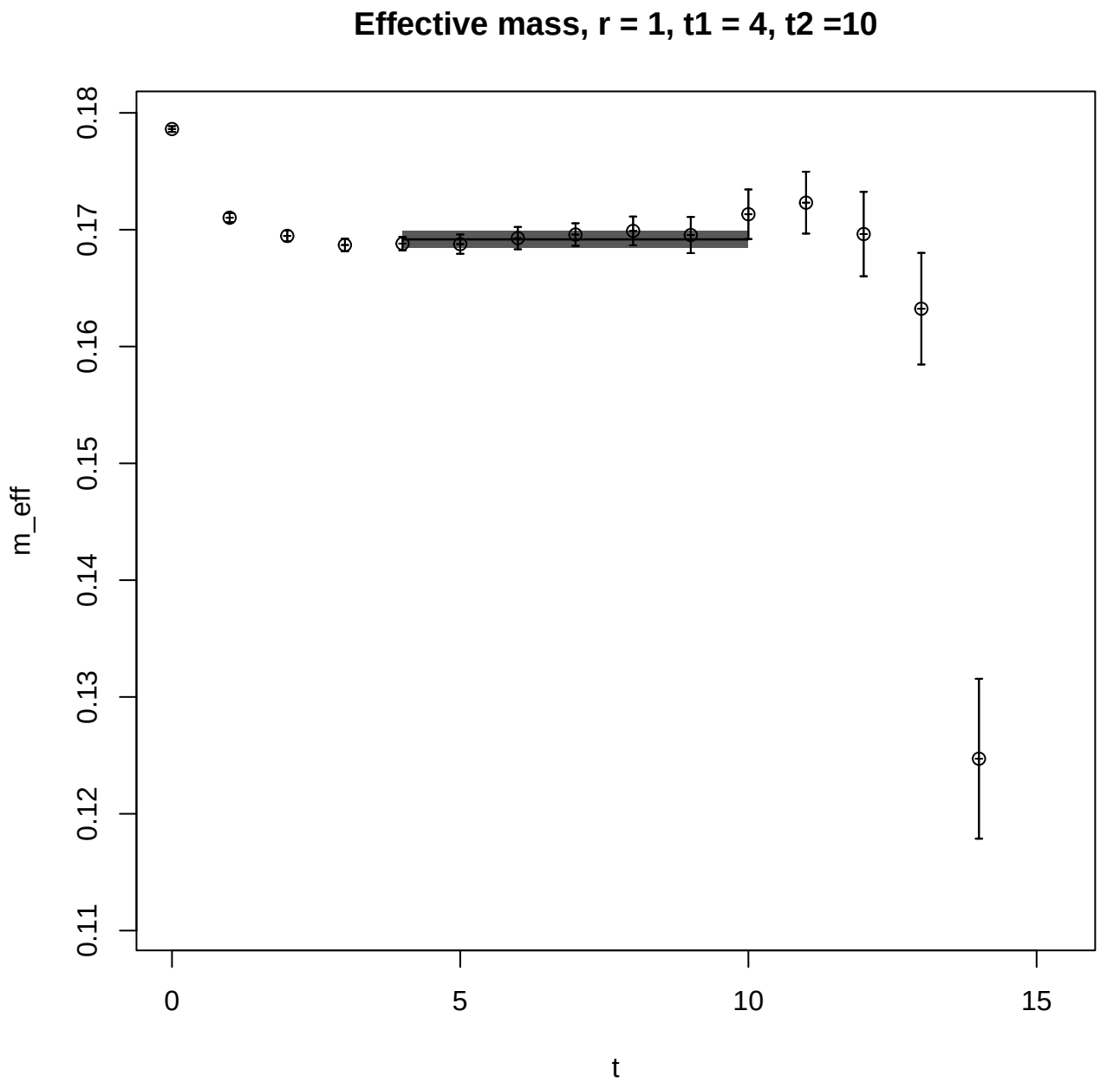
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =10
m_eff
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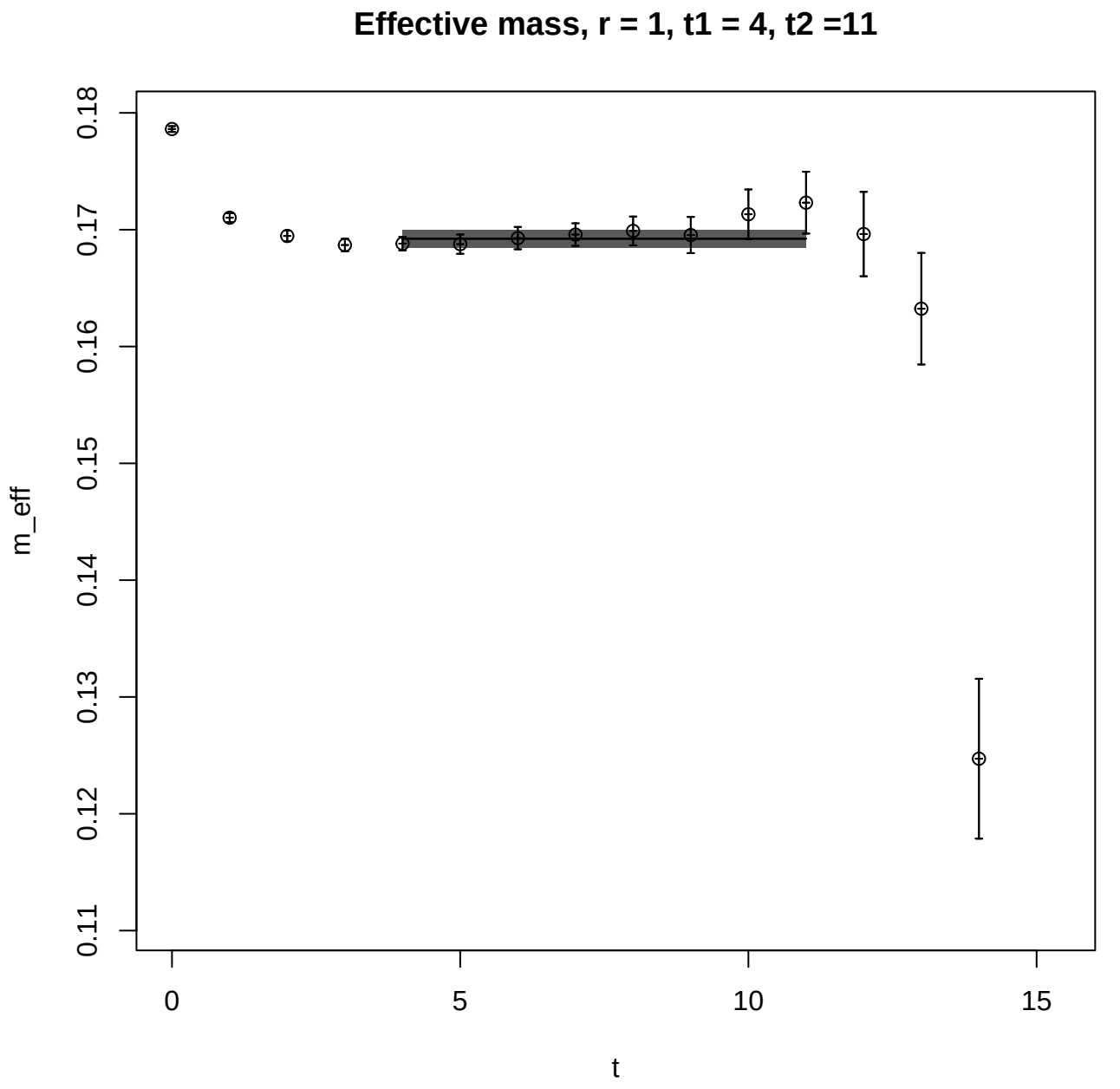
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =11
m_eff
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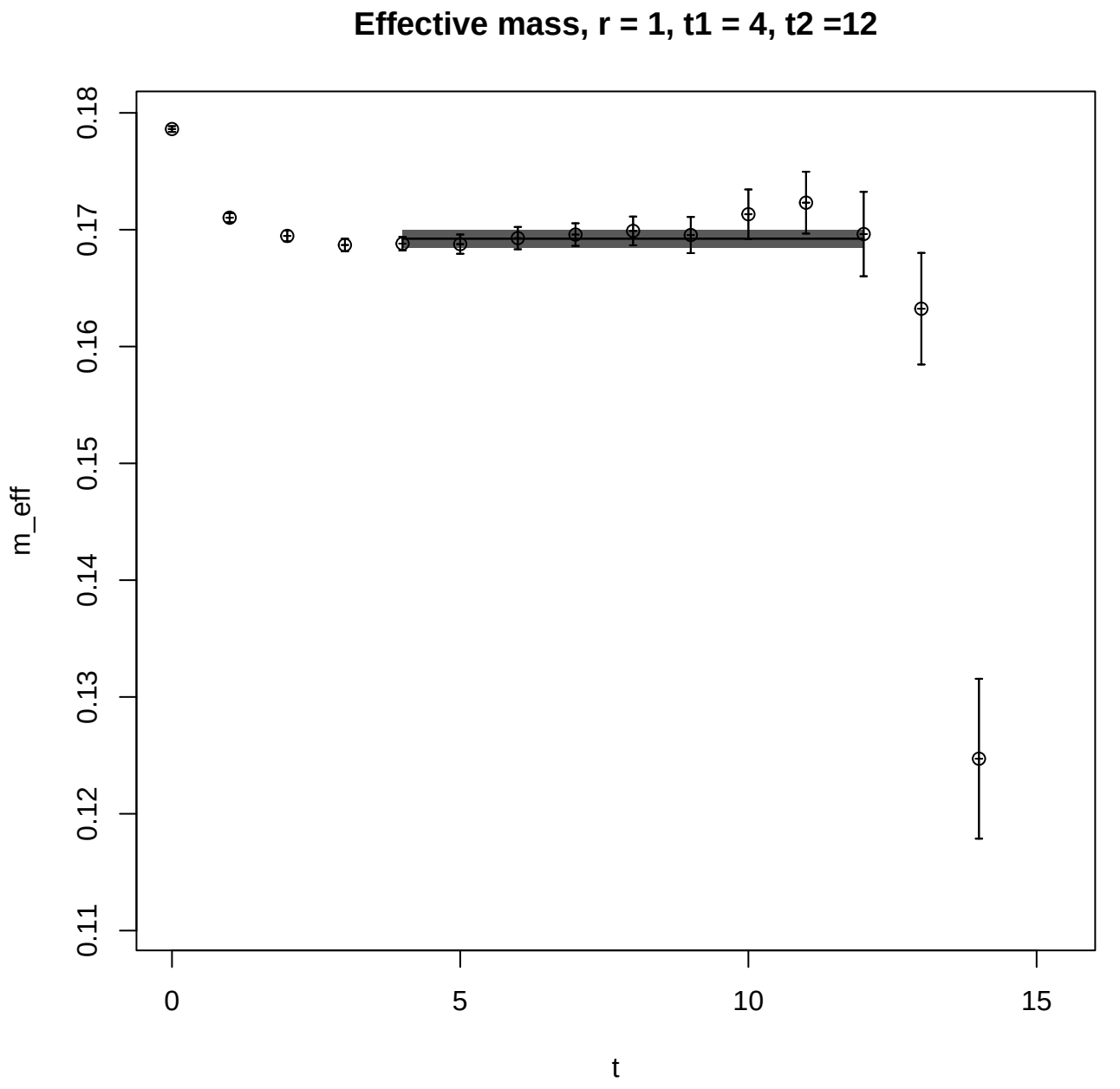
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =12
m_eff
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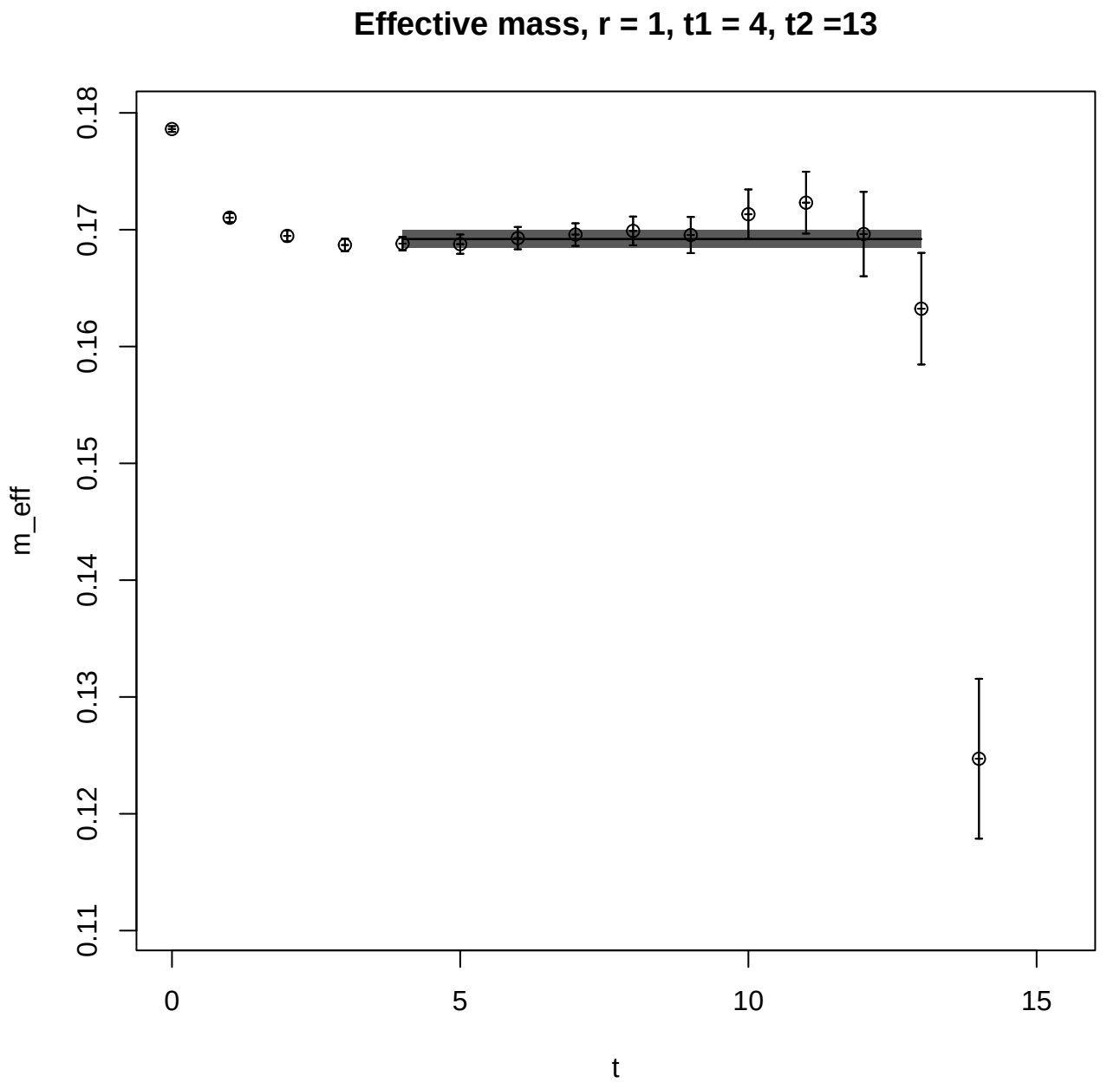
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 4, t2 =13
m_eff
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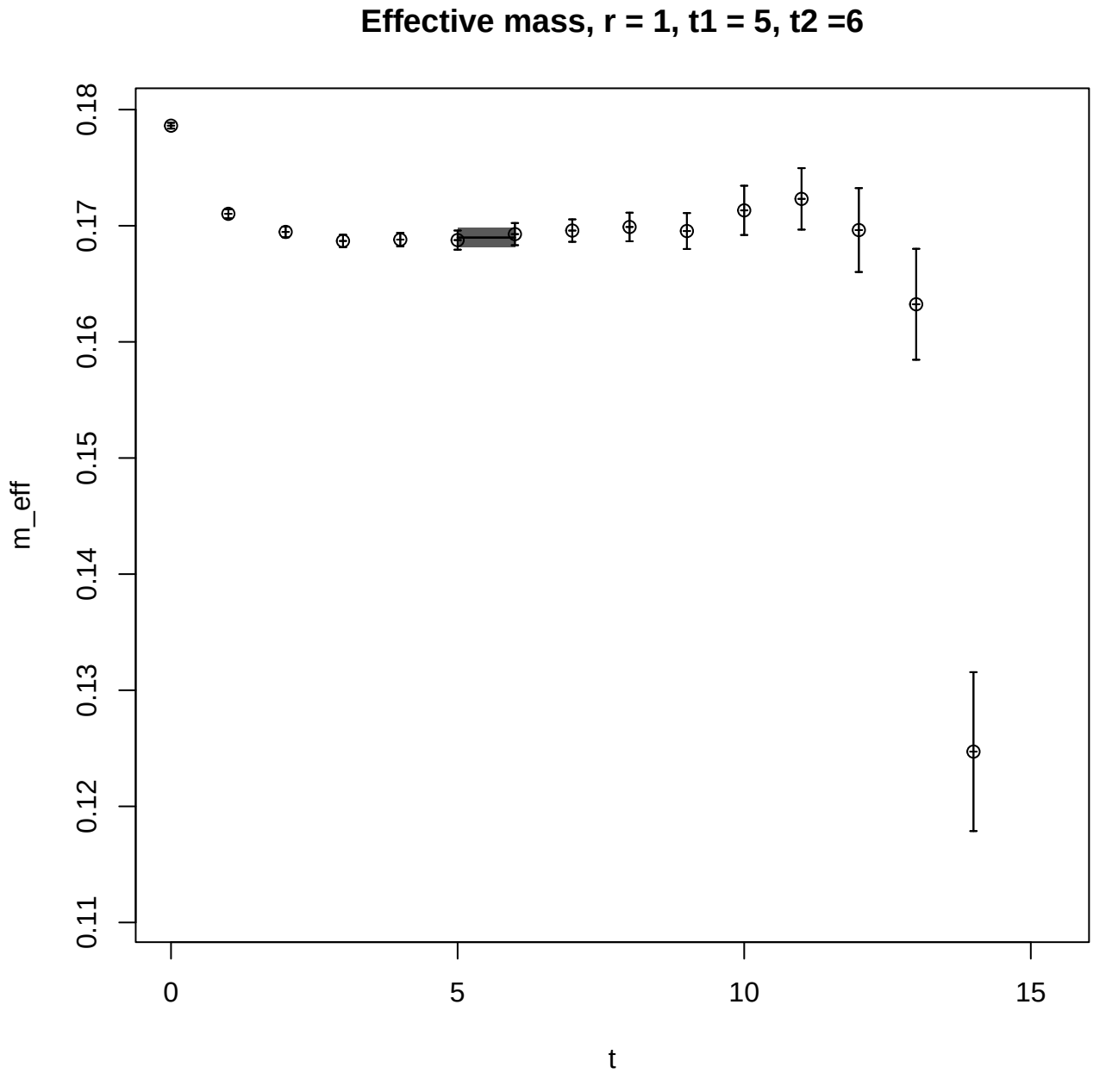
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =6
m_eff
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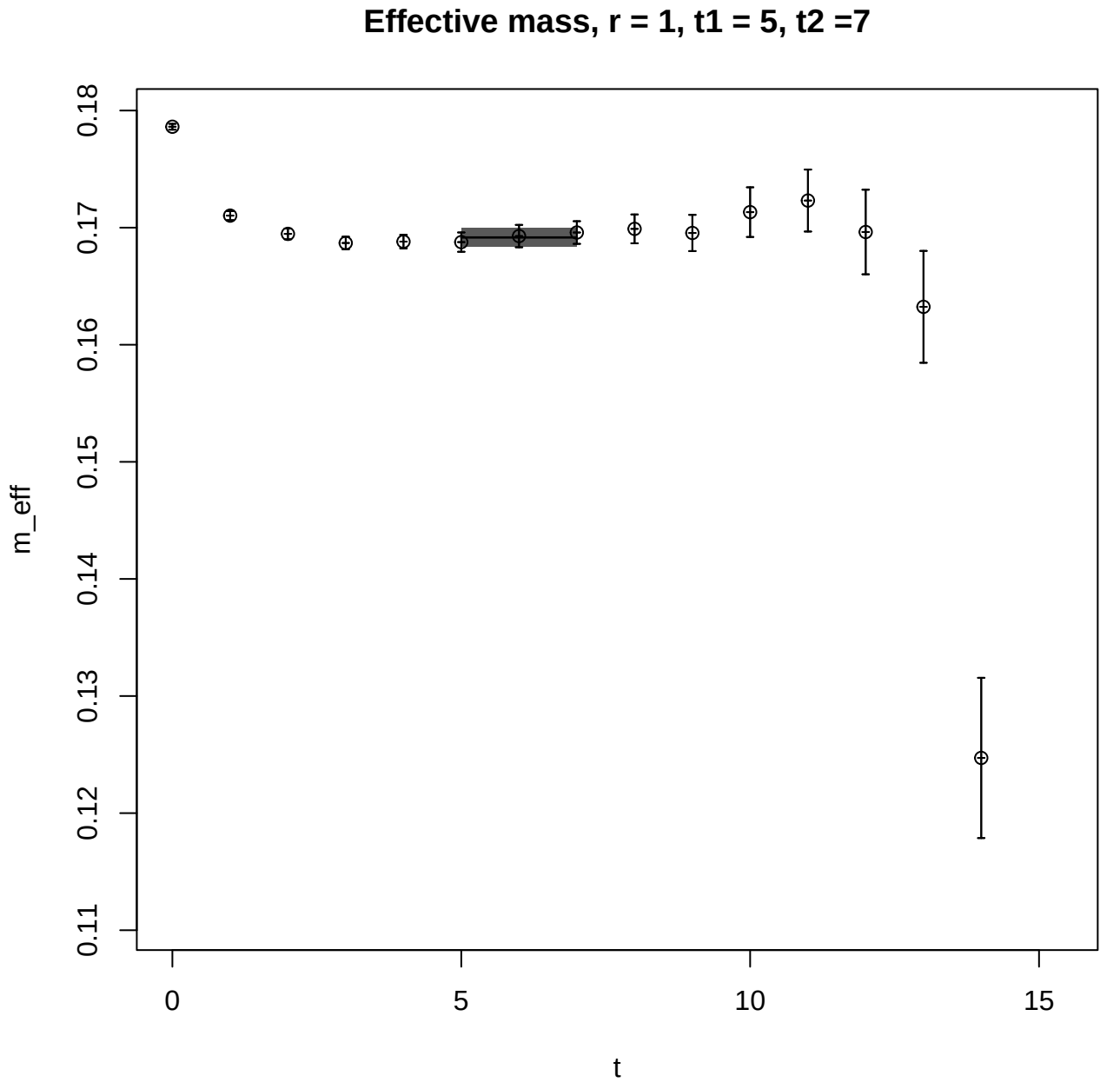
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =7
m_eff
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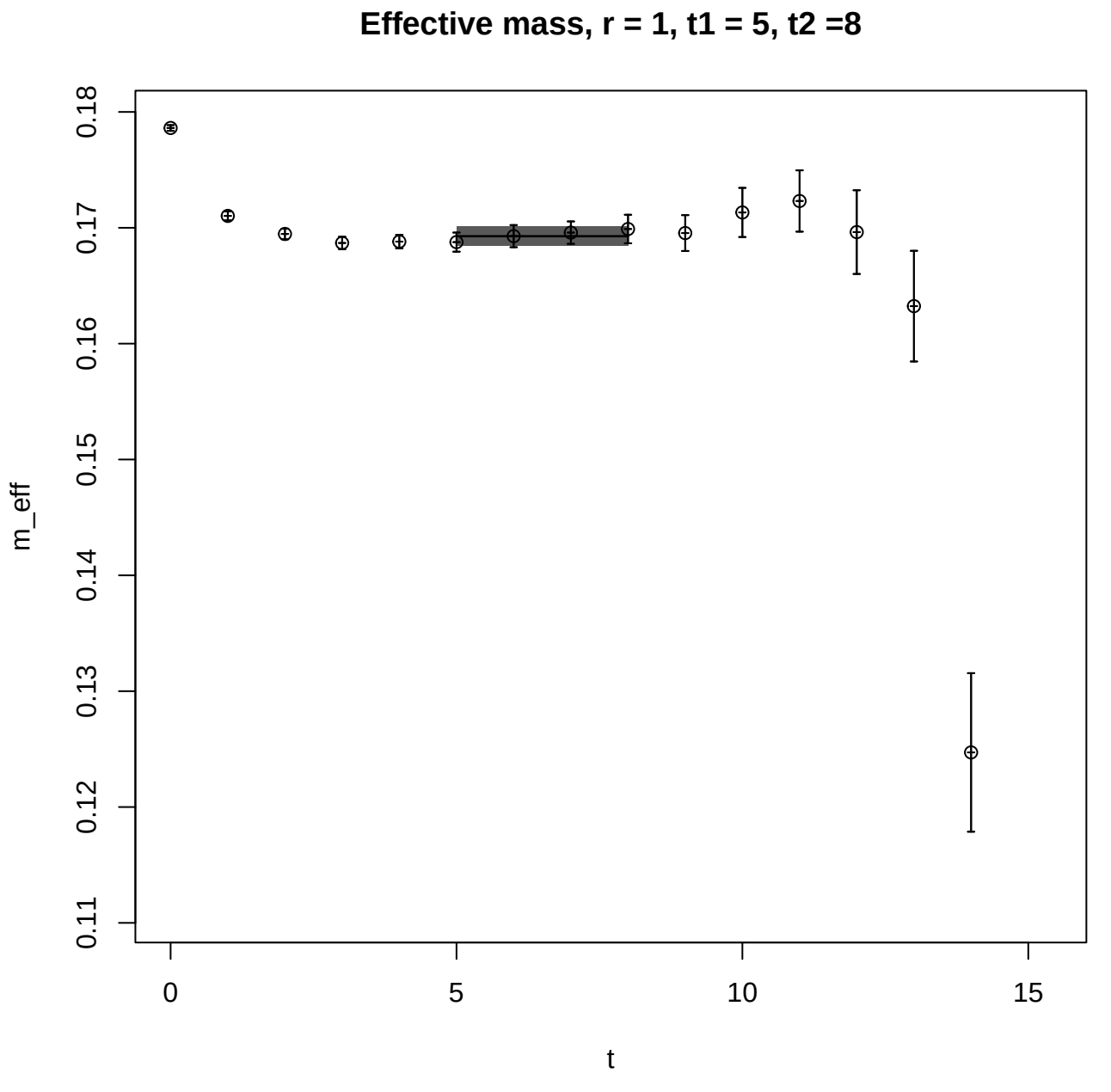
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =8
m_eff
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0.12
0.13
0.14
0.15
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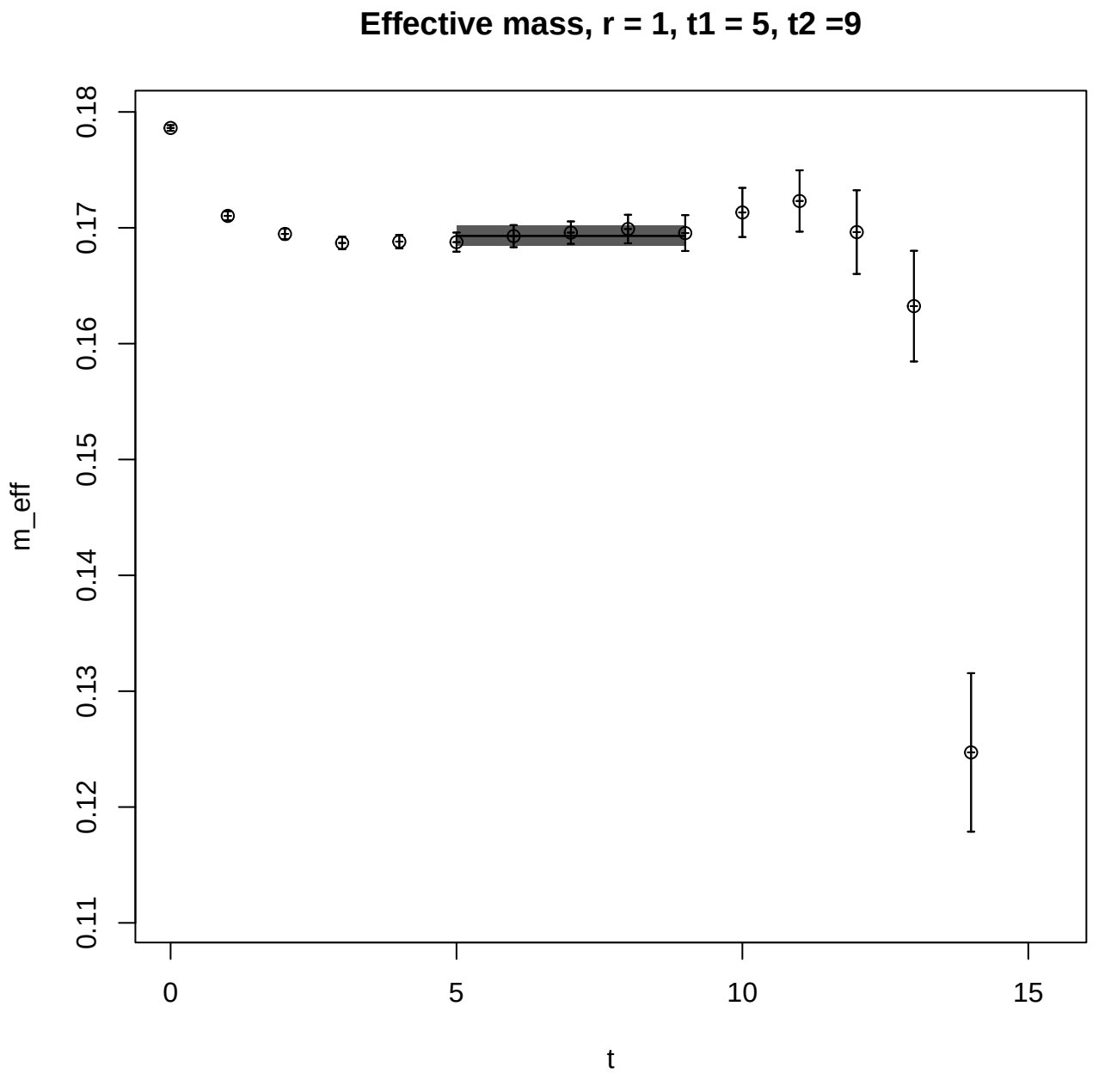
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =9
m_eff
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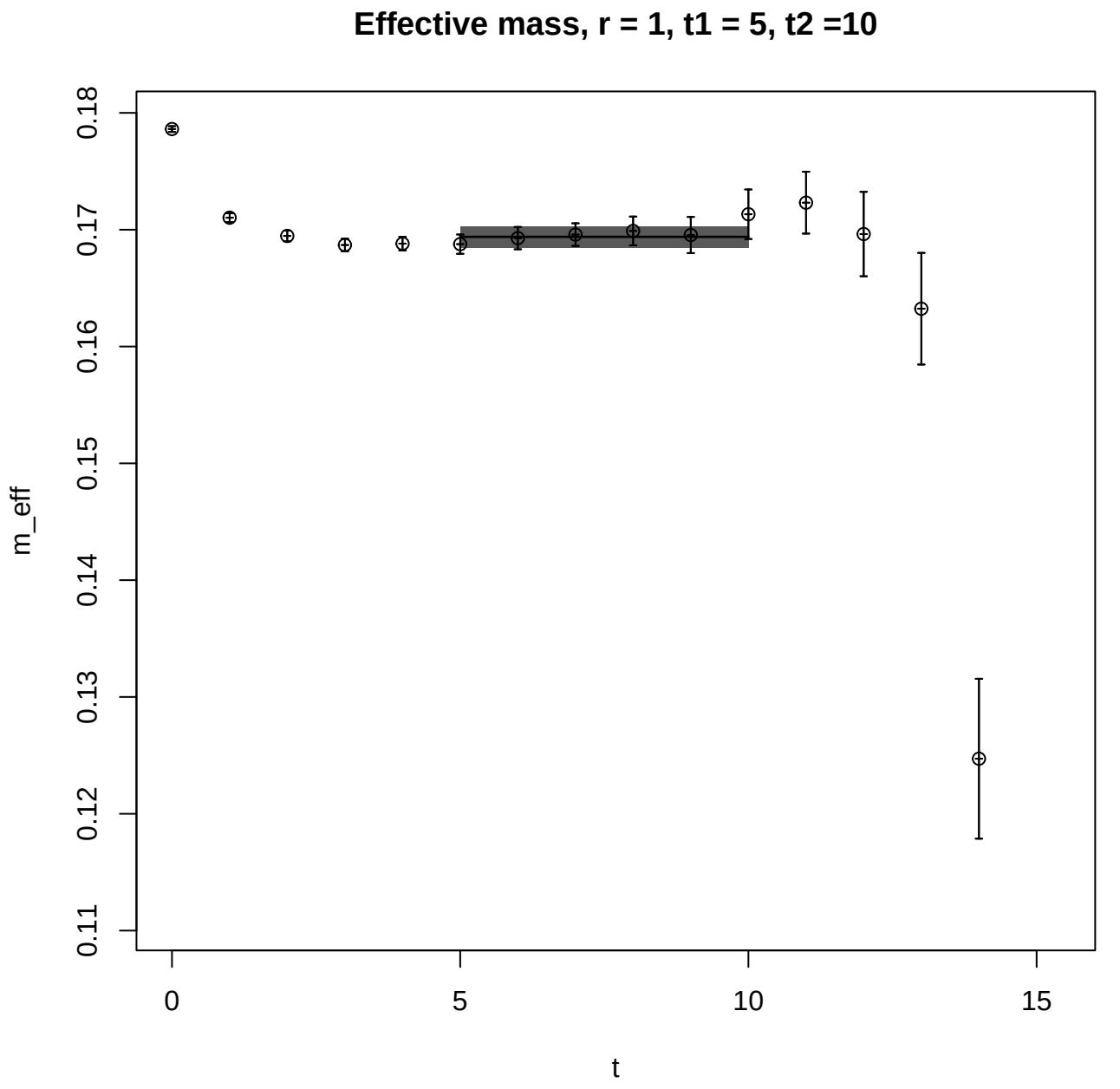
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 5, t2 =10
m_eff
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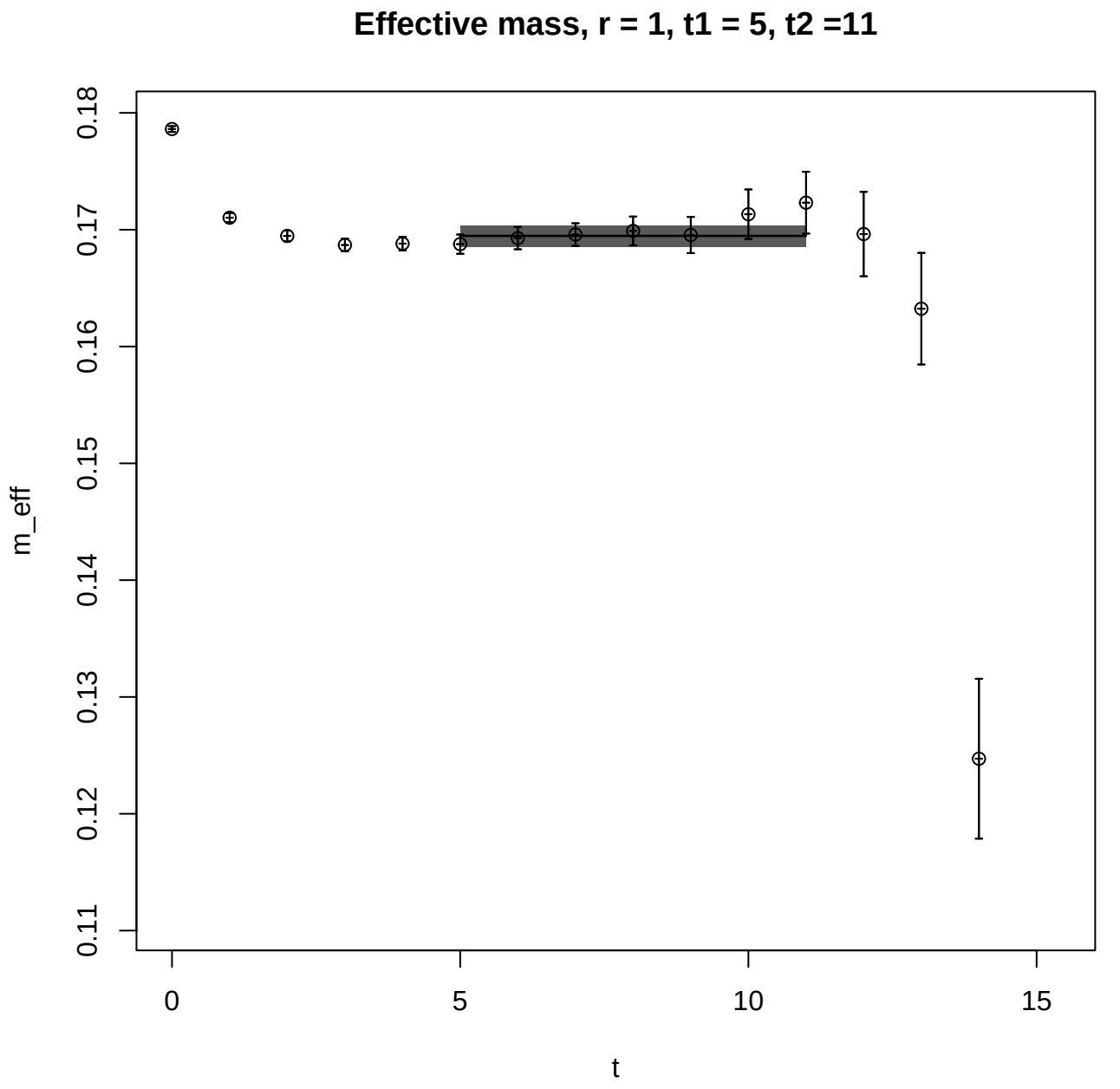
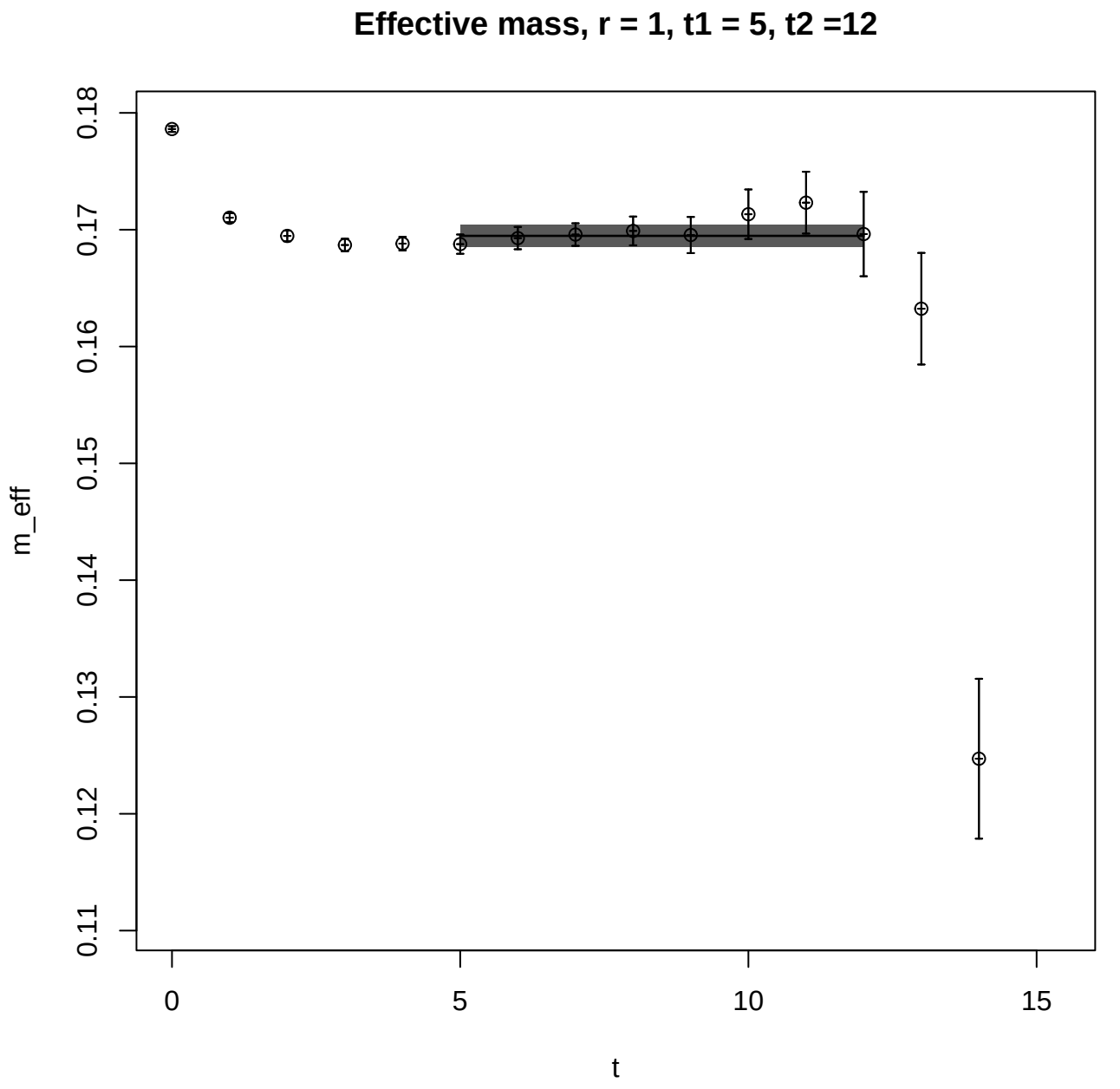
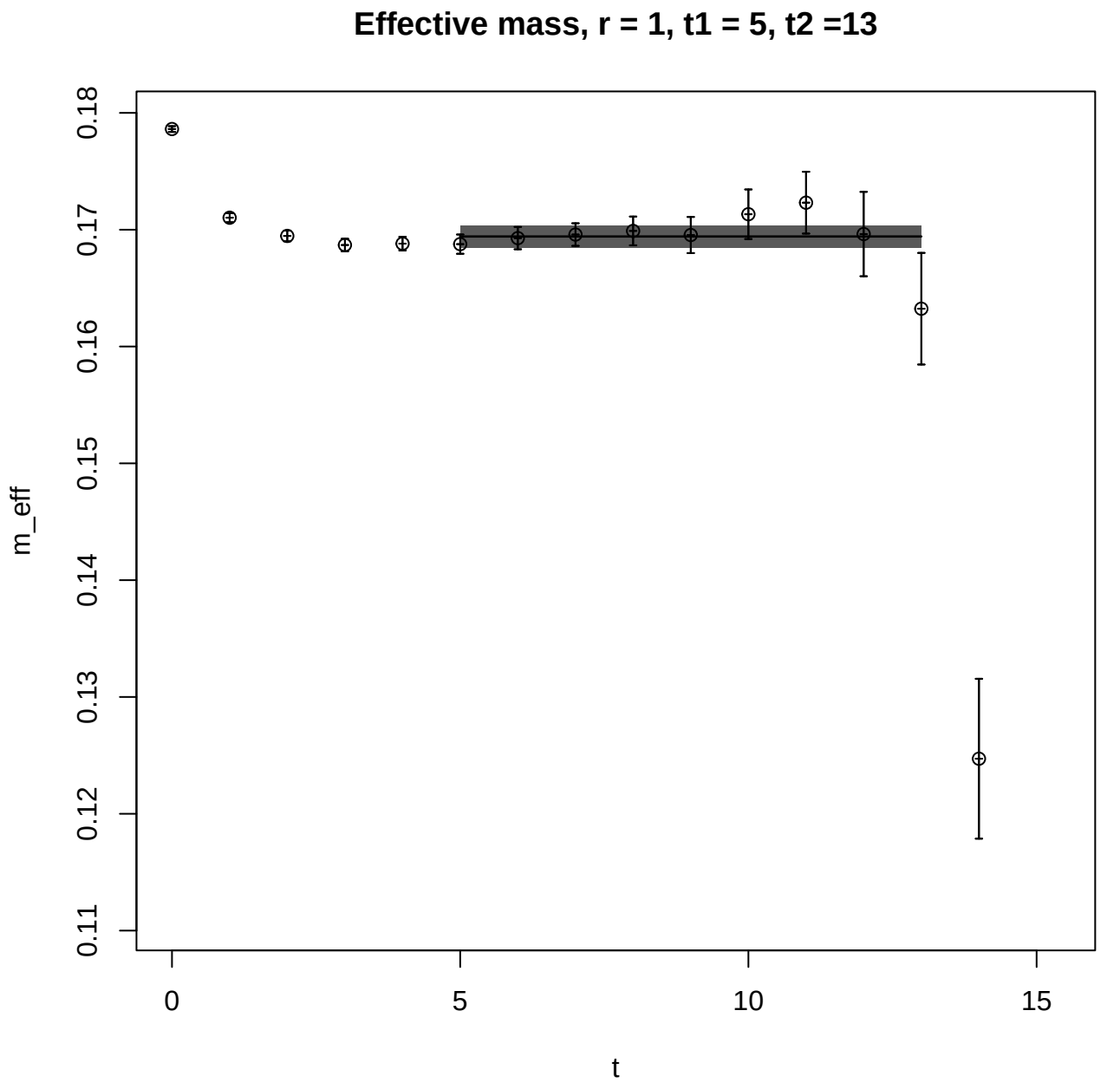
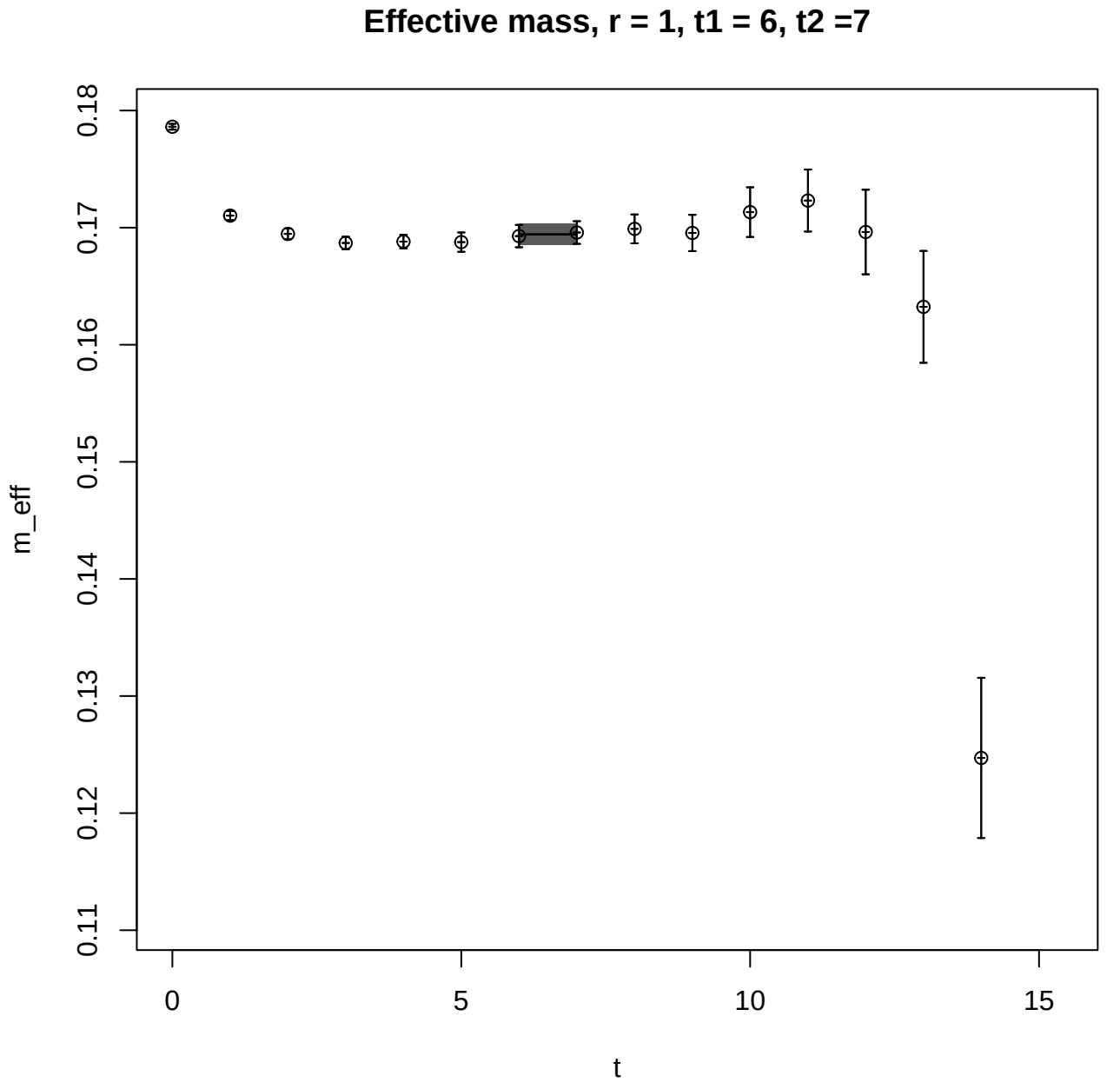
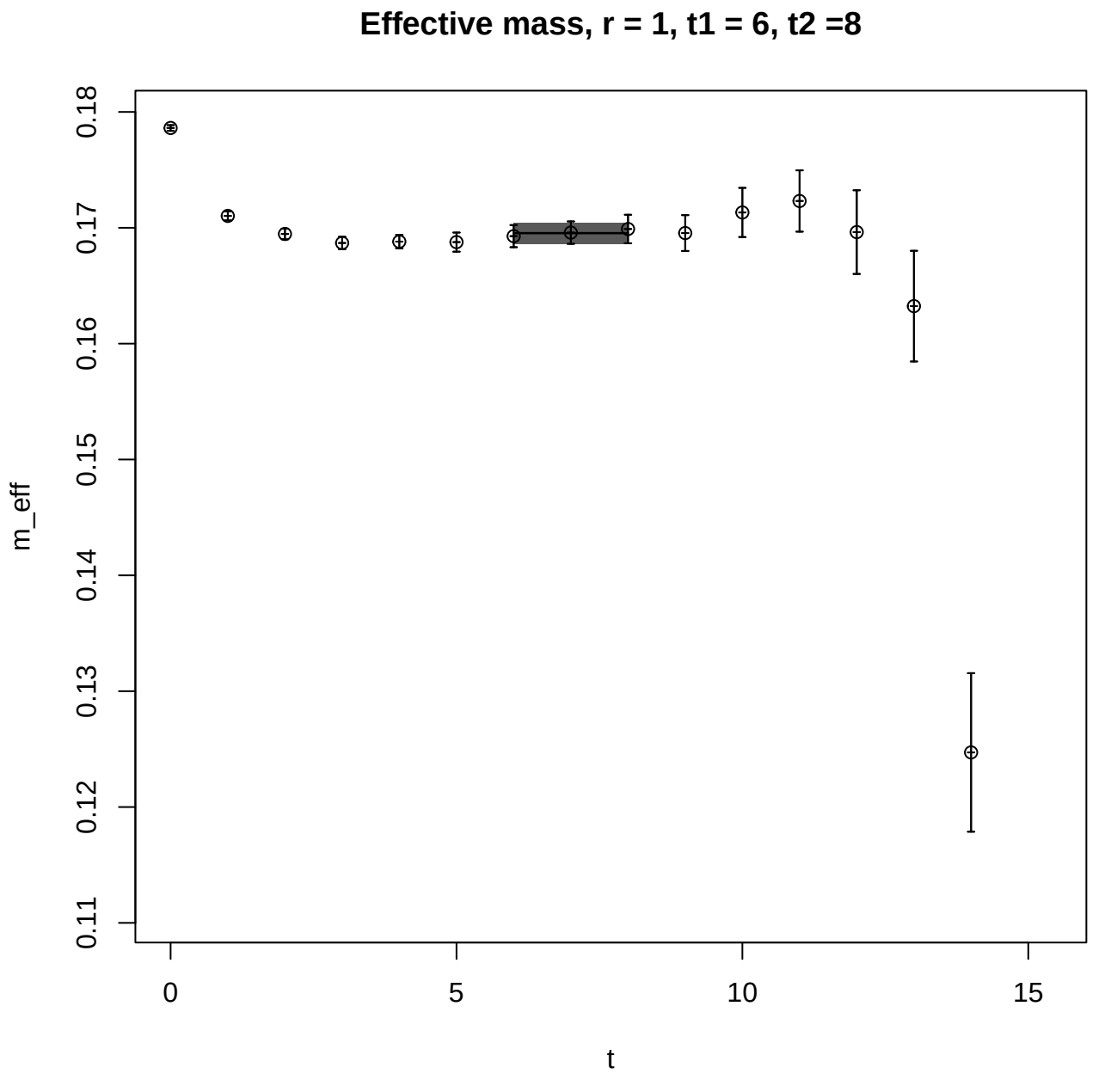
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

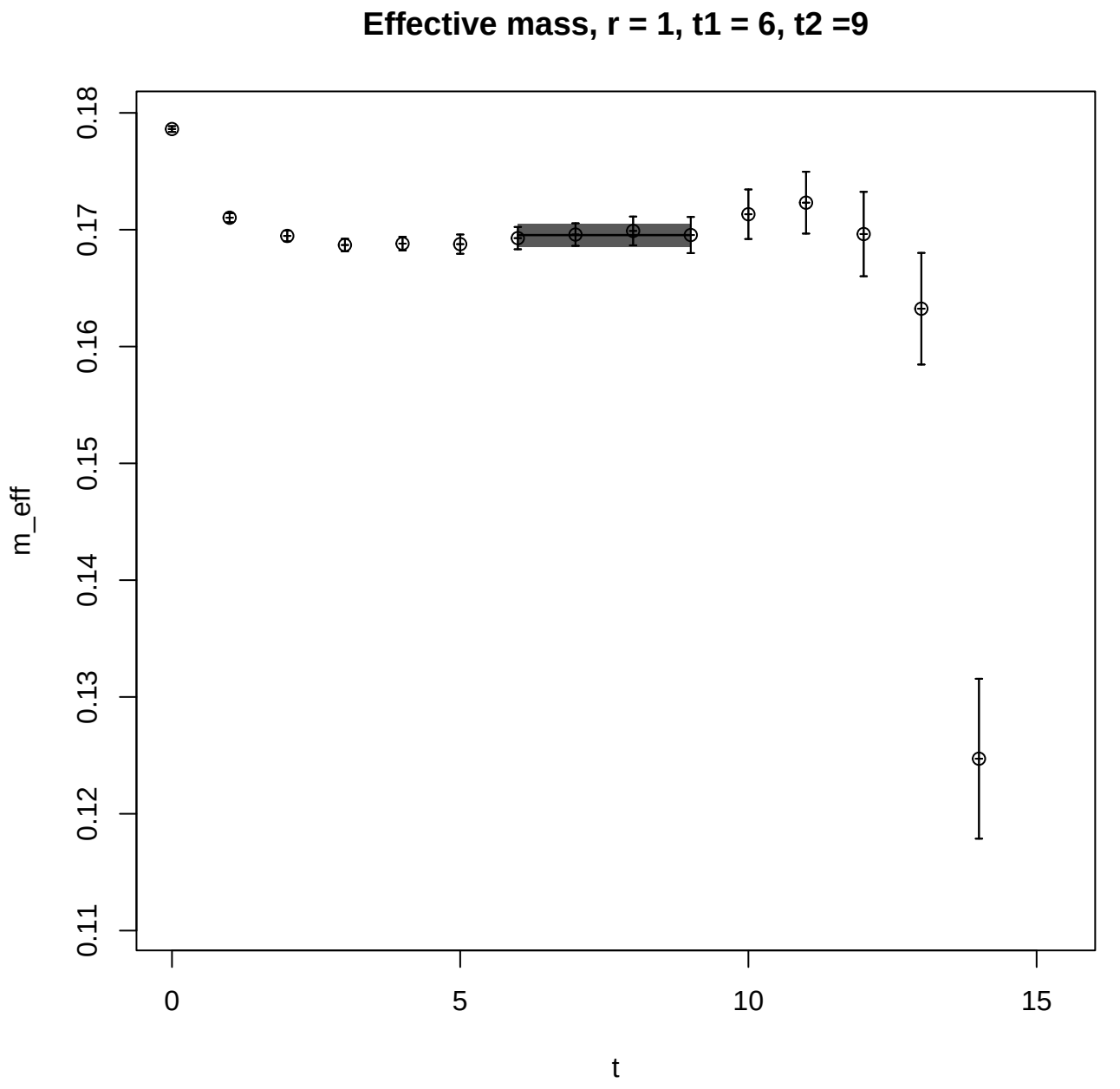

Effective mass, r = 1, t1 = 5, t2 =11
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

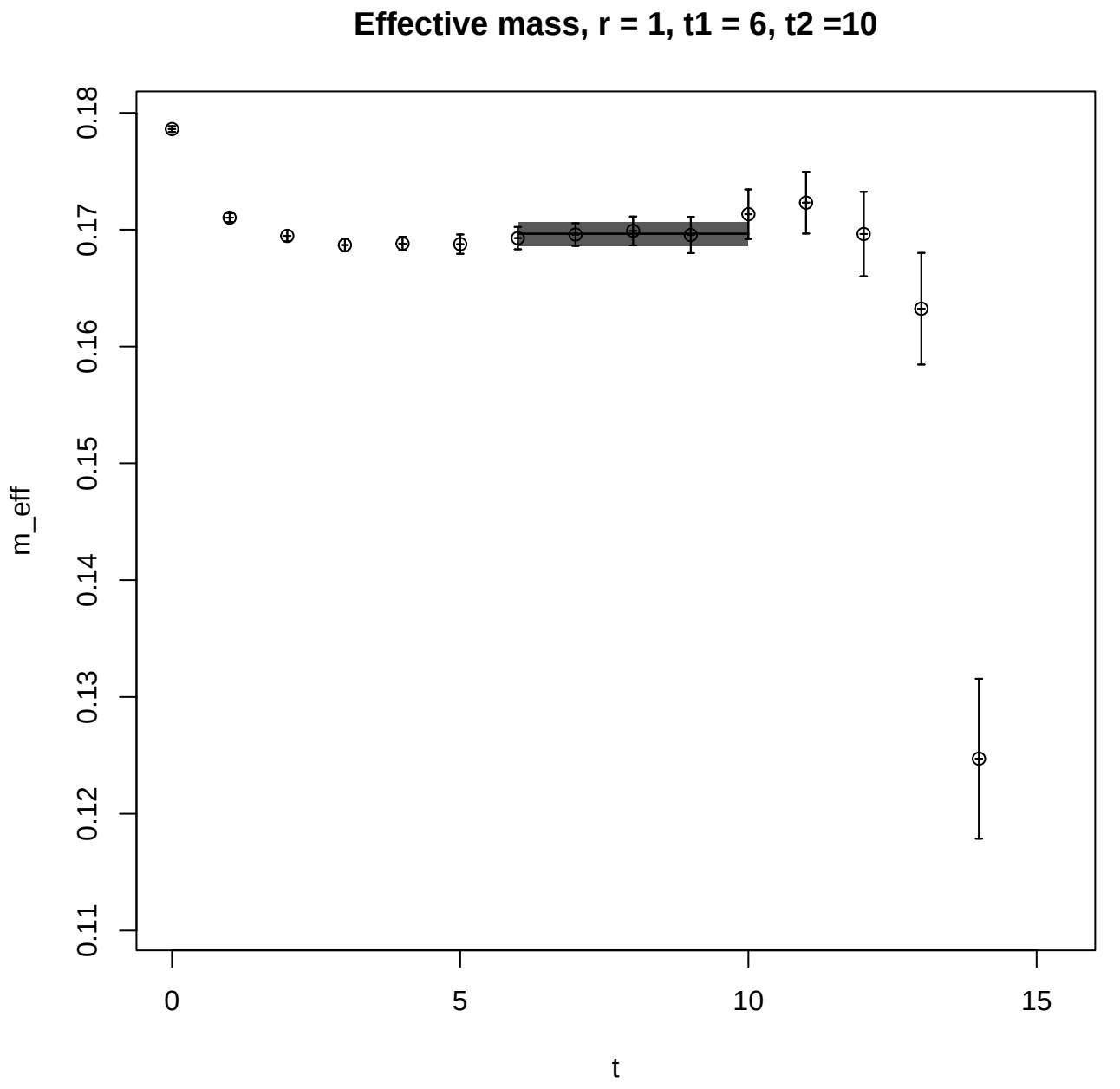

Effective mass, r = 1, t1 = 5, t2 =12
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

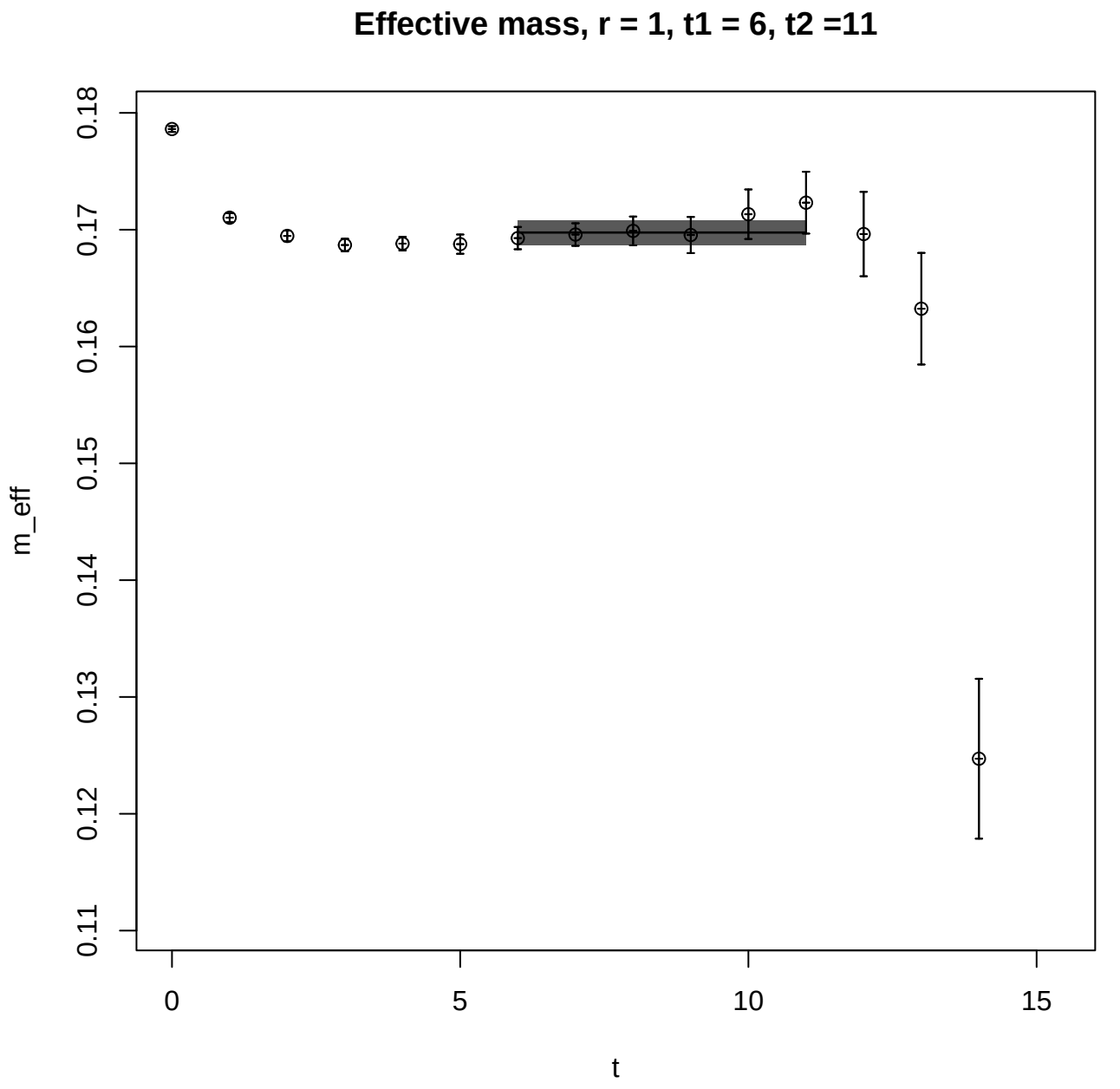

Effective mass, r = 1, t1 = 5, t2 =13
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

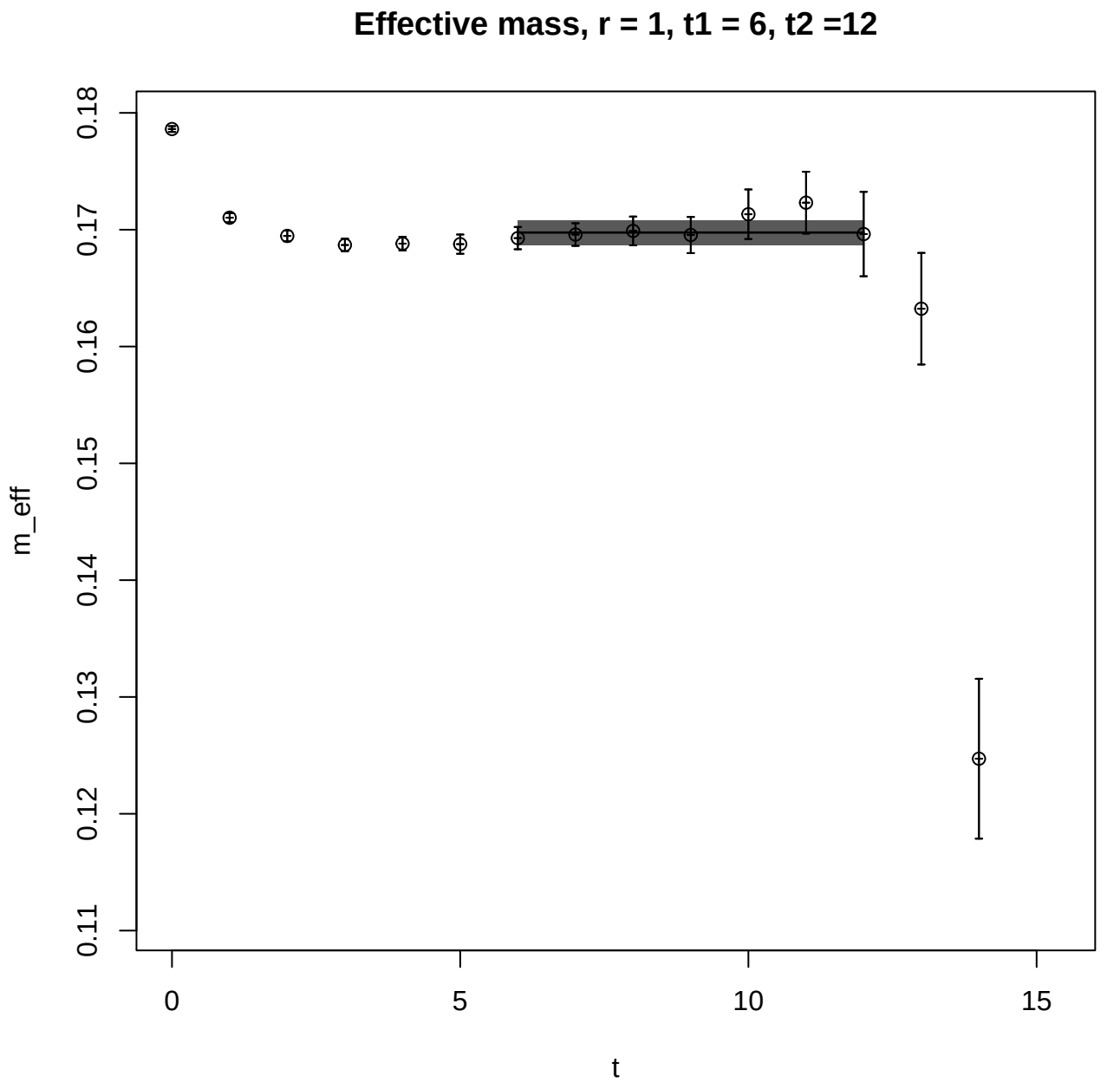

Effective mass, r = 1, t1 = 6, t2 =7
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

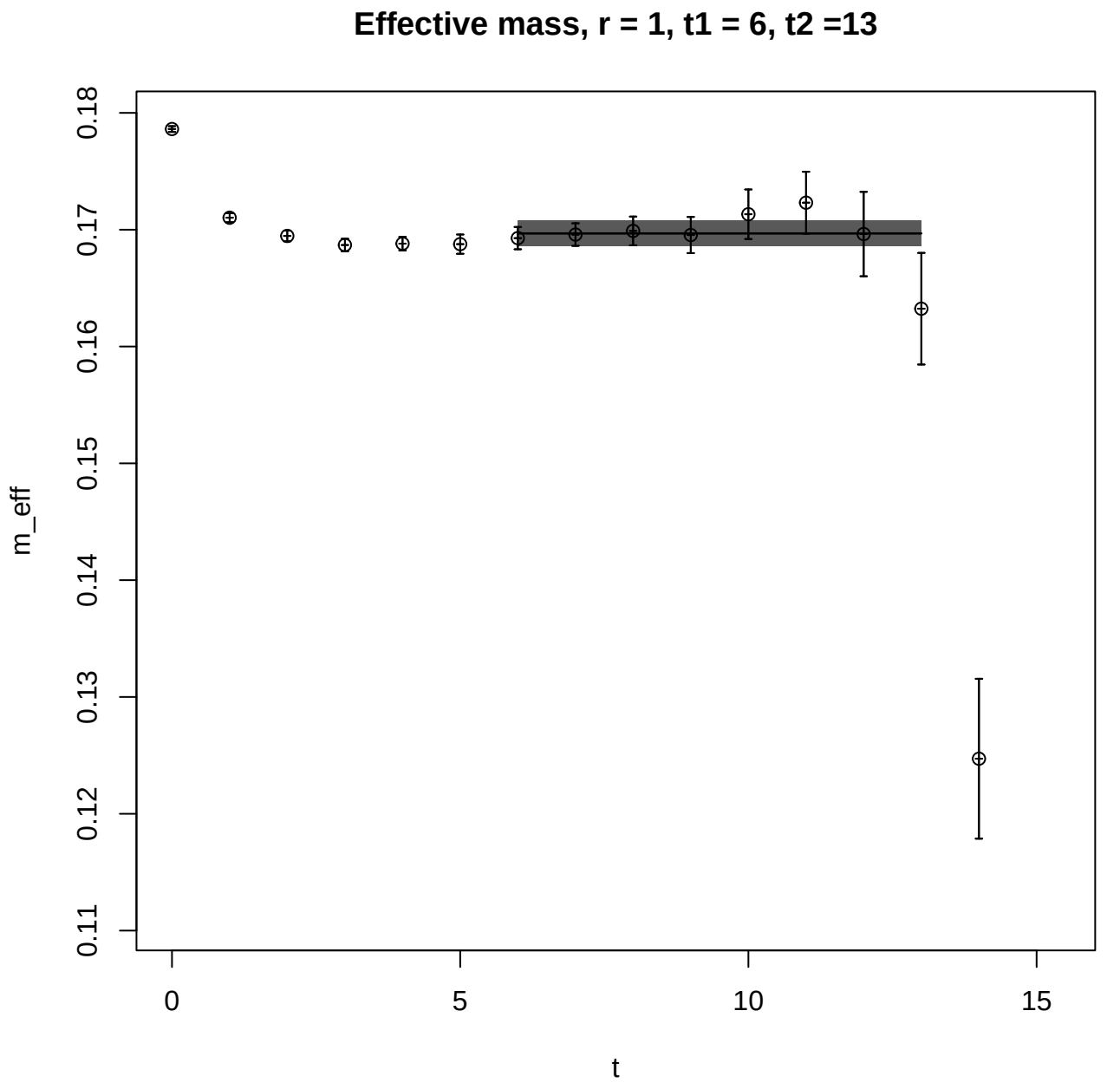

Effective mass, r = 1, t1 = 6, t2 =8
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

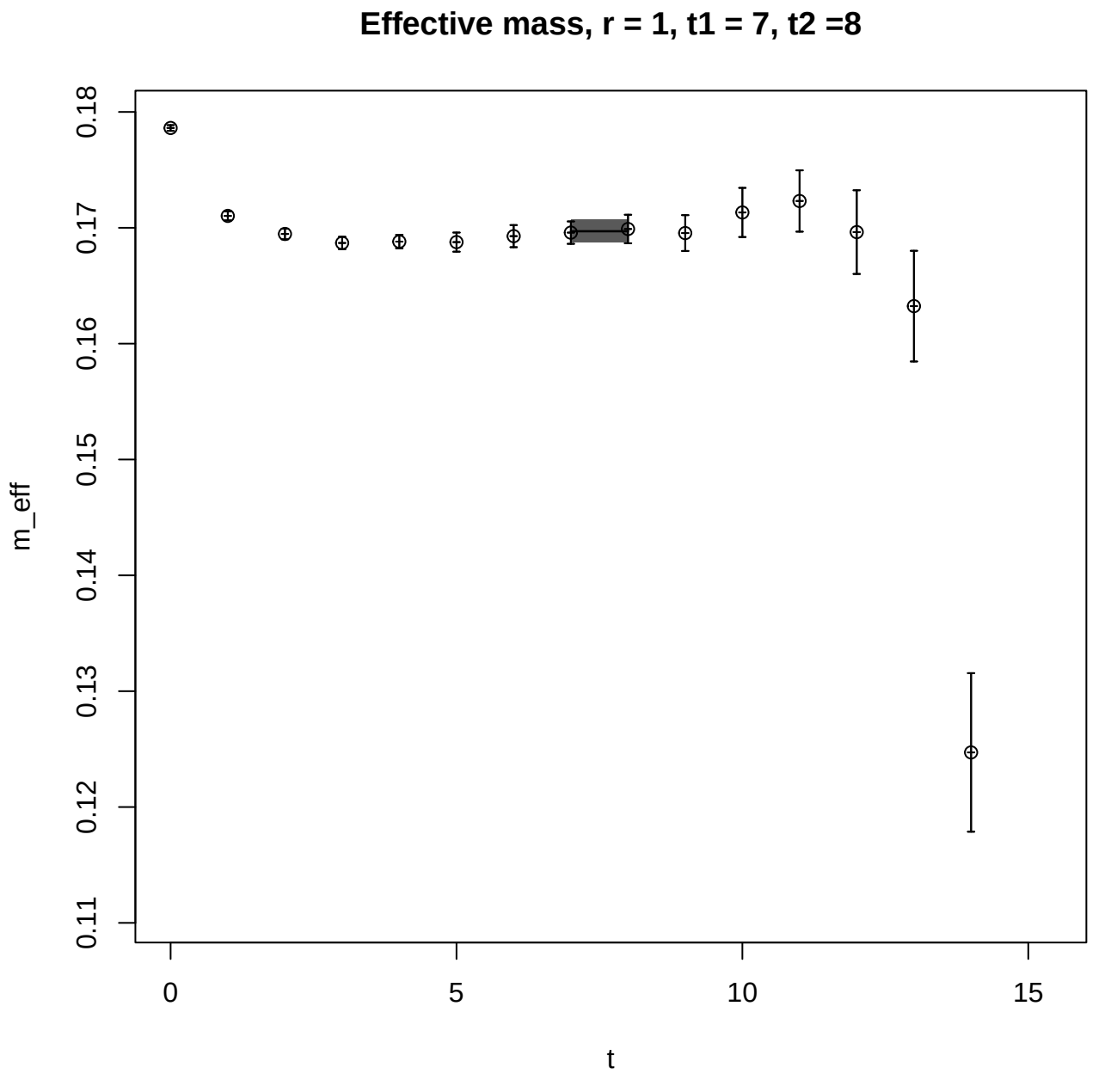

Effective mass, r = 1, t1 = 6, t2 =9
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

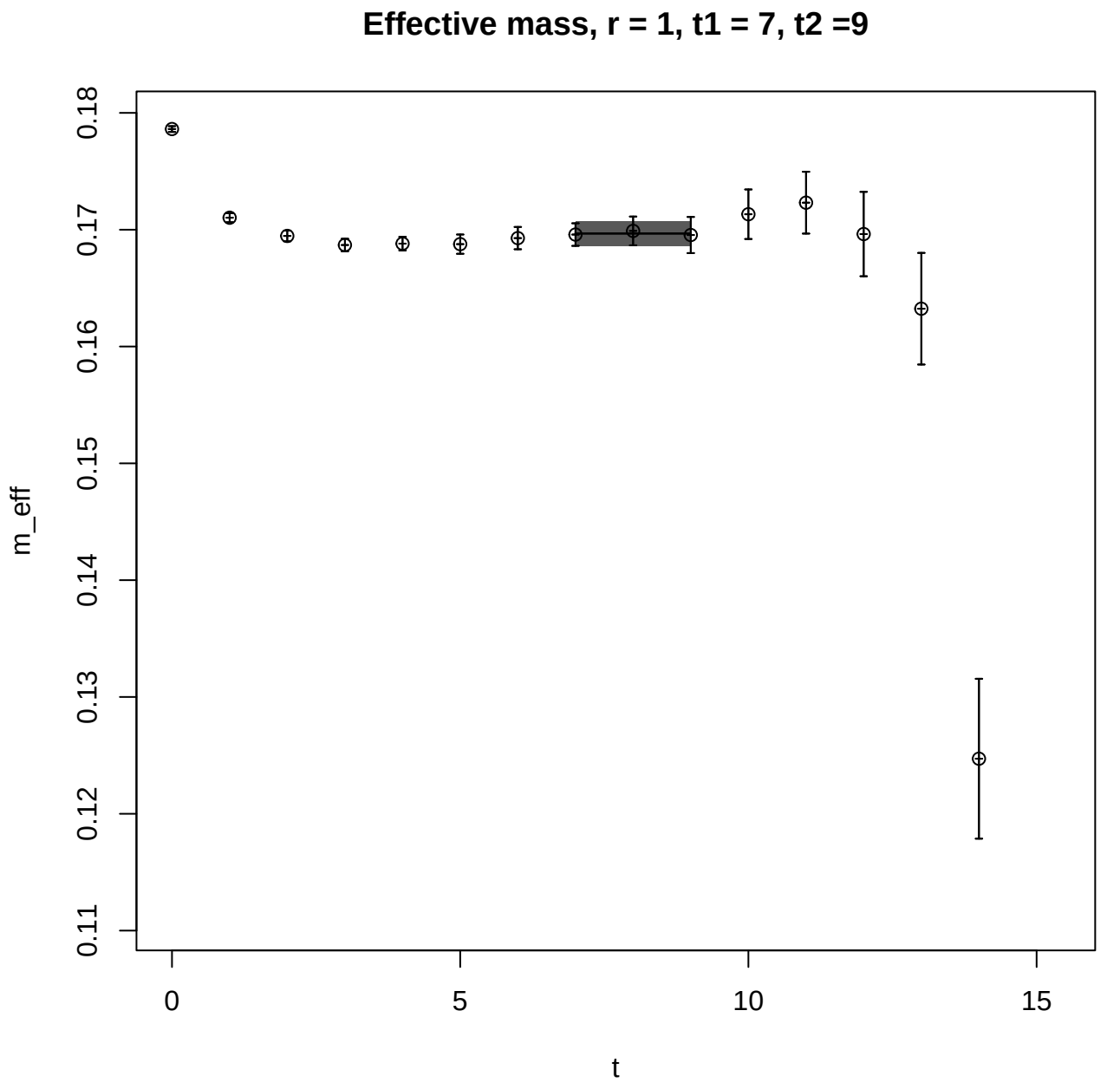

Effective mass, r = 1, t1 = 6, t2 =10
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

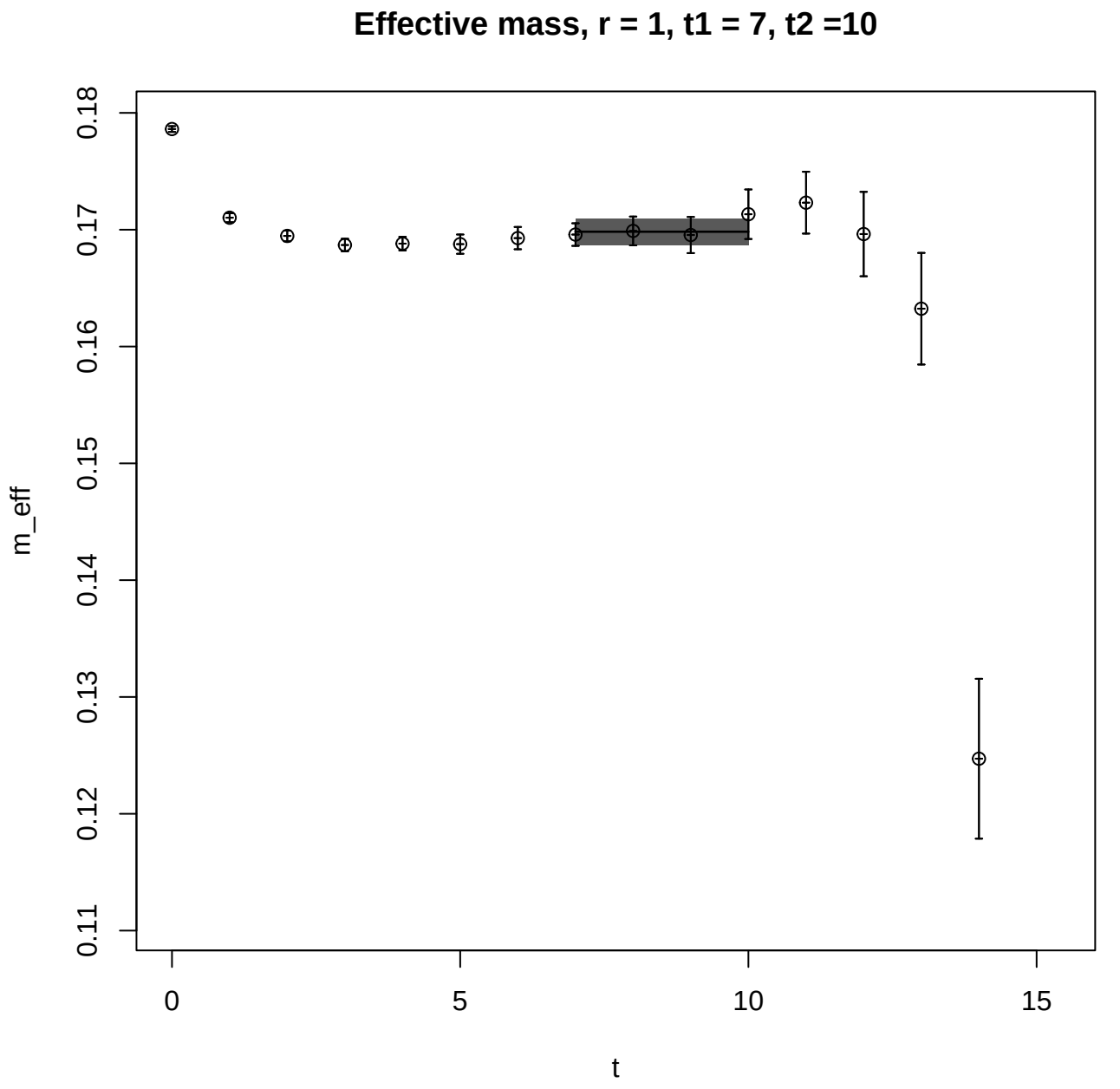

Effective mass, r = 1, t1 = 6, t2 =11
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

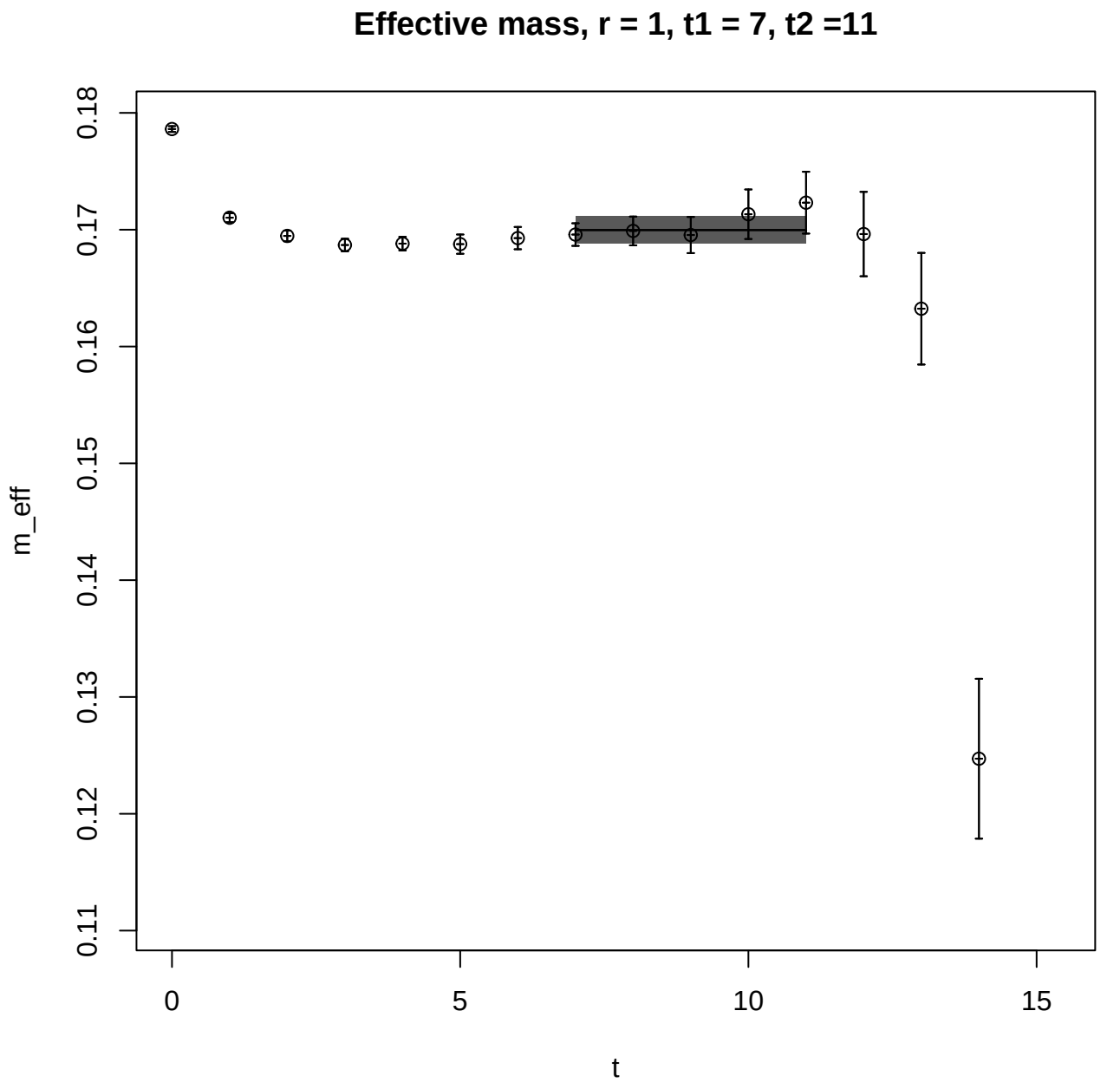

Effective mass, r = 1, t1 = 6, t2 =12
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

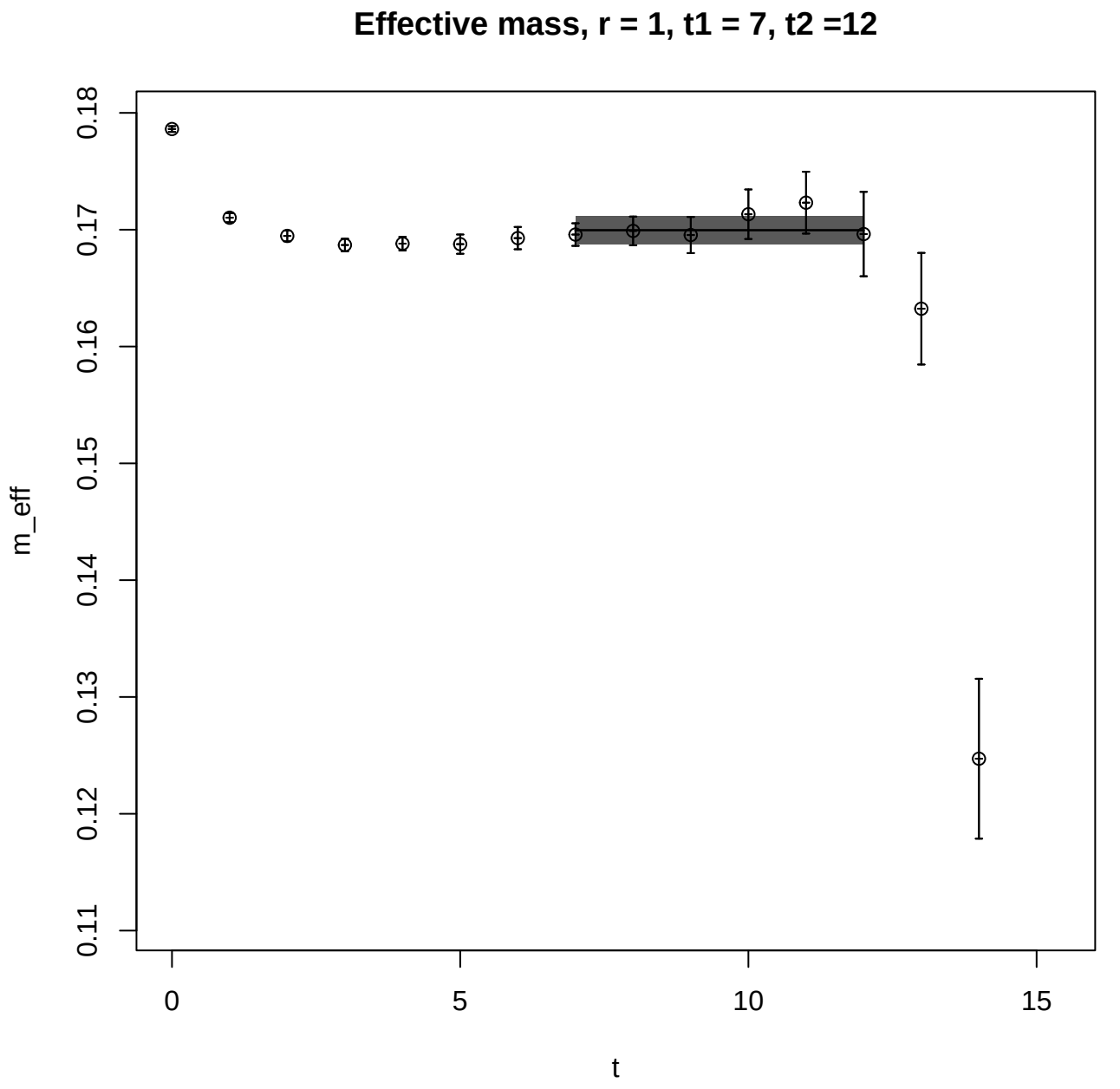

Effective mass, r = 1, t1 = 6, t2 =13
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

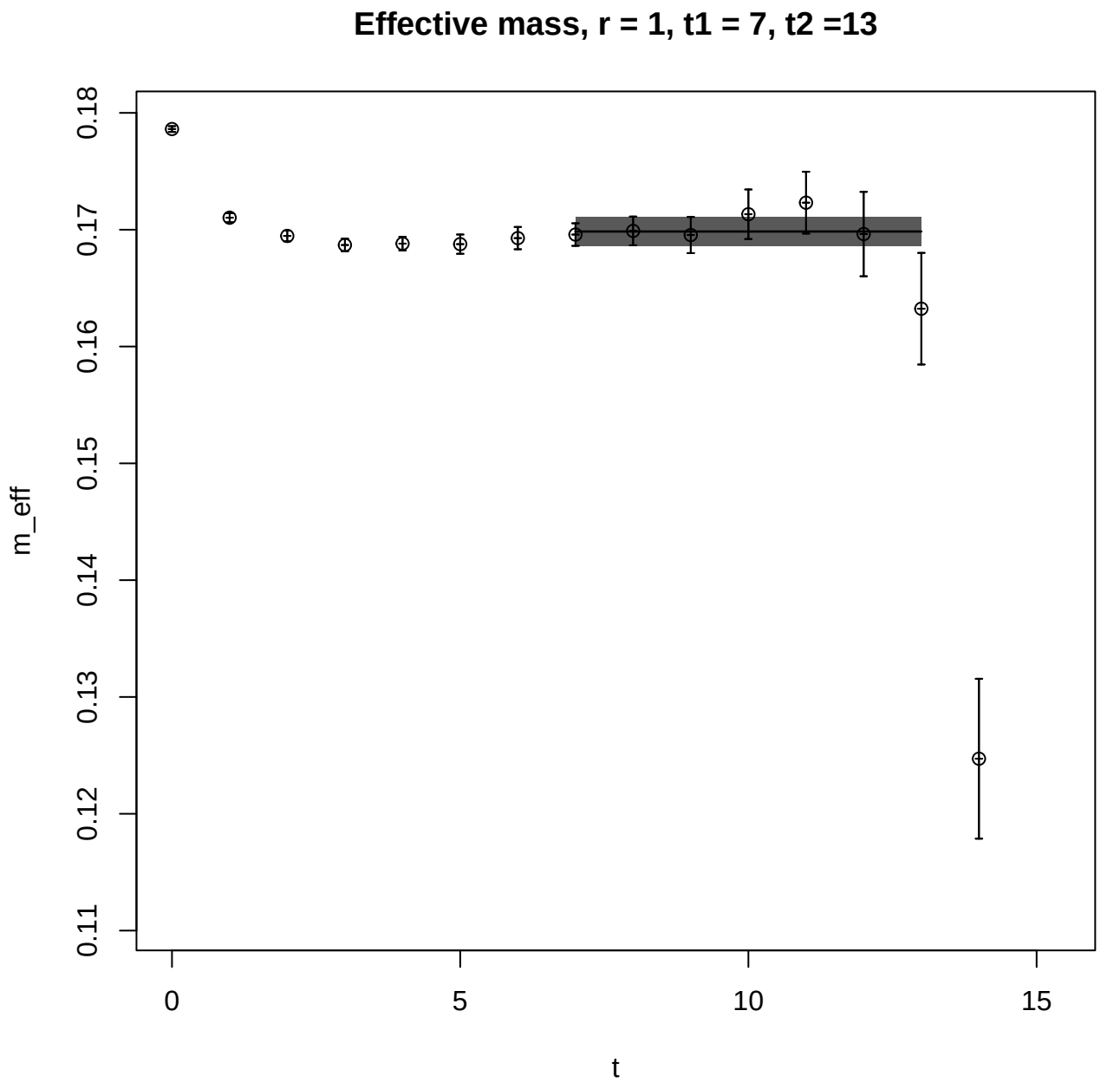

Effective mass, r = 1, t1 = 7, t2 =8
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

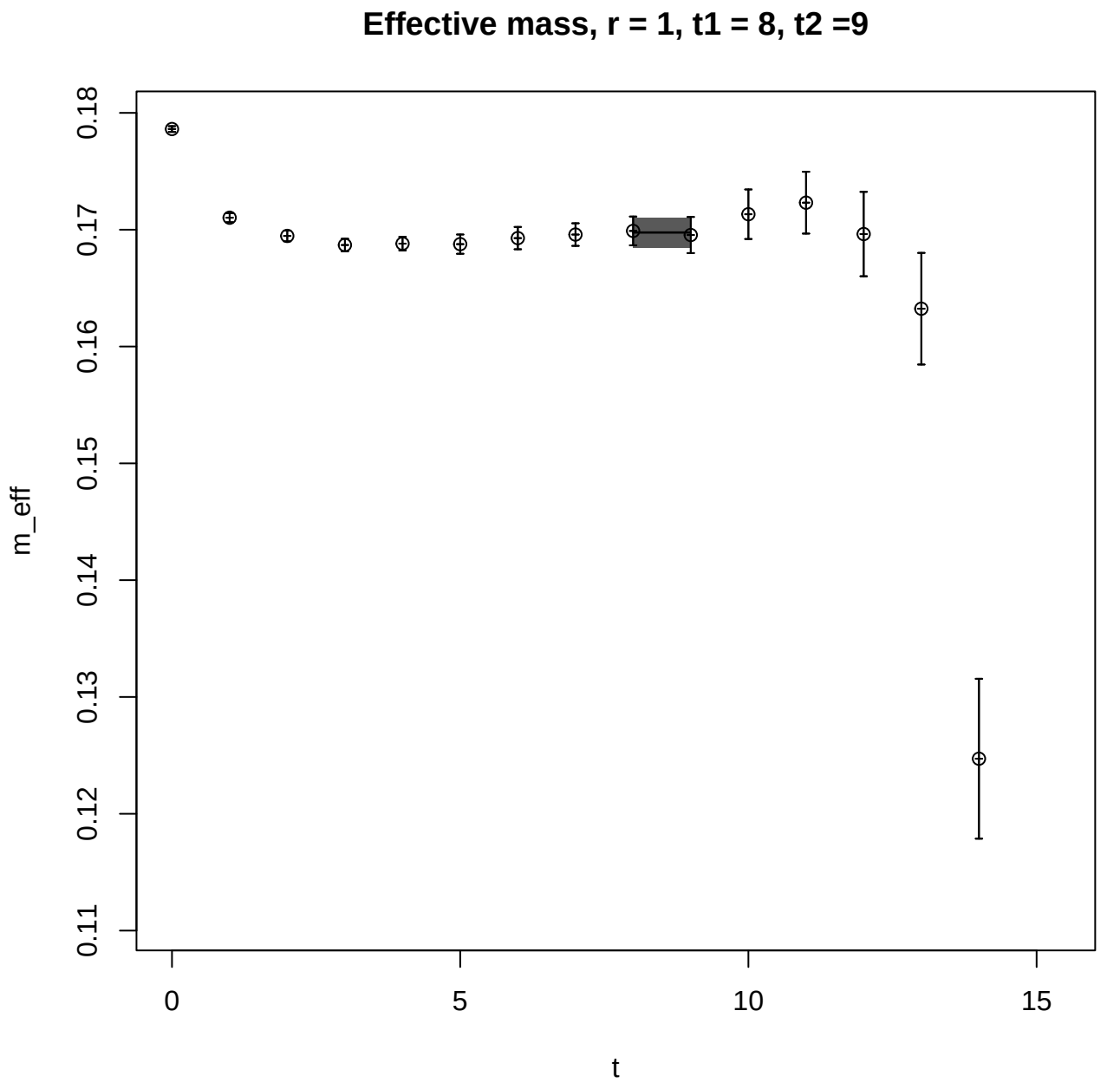

Effective mass, r = 1, t1 = 7, t2 =9
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =10
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =11
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =12
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 7, t2 =13
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =9
m_eff
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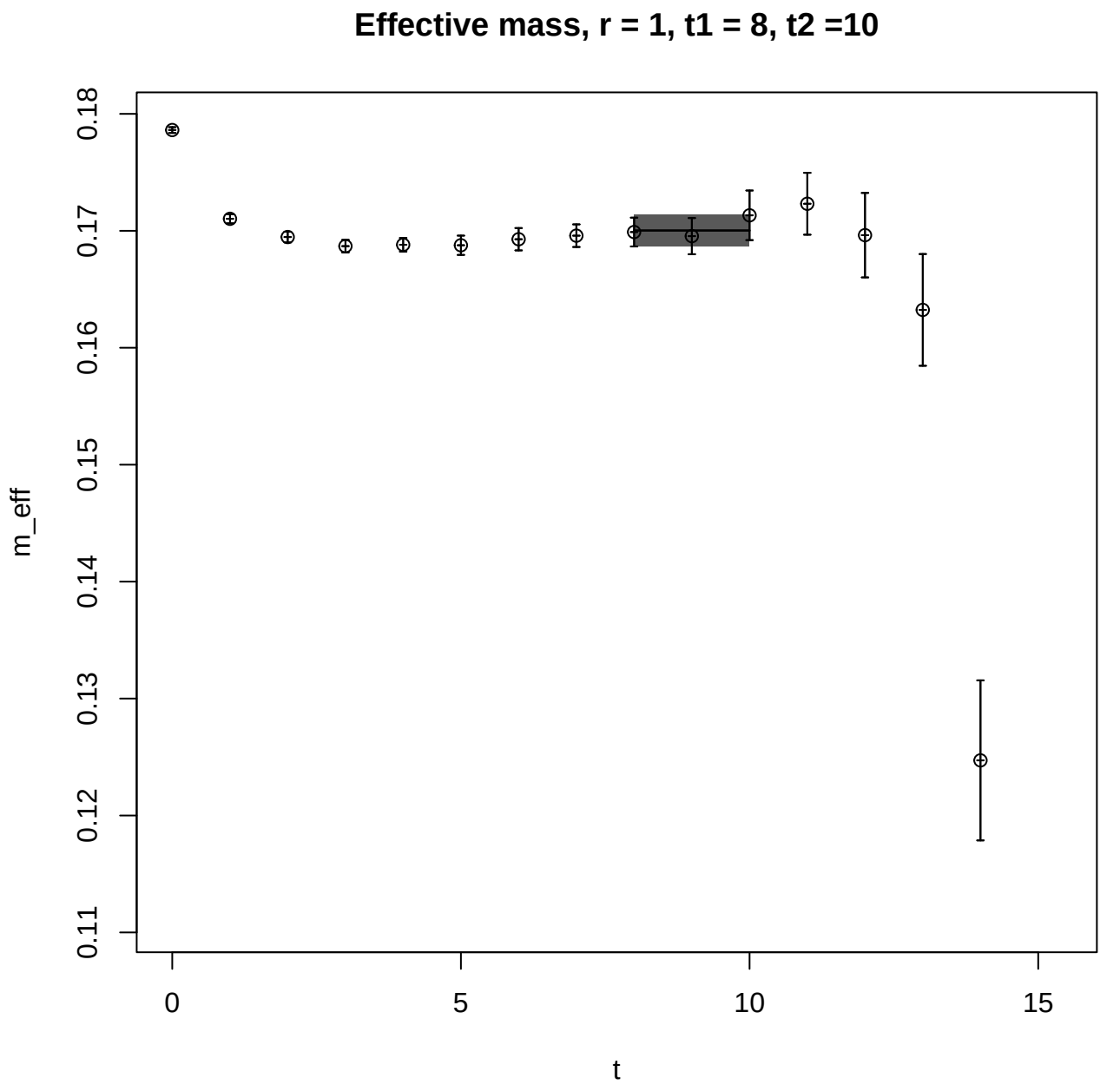
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =10
m_eff
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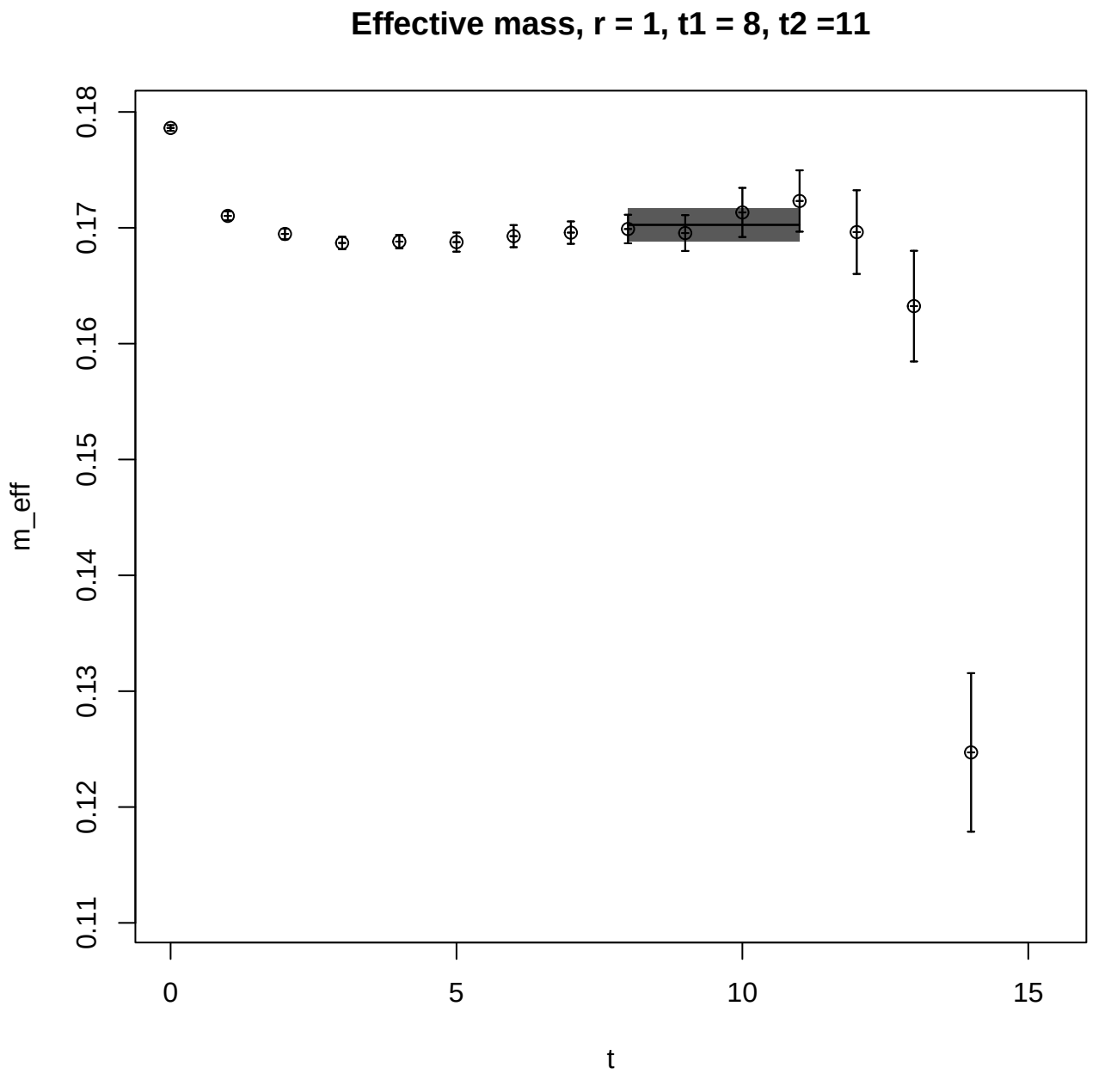
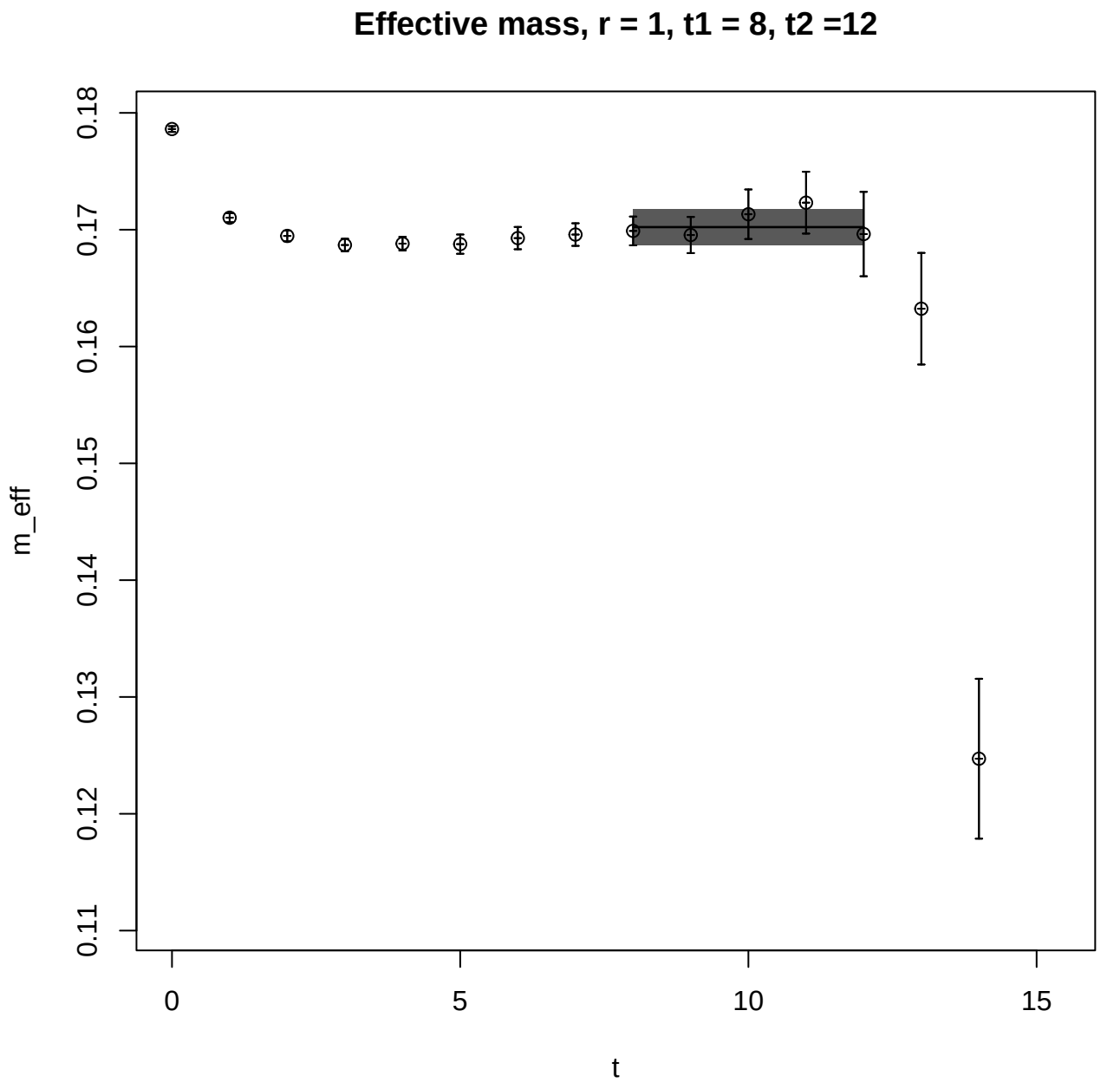
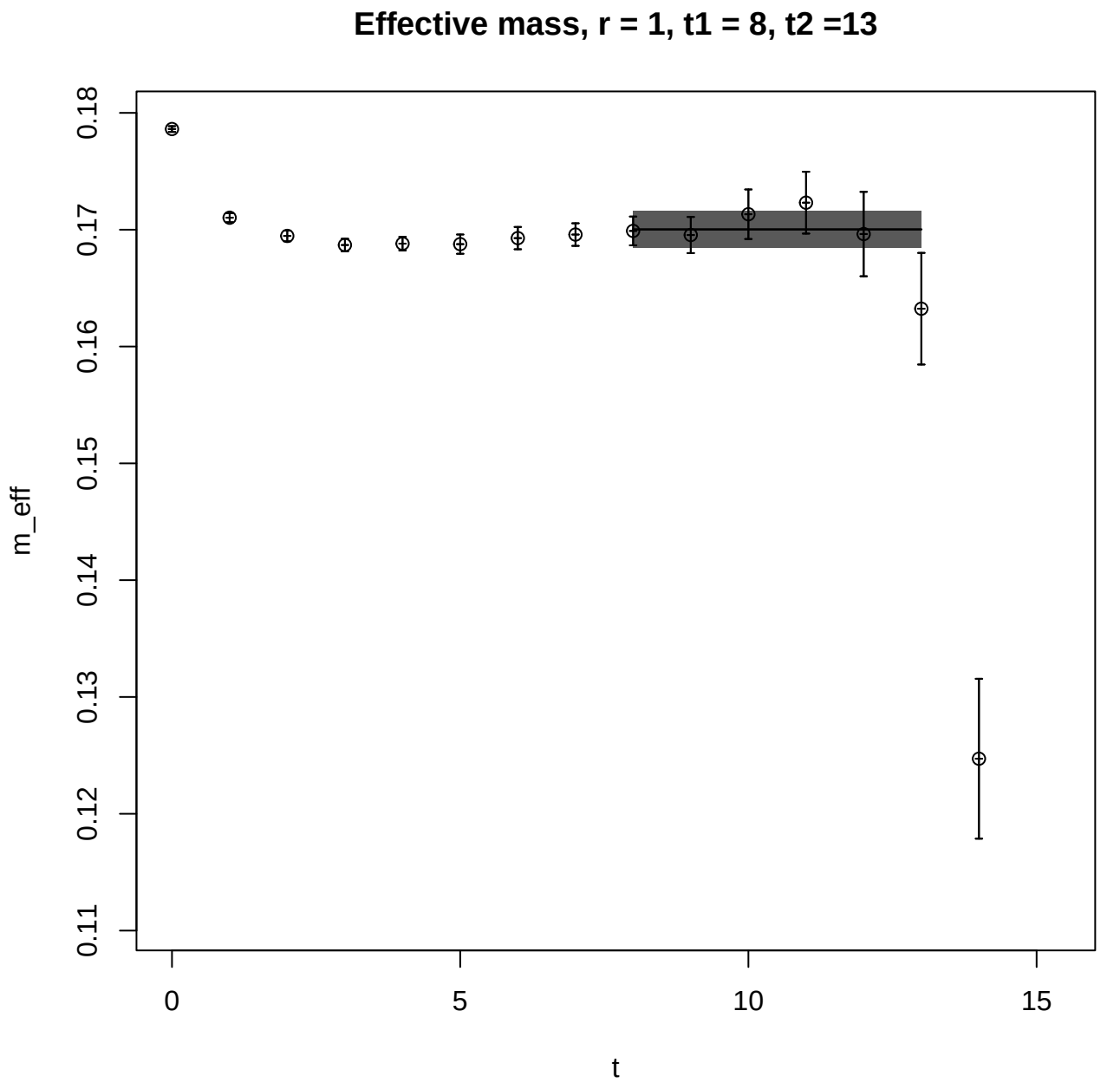
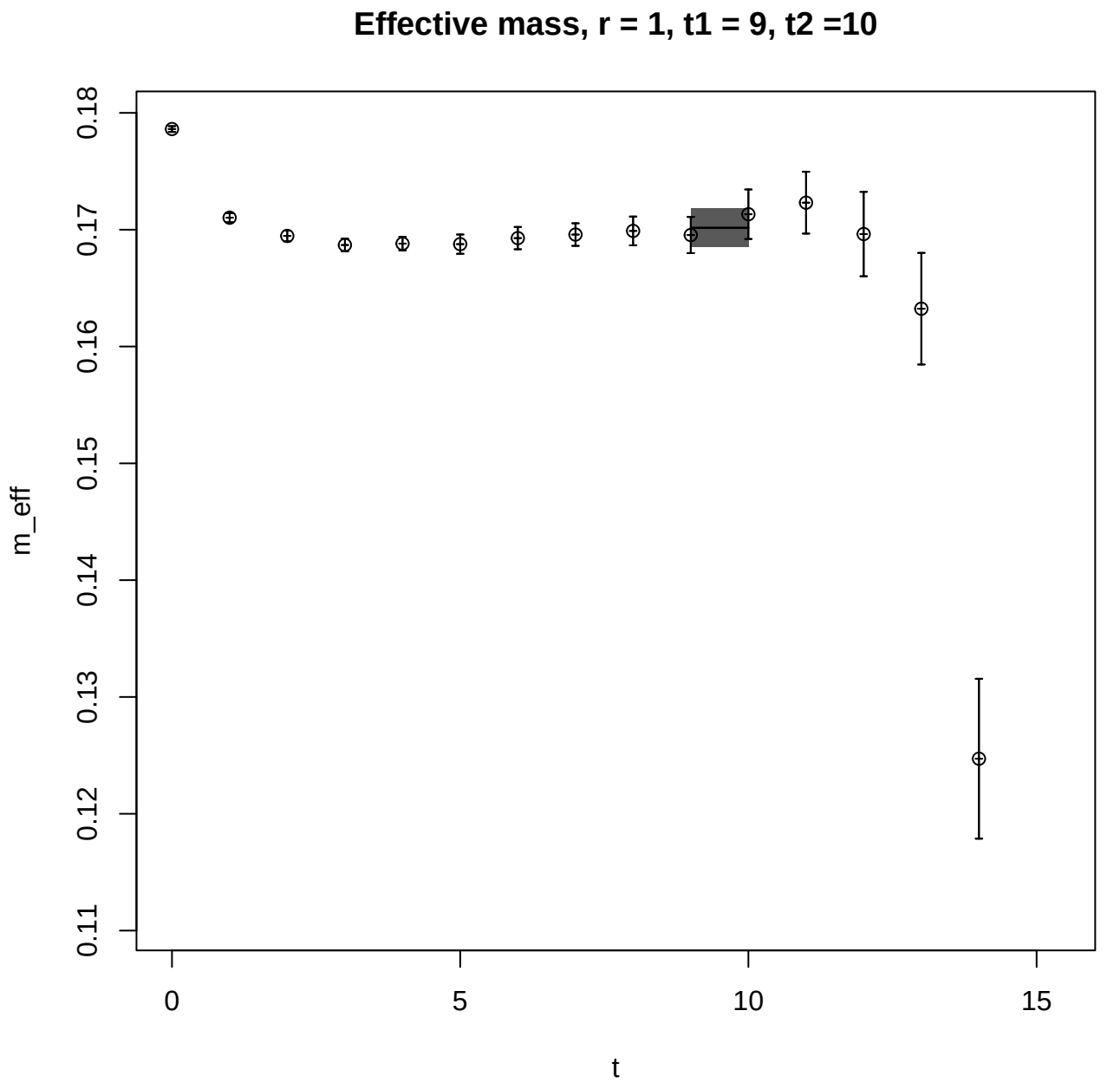
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =11
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =12
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 8, t2 =13
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =10
m_eff
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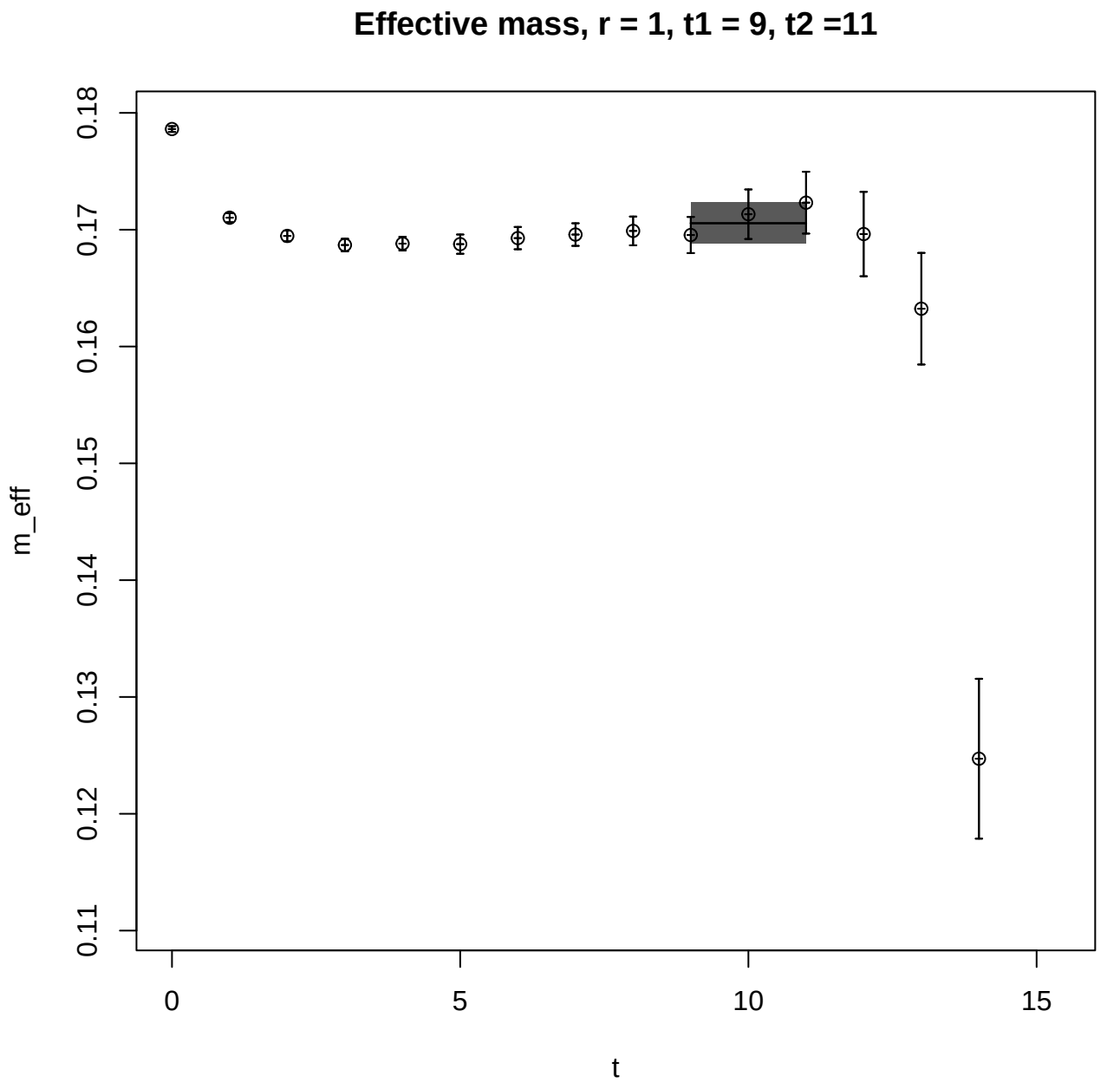
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =11
m_eff
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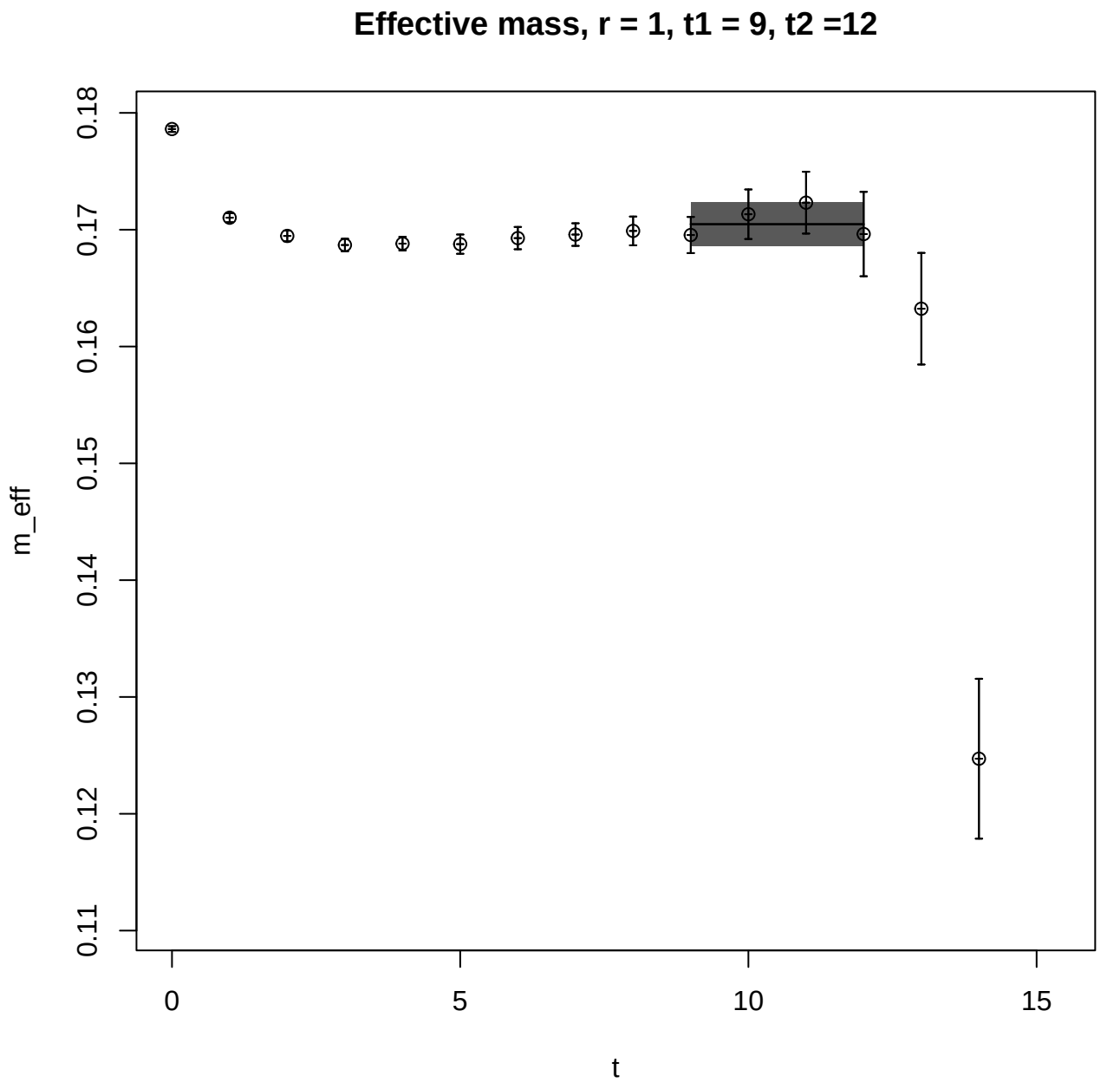
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =12
m_eff
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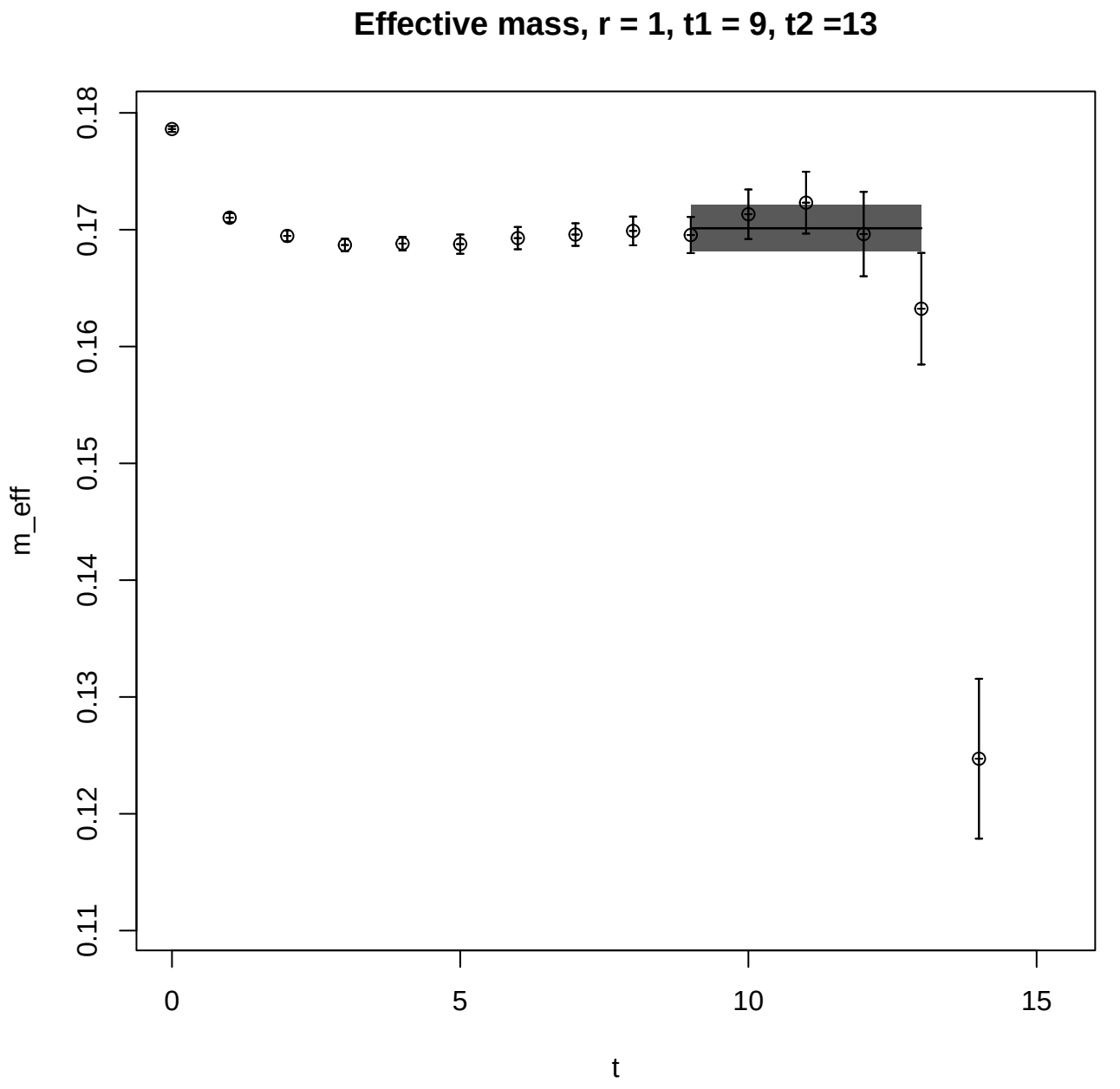
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t


Effective mass, r = 1, t1 = 9, t2 =13
m_eff
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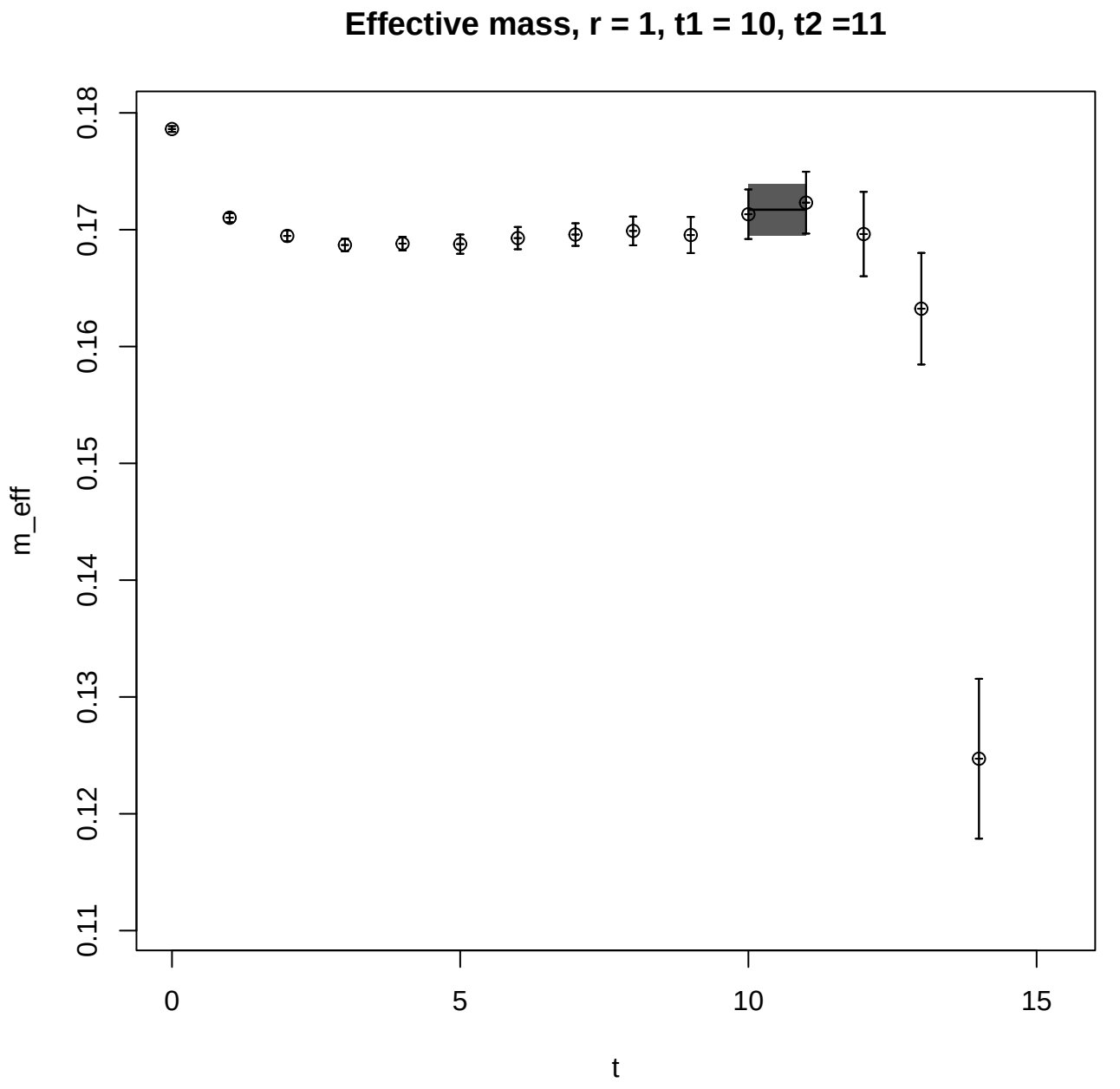
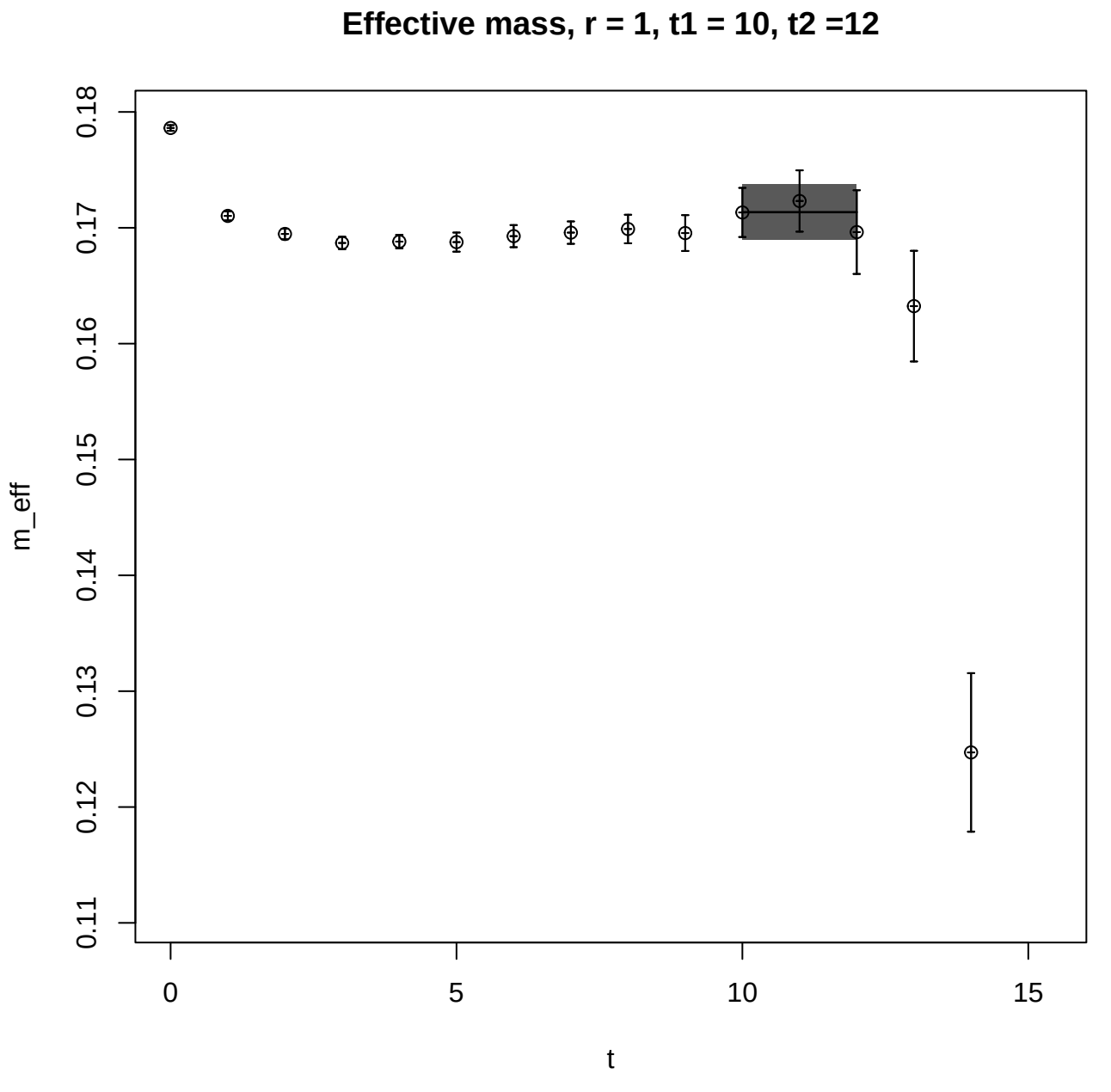
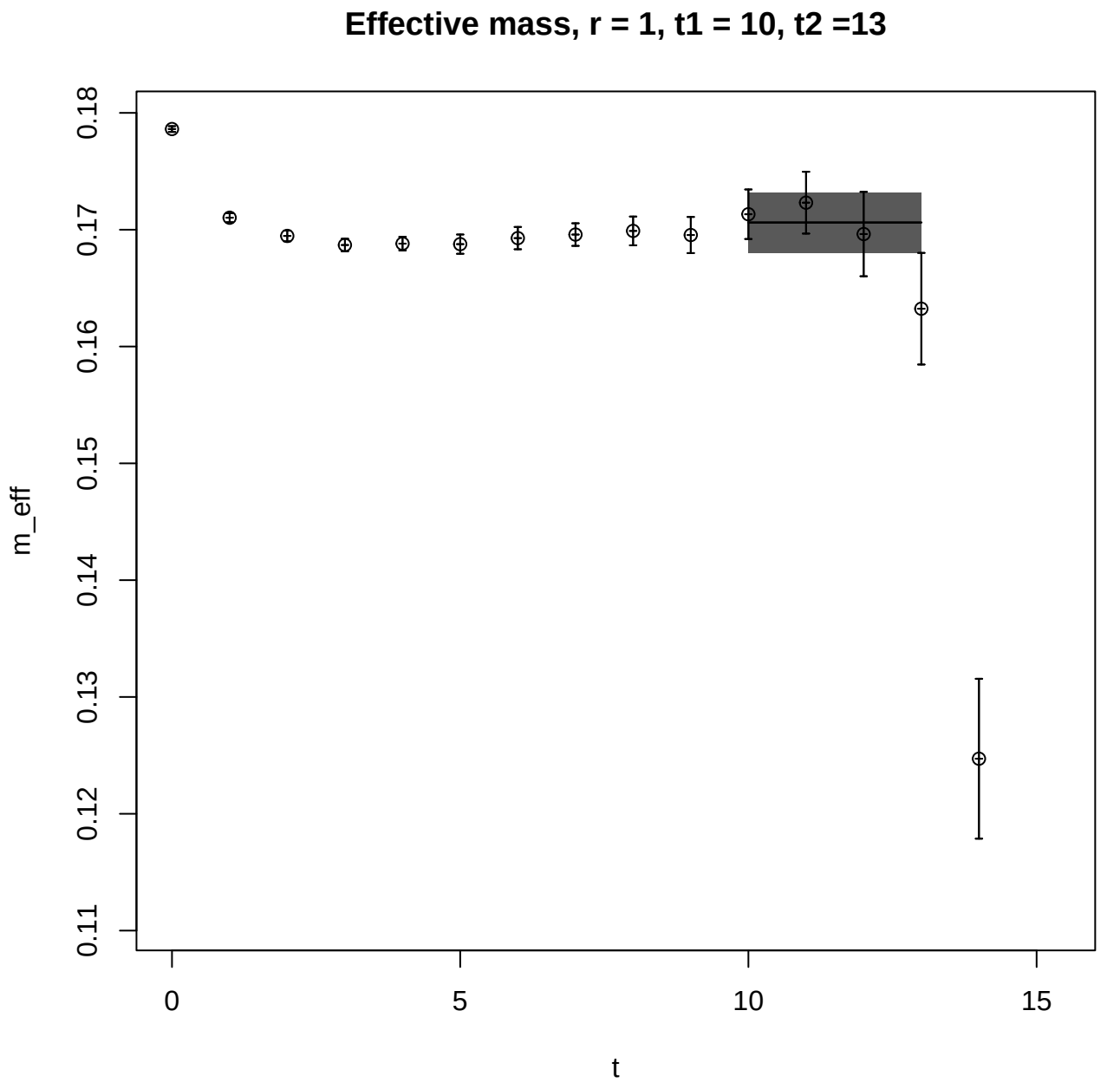
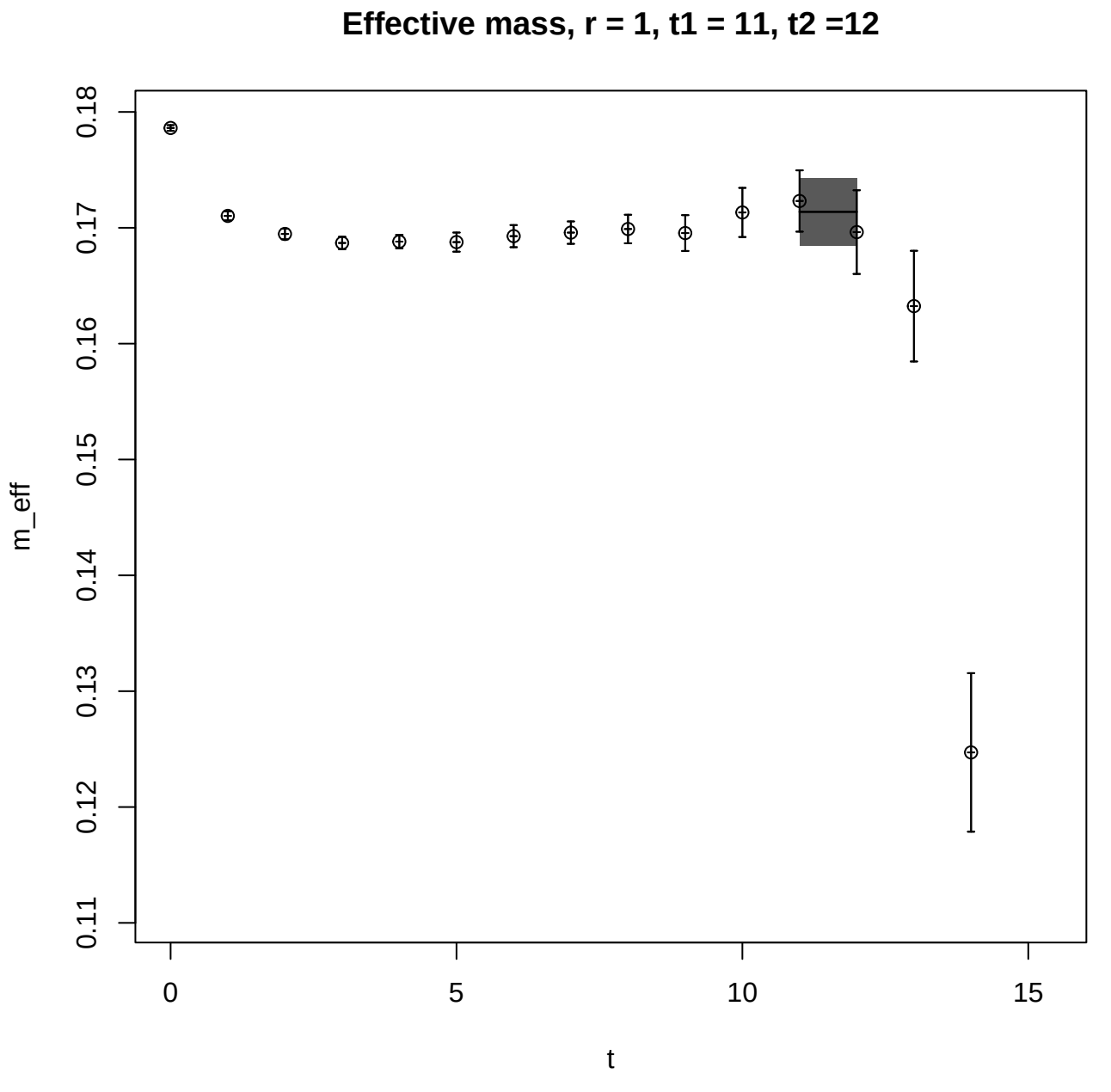
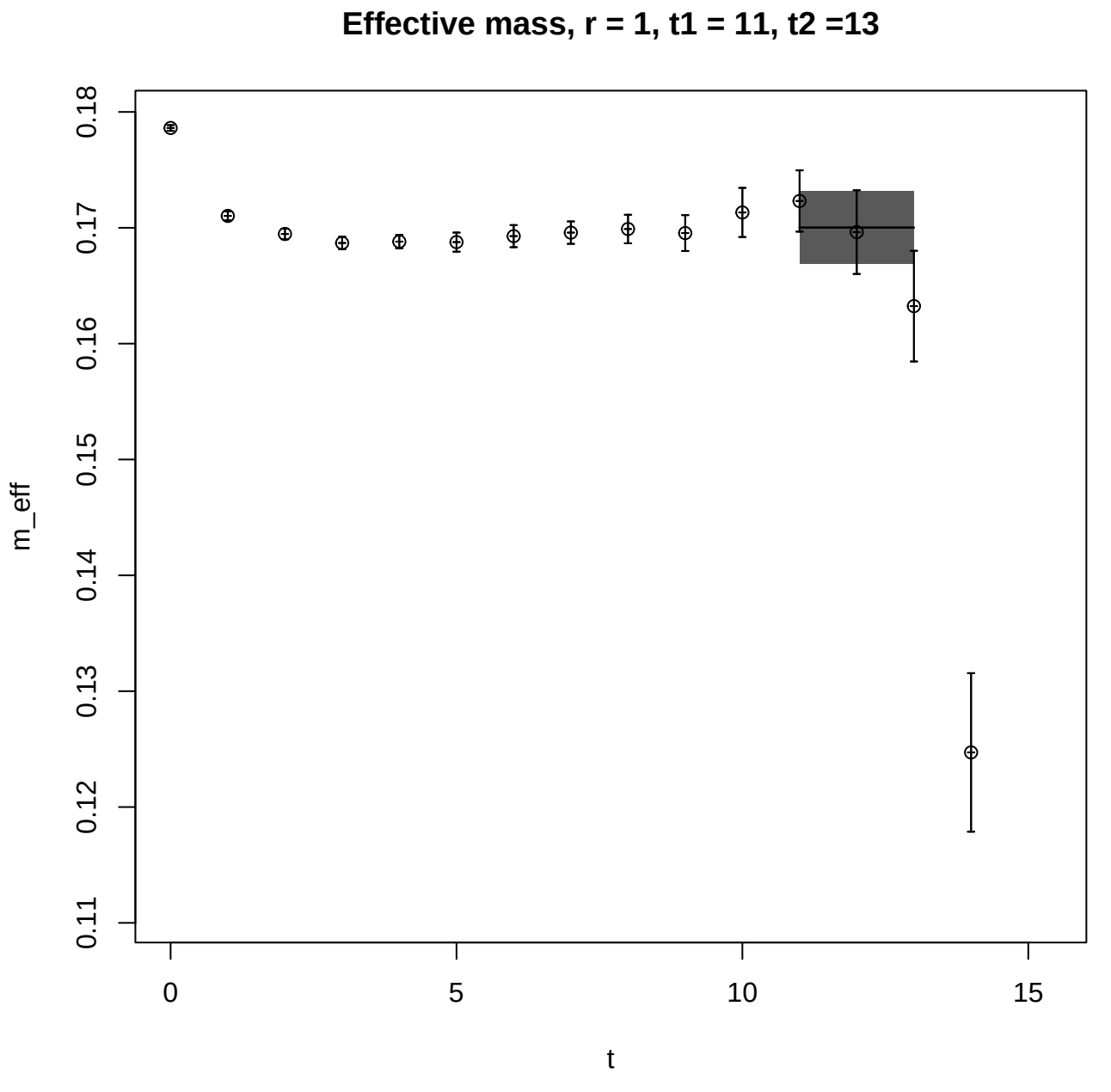
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

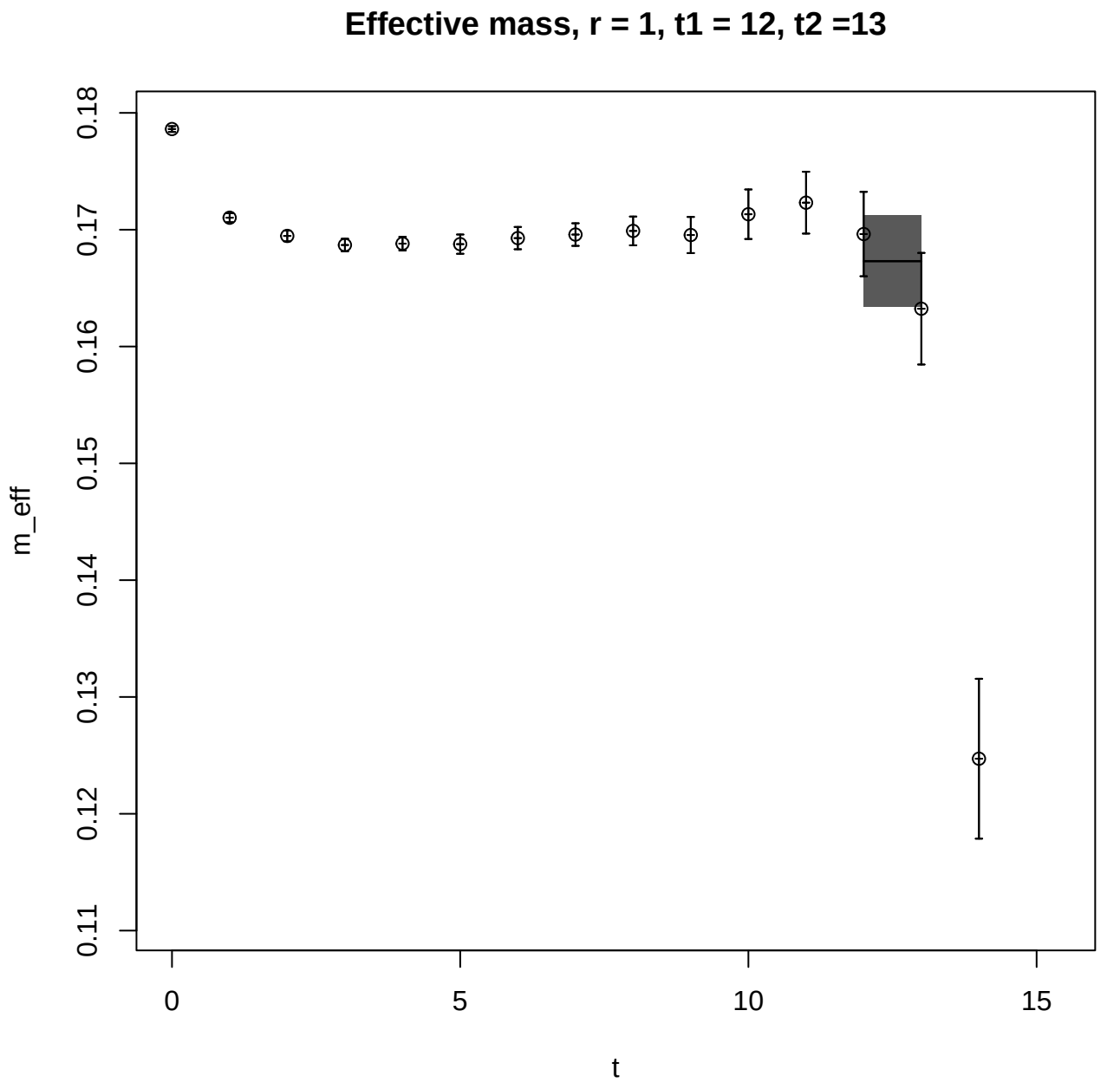

Effective mass, r = 1, t1 = 10, t2 =11
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

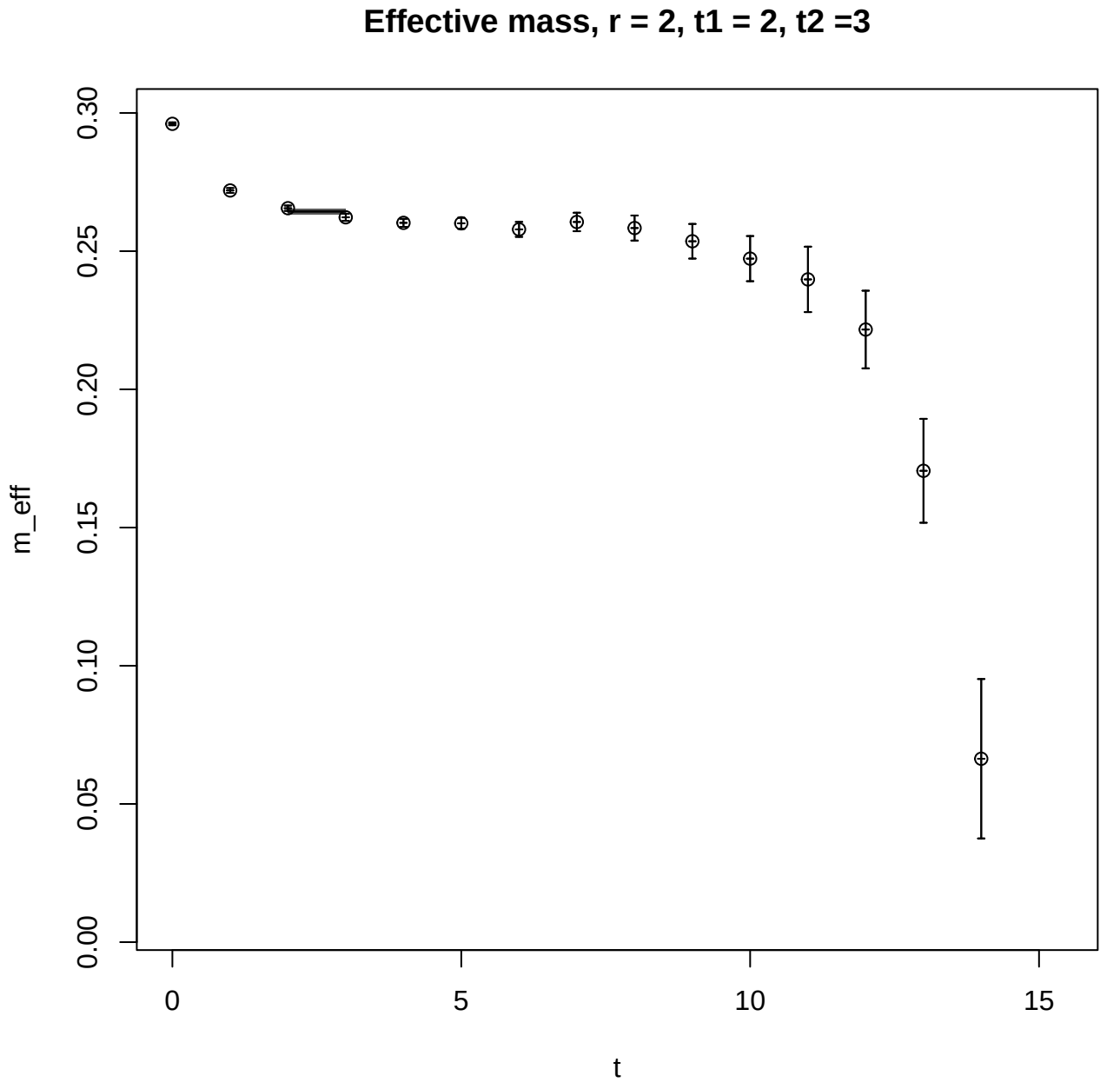

Effective mass, r = 1, t1 = 10, t2 =12
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

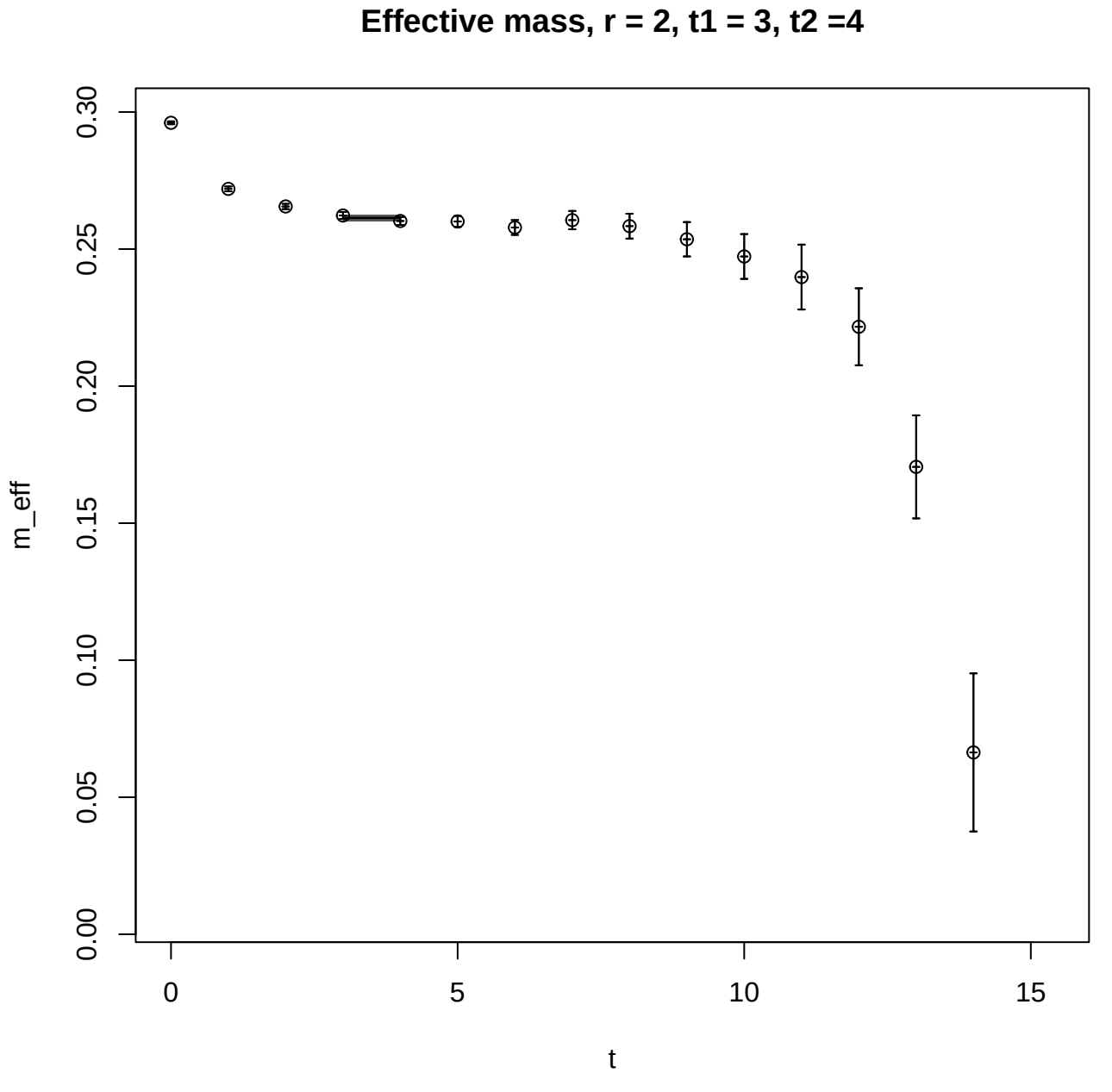

Effective mass, r = 1, t1 = 10, t2 =13
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

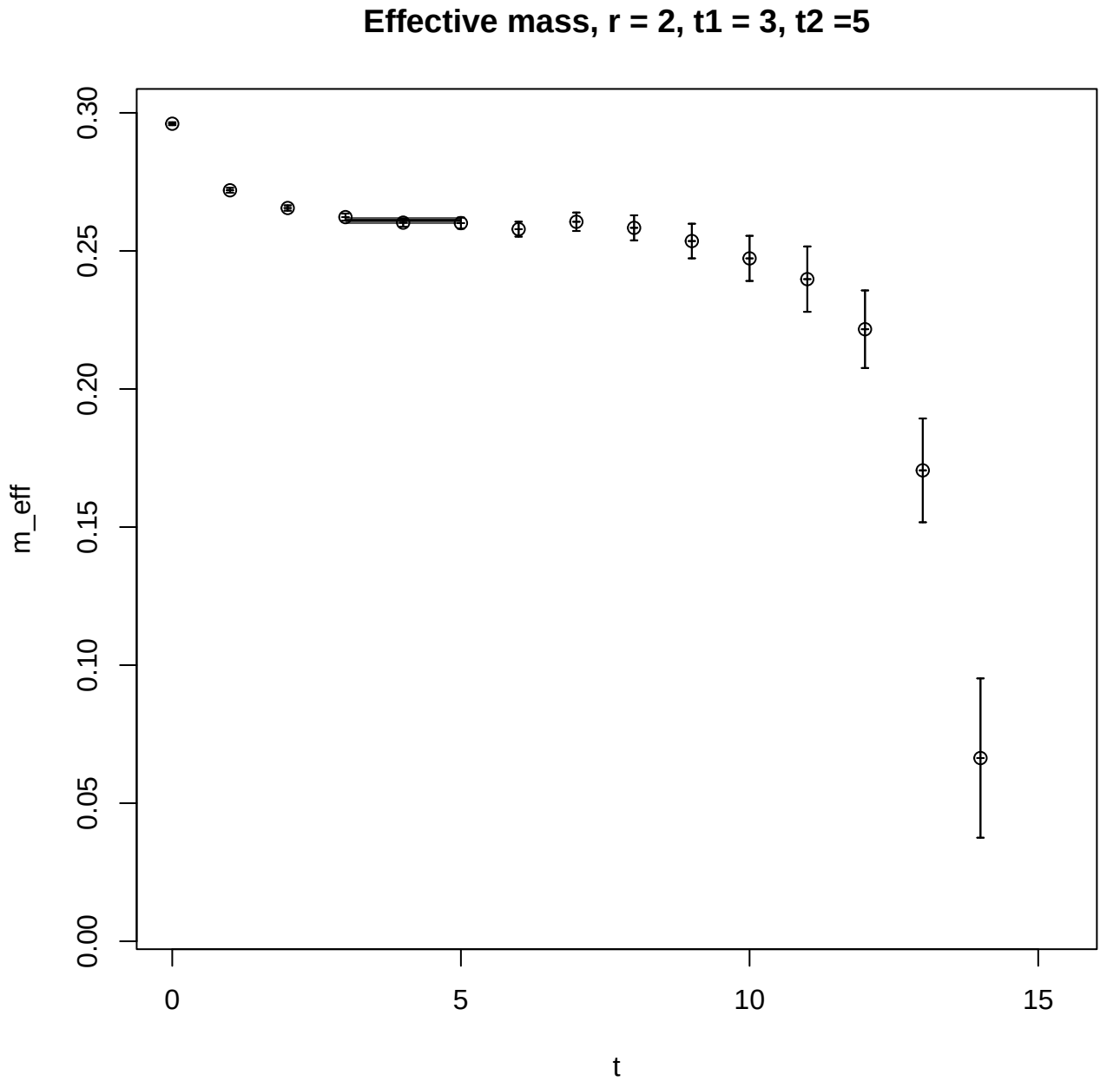

Effective mass, r = 1, t1 = 11, t2 =12
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

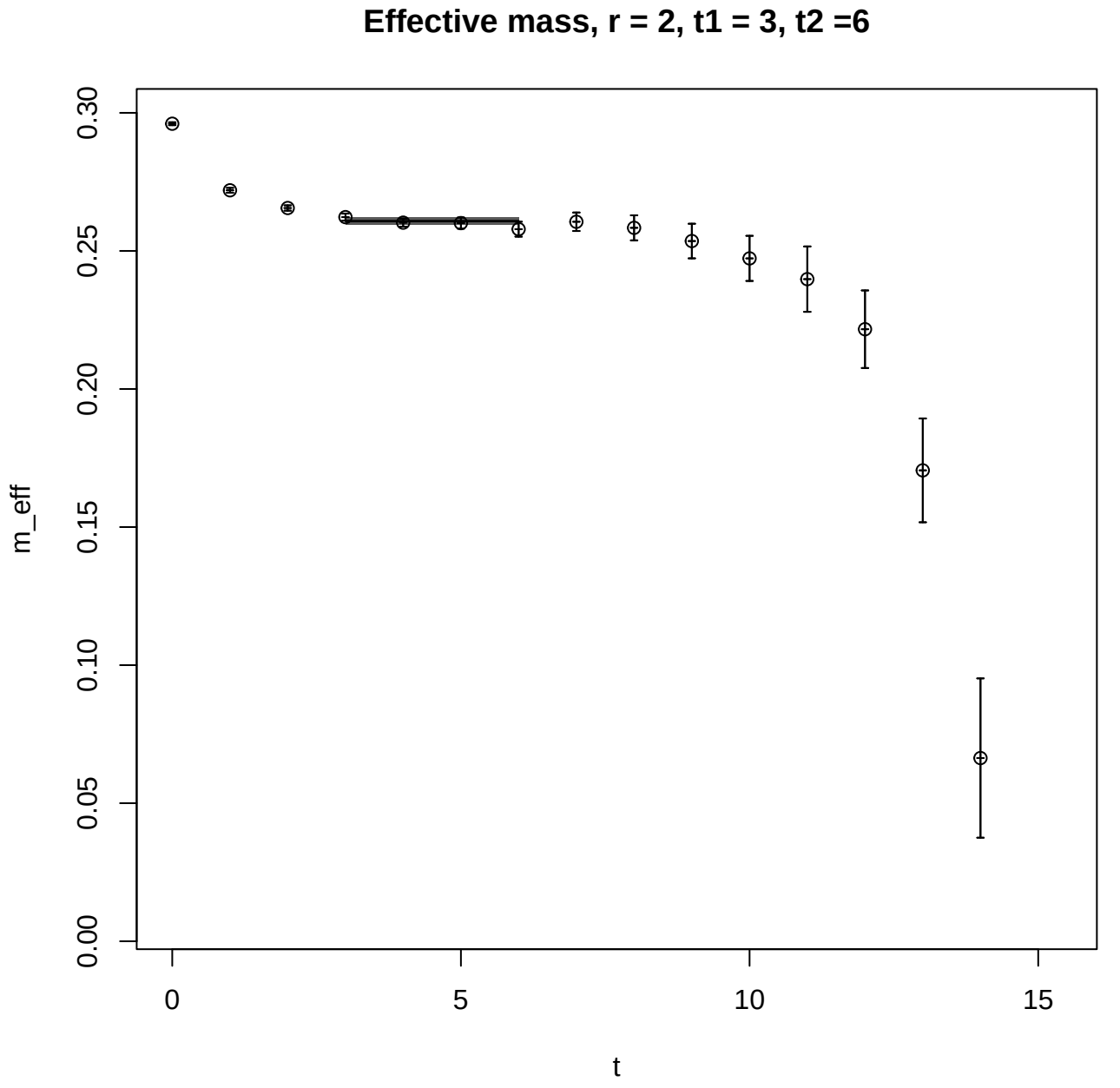

Effective mass, r = 1, t1 = 11, t2 =13
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

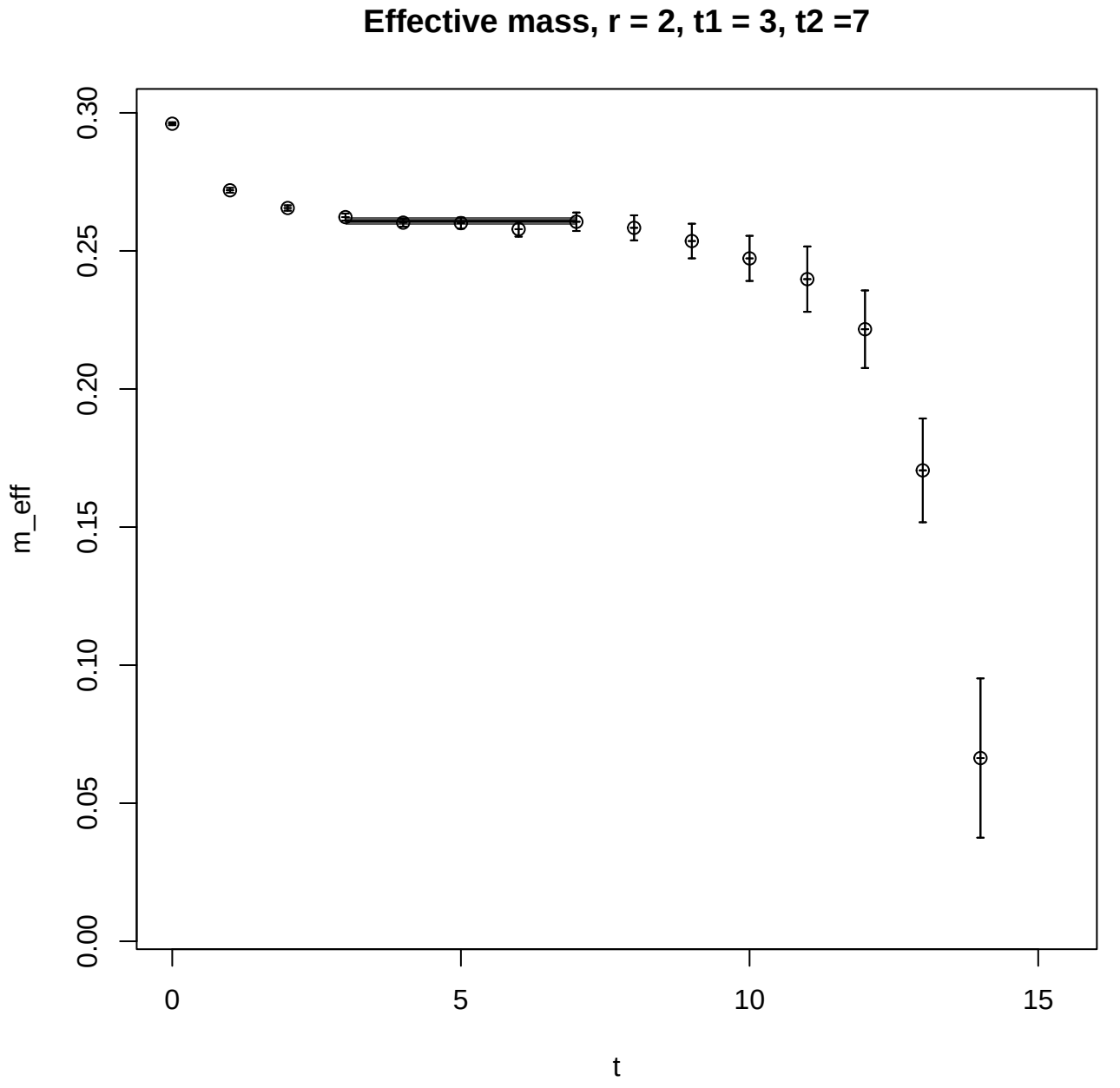

Effective mass, r = 1, t1 = 12, t2 =13
m_eff
0.11
0.12
0.13
0.14
0.15
0.16
0.17
0.18
0
5
10
15
t

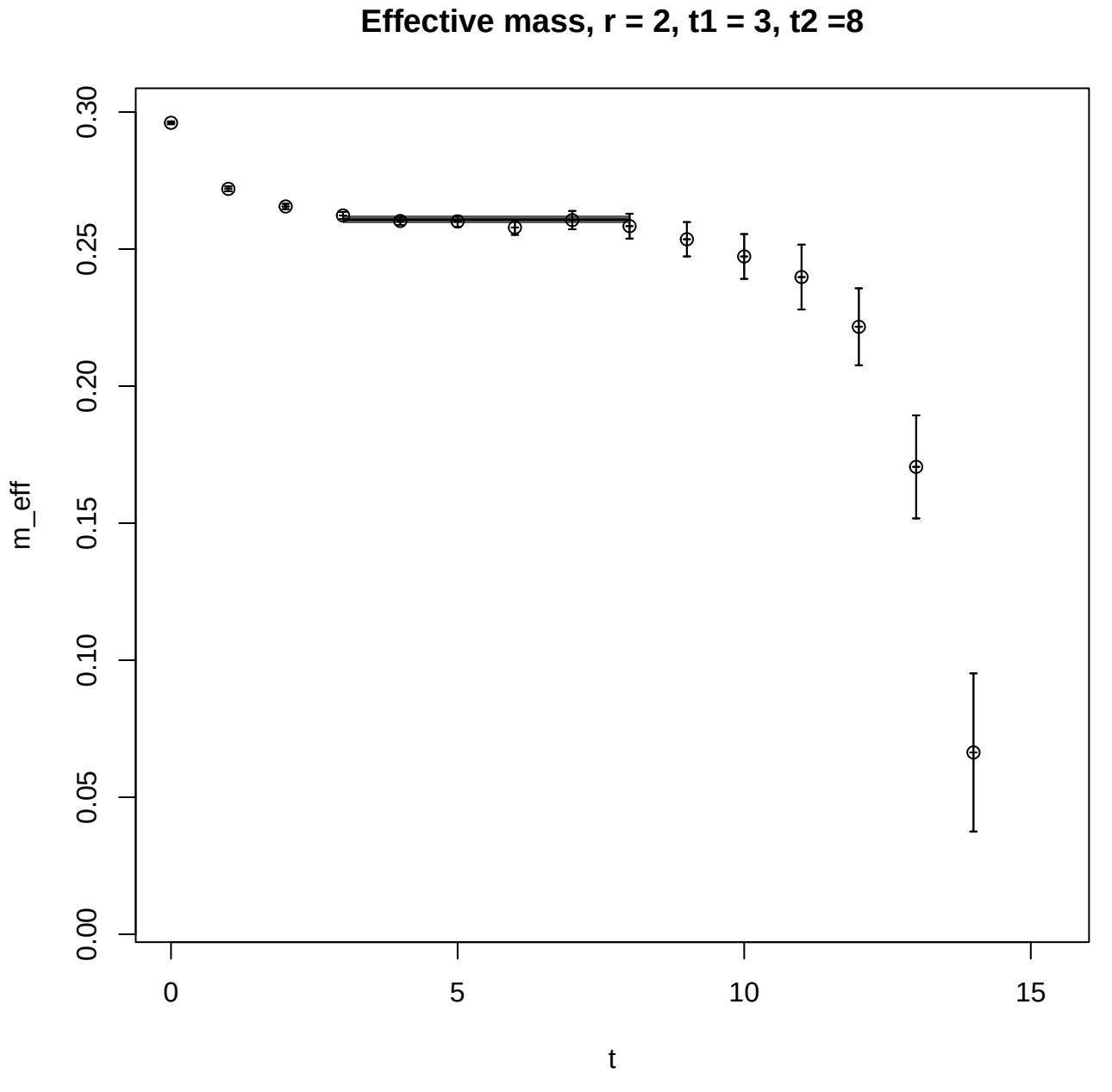

Effective mass, r = 2, t1 = 2, t2 =3
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t

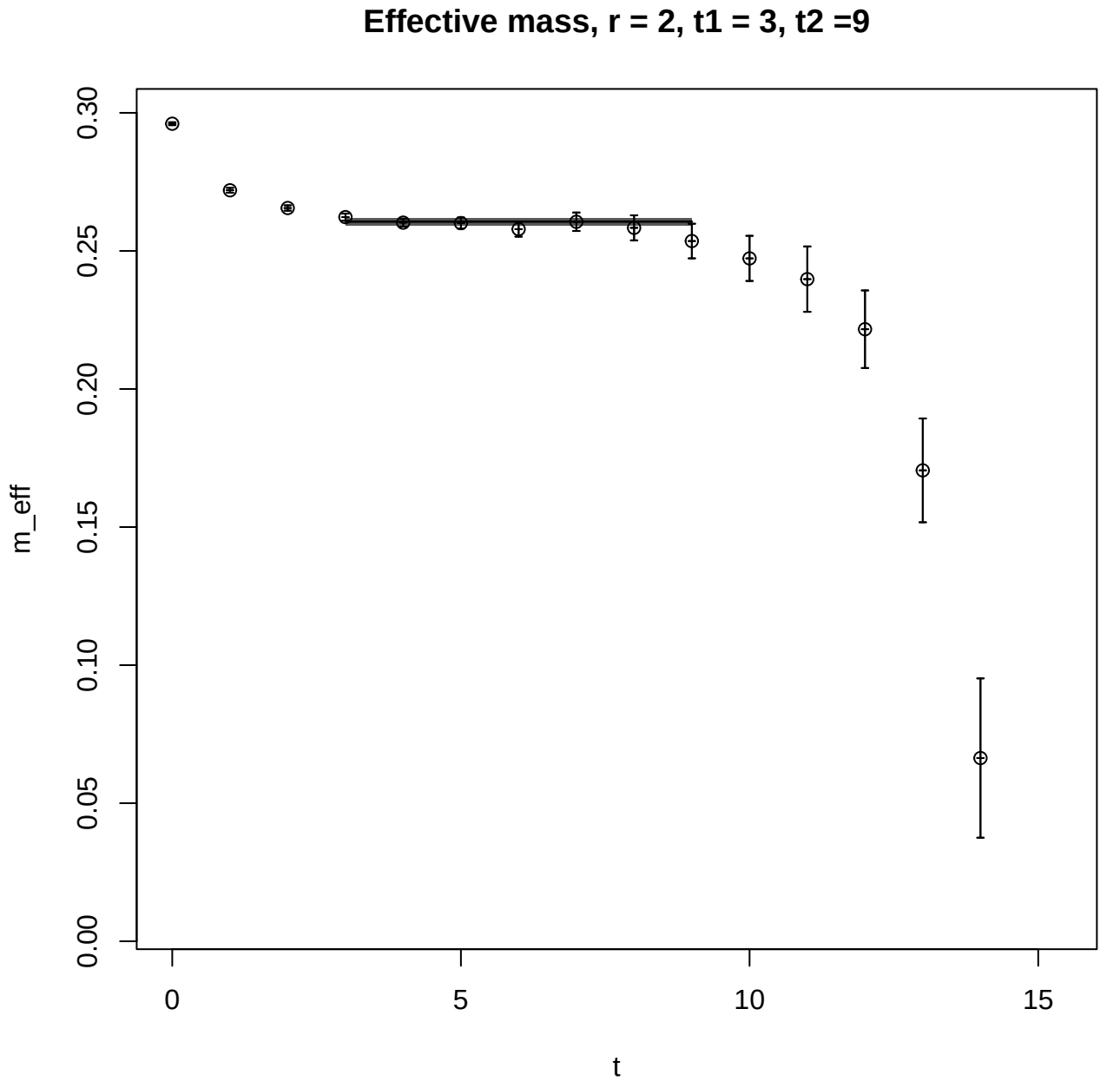

Effective mass, r = 2, t1 = 3, t2 =4
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =5
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =6
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =7
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =8
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =9
m_eff
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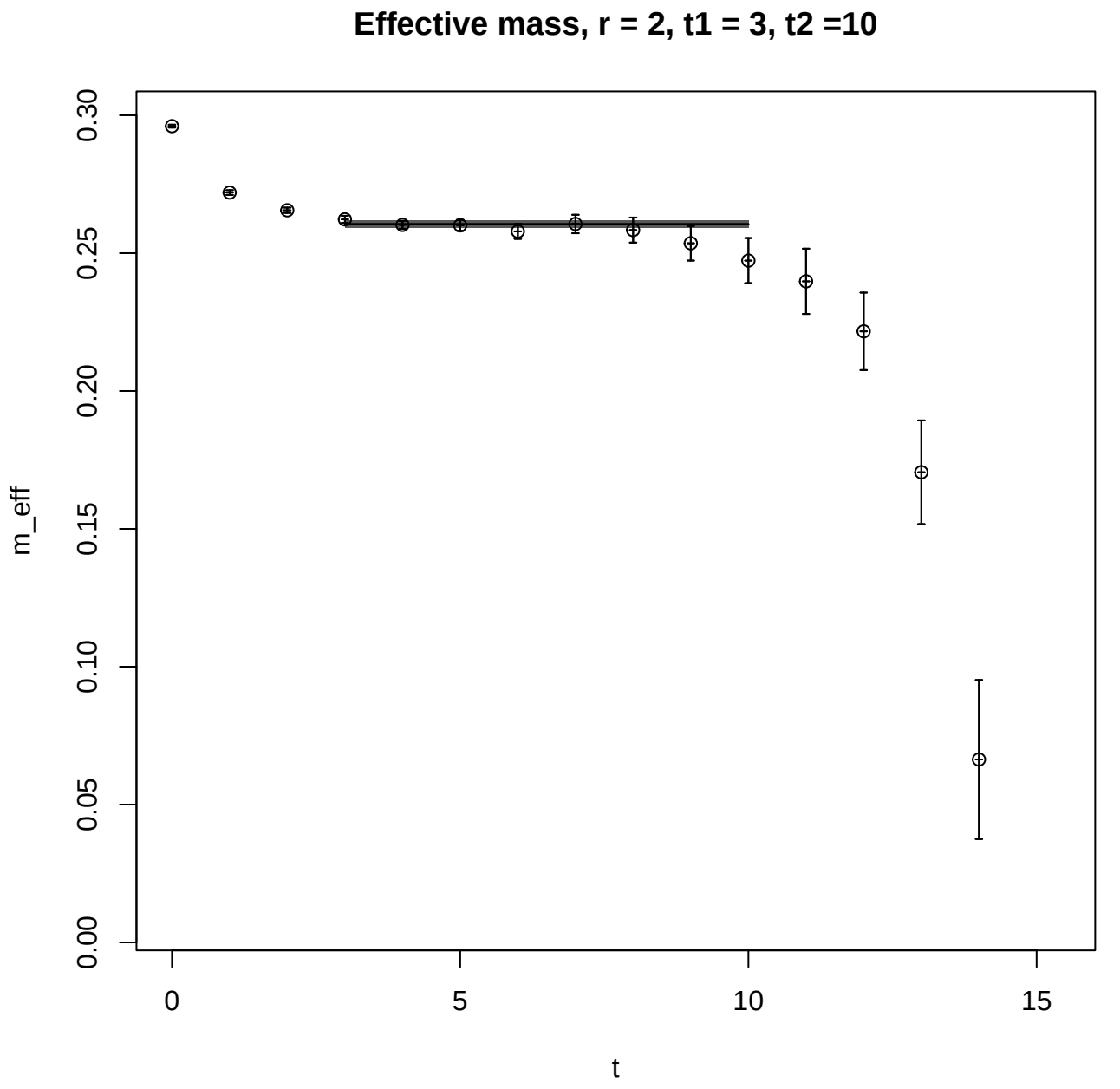
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 3, t2 =10
m_eff
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0.10
0.15
0.20
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0.30
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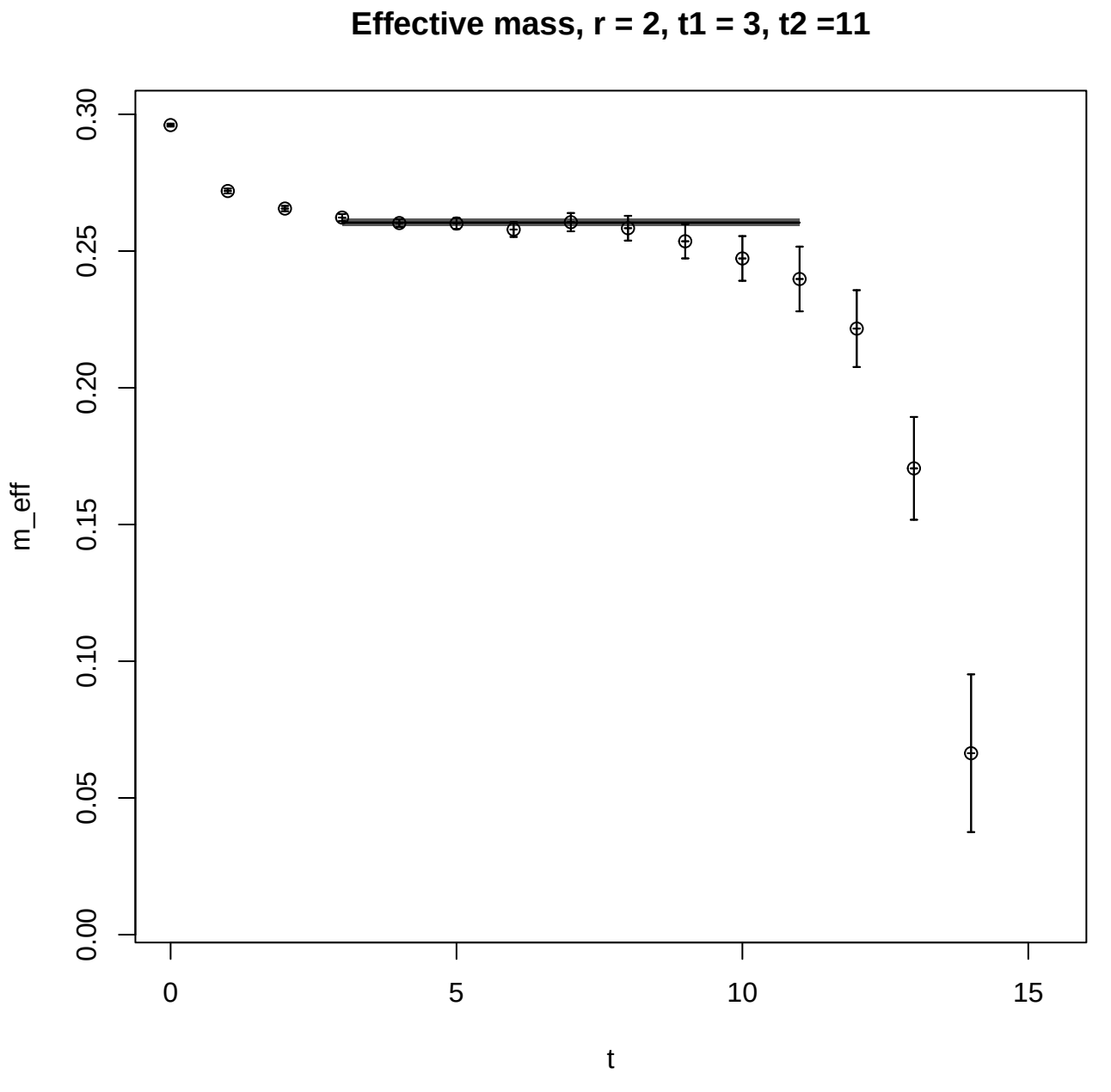
15
t


Effective mass, r = 2, t1 = 3, t2 =11
m_eff
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0.10
0.15
0.20
0.25
0.30
0
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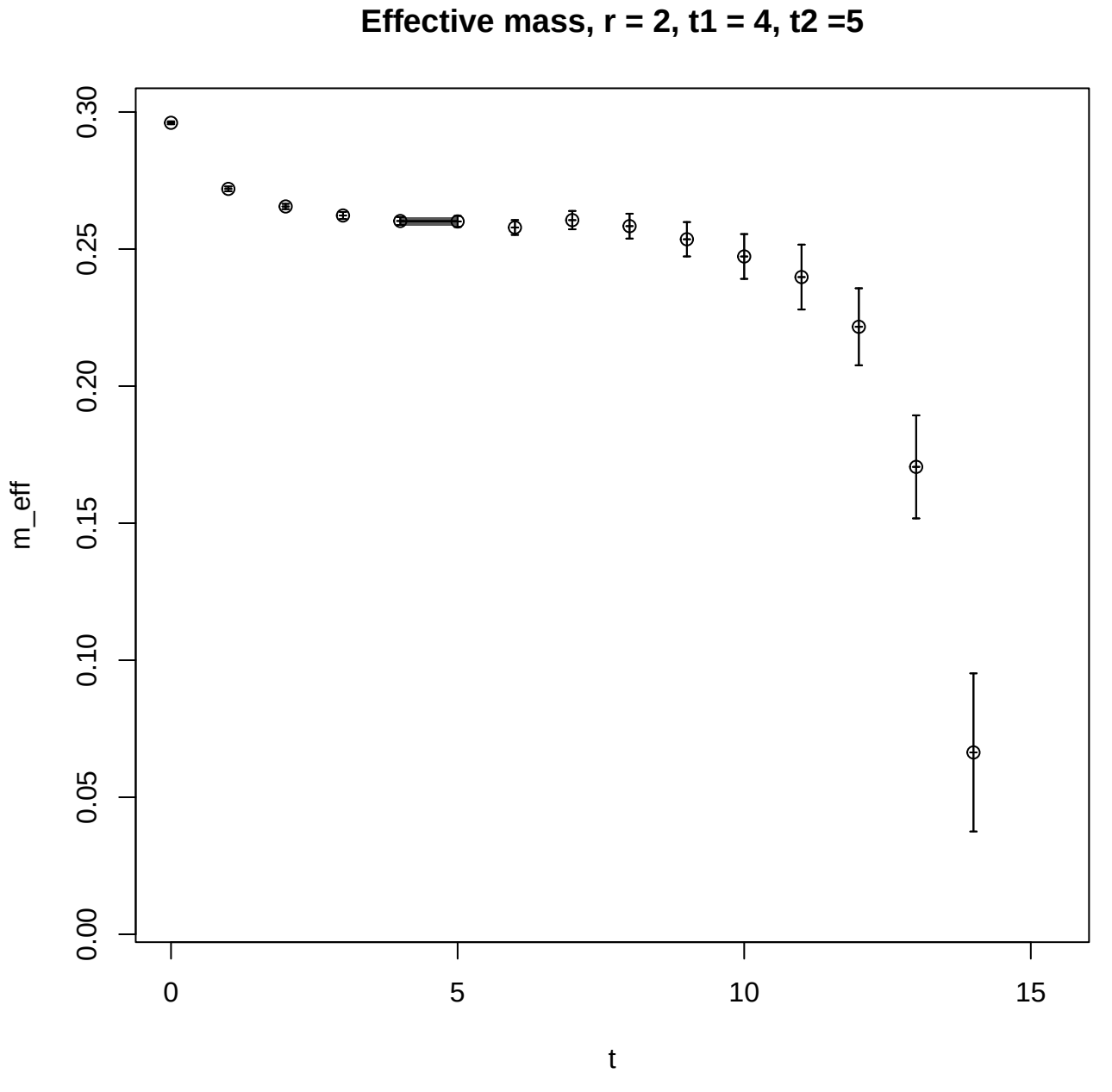
15
t


Effective mass, r = 2, t1 = 4, t2 =5
m_eff
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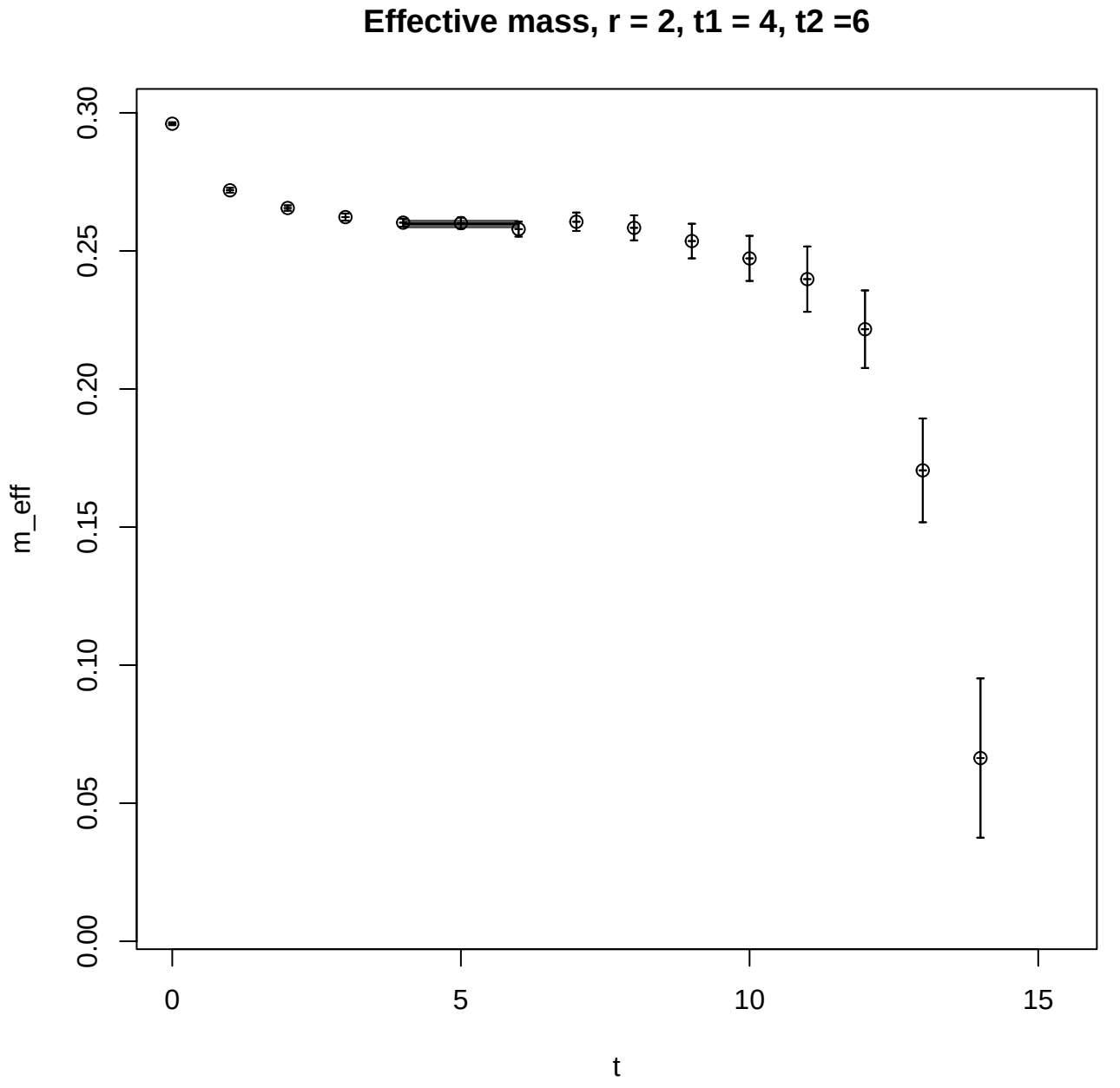
5
10
15
t


Effective mass, r = 2, t1 = 4, t2 =6
m_eff
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0.20
0.25
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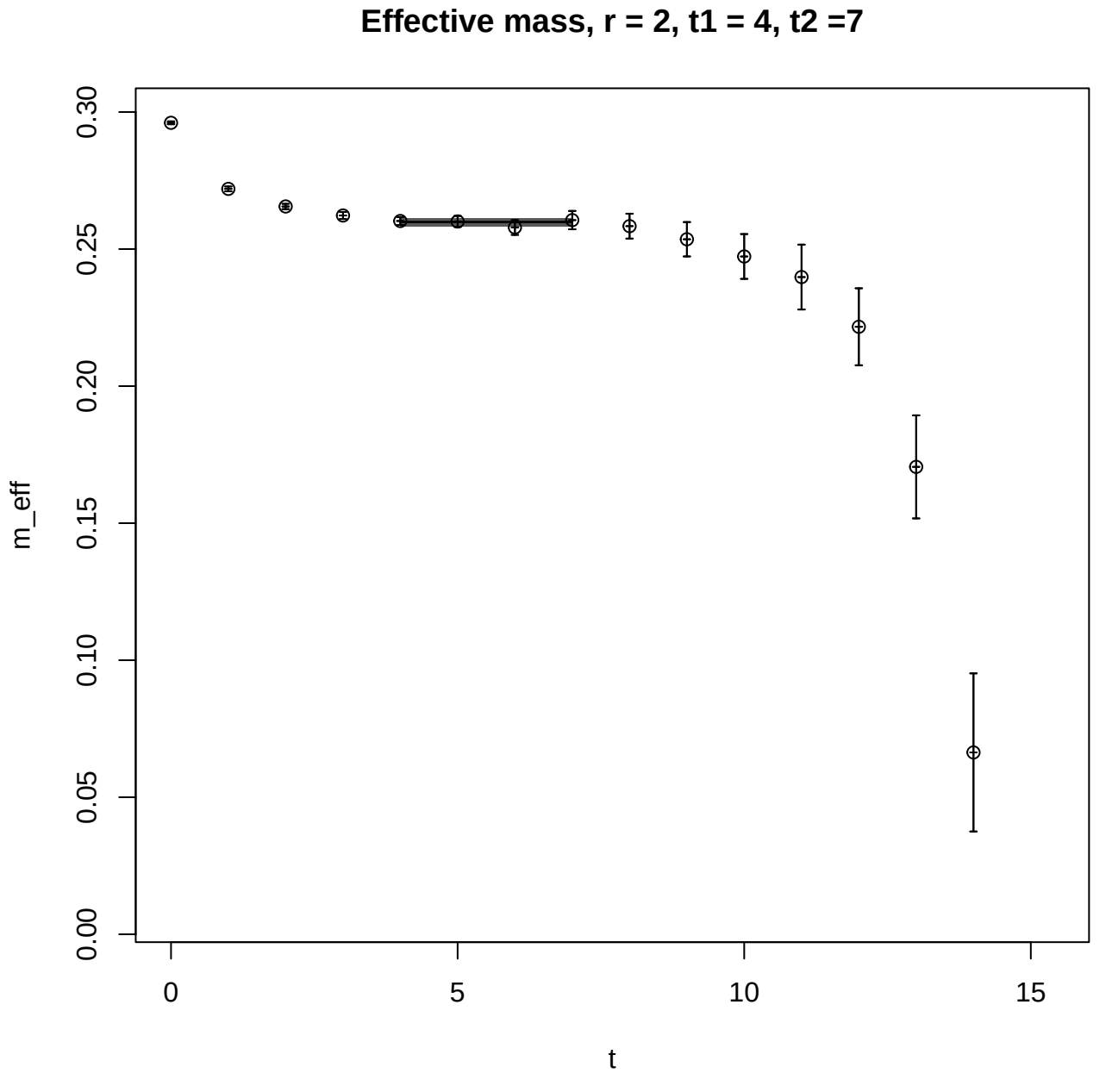
15
t


Effective mass, r = 2, t1 = 4, t2 =7
m_eff
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0.20
0.25
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5
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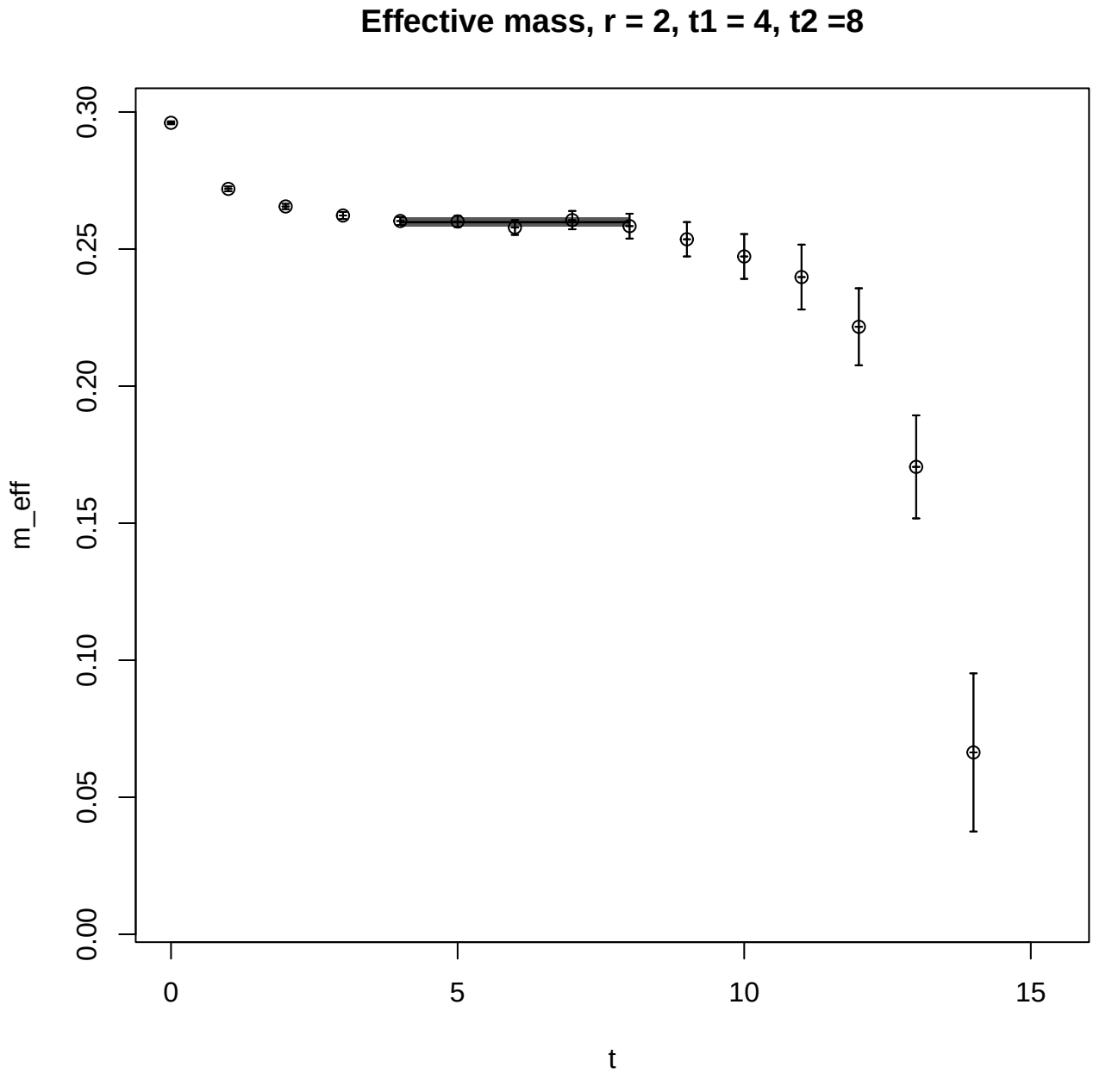
15
t


Effective mass, r = 2, t1 = 4, t2 =8
m_eff
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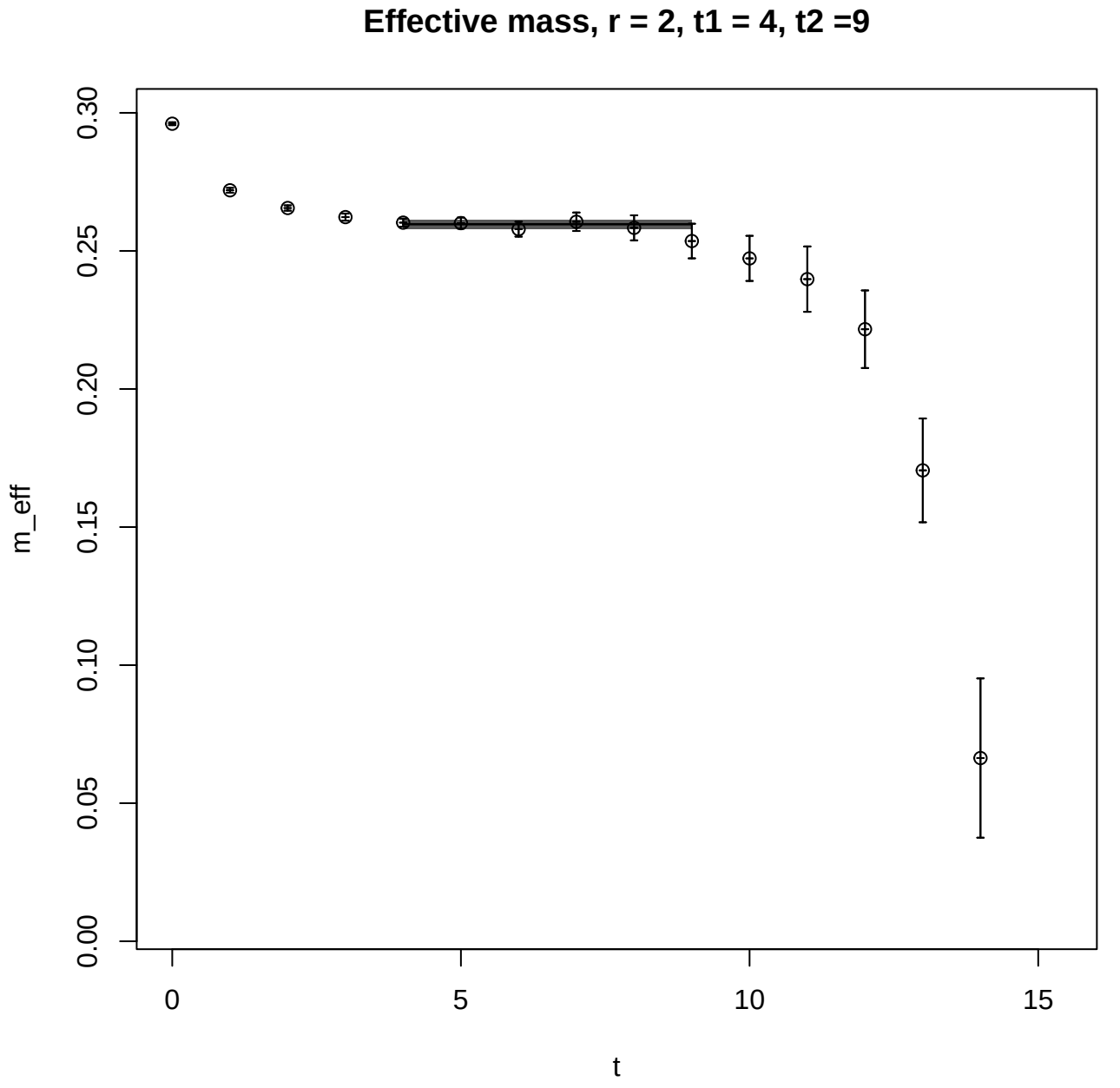
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 4, t2 =9
m_eff
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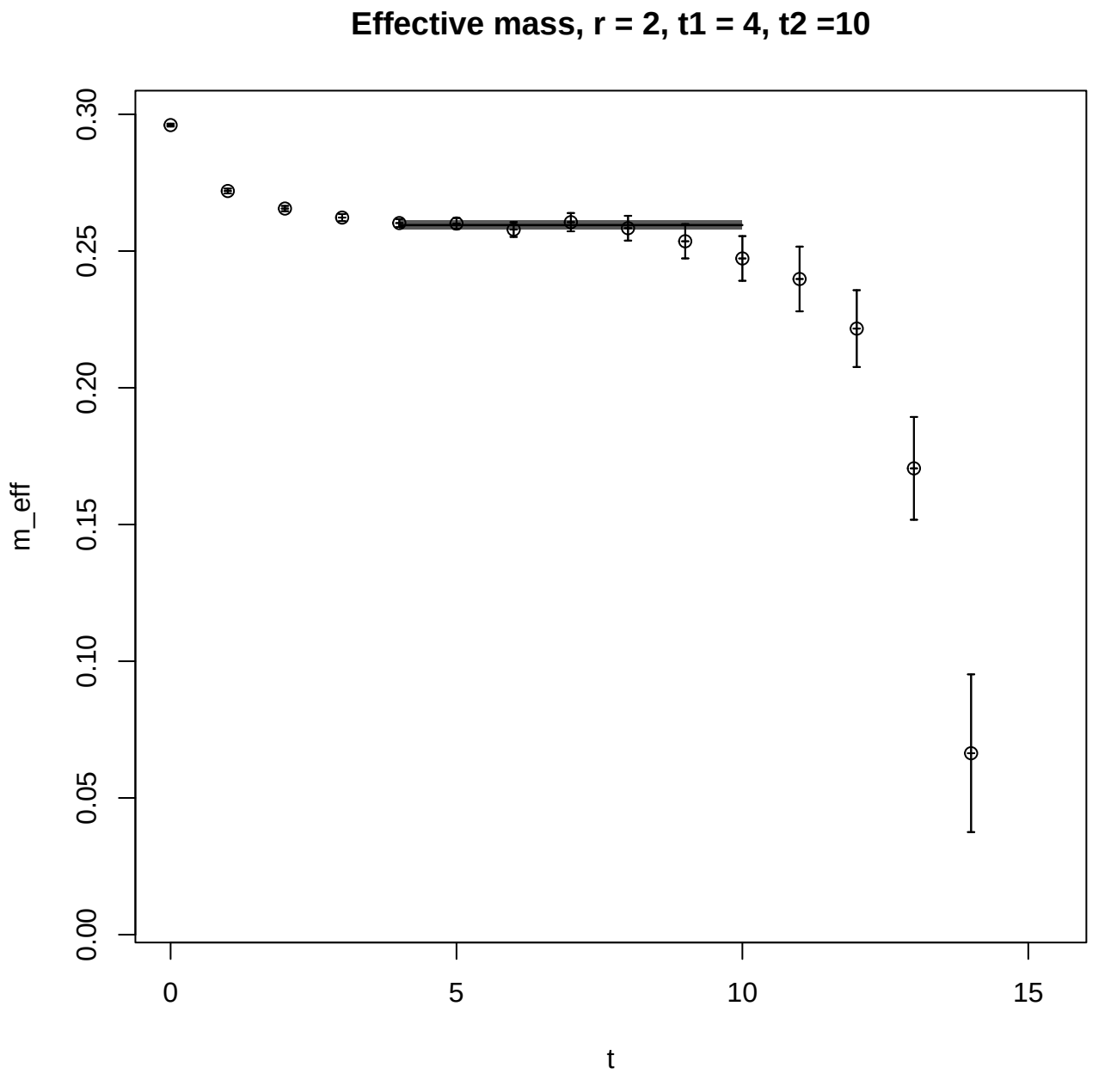
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 4, t2 =10
m_eff
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0.05
0.10
0.15
0.20
0.25
0.30
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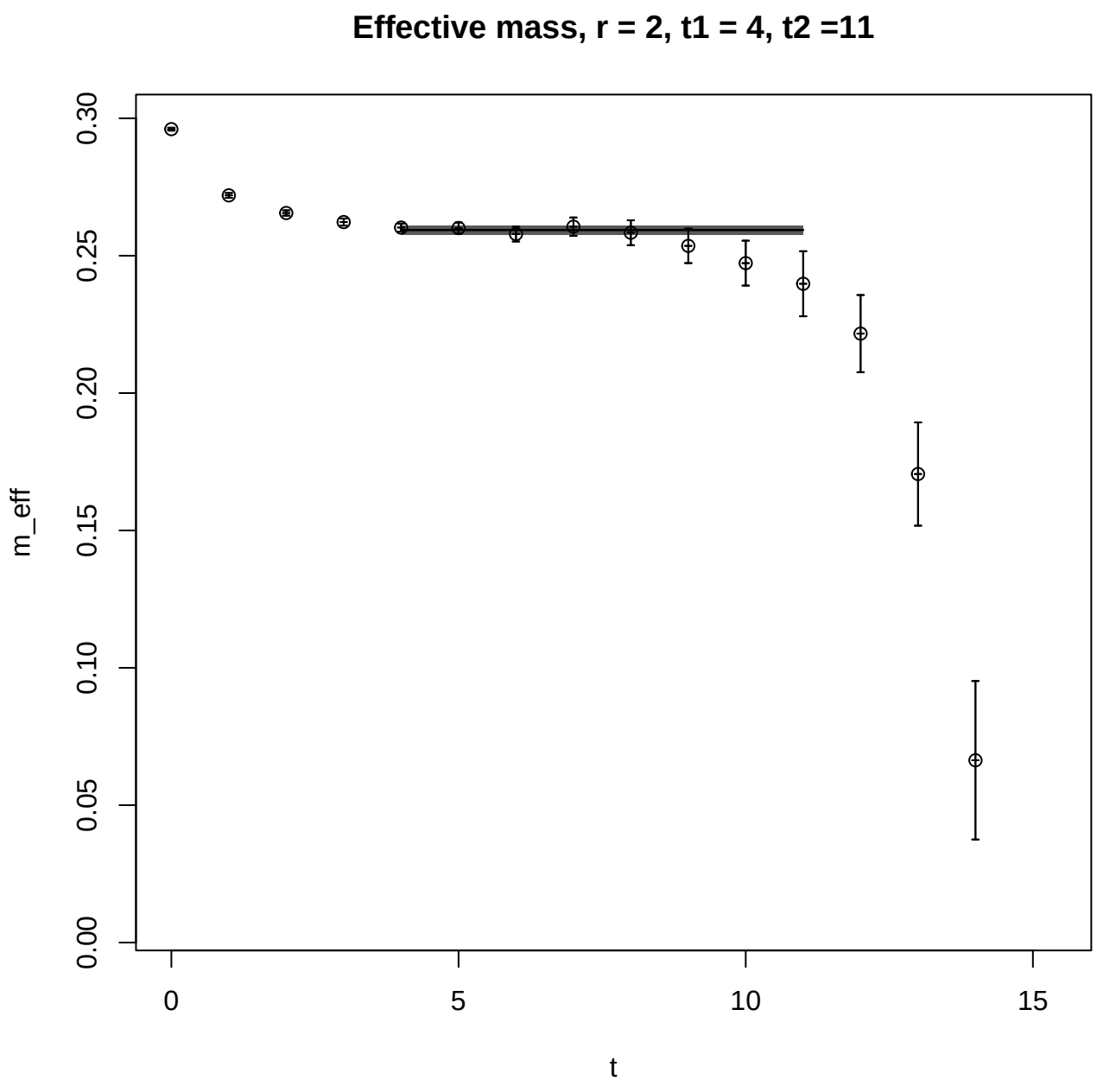
5
10
15
t


Effective mass, r = 2, t1 = 4, t2 =11
m_eff
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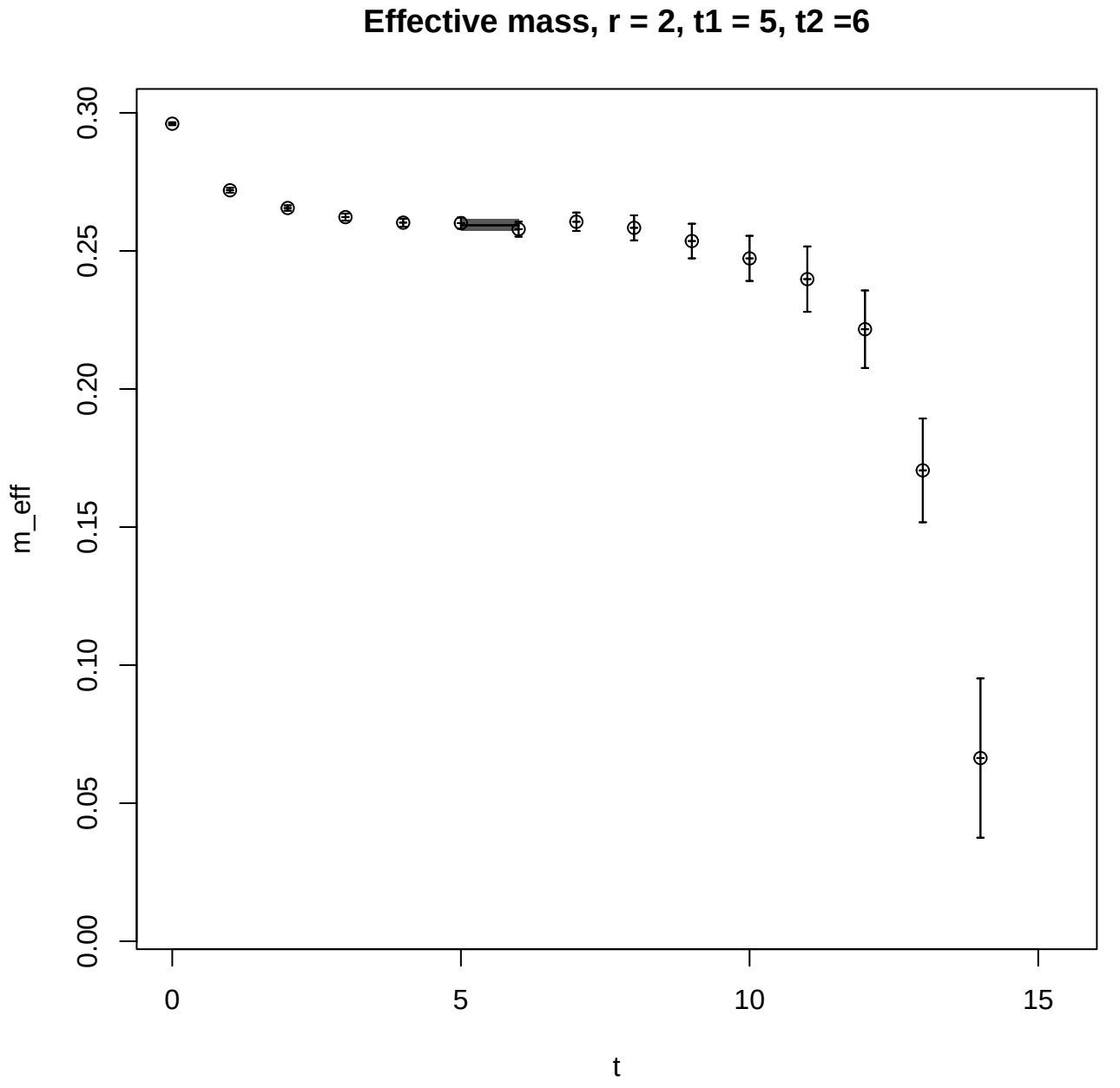
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =6
m_eff
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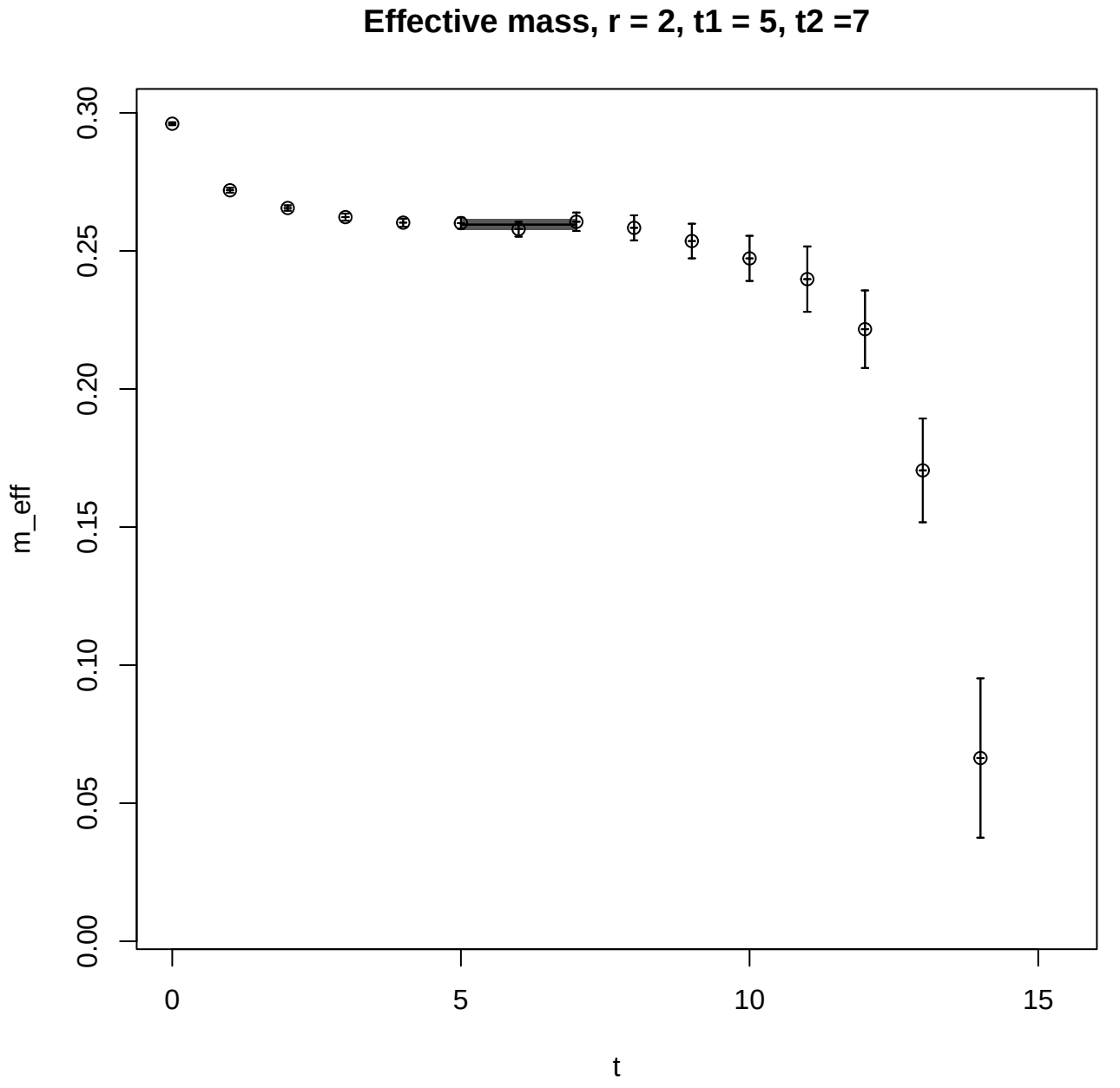
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =7
m_eff
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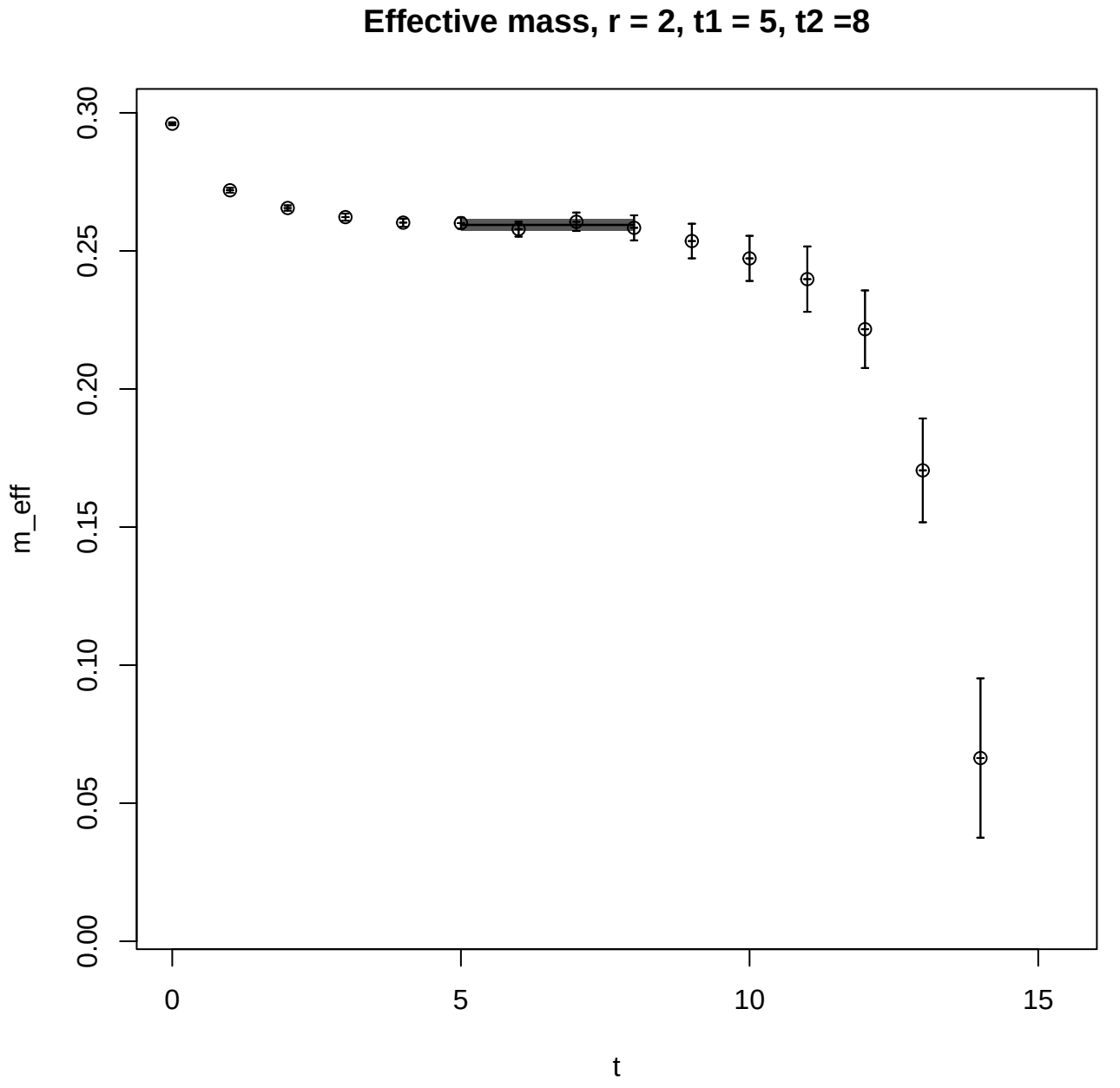
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =8
m_eff
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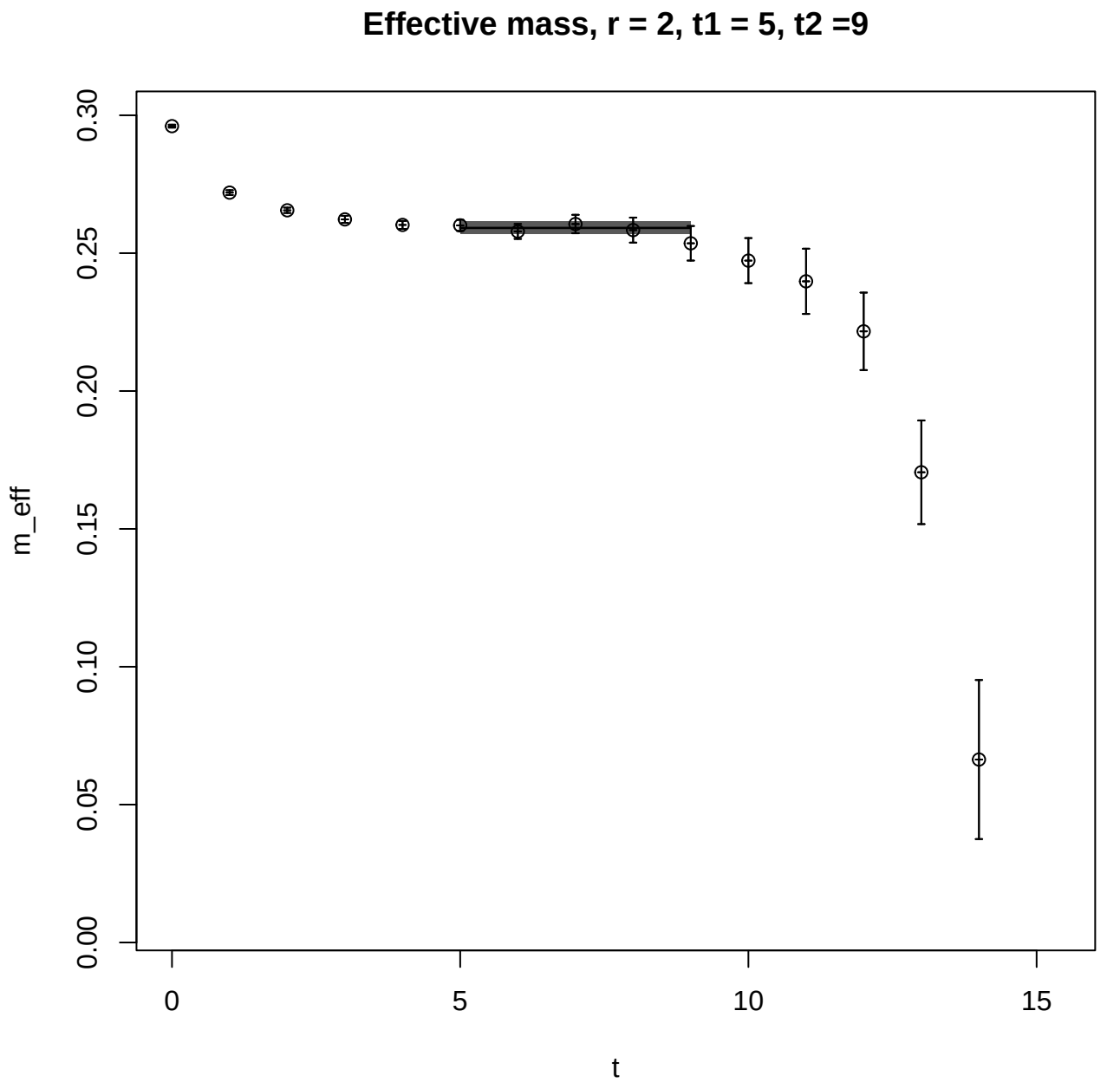
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =9
m_eff
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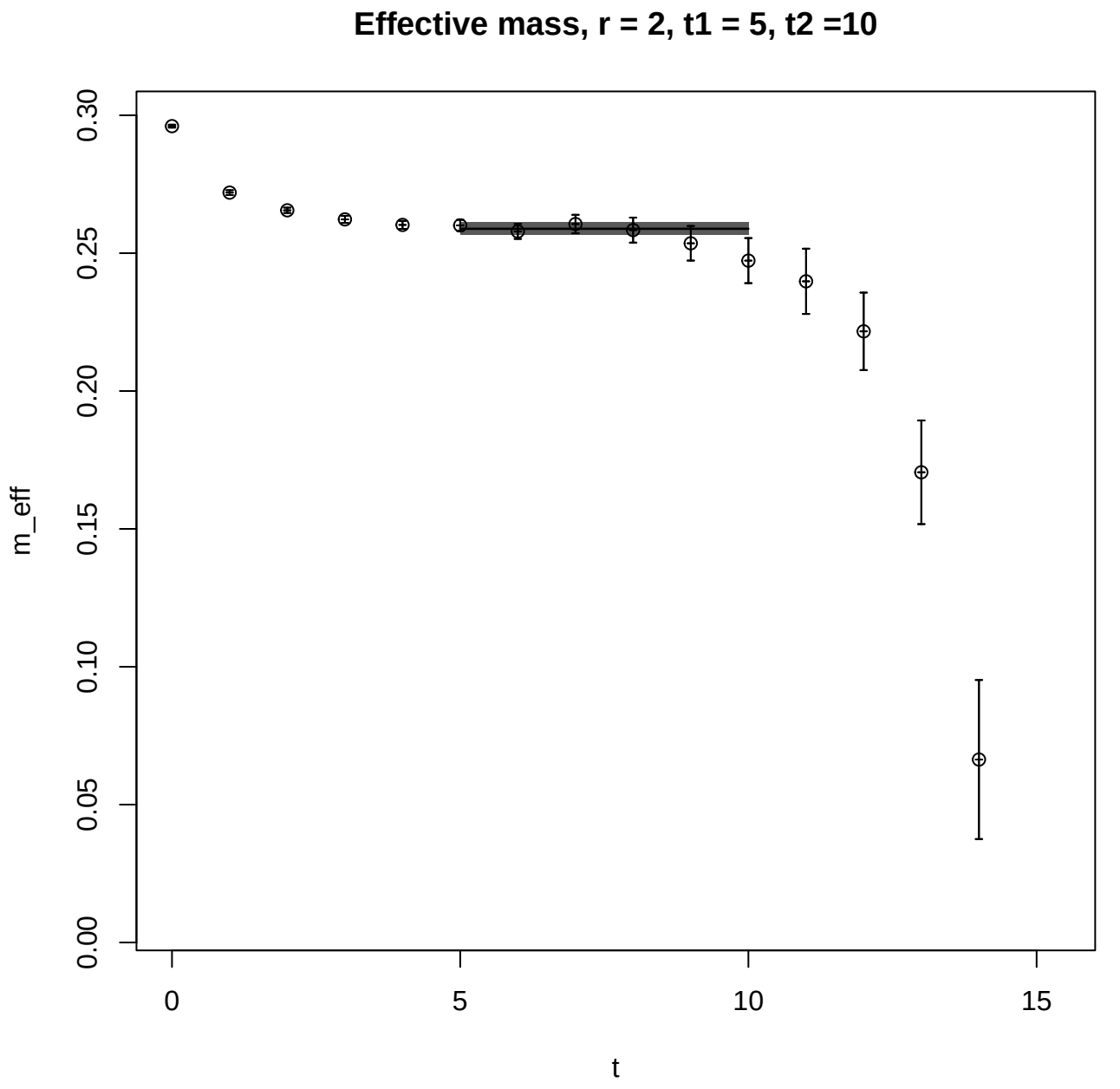
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =10
m_eff
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0.05
0.10
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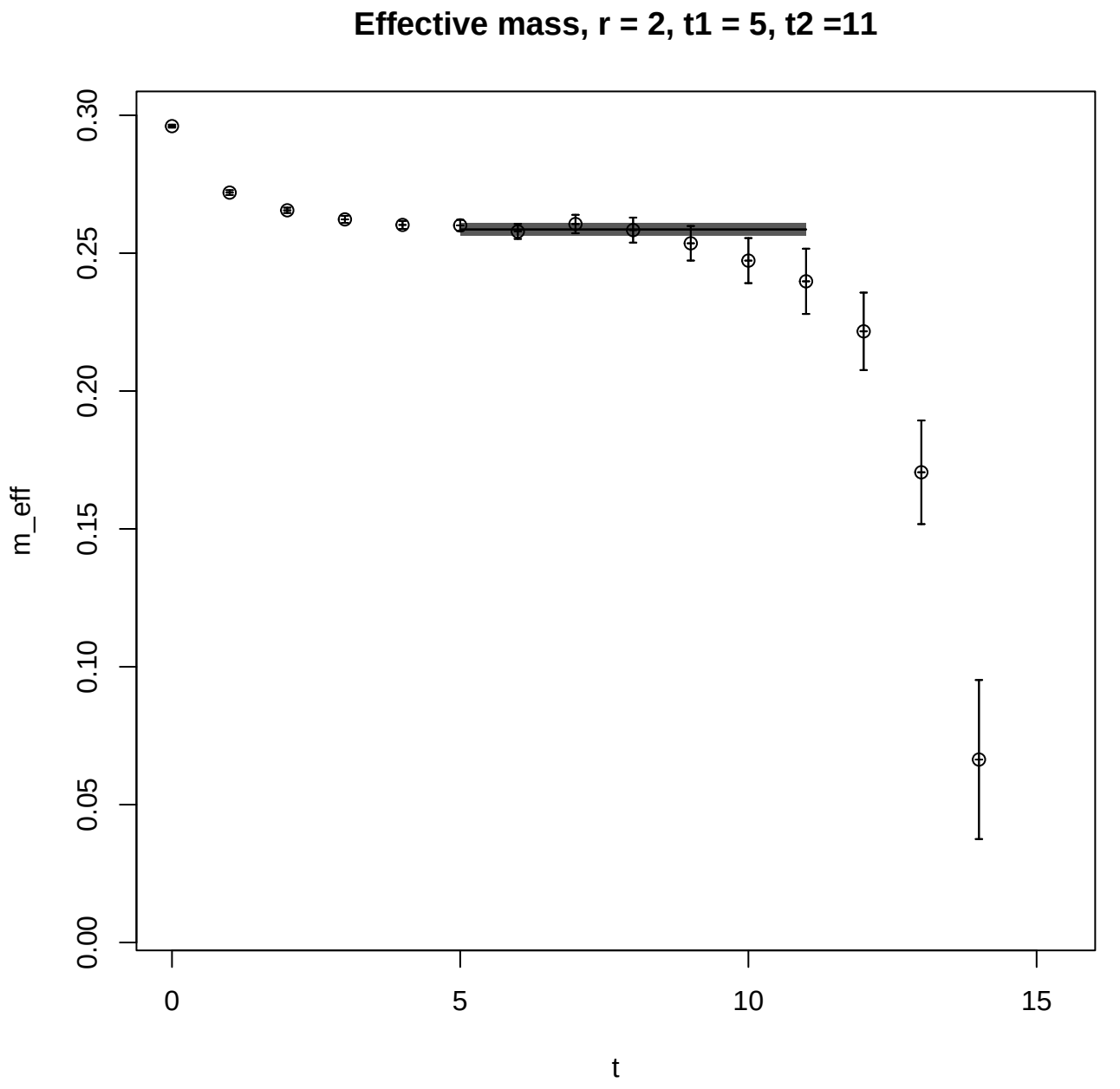
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 5, t2 =11
m_eff
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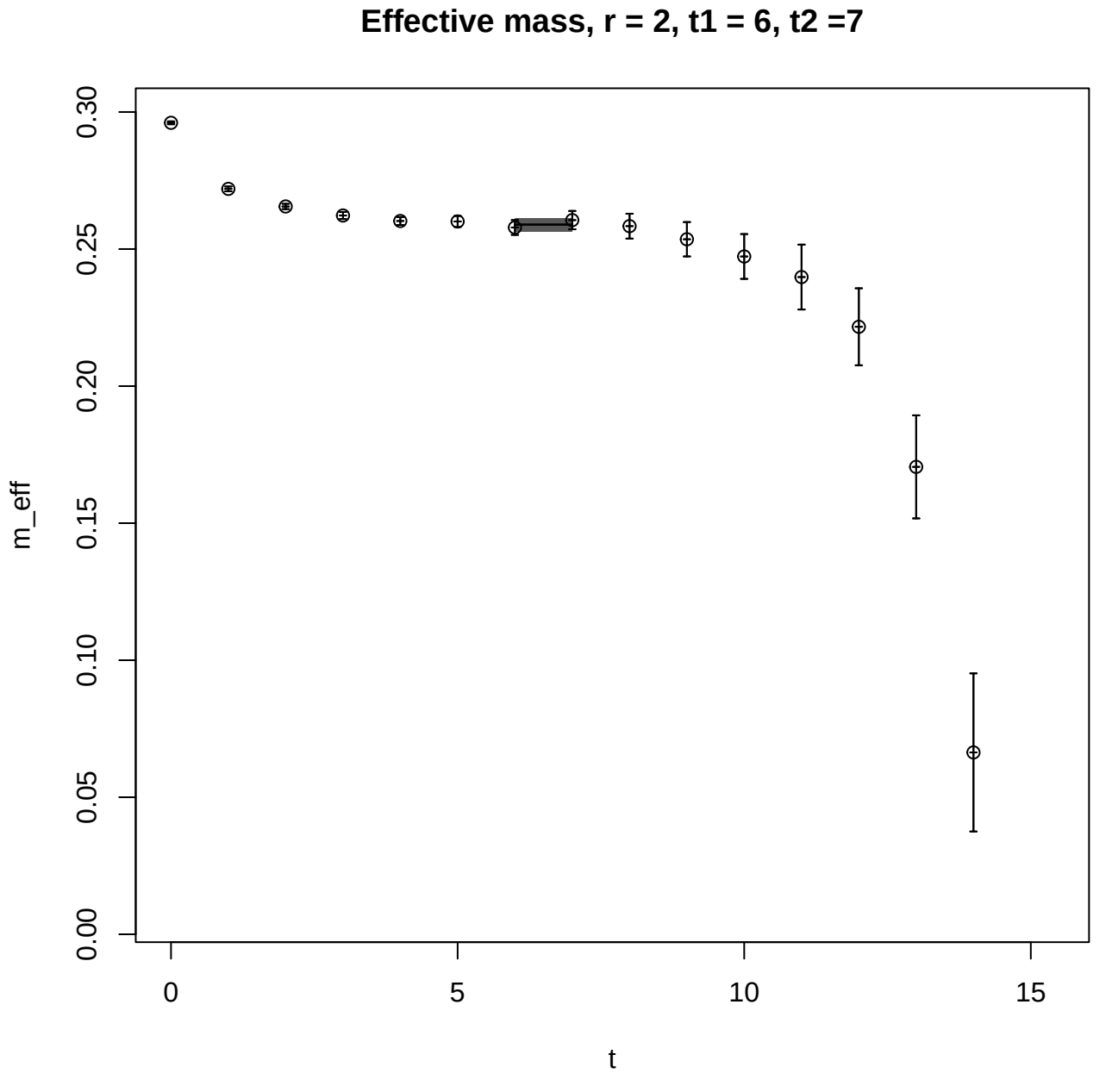
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =7
m_eff
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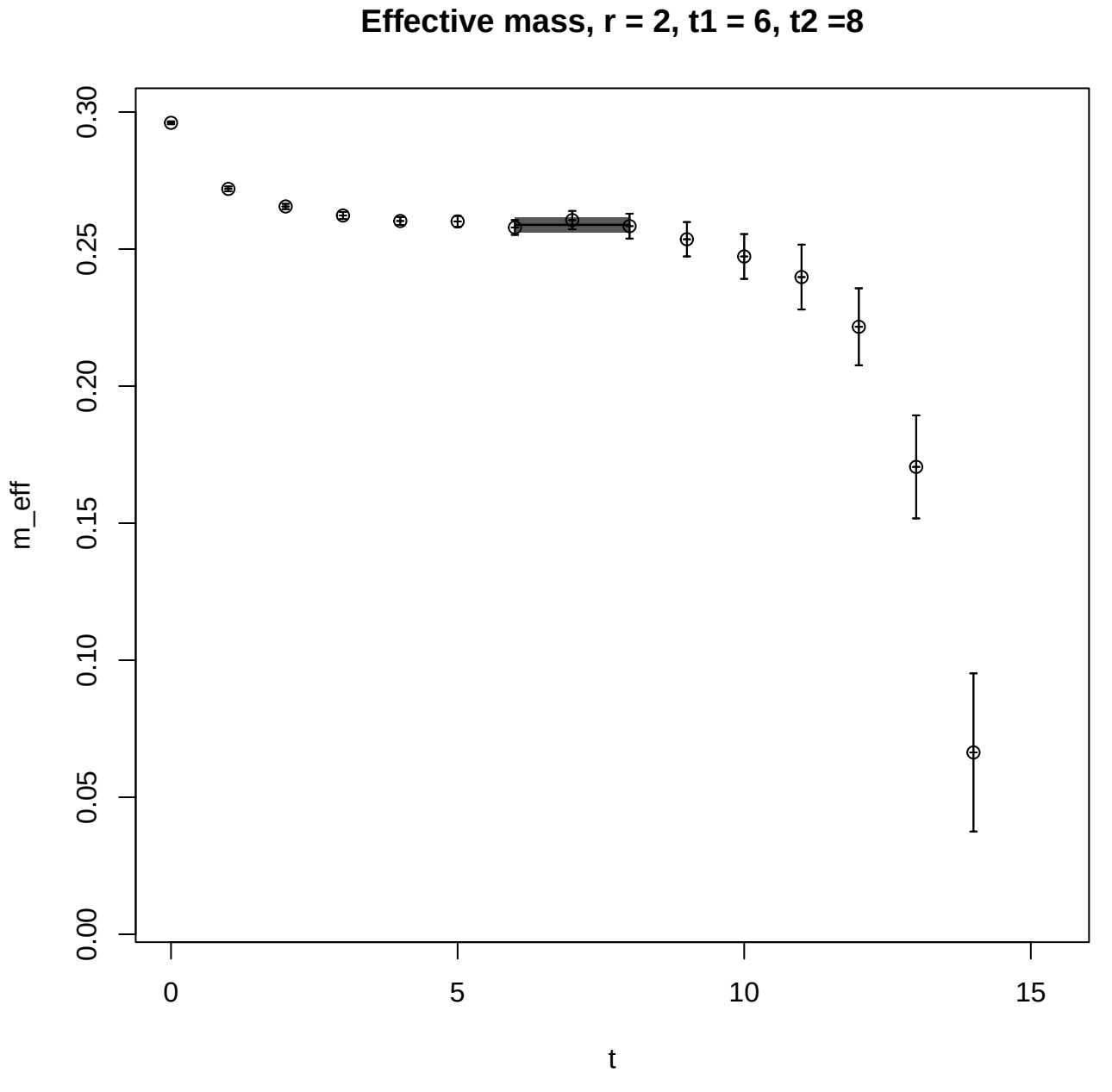
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =8
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0.25
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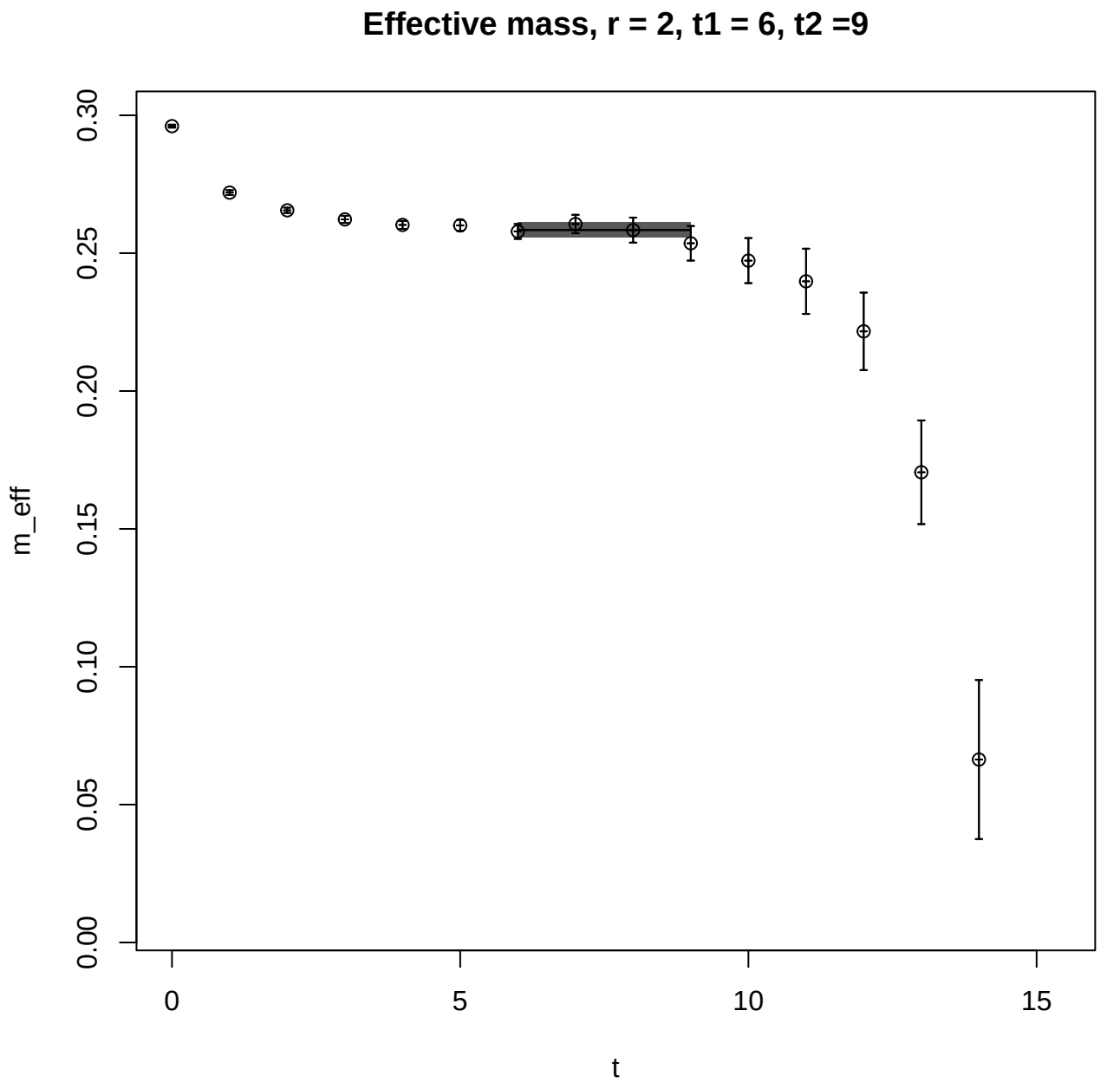
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =9
m_eff
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0.05
0.10
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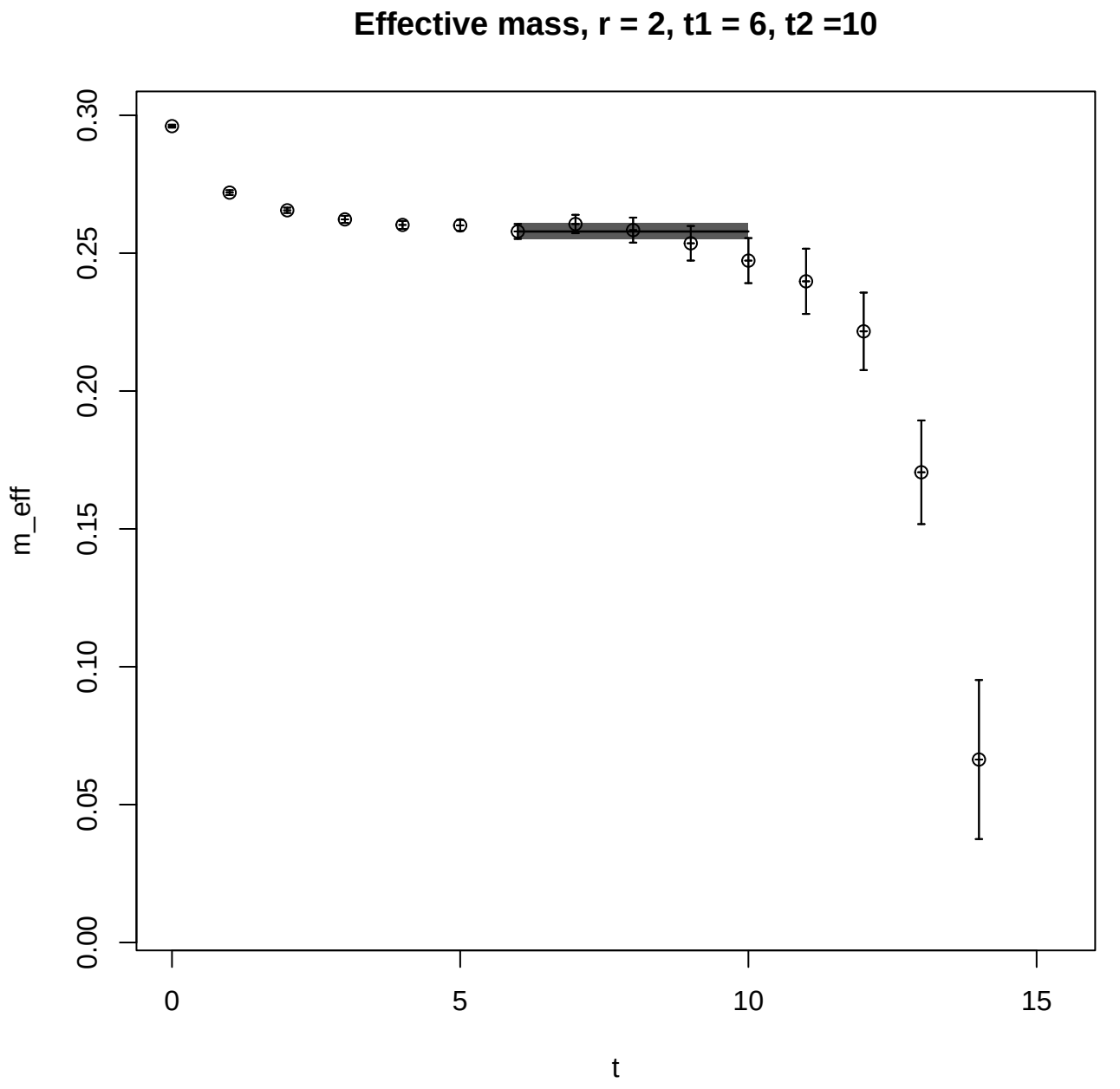
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =10
m_eff
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0.05
0.10
0.15
0.20
0.25
0.30
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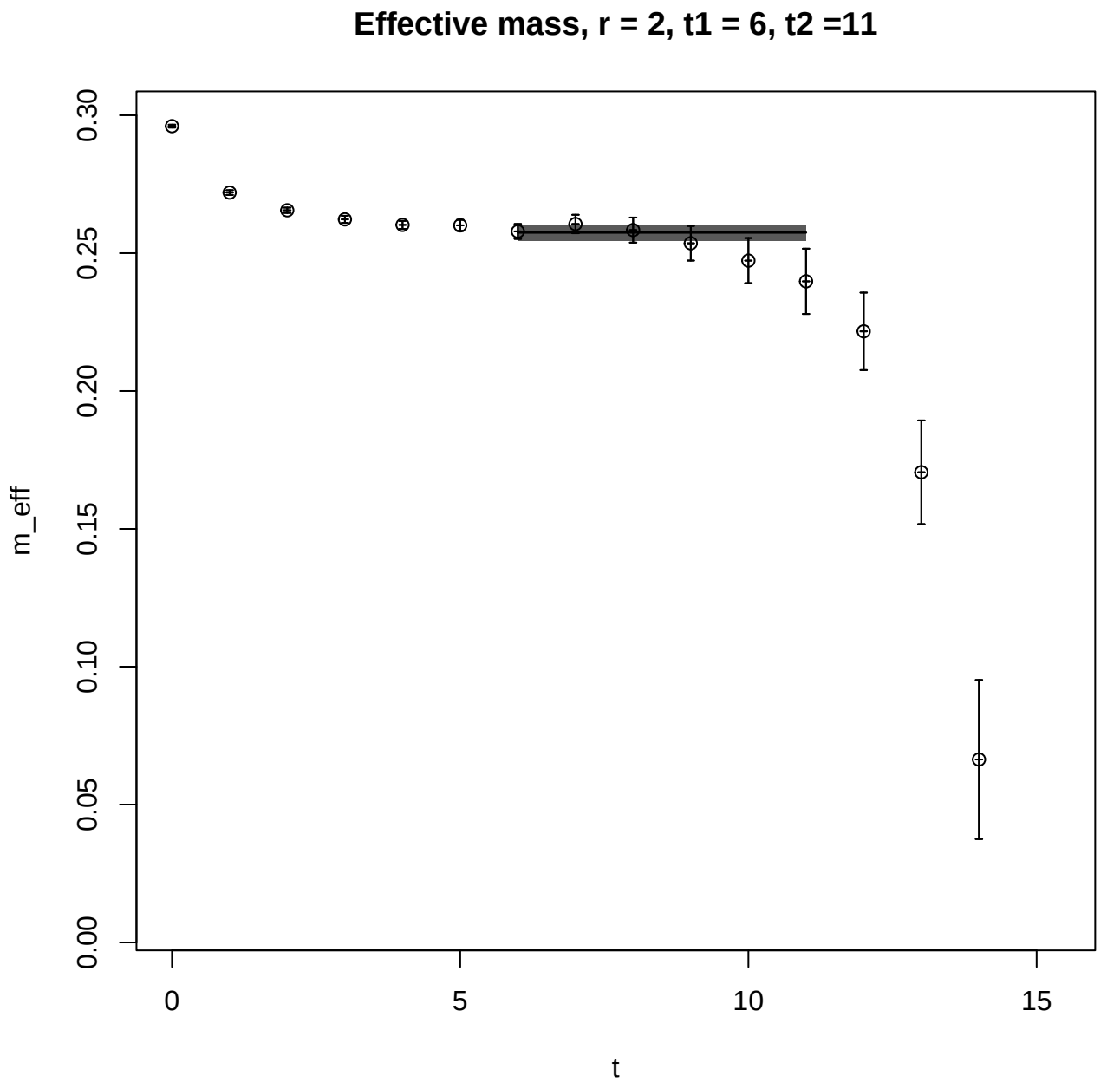
5
10
15
t


Effective mass, r = 2, t1 = 6, t2 =11
m_eff
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0.05
0.10
0.15
0.20
0.25
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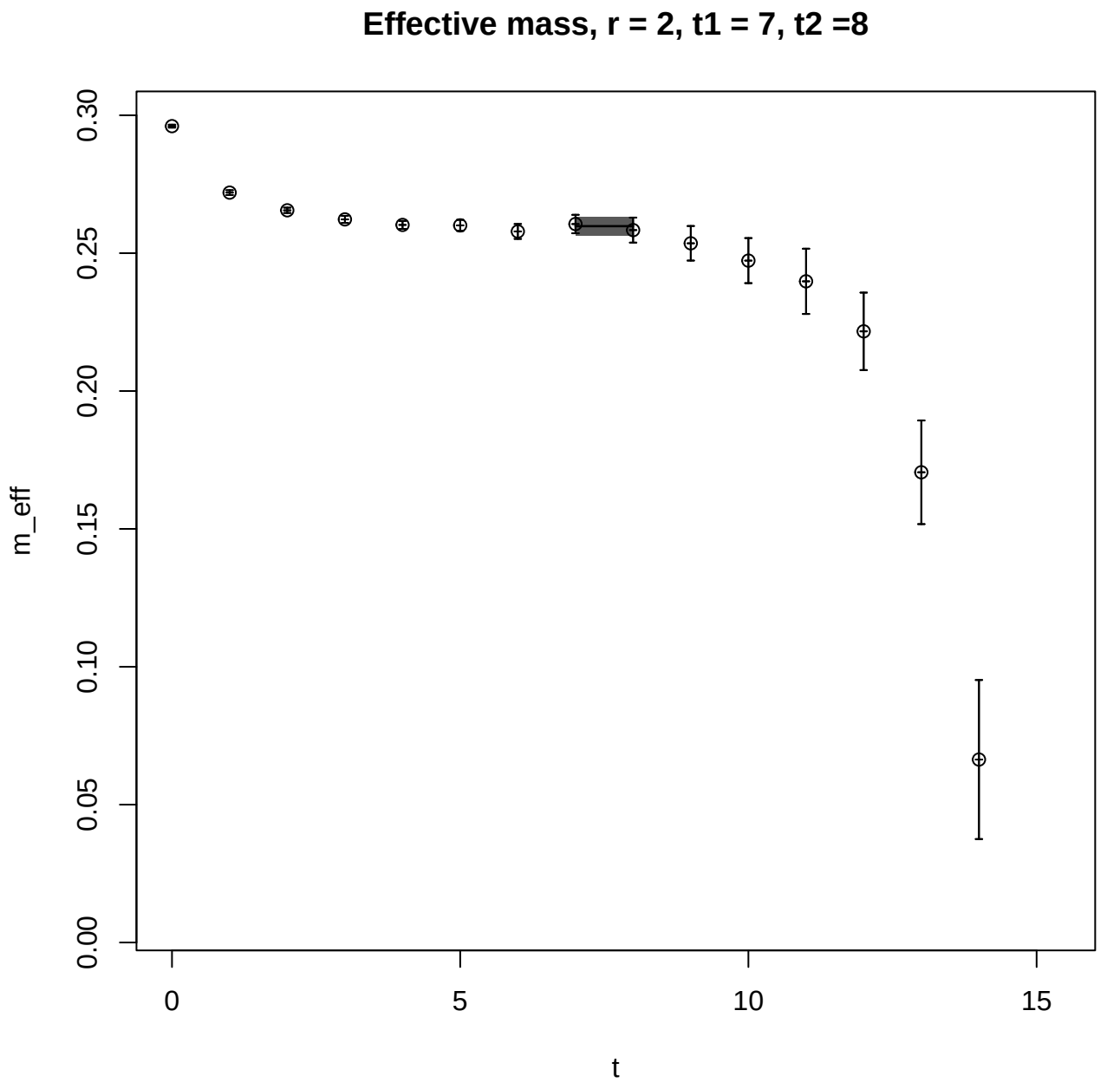
15
t


Effective mass, r = 2, t1 = 7, t2 =8
m_eff
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0.10
0.15
0.20
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0.30
0
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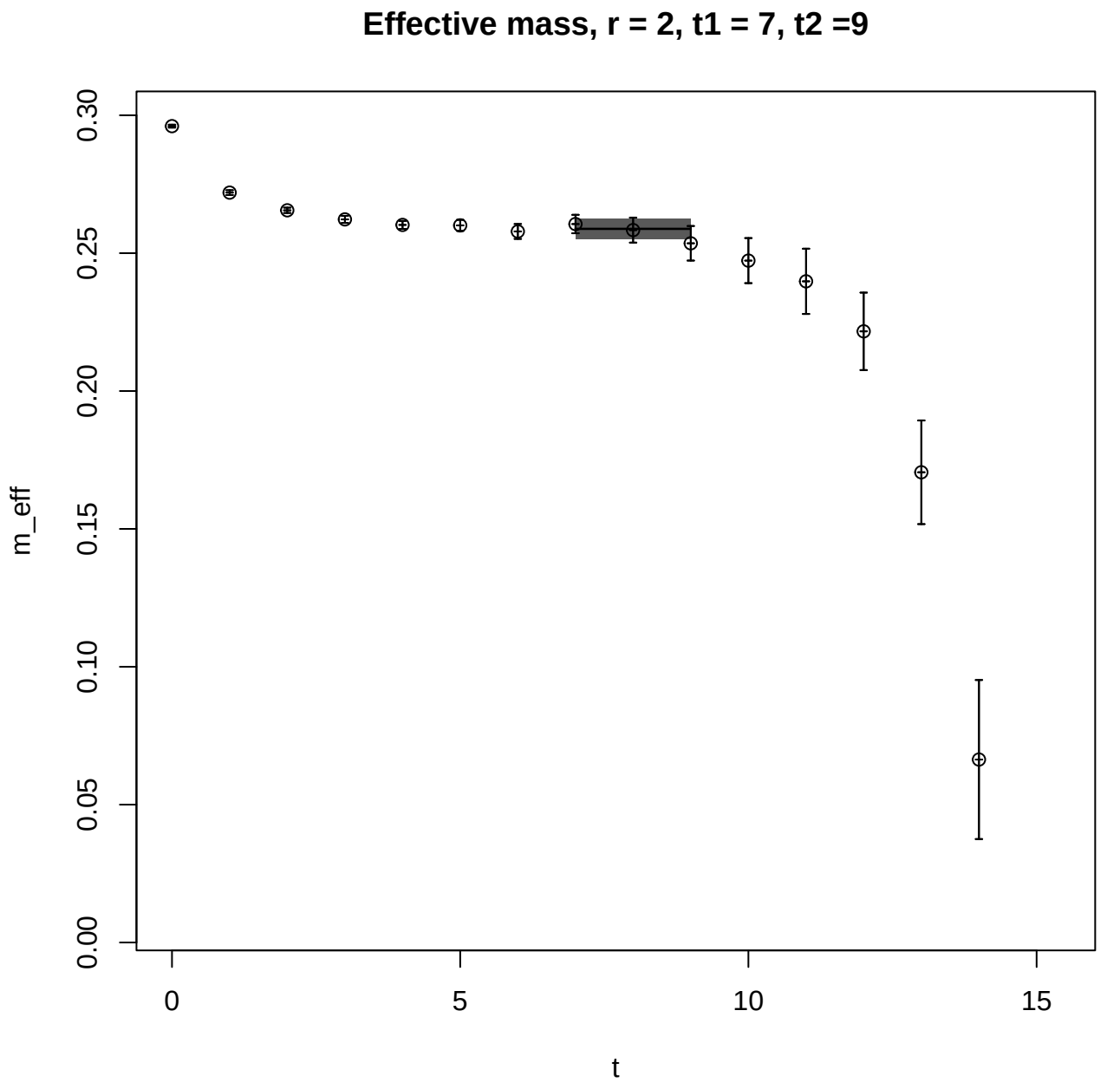
15
t


Effective mass, r = 2, t1 = 7, t2 =9
m_eff
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0.10
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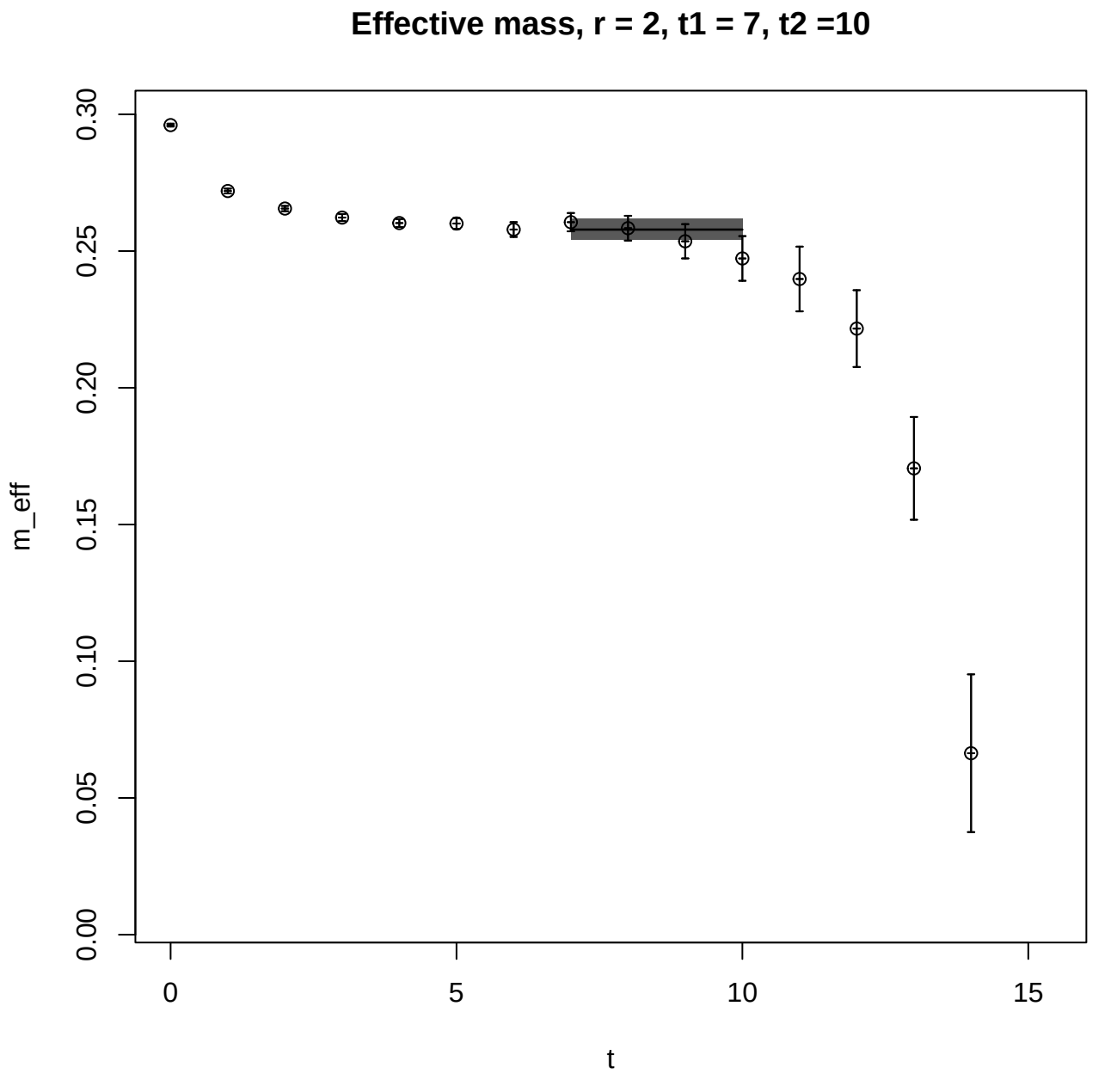
t


Effective mass, r = 2, t1 = 7, t2 =10
m_eff
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0.05
0.10
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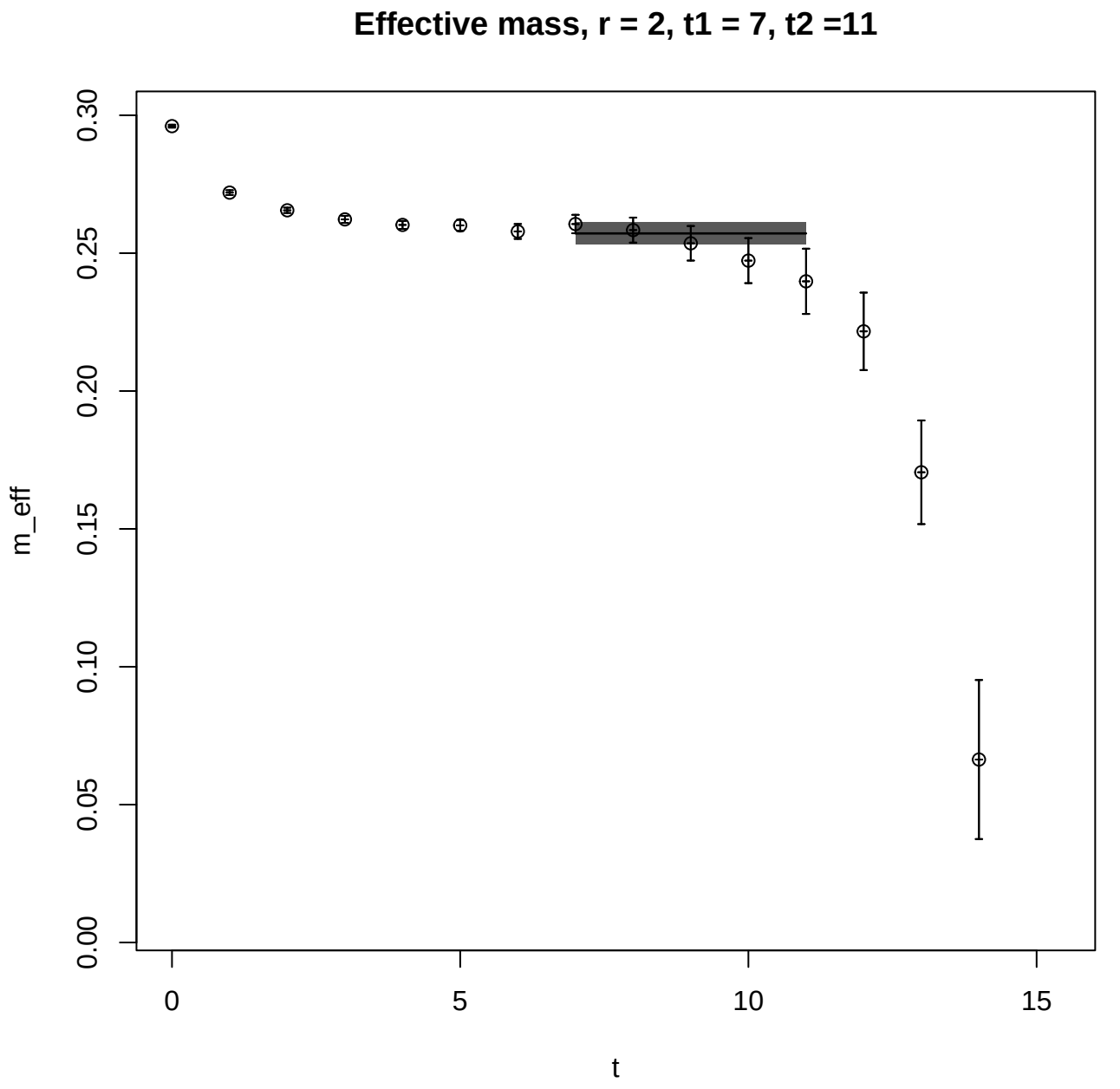
0
5
10
15
t


Effective mass, r = 2, t1 = 7, t2 =11
m_eff
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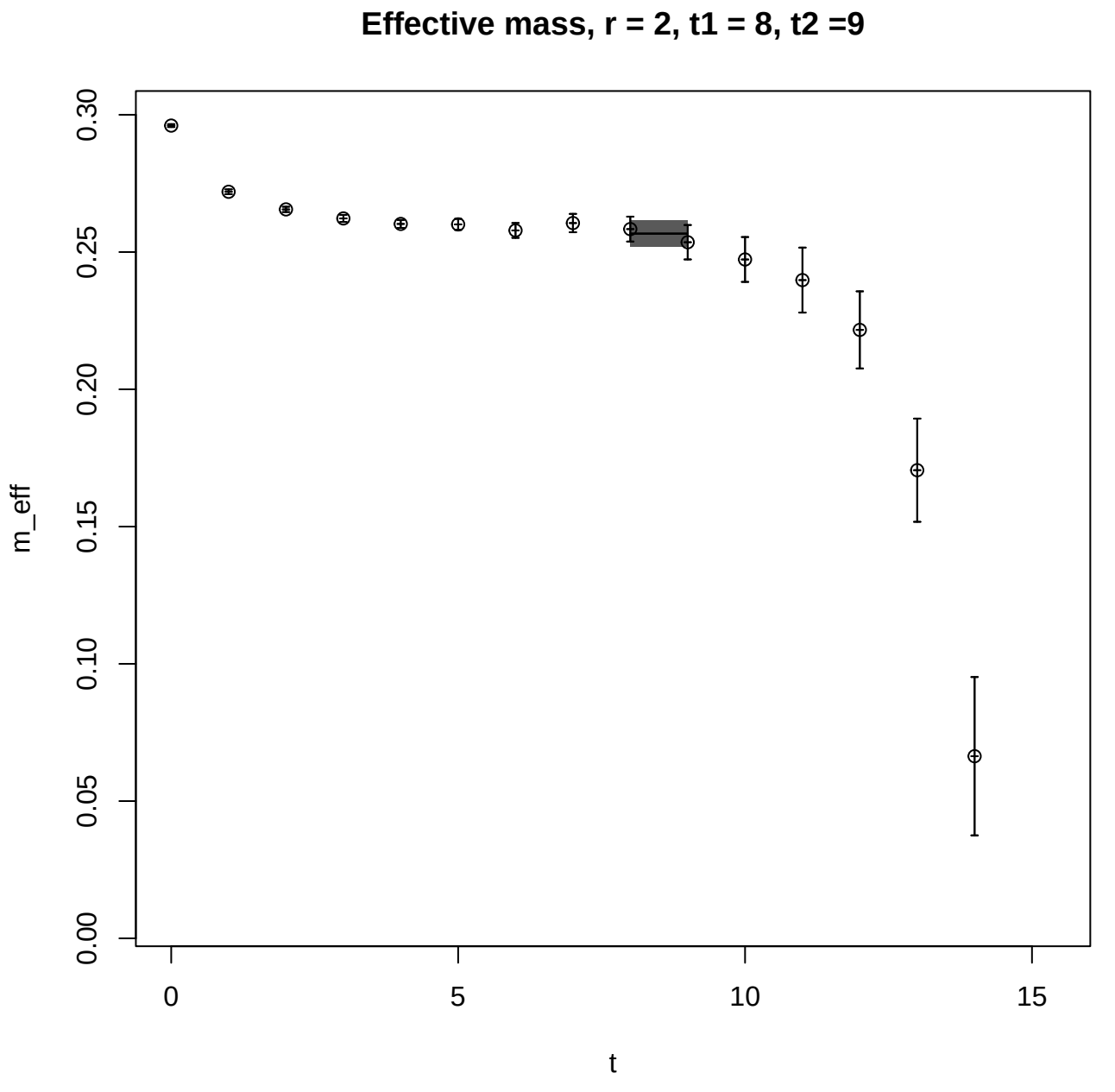
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =9
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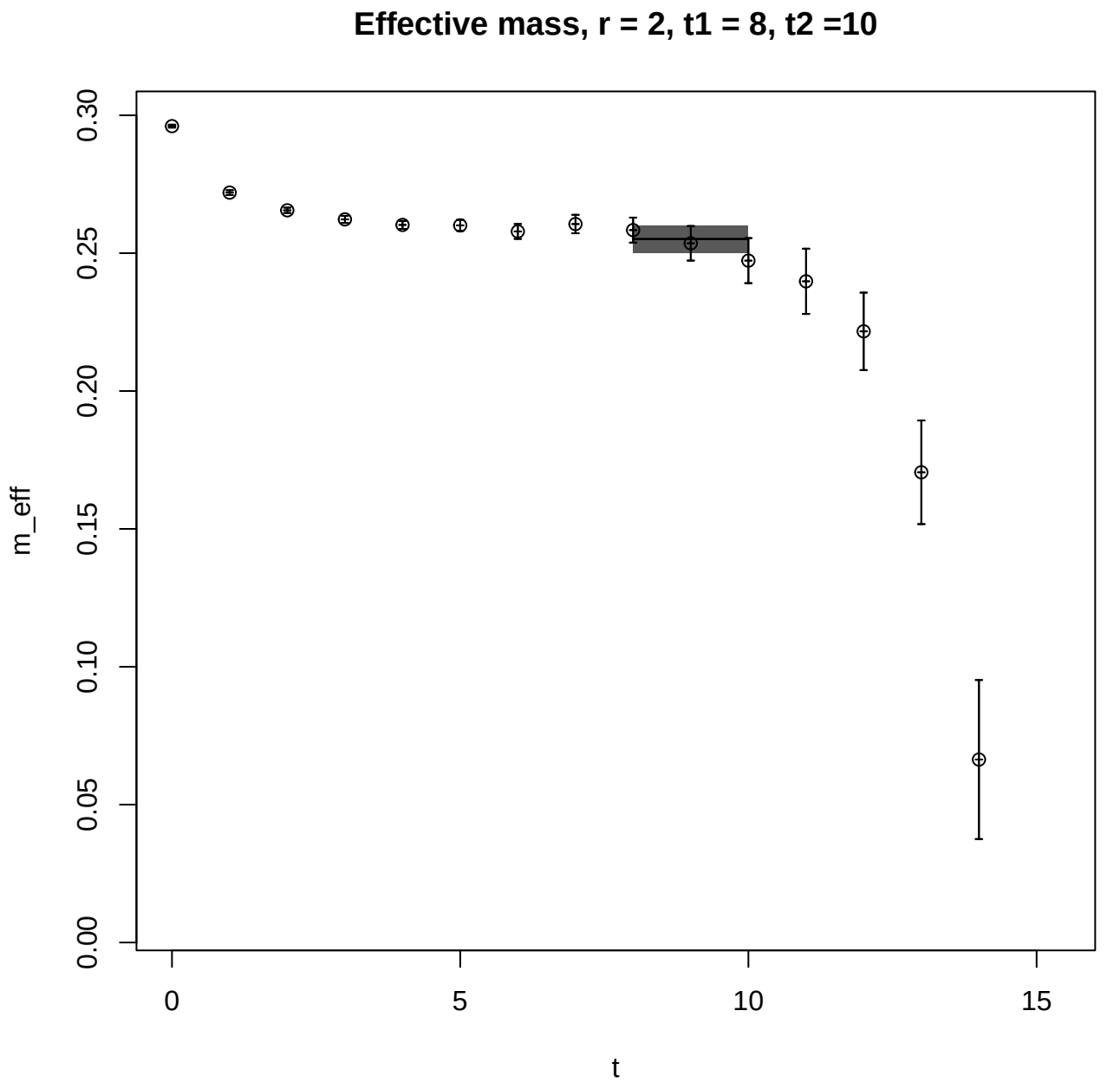
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =10
m_eff
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0.05
0.10
0.15
0.20
0.25
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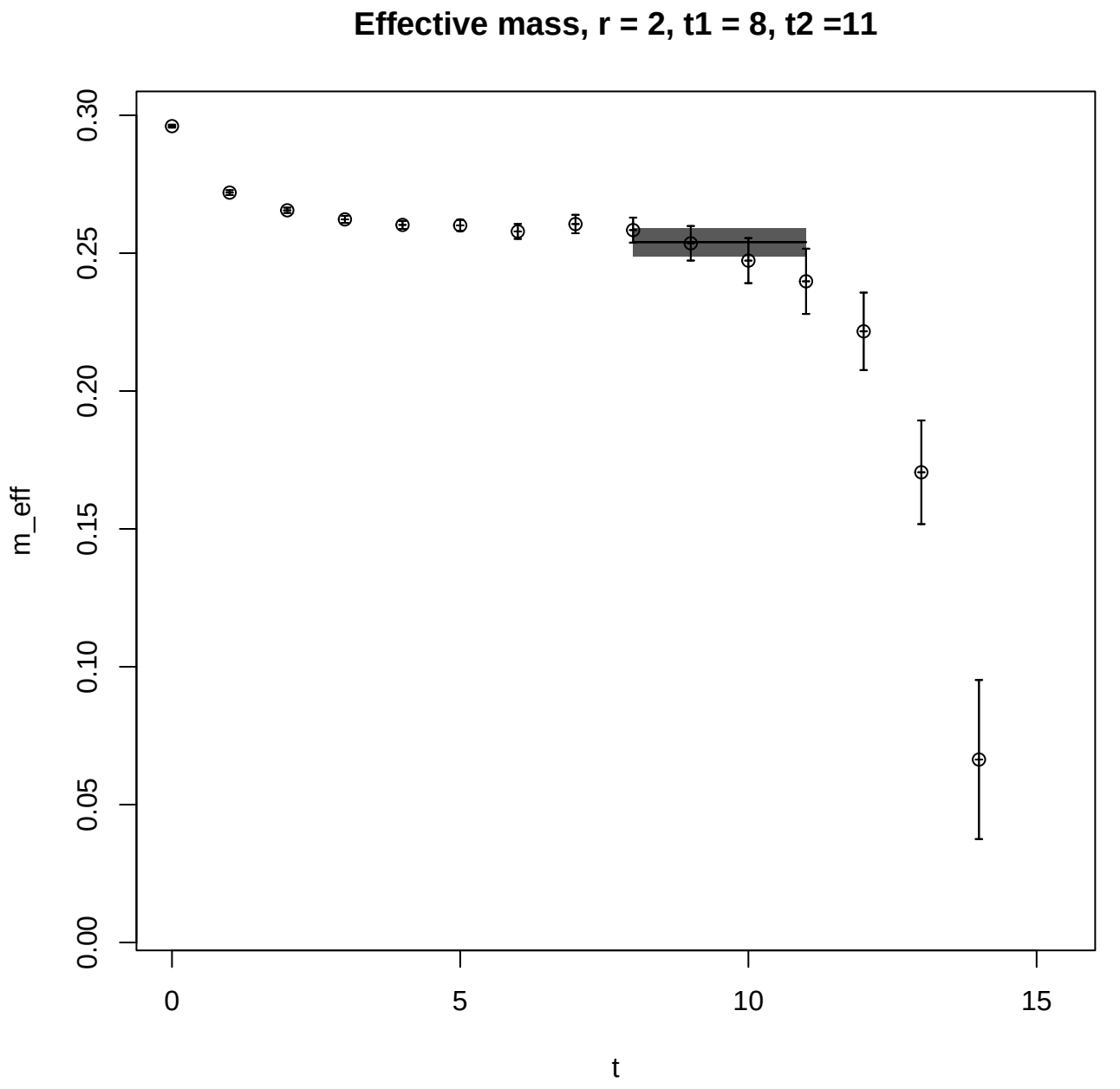
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =11
m_eff
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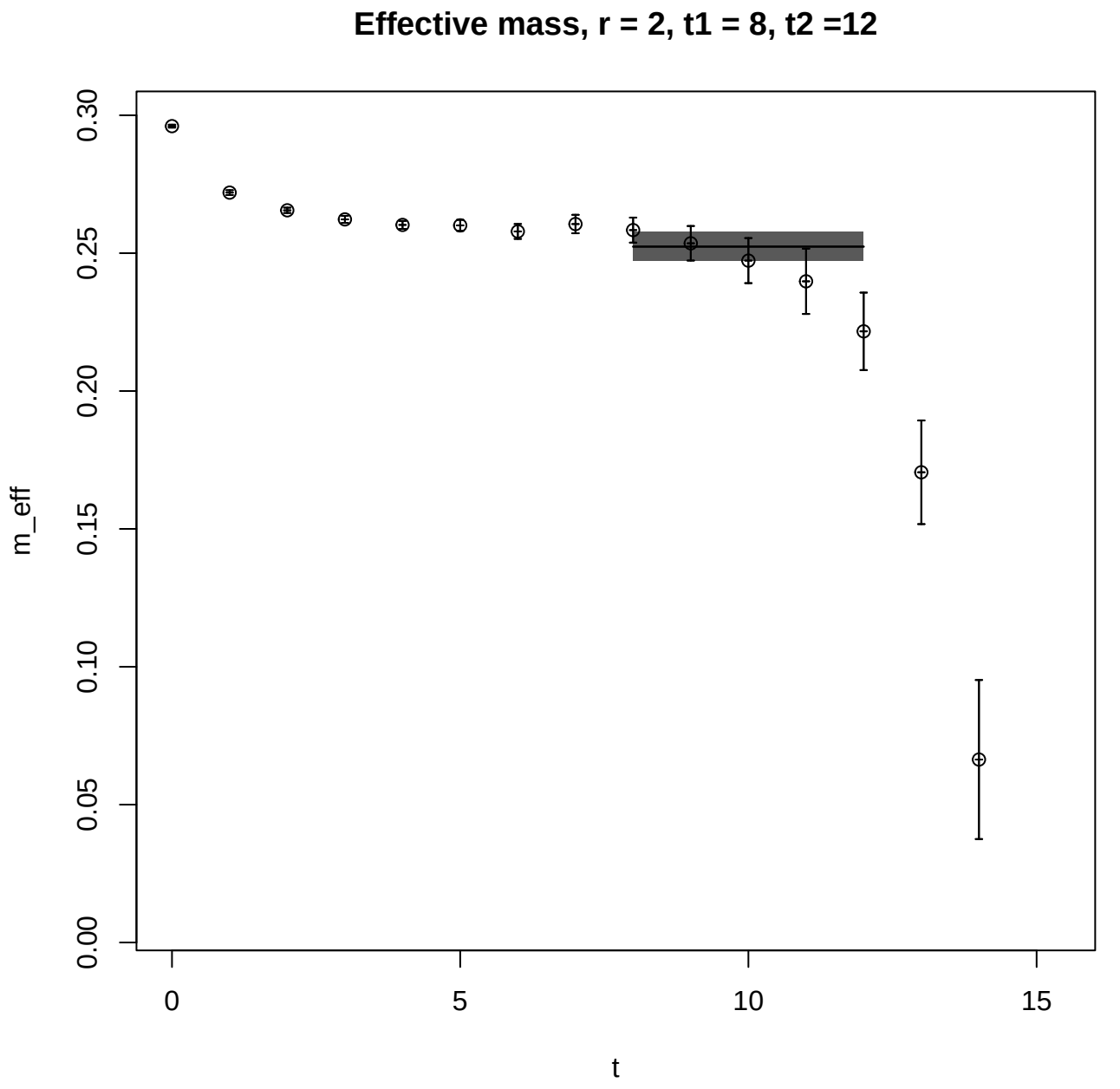
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 8, t2 =12
m_eff
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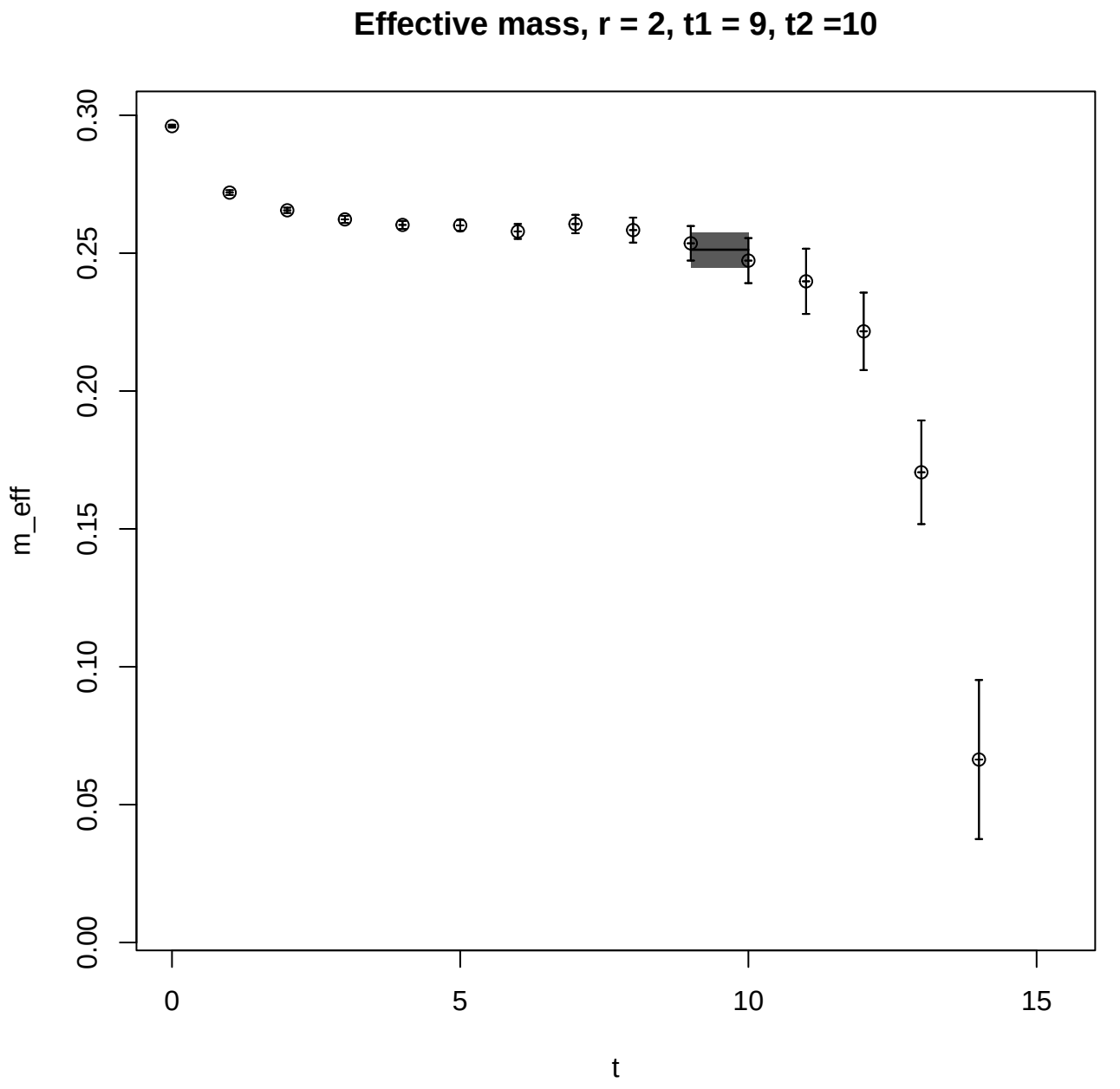
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 9, t2 =10
m_eff
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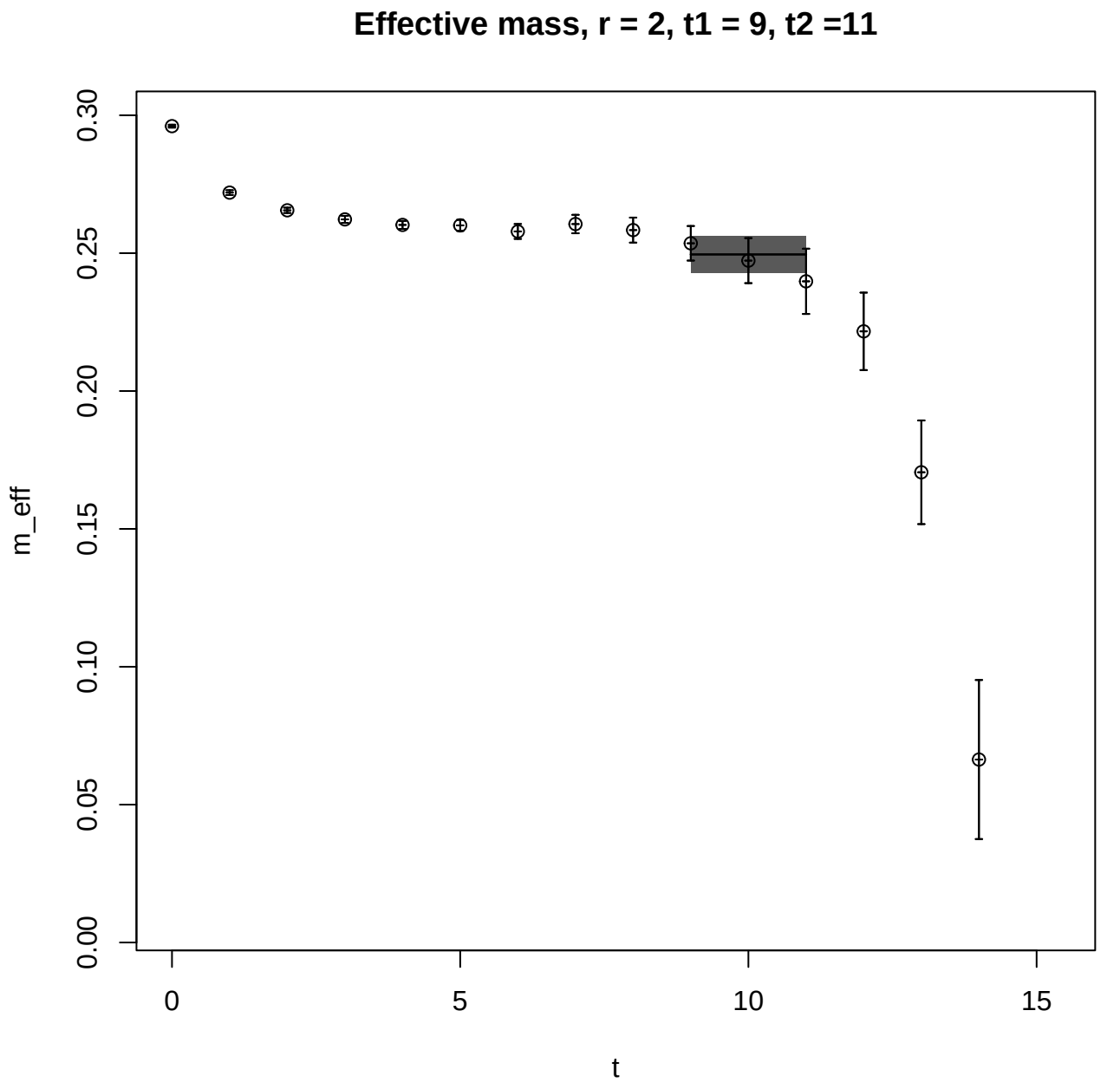
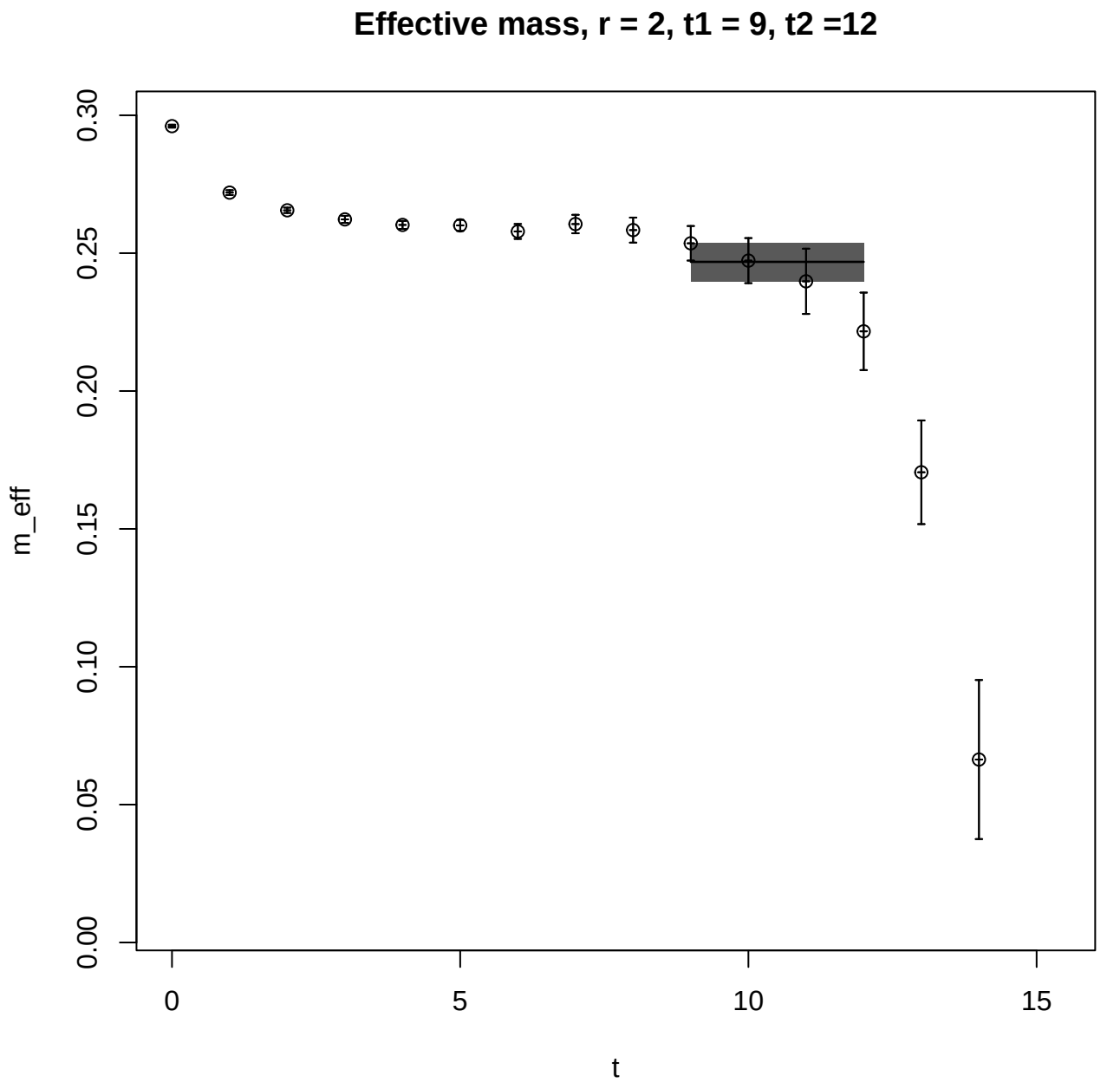
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 9, t2 =11
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 9, t2 =12
m_eff
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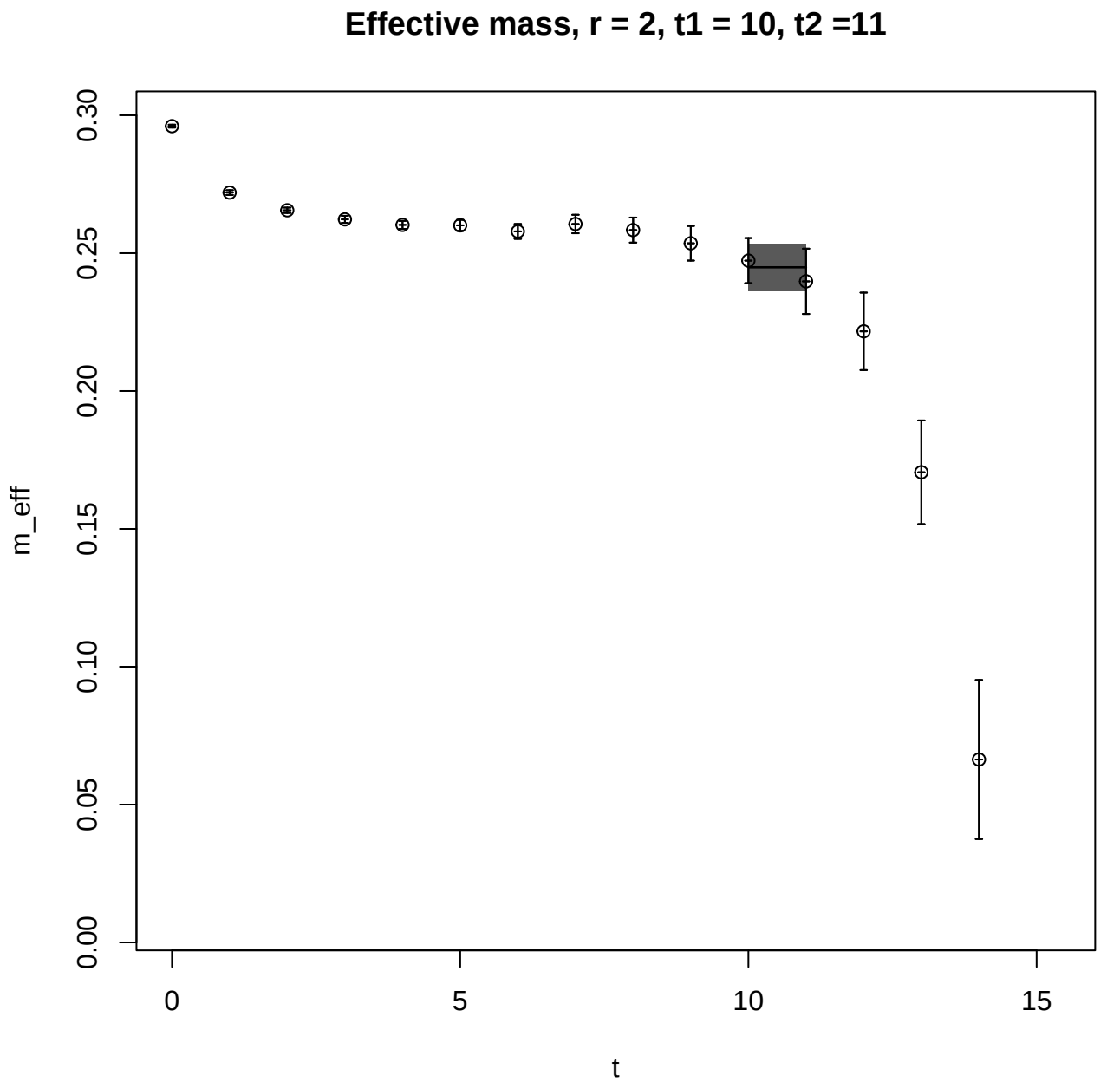
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 10, t2 =11
m_eff
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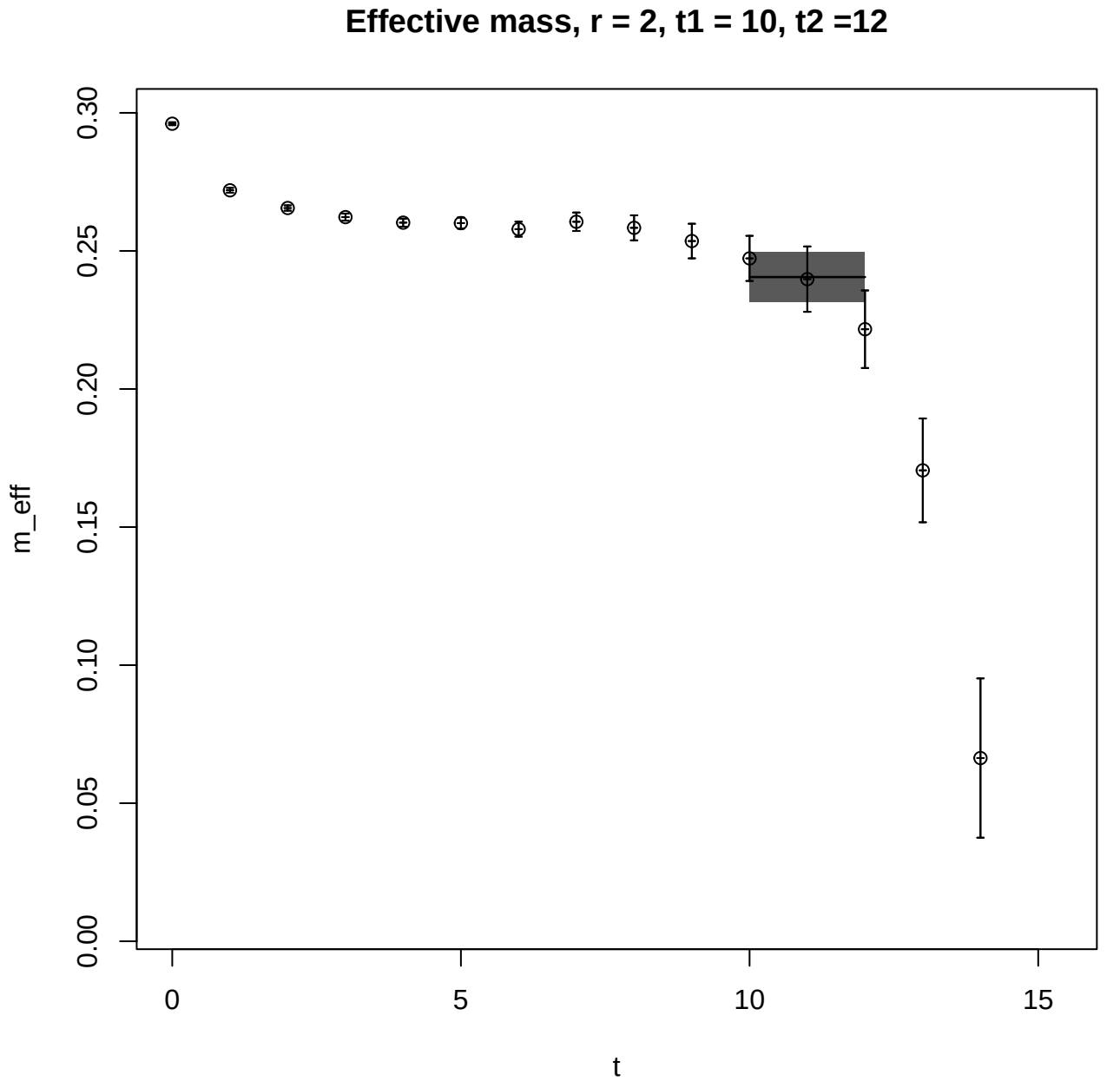
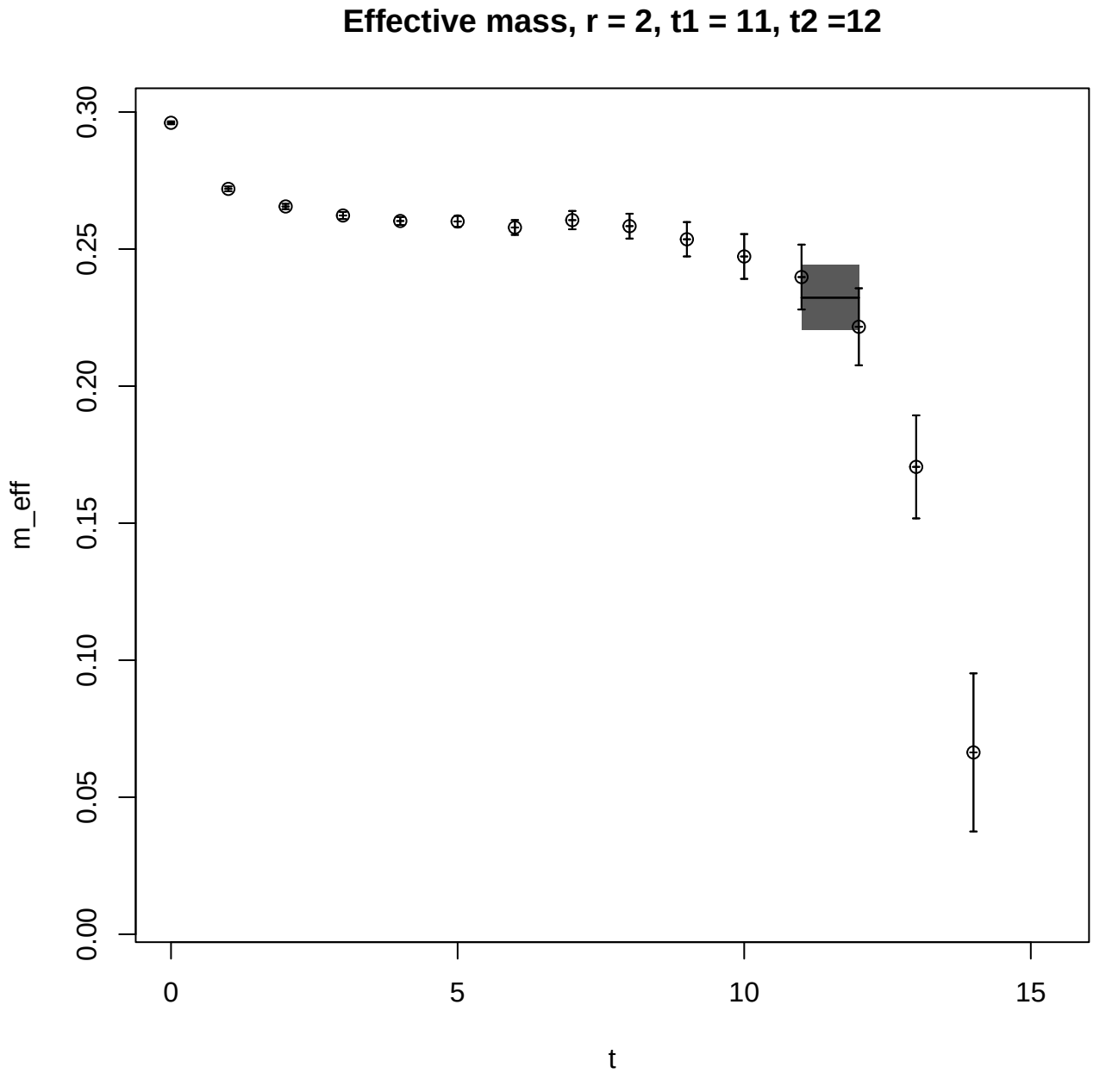
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 10, t2 =12
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 11, t2 =12
m_eff
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0.15
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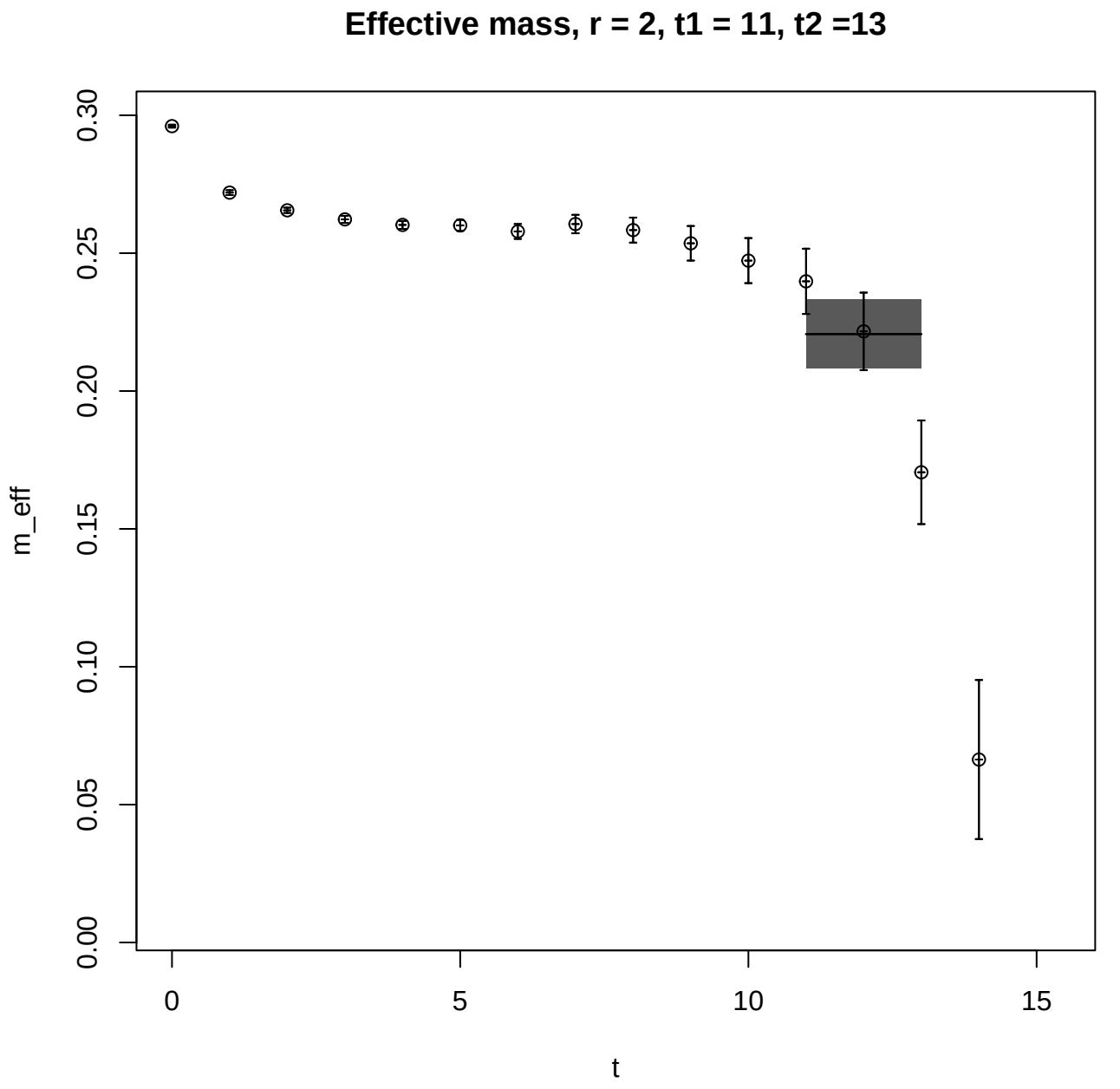
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 11, t2 =13
m_eff
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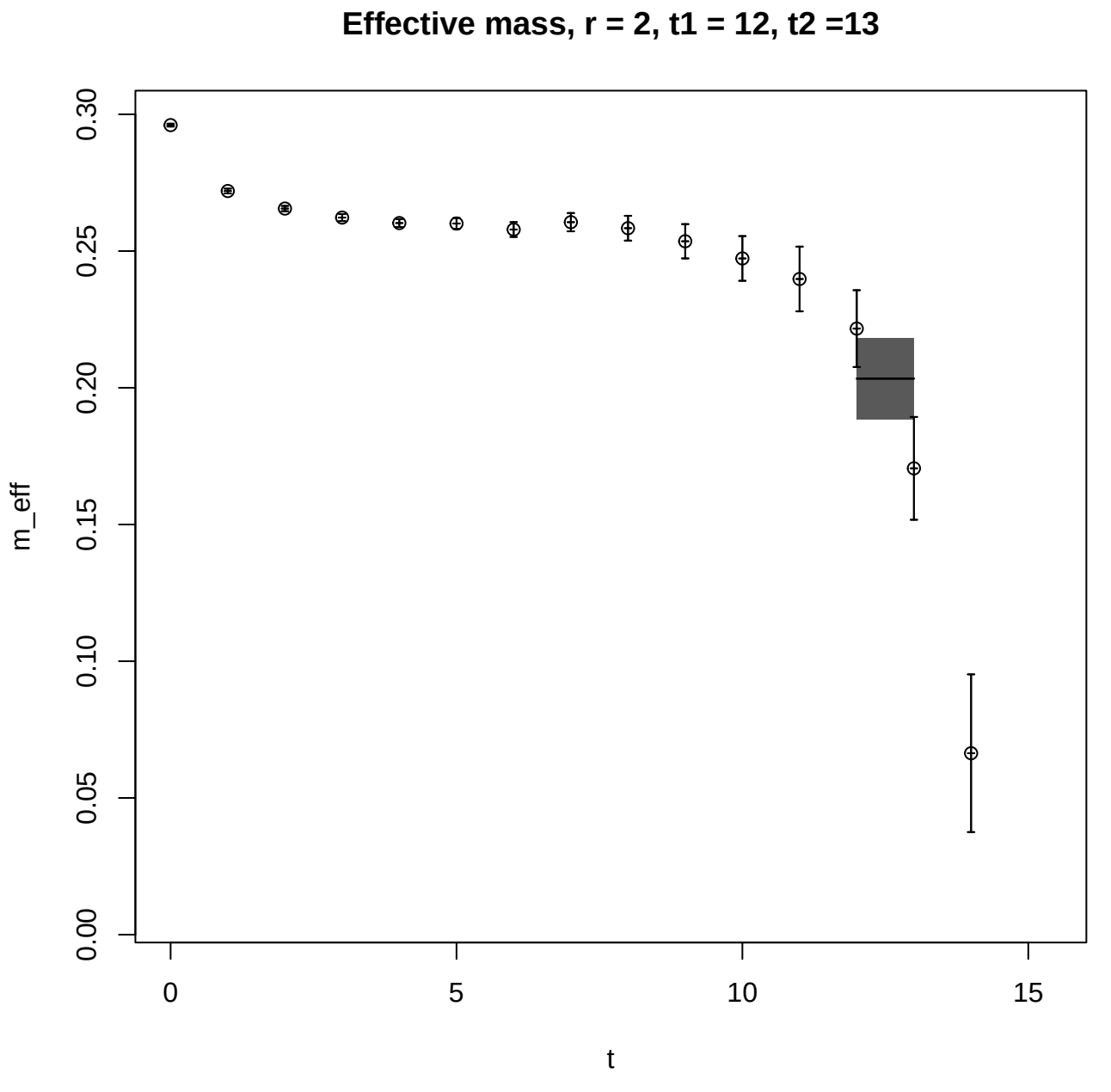
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t


Effective mass, r = 2, t1 = 12, t2 =13
m_eff
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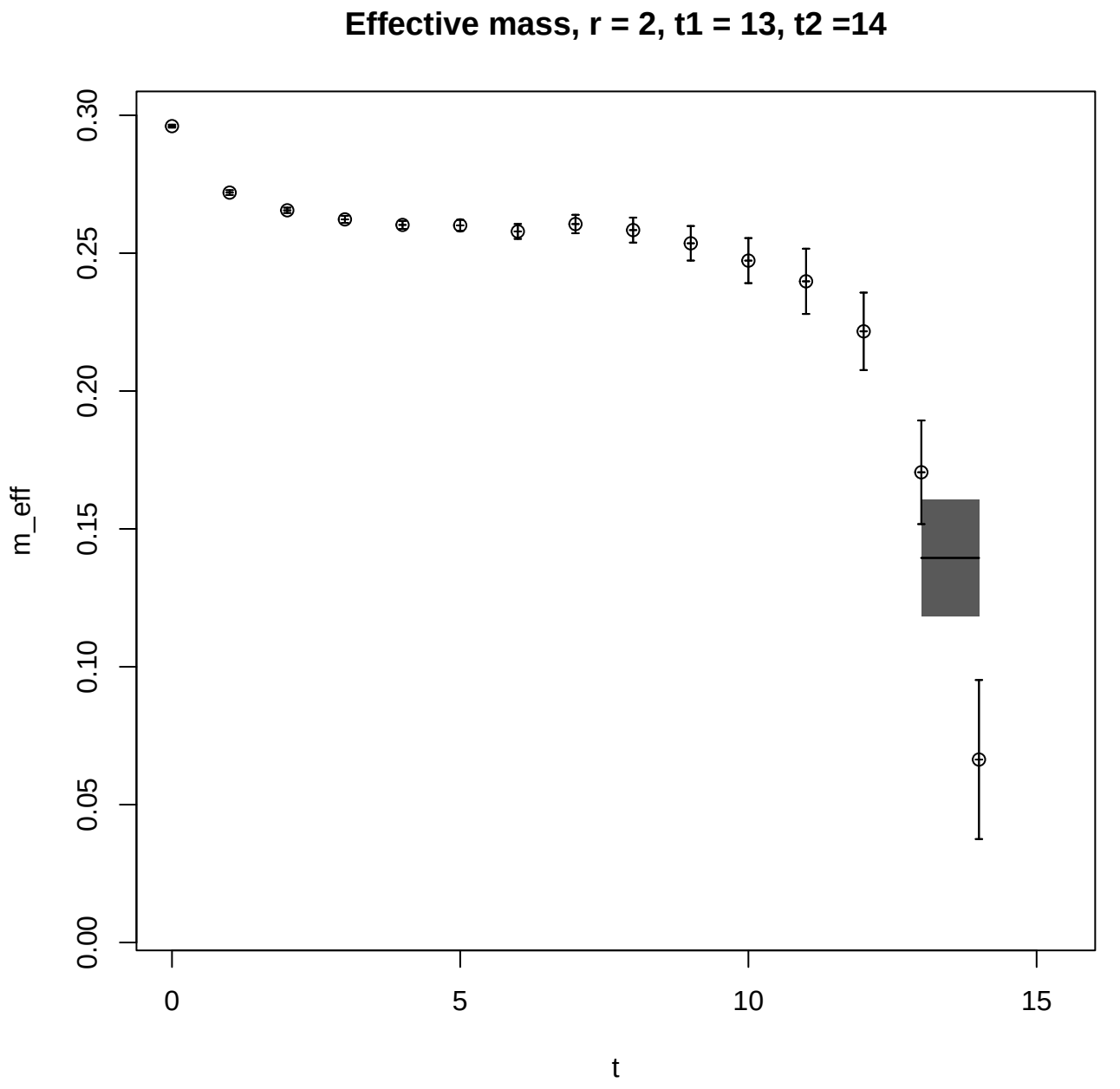
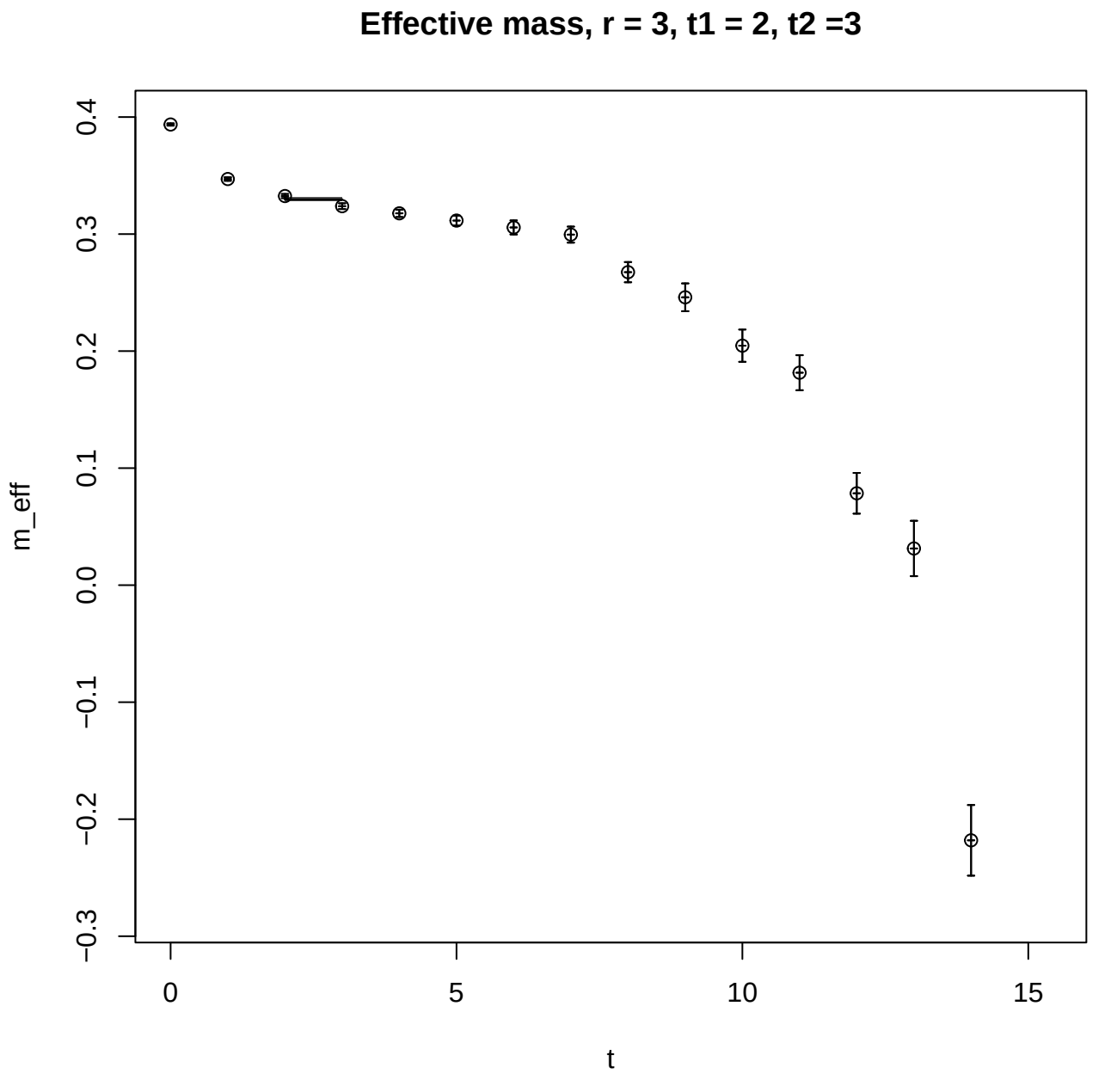
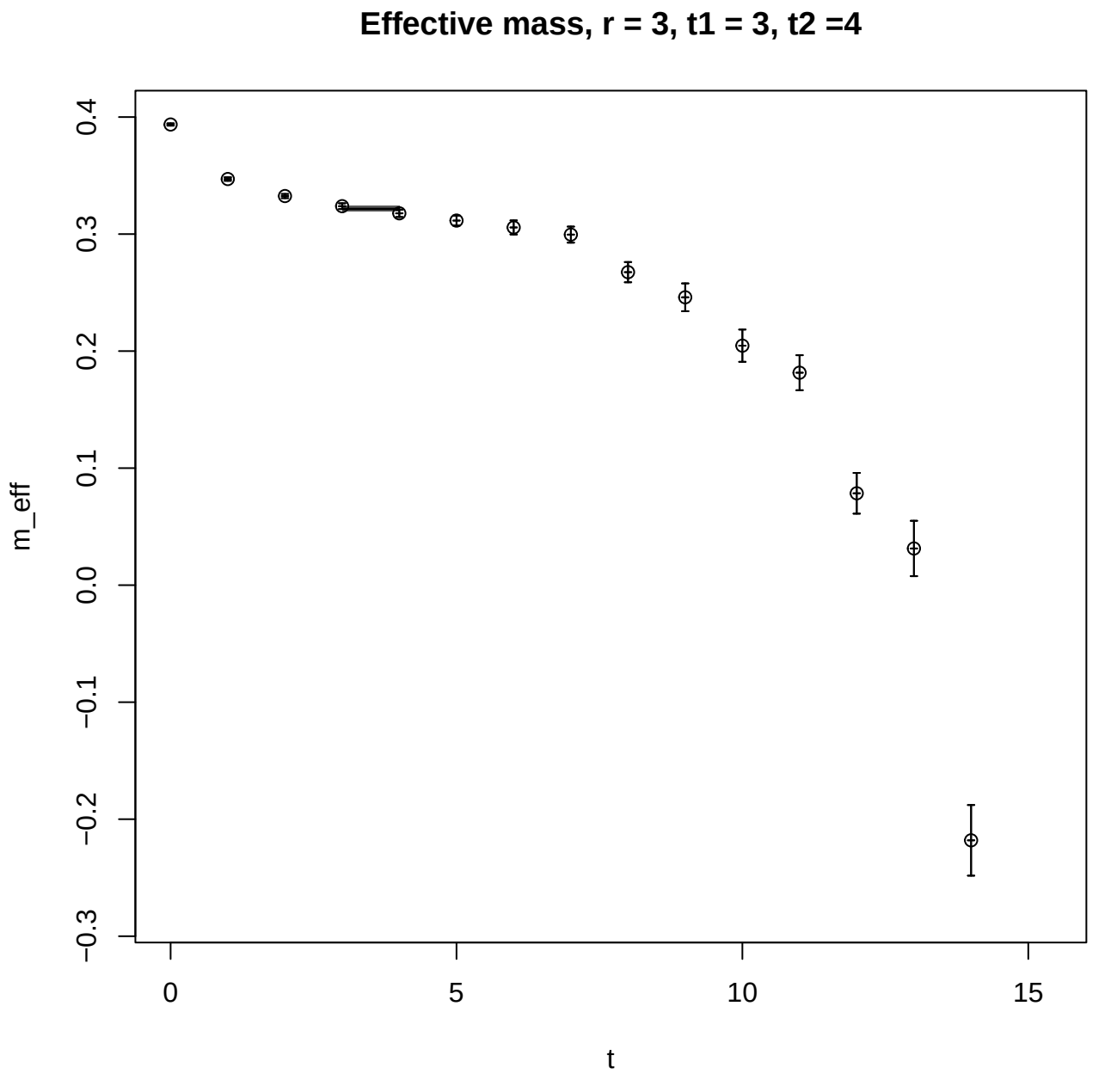
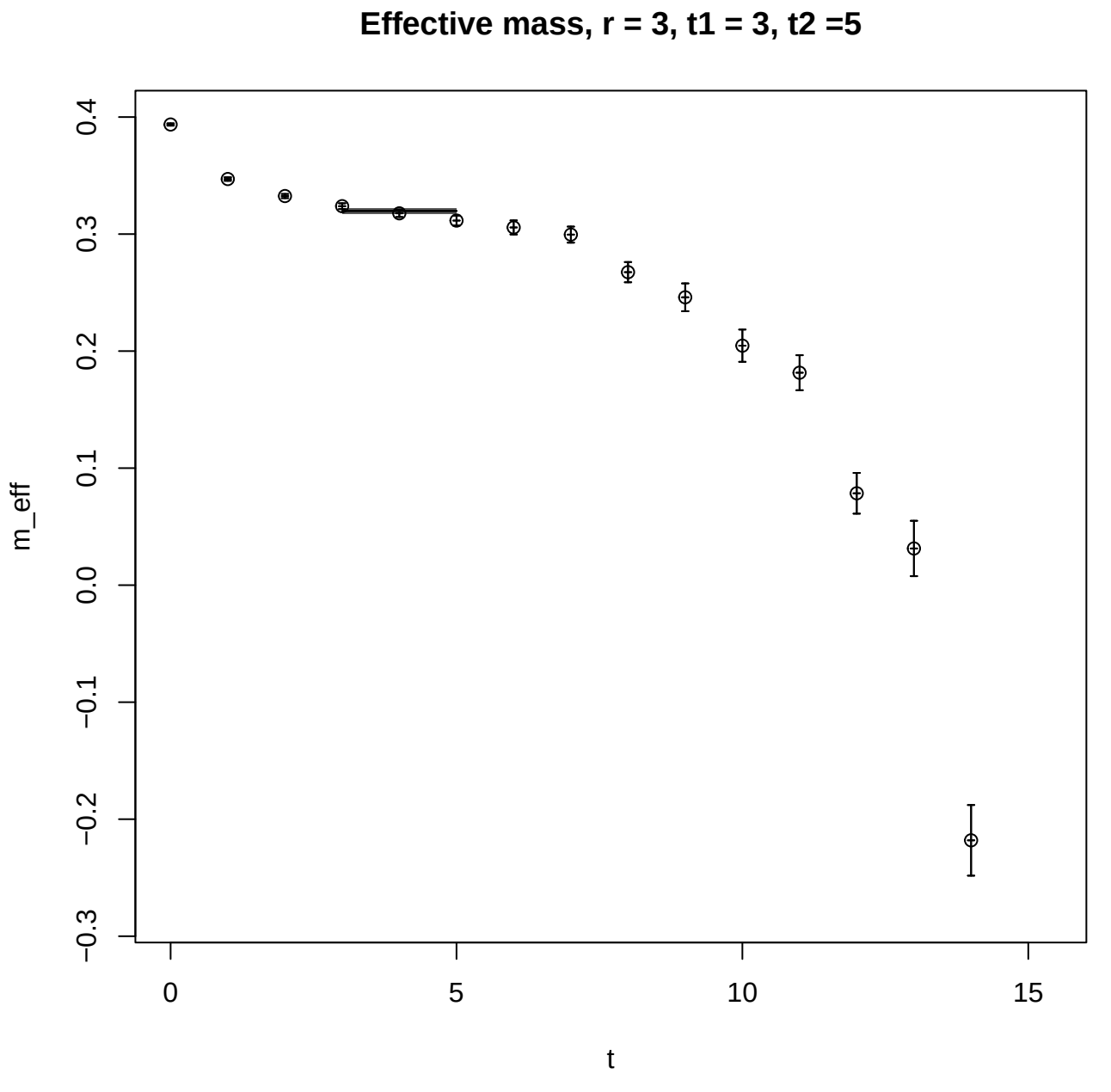
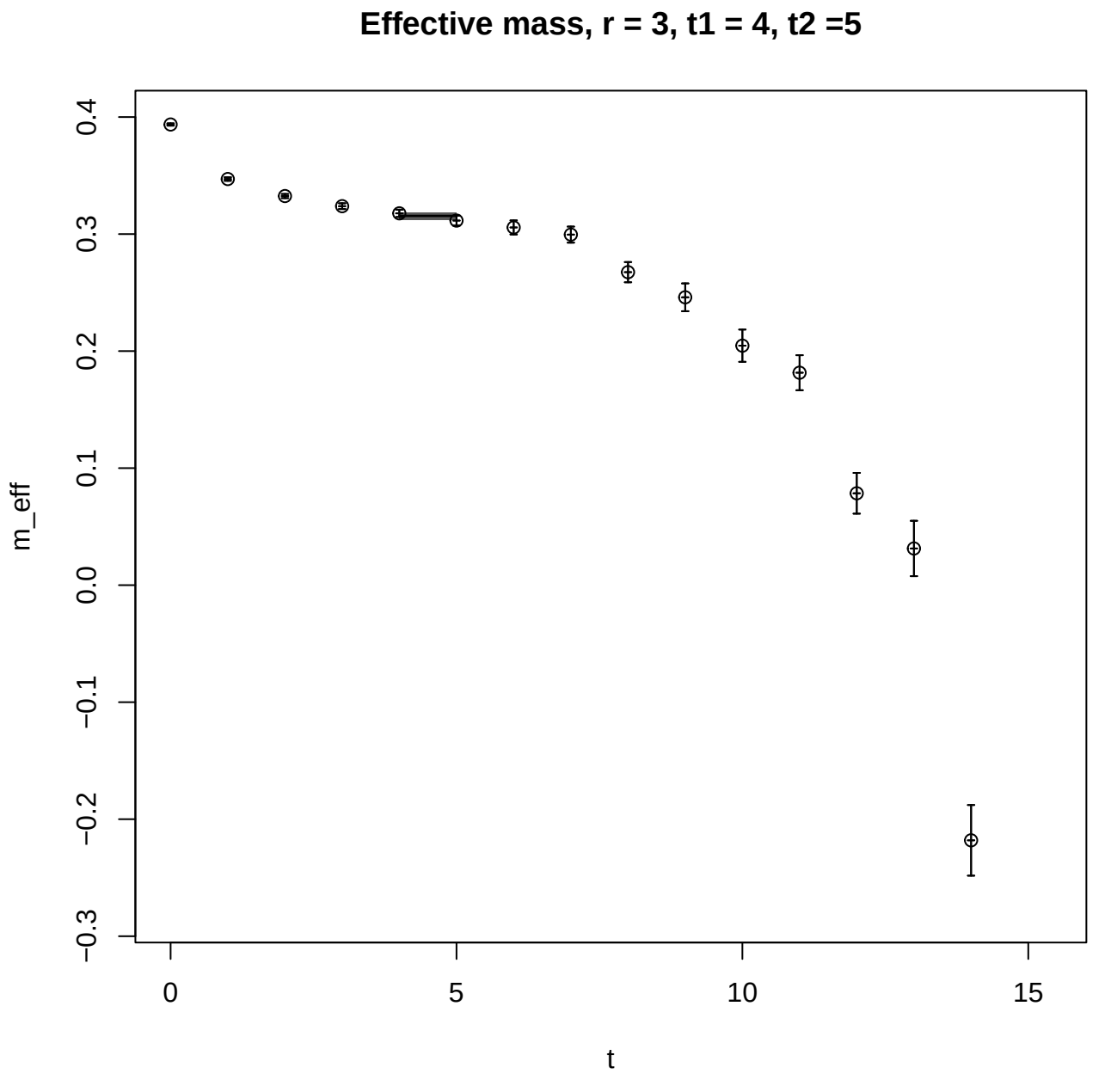
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t

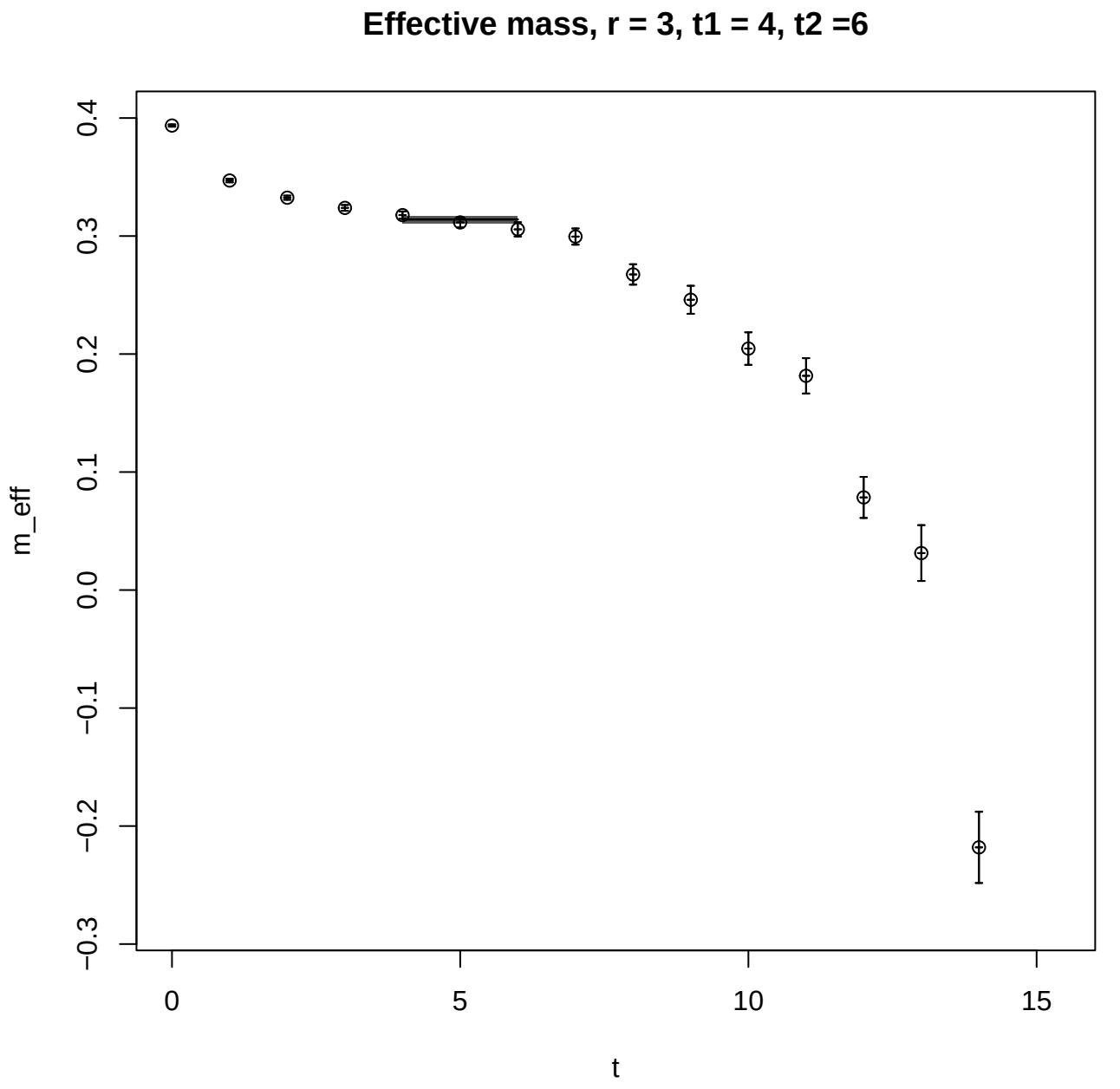

Effective mass, r = 2, t1 = 13, t2 =14
m_eff
0.00
0.05
0.10
0.15
0.20
0.25
0.30
0
5
10
15
t

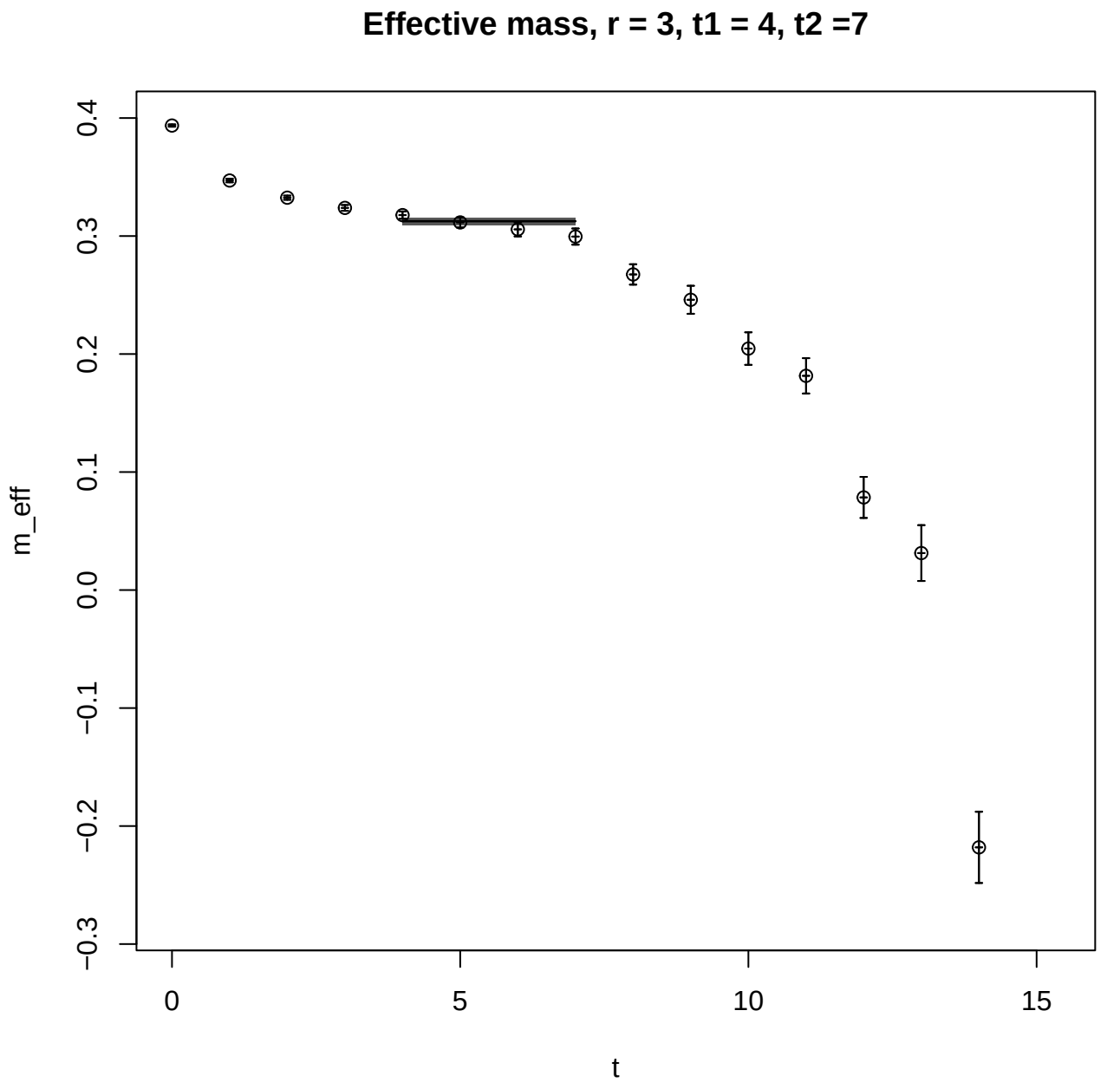

Effective mass, r = 3, t1 = 2, t2 =3
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t

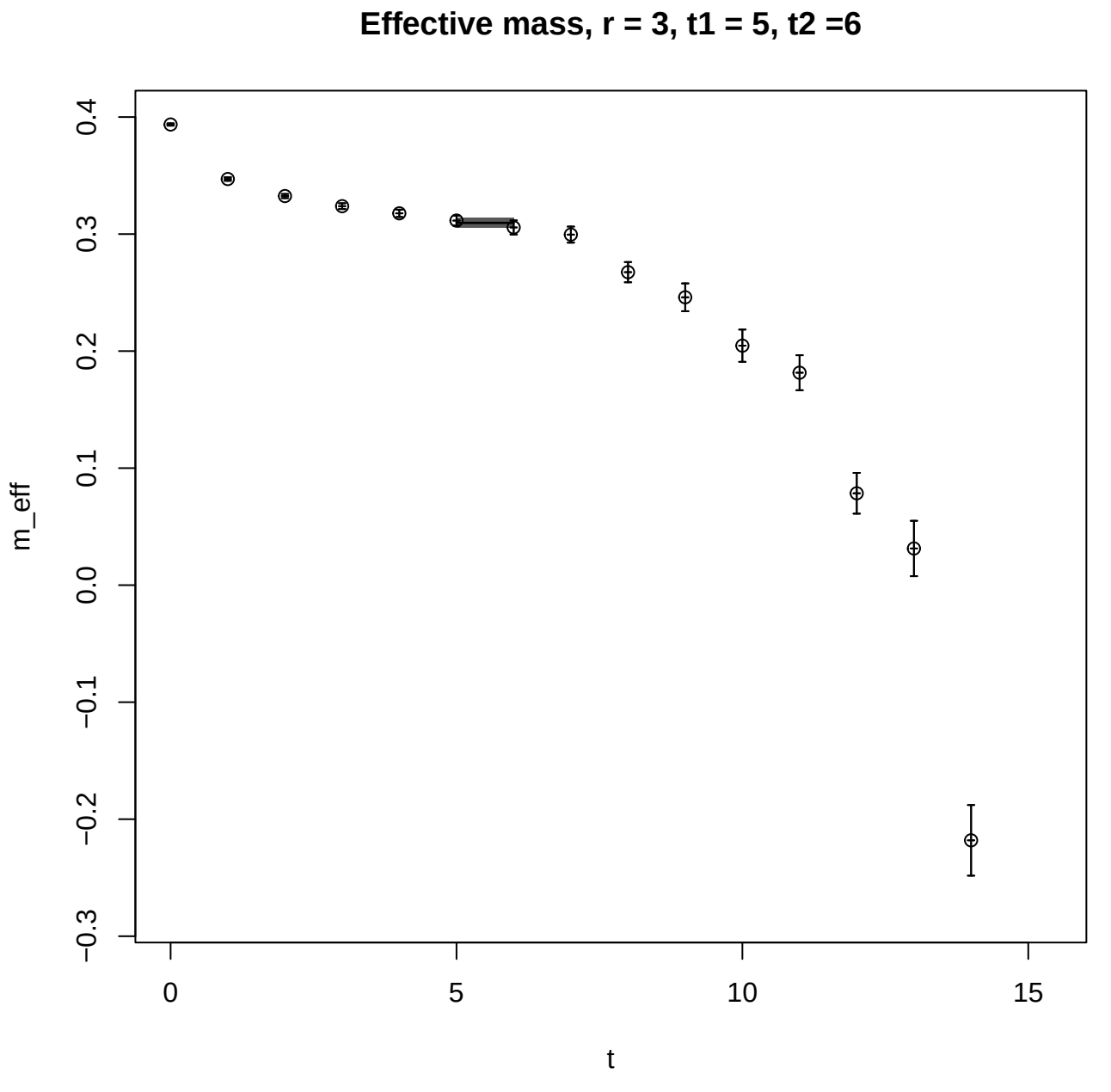

Effective mass, r = 3, t1 = 3, t2 =4
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t

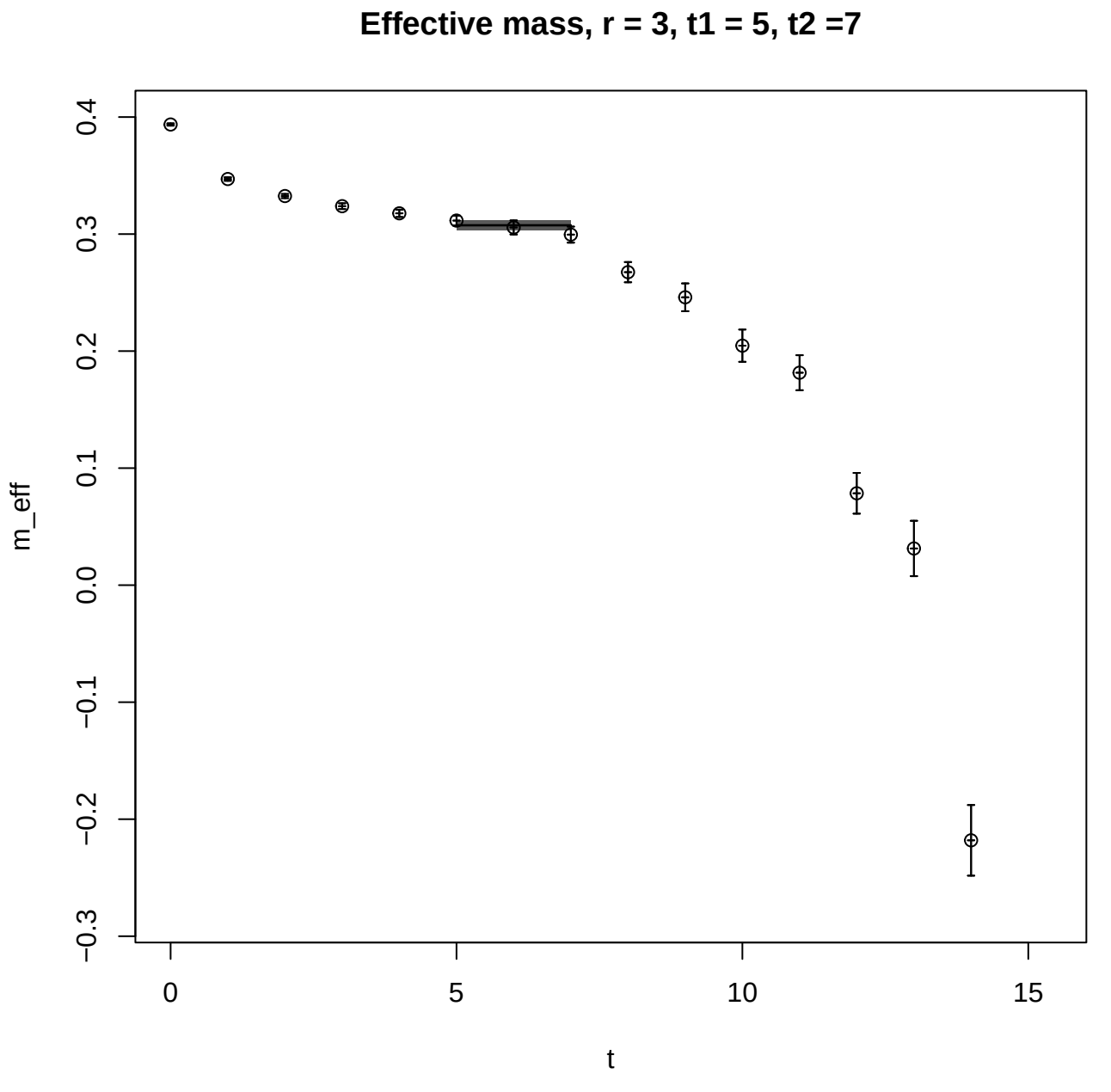

Effective mass, r = 3, t1 = 3, t2 =5
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t

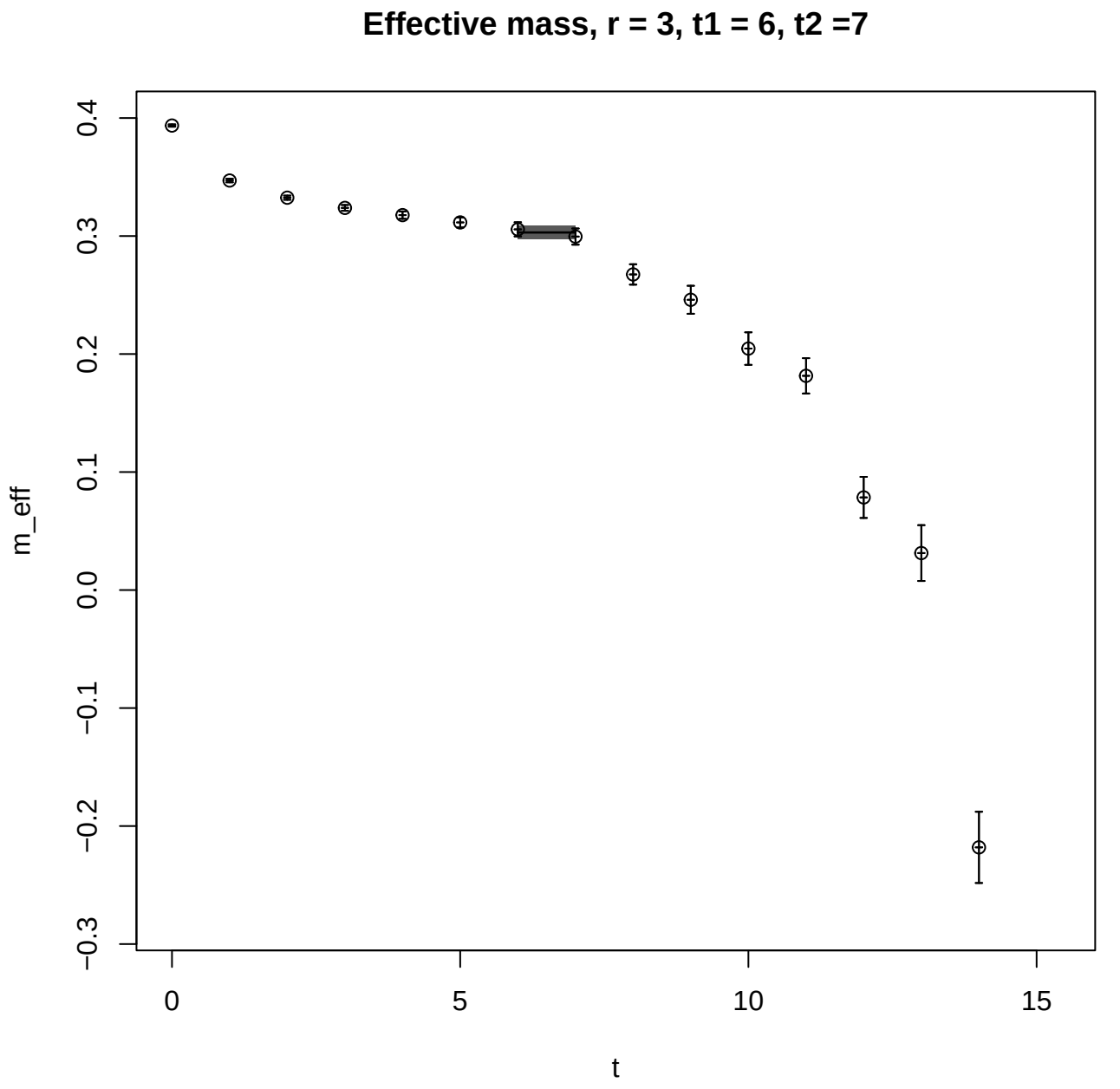

Effective mass, r = 3, t1 = 4, t2 =5
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t

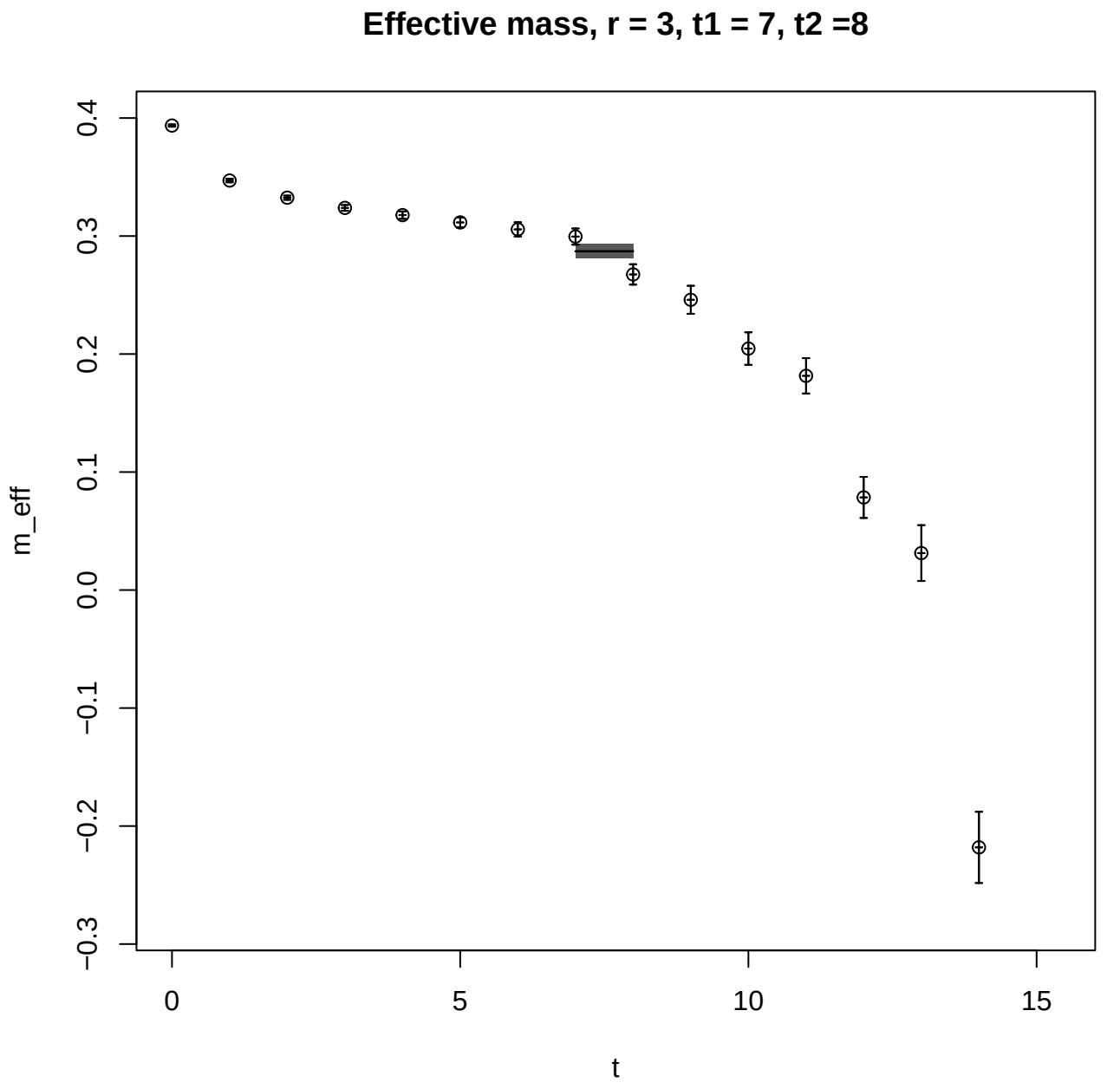

Effective mass, r = 3, t1 = 4, t2 =6
m_eff
t
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15

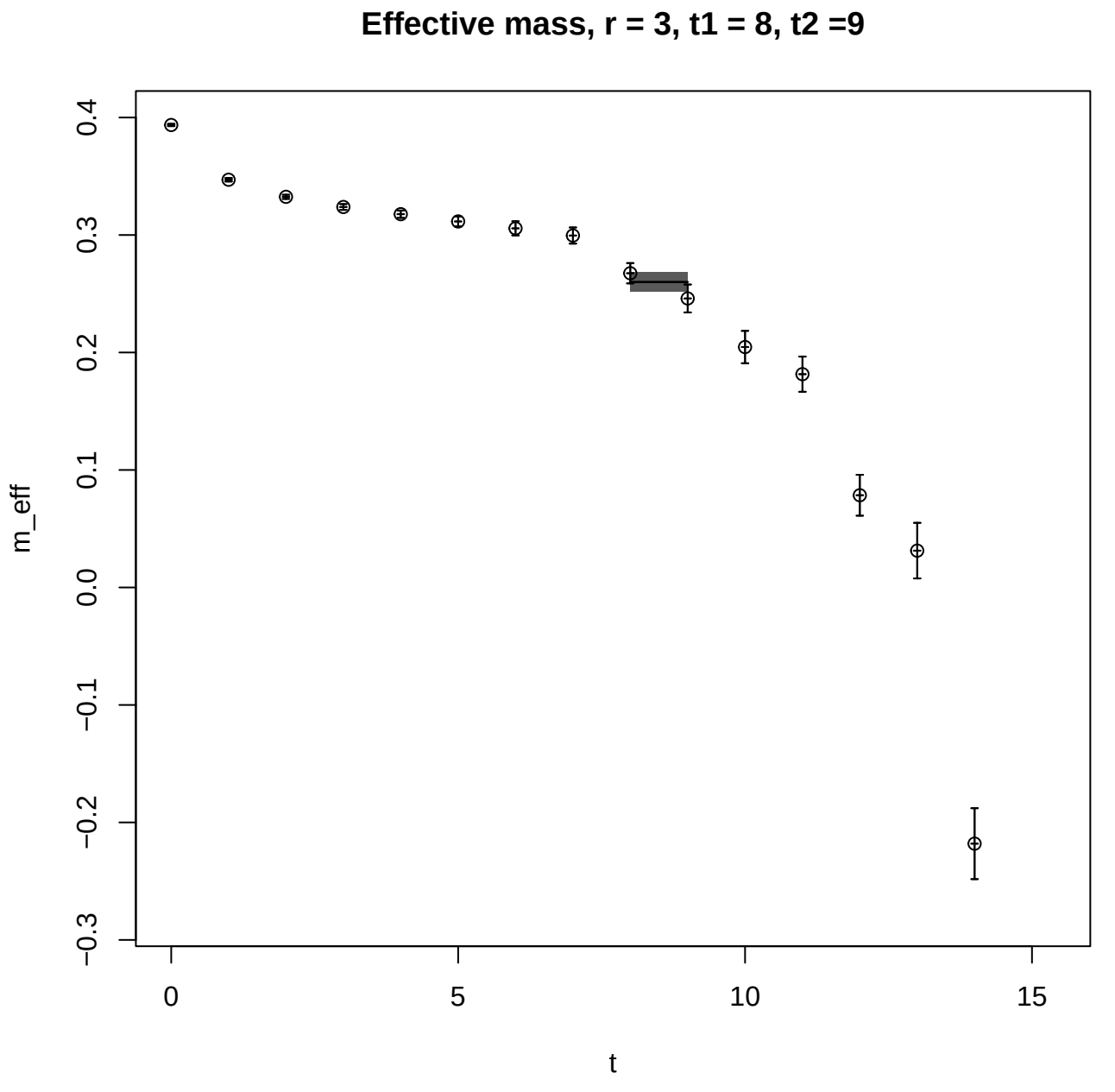

Effective mass, r = 3, t1 = 4, t2 =7
m_eff
t
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15

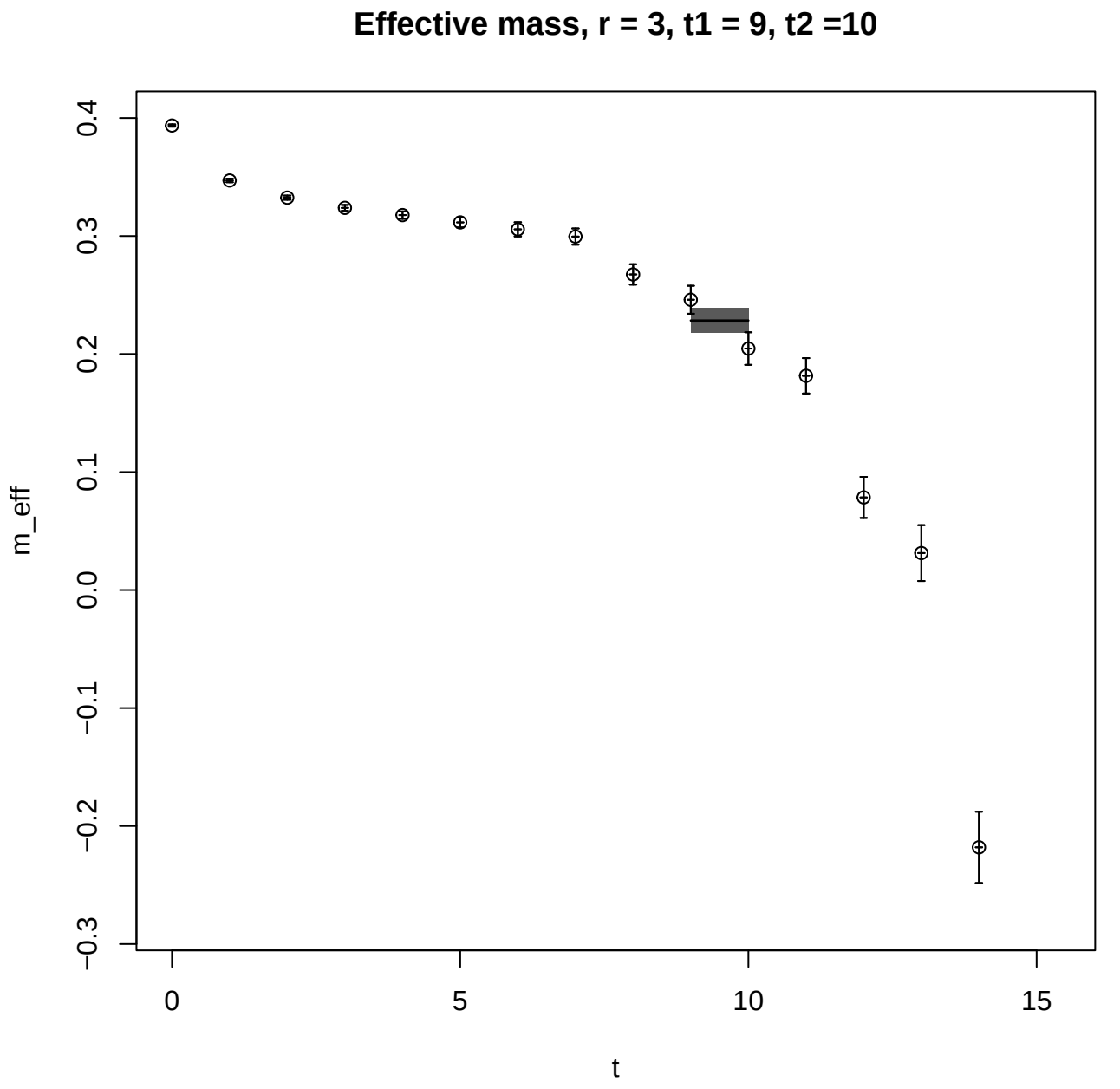

Effective mass, r = 3, t1 = 5, t2 =6
m_eff
t
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15

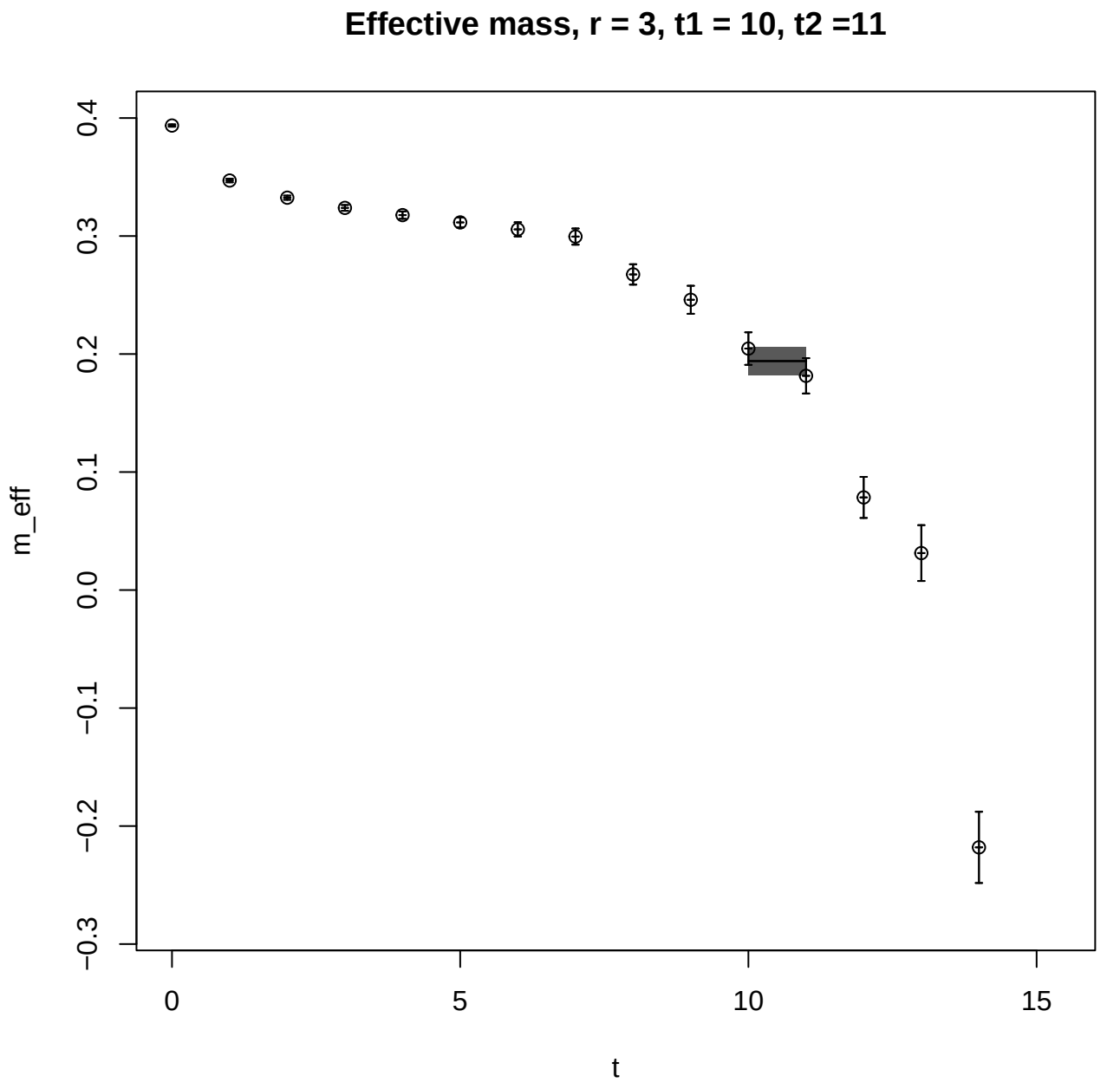

Effective mass, r = 3, t1 = 5, t2 =7
m_eff
t
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15

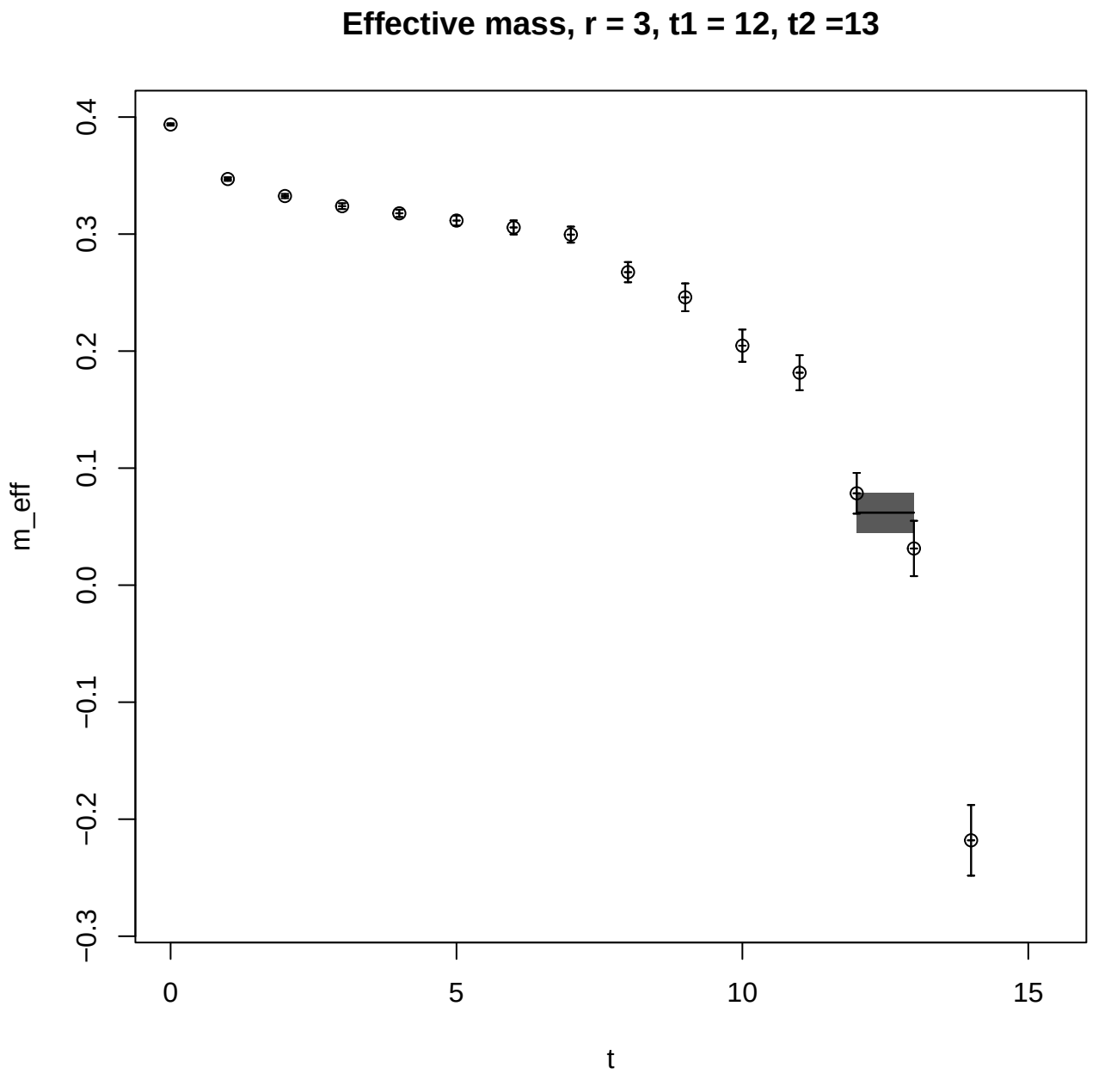

Effective mass, r = 3, t1 = 6, t2 =7
m_eff
t
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15

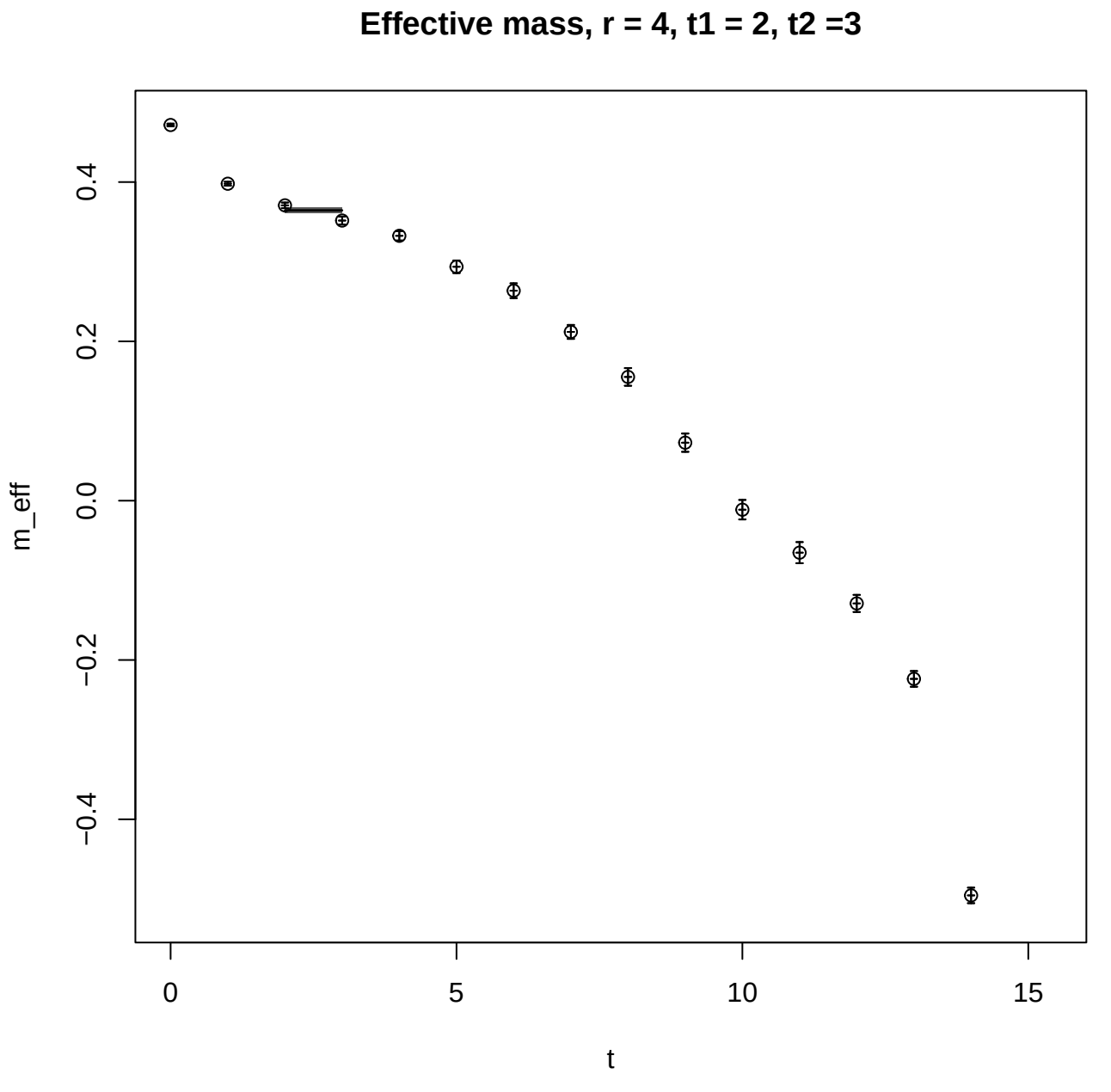

Effective mass, r = 3, t1 = 7, t2 =8
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t

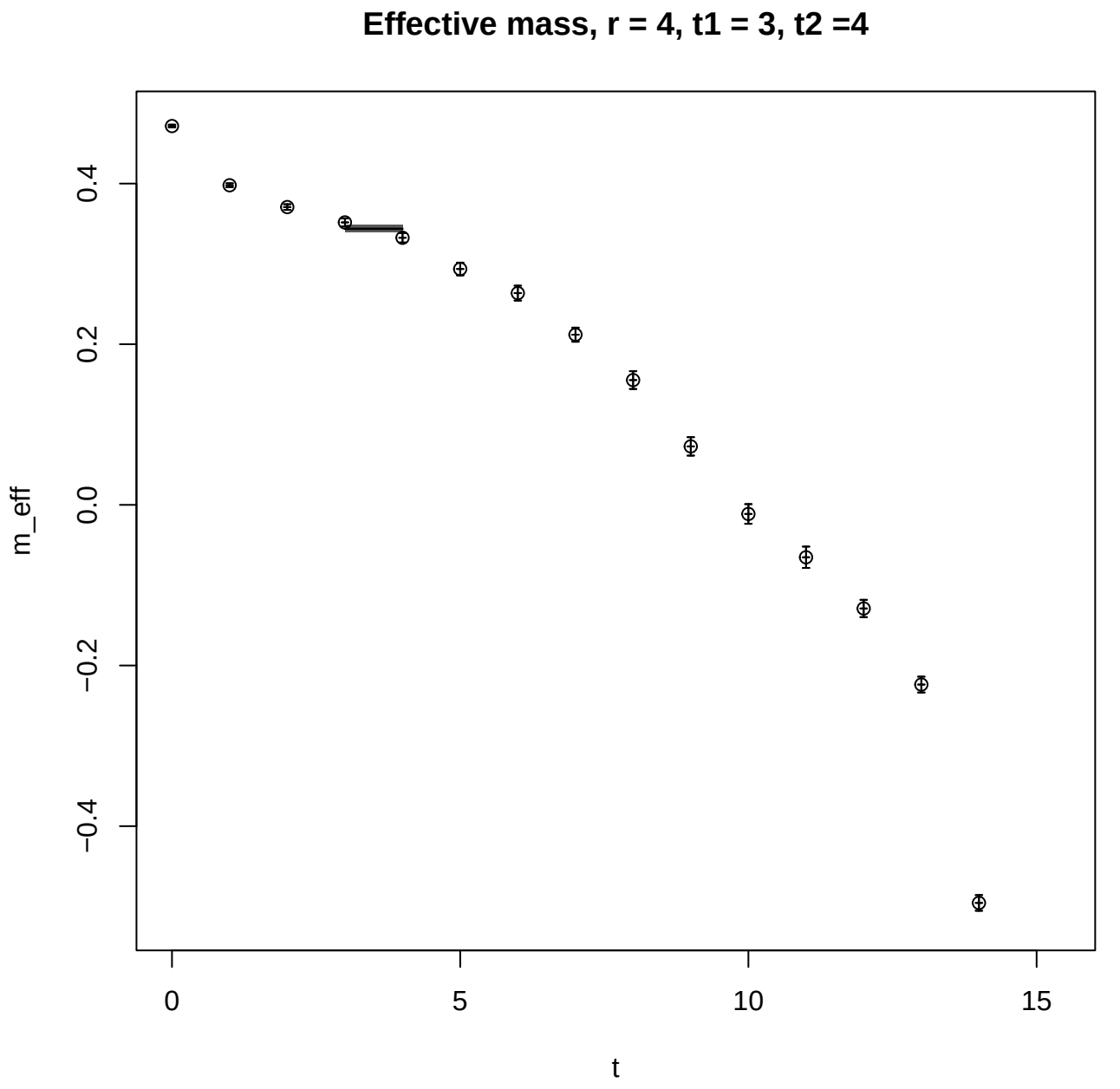

Effective mass, r = 3, t1 = 8, t2 =9
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t

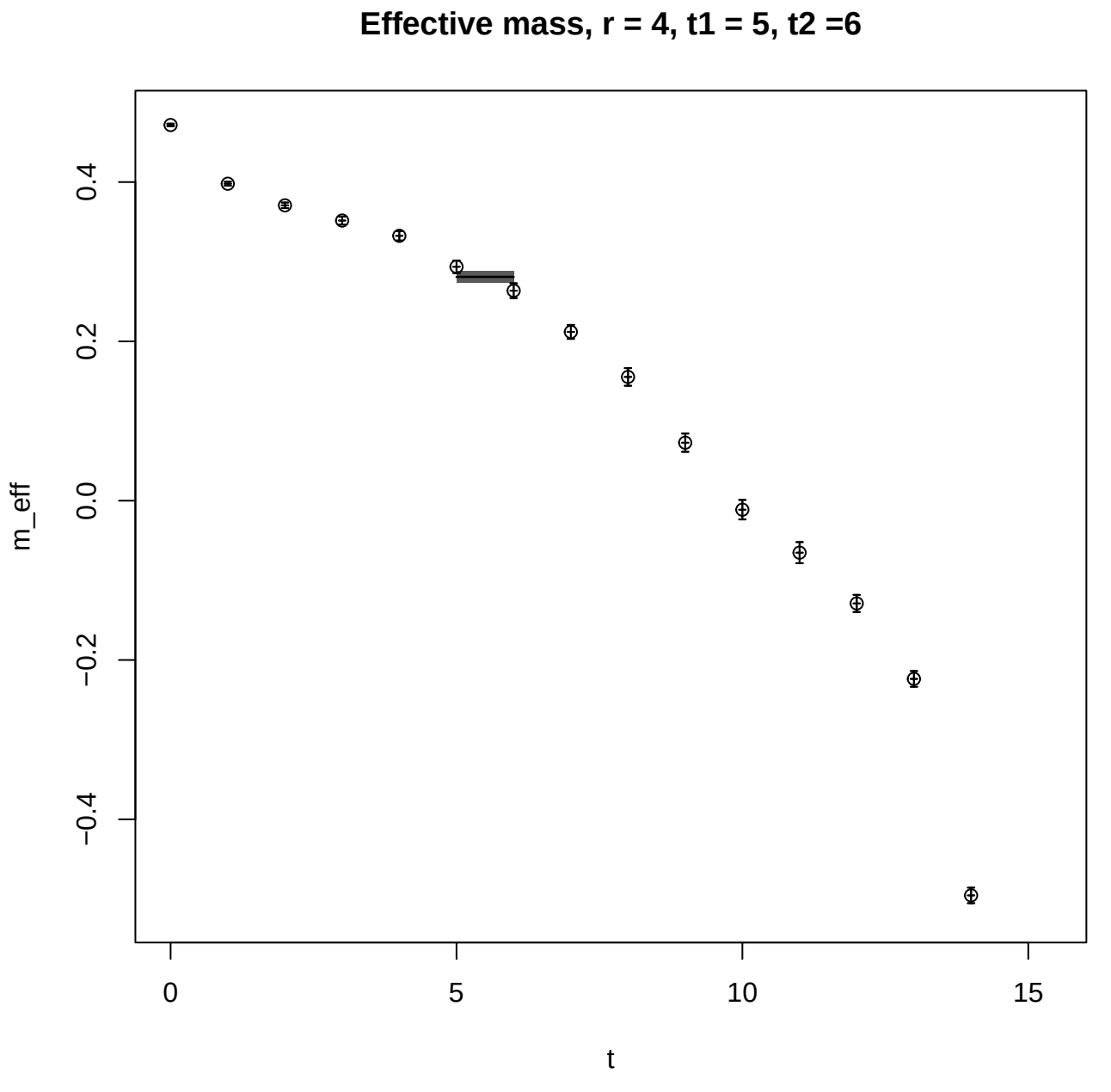

Effective mass, r = 3, t1 = 9, t2 =10
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t


Effective mass, r = 3, t1 = 10, t2 =11
m_eff
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15
t


Effective mass, r = 3, t1 = 12, t2 =13
m_eff
t
0.4
0.3
0.2
0.1
0.0
−0.1
−0.2
−0.3
0
5
10
15


Effective mass, r = 4, t1 = 2, t2 =3
m_eff
0.4
0.2
0.0
−0.2
−0.4
0
5
10
15
t


Effective mass, r = 4, t1 = 3, t2 =4
m_eff
0.4
0.2
0.0
−0.2
−0.4
0
5
10
15
t


Effective mass, r = 4, t1 = 5, t2 =6
m_eff
0.4
0.2
0.0
−0.2
−0.4
0
5
10
15
t